Gavin's Lectures on Physics Combined Edition (English)



9 Courses in Physics Sequence

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Gavin's Lectures on Physics - Optics



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1 Geometric Optics

1.1 Fermat's Principle

Fermat's principle states that the actual path taken by light between two points is the one for which the optical path length is stationary:

$$0 = \delta \int_S \mathrm{d}t = \frac{1}{c}\,\delta \int_S n\,\mathrm{d}s = \frac{1}{c}\,\delta L, \tag{1}$$

where the optical path length is defined as

$$L = \int_S n\,\mathrm{d}s. \tag{2}$$

1.2 Snell's Law

Consider the geometry with distances

$$l_1 = \frac{1}{\cos\theta_1}, \qquad l_2 = \frac{1}{\cos\theta_2}, \tag{1}$$

and

$$L = \tan\theta_1 + \tan\theta_2. \tag{2}$$

The travel time is

$$t = \frac{1}{c}\left(l_1 n_1 + l_2 n_2\right). \tag{3}$$

From (2) we have

$$\delta\theta_2 = -\frac{\cos^2\theta_2}{\cos^2\theta_1}\,\delta\theta_1. \tag{3}$$

Applying Fermat's principle:

$$\begin{aligned} 0 = \delta \int_S \mathrm{d}t &= \frac{1}{c}\left(n_1\delta l_1 + n_2\delta l_2\right) \\ &= \frac{1}{c}\left(n_1\delta\frac{1}{\cos\theta_1} + n_2\delta\frac{1}{\cos\theta_2}\right) \\ &= \frac{1}{c}\left(n_1\frac{\sin\theta_1}{\cos^2\theta_1}\delta\theta_1 + n_2\frac{\sin\theta_2}{\cos^2\theta_2}\delta\theta_2\right) \\ &= \frac{1}{c}\left(n_1\sin\theta_1 - n_2\sin\theta_2\right)\cdot\frac{1}{\cos^2\theta_1}\cdot\delta\theta_1. \end{aligned} \tag{4}$$

Thus we obtain Snell's law:

$$n_1\sin\theta_1 = n_2\sin\theta_2. \tag{5}$$

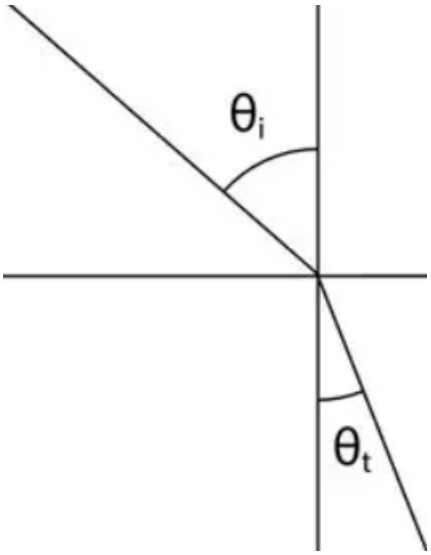


Figure 1: Reflection and refraction of a light ray at a dielectric interface, illustrating Snell's law and the law of reflection.

2 Interference

2.1 Types of Interference

Wave-front division : Splits one wavefront into coherent sub-sources that interfere in space.

Examples: Young' s double slit, Fresnel biprism, Lloyd' s mirror.

Amplitude division : Splits beam amplitude via film reflections; interference comes from optical path difference.

Examples: Equal-inclination interference, wedge film interference, Newton' s rings.

2.2 Intensity of Interference

The real wave is expressed as

$$u = u_0 \cos\left(\omega t - \mathbf{k} \cdot \mathbf{r}\right). \tag{6}$$

Using complex representation:

$$u = u_0 e^{i(\omega t - \mathbf{k}\cdot\mathbf{r})}, \qquad u_0 = |u_0| e^{i\phi}, \tag{7}$$

and the conjugate

$$u^* = u_0^* e^{-i(\omega t - \mathbf{k}\cdot\mathbf{r})}, \qquad u_0^* = |u_0| e^{-i\phi}. \tag{8}$$

The intensity (irradiance) is proportional to

$$I \propto \langle u u^* \rangle. \tag{9}$$

2.3 Superposition of Waves

The superposition at point P is

$$u_P = u_A + u_B, \tag{10}$$

with

$$u_A = u_{A0} e^{i(\omega t - k_A r_A)}, \qquad u_B = u_{B0} e^{i(\omega t - k_B r_B)}. \tag{11}$$

The intensity expansion gives

$$I_P \propto \langle u_P u_P^* \rangle = \langle u_A u_A^* \rangle + \langle u_B u_B^* \rangle + \langle u_A u_B^* + u_A^* u_B \rangle. \tag{12}$$

2.3.1 Incoherent case

If $k_A \neq k_B$, $\omega_A \neq \omega_B$ (unrelated sources), then

$$\langle u_A u_B^* \rangle = \langle u_A \rangle \langle u_B^* \rangle = 0, \tag{13}$$

hence

$$I_P = I_A + I_B. \tag{14}$$

2.3.2 Coherent case

If $k_A = k_B$, $\omega_A = \omega_B$, then

$$I_P = I_A + I_B + 2\sqrt{I_A I_B} \cos\left(\phi_A - \phi_B\right). \tag{15}$$

2.4 Interference between Two Coherent Point Sources

The field at point P is

$$u_P = \frac{u_{0A}}{r_A} e^{i(\omega t - k r_A)} + \frac{u_{0B}}{r_B} e^{i(\omega t - k r_B)}. \tag{16}$$

In the far-field regime ($r_A, r_B \gg \lambda$), we use the approximations

$$r_A \approx r - \frac{x \sin\theta}{2}, \qquad r_B \approx r + \frac{x \sin\theta}{2}. \tag{17}$$

Assuming equal amplitudes $u_{0A} = u_{0B} = u_0$, we get

$$\begin{aligned} u_P &= \frac{u_0}{r} e^{i(\omega t - kr)} \left[e^{ik\frac{x\sin\theta}{2}} + e^{-ik\frac{x\sin\theta}{2}} \right] \\ &= 2\frac{u_0}{r} e^{i(\omega t - kr)} \cos\left(\frac{\delta}{2}\right), \end{aligned} \tag{18}$$

with phase difference

$$\delta = kx \sin\theta. \tag{19}$$

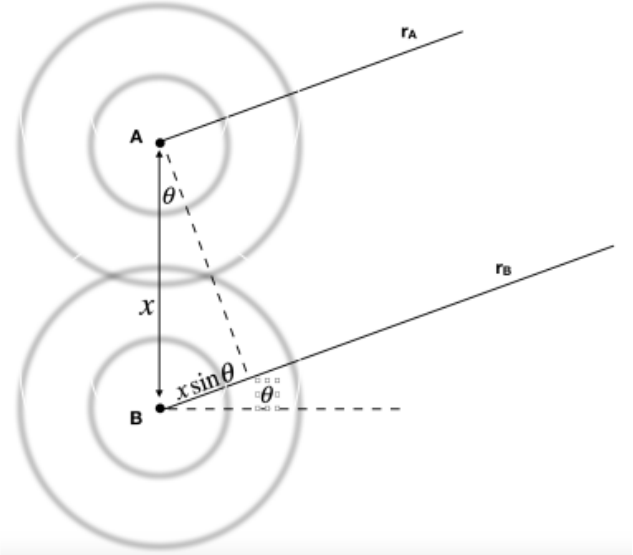


Figure 2: Geometry of two-point-source interference: two coherent sources S_1 and S_2 emit spherical waves that interfere at point P on a distant screen.

2.5 Real Field and Multi-Source Interference

The real physical field is

$$u_{P,\mathrm{Re}} = \mathrm{Re}(u_P) = 2\frac{u_0}{r}\cos(\omega t - kr)\cos\left(\frac{\delta}{2}\right). \tag{20}$$

Time-averaged intensity:

$$\begin{aligned} I \propto \langle u_{P,\mathrm{Re}}^2 \rangle &= 4\frac{u_0^2}{r^2}\cos^2\left(\frac{\delta}{2}\right)\langle \cos^2(\omega t - kr)\rangle \\ &= 2\frac{u_0^2}{r^2}\cos^2\frac{\delta}{2} = \frac{u_0^2}{r^2}(1+\cos\delta). \end{aligned} \tag{21}$$

Thus

$$I = I_0\cos^2\left(\frac{kx\sin\theta}{2}\right). \tag{22}$$

For incoherent sources, $\langle\cos\delta\rangle = 0$, and $I \propto \frac{u_0^2}{r^2}$.

2.6 Interference between Multiple Coherent Point Sources (Point-Source Model)

For N equally spaced point sources (no slit width),

$$u \propto \sum_{m=0}^{N-1} e^{im\delta}. \tag{23}$$

Summing the geometric series:

$$\begin{aligned} u &= \frac{e^{iN\delta} - 1}{e^{i\delta} - 1} \\ &= e^{i(N-1)\delta/2} \, \frac{\sin\left(\frac{N\delta}{2}\right)}{\sin\left(\frac{\delta}{2}\right)}. \end{aligned} \tag{24}$$

The intensity is

$$I(\delta) \propto \langle uu^* \rangle = \frac{I_0}{N^2} \, \frac{\sin^2\left(\frac{N\delta}{2}\right)}{\sin^2\left(\frac{\delta}{2}\right)}. \tag{25}$$

Maxima occur when $\delta = 2n\pi$, yielding $I_{\max} = I_0$. This result will be used later in the diffraction grating analysis (Section 4.6), where finite slit widths are taken into account.

3 Diffraction

3.1 Green's Formula

The vector calculus identity is

$$\nabla \cdot (\psi \nabla \varphi) = \nabla \psi \cdot \nabla \varphi + \psi \nabla^2 \varphi. \tag{26}$$

Subtracting the symmetric form gives Green's second identity:

$$\int_V \left(\psi \nabla^2 \varphi - \varphi \nabla^2 \psi\right) \mathrm{d}V = \oint_S \left(\psi \frac{\partial \varphi}{\partial n} - \varphi \frac{\partial \psi}{\partial n}\right) \mathrm{d}s = \oint_S \left(\psi \nabla \varphi - \varphi \nabla \psi\right) \cdot \mathrm{d}\mathbf{S}. \tag{27}$$

3.2 Kirchhoff Formula

With $k = \frac{\omega}{c}$, the wave equations for the electromagnetic field are

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right) \mathbf{E} = 0, \qquad \left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right) \mathbf{B} = 0. \tag{28}$$

Monochromatic plane wave solutions:

$$\mathbf{E} = E_0 \cos\left(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta\right) \hat{n}, \tag{29}$$

$$\mathbf{B} = \frac{1}{c} E_0 \cos\left(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta\right) \left(\hat{k} \times \hat{n}\right). \tag{30}$$

Reduction to the Helmholtz equation:

$$\left(\nabla^2 + k^2\right) \psi = 0, \tag{31}$$

where ψ represents any Cartesian component of $\mathbf{E}$ or $\mathbf{B}$.

3.3 Green Function of the Helmholtz Equation

Let $G(\mathbf{R})$ be the Green function satisfying

$$\left(\nabla^2 + k^2\right) G(\mathbf{R}) = -\delta(\mathbf{r}_s - \mathbf{r}_0) = -\delta(\mathbf{R}), \tag{32}$$

with $\mathbf{R} = \mathbf{r}_s - \mathbf{r}_0$. For spherical symmetry, $G = G(R)$, and the radial ODE is

$$\frac{1}{R^2}\frac{\mathrm{d}}{\mathrm{d}R}\left(R^2 \frac{\mathrm{d}G}{\mathrm{d}R}\right) + k^2 G = 0. \tag{33}$$

The outgoing solution is

$$G(R) = \frac{e^{ikR}}{4\pi R}. \tag{34}$$

Applying Green's second identity to ψ and G:

$$G\left(\nabla^2 + k^2\right) \psi = 0, \tag{35}$$

$$\psi\left(\nabla^2 + k^2\right) G = -\psi(\mathbf{r}_s)\delta(\mathbf{r}_s - \mathbf{r}_0). \tag{36}$$

Thus

$$\begin{aligned}\psi(\mathbf{r}_0) &= \int \psi(\mathbf{r}_s)\delta(\mathbf{r}_s - \mathbf{r}_0)\,\mathrm{d}V \\ &= \int \left(G\nabla^2\psi - \psi\nabla^2 G\right)\mathrm{d}V \\ &= \oint_S \left(G\frac{\partial\psi}{\partial n} - \psi\frac{\partial G}{\partial n}\right)\mathrm{d}s. \end{aligned} \tag{37}$$

This yields the full Kirchhoff integral:

$$\psi(\mathbf{r}_0) = \oint_S \left[G(\mathbf{r},\mathbf{r}_0)\nabla\psi(\mathbf{r}_s) - \psi(\mathbf{r}_s)\nabla G(\mathbf{r},\mathbf{r}_0)\right]\cdot\mathrm{d}\mathbf{S} = \frac{1}{4\pi}\oint_S \left[\frac{e^{ikr}}{r}\nabla\psi - \psi\nabla\frac{e^{ikr}}{r}\right]\cdot\mathrm{d}\mathbf{S}. \tag{38}$$

The gradient of the Green function is

$$\nabla G = \nabla\frac{e^{ikr}}{4\pi r} = \frac{\hat{r}}{4\pi}\frac{\partial}{\partial r}\left(\frac{e^{ikr}}{r}\right) = \frac{e^{ikr}}{4\pi r}\left(ik - \frac{1}{r}\right)\hat{r}. \tag{39}$$

In the far-field limit ($kr \gg 1$):

$$\nabla G \approx \frac{e^{ikr}}{4\pi r}\,ik\,\hat{r} = ikG\,\hat{r}, \qquad \mathrm{d}\mathbf{S} = \hat{n}\,\mathrm{d}s. \tag{40}$$

3.4 Huygens–Fresnel Diffraction

For a spherical source,

$$\psi(\mathbf{r}) = \frac{Ae^{ik|\mathbf{r}-\mathbf{r}_{\text{source}}|}}{|\mathbf{r}-\mathbf{r}_{\text{source}}|} = \frac{Ae^{ikr_s}}{r_s}. \tag{41}$$

In the far field, $\nabla\psi \approx ik\psi\,\hat{r}_s$.

Huygens–Fresnel Diffraction Formula Starting from the Kirchhoff integral:

$$\begin{aligned}\psi(\mathbf{r}_0) &= \oint_S (G\nabla\psi - \psi\nabla G)\cdot\hat{n}\,\mathrm{d}s \\ &\approx \oint_S \left[ik\left(G\psi\,\hat{r}_s + \psi G\,\hat{r}\right)\cdot\hat{n}\right]\mathrm{d}s \\ &= \frac{ik}{4\pi}\oint_S \psi(\mathbf{r})\frac{e^{ikr}}{r}\left(\cos\theta_s + \cos\theta\right)\mathrm{d}s \\ &= \frac{1}{i\lambda}\oint_S \psi(\mathbf{r})\frac{e^{ikr}}{r}\,\frac{\cos\theta + \cos\theta_s}{2}\,\mathrm{d}s. \end{aligned} \tag{42}$$

For normal incidence ($\theta_s = 0$, $\hat{r}_s = \hat{n}$, $\hat{r}\cdot\hat{n} = \cos\theta$):

$$\psi(\mathbf{r}_0) = \frac{1}{i\lambda}\oint_S \psi(\mathbf{r})\frac{e^{ikr}}{r}\,\frac{1+\cos\theta}{2}\,\mathrm{d}s. \tag{43}$$

In compact notation:

$$\psi(a) = \frac{1}{i\lambda}\oint_S \psi(\rho)\frac{e^{ikr}}{r}\left(\frac{1+\cos\theta}{2}\right)\mathrm{d}s. \tag{44}$$

For a spherical source with boundary conditions

$$\psi_P = \psi(a) = \psi_0 \frac{e^{ika}}{a}, \qquad \psi(\rho) = \psi_0 \frac{e^{ik\rho}}{\rho}, \qquad \psi_A = \psi_0, \tag{45}$$

and in the paraxial limit ($a \gg \rho$, $\alpha \approx \theta$), we have

$$\frac{1}{r} \approx \frac{1}{a}, \qquad \cos\theta \approx \cos\alpha = \frac{x}{\rho}, \tag{46}$$

with the geometrical substitutions

$$x = \rho\cos\alpha, \qquad r = a - x, \qquad \mathrm{d}s = 2\pi\rho\sin\alpha \cdot \rho\,\mathrm{d}\alpha = -2\pi\rho\,\mathrm{d}x. \tag{47}$$

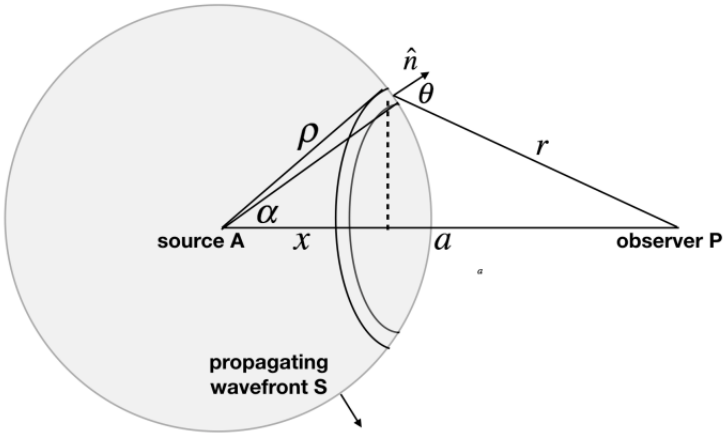


Figure 3: Huygens–Fresnel diffraction principle: each point on a wavefront acts as a secondary spherical wavelet source; the envelope of these wavelets forms the new wavefront.

Proof of the Huygens–Fresnel Integral

$$\begin{aligned}
\psi(a) &= \frac{1}{i\lambda} \oint \psi(\rho) \frac{e^{ikr}}{r} \frac{1+\cos\theta}{2}\,\mathrm{d}s \\
&\approx \frac{1}{i\lambda} \int_{-\rho}^{\rho} \frac{\psi_0 e^{ik\rho}}{\rho} \cdot \frac{e^{ik(a-x)}}{a} \cdot \frac{1+x/\rho}{2} \cdot 2\pi\rho\,\mathrm{d}x \\
&= \frac{\pi}{i\lambda} \cdot \psi_0 \frac{e^{ik(\rho+a)}}{\rho\, a} \int_{-\rho}^{\rho} e^{-ikx}(x+\rho)\,\mathrm{d}x \\
&= \frac{\pi}{i\lambda} \psi_0 \frac{e^{ik(\rho+a)}}{\rho\, a} \cdot \left(-\frac{1}{ik}\right) \cdot 2\rho \left(e^{-ik\rho} - \frac{\sin k\rho}{k\rho}\right).
\end{aligned} \tag{48}$$

In the limit $k\rho \gg 1$:

$$\psi(a) \approx \frac{\pi}{i\lambda} \psi_0 \frac{e^{ik(\rho+a)}}{\rho\, a} \cdot \left(-\frac{2\rho}{ik}\right) e^{-ik\rho} = \psi_0 \frac{e^{ika}}{a}. \tag{49}$$

3.5 Fresnel Number

The nature of diffraction observed at a given distance is determined by the Fresnel number, which quantifies the relative importance of the quadratic phase curvature across the aperture. Using the binomial expansion for small $x \ll r_0$:

$$\sqrt{1+x} \approx 1 + \frac{1}{2}x - \frac{1}{8}x^2. \tag{50}$$

The path length from a point in the aperture to the observation point is expanded as

$$r = \left(r_0^2 + x^2 - 2ux\right)^{1/2} \approx r_0 \left(1 - \frac{ux}{r_0^2} + \frac{x^2}{2r_0^2}\right). \tag{51}$$

Thus the phase factor becomes

$$\psi_P \propto e^{ikr} = e^{ikr_0} \cdot e^{-ikux/r_0} \cdot e^{ikx^2/(2r_0)}. \tag{52}$$

The Fresnel number is defined as

$$F = \frac{x^2}{\lambda r_0} = \frac{1}{2\pi}\frac{kx^2}{r_0}. \tag{53}$$

The distinction between the two diffraction regimes is based on the value of F:

- **Fraunhofer (far-field) diffraction** occurs when $F \ll 1$. In this limit, the quadratic phase term $e^{ikx^2/(2r_0)}$ is negligible across the aperture ($r_A \approx r_B$), and the phase variation reduces to a linear function of x: $\delta \propto x$. The diffraction pattern is simply the Fourier transform of the aperture function.
- **Fresnel (near-field) diffraction** occurs when $F \sim 1$ or $F \gtrsim 1$. The quadratic phase term cannot be neglected, and the propagation must be treated using the Fresnel approximation (convolution with a quadratic phase factor), leading to more complex patterns such as those described by the Cornu spiral.

3.6 Fraunhofer Diffraction

3.6.1 Geometric setup and Fraunhofer approximation

Let the aperture have dimensions $a \times b$ in the x–y plane, and let the observation point be at (u, v) with distance R from the origin. Define $r_0^2 = R^2 + u^2 + v^2$. Then

$$r^2 = R^2 + (u-x)^2 + (v-y)^2 = r_0^2 - 2ux - 2vy + x^2 + y^2. \tag{54}$$

Using the binomial approximation under the Fraunhofer condition ($F \ll 1$):

$$\begin{aligned} r &= r_0\sqrt{1 - 2\frac{ux+vy}{r_0^2} + \frac{x^2+y^2}{r_0^2}} \\ &\approx r_0\left(1 - \frac{ux+vy}{r_0^2}\right) = r_0 - \frac{ux+vy}{r_0}. \end{aligned} \tag{55}$$

3.6.2 Fraunhofer Diffraction from a Rectangular Aperture

The elementary field contribution is

$$\mathrm{d}u_P \propto e^{ikr}\,\mathrm{d}x\,\mathrm{d}y = e^{ikr_0}e^{-ik\frac{ux+vy}{r_0}}\,\mathrm{d}x\,\mathrm{d}y. \tag{56}$$

Integrating over the aperture:

$$\begin{aligned} u_P &\propto e^{ikr_0}\int_{-a/2}^{a/2}\mathrm{d}x\int_{-b/2}^{b/2}\mathrm{d}y\; e^{-ik(ux+vy)/r_0} \\ &\propto e^{ikr_0}\,ab\cdot\frac{\sin\left(\frac{ka}{2}\frac{u}{r_0}\right)}{\frac{kau}{2r_0}}\cdot\frac{\sin\left(\frac{kb}{2}\frac{v}{r_0}\right)}{\frac{kbv}{2r_0}} \\ &\propto e^{ikr_0}\,ab\cdot\mathrm{sinc}\left(\frac{ka}{2}\frac{u}{r_0}\right)\mathrm{sinc}\left(\frac{kb}{2}\frac{v}{r_0}\right), \end{aligned} \tag{57}$$

where $\mathrm{sinc}(x) = \frac{\sin x}{x}$. Thus the intensity is

$$I(u,v) \propto |u_P|^2 \propto a^2b^2\mathrm{sinc}^2\left(\frac{ka}{2}\frac{u}{r_0}\right)\mathrm{sinc}^2\left(\frac{kb}{2}\frac{v}{r_0}\right). \tag{58}$$

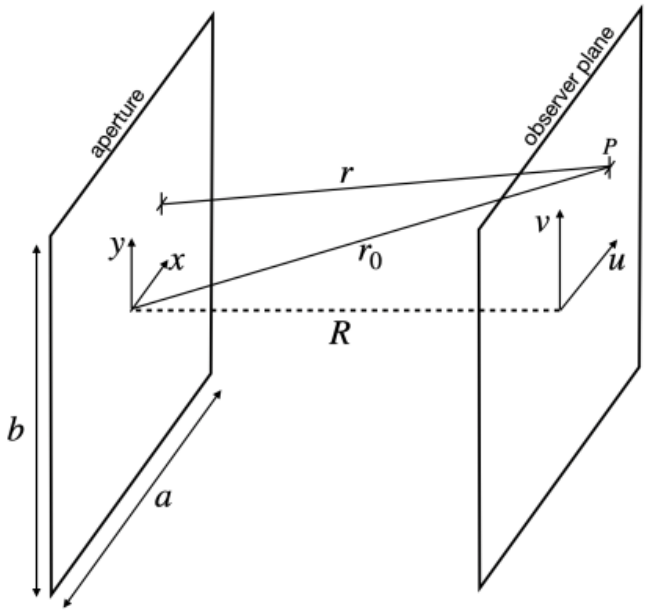


Figure 4: Fraunhofer diffraction through a rectangular aperture: the far-field intensity pattern is the product of two sinc functions, corresponding to diffraction along the two orthogonal axes.

3.7 Fresnel Diffraction

3.7.1 Geometric setup and Fresnel approximation

Consider a narrow slit of half-width a located in the y-direction, illuminated by a plane wave of wavelength λ ($k = 2\pi/\lambda$). The observation point P lies on a screen at distance z from the aperture, with transverse coordinate u. For the axial point Q, we have

$u = 0$. Under the standard Fresnel approximation ($z \gg a, |u|$), the distance r from an infinitesimal aperture element at y to the observation point is expanded as:

$$r = \sqrt{z^2 + (u-y)^2} \approx r_0 - \frac{uy}{r_0} + \frac{y^2}{2r_0}, \quad \text{where } r_0 = \sqrt{z^2+u^2}. \tag{59}$$

For the axial point Q ($u = 0$), this simplifies to $r_0 = z$, thus

$$r \approx z + \frac{y^2}{2z}. \tag{60}$$

3.7.2 Amplitude on the axial point

The diffracted field amplitude at the axial point Q is proportional to the integral of the phase over the slit. Let the slit have total width a and be centered on the y-axis, so the integration runs from $-a/2$ to $a/2$. Under the Fresnel approximation, the distance r from a point at y to Q is $r \approx z + y^2/(2z)$. Thus,

$$u_Q(z) \propto \int_{-a/2}^{a/2} e^{ikr}\, dy \propto e^{ikz} \int_{-a/2}^{a/2} e^{i\frac{ky^2}{2z}} dy = e^{ikz} \int_{-a/2}^{a/2} e^{i\frac{\pi y^2}{\lambda z}} dy. \tag{61}$$

To normalize, introduce the dimensionless variable

$$t = y\sqrt{\frac{2}{\lambda z}}, \qquad v = \frac{a}{2}\sqrt{\frac{2}{\lambda z}} \tag{62}$$

Substituting, the integral becomes

$$u_Q(z) \propto \int_{-v}^{v} e^{i\pi t^2/2}\, dt. \tag{63}$$

3.7.3 Cornu spiral

The Cornu spiral (also known as the clothoid) is the parametric curve in the complex plane defined by the Fresnel function:

$$Z(v) = C(v) + i\, S(v) = \int_0^v e^{i\pi t^2/2}\, dt, \tag{64}$$

where the coordinates are explicitly given by the Fresnel integrals:

$$C(v) = \int_0^v \cos\left(\frac{\pi t^2}{2}\right) dt, \qquad S(v) = \int_0^v \sin\left(\frac{\pi t^2}{2}\right) dt. \tag{65}$$

The parameter v represents the arc length along the curve (as will be shown below). The spiral winds towards two distinct focal points in the complex plane as $v \to \pm\infty$.

Property 1: Asymptotic limits The Cornu spiral converges to two finite points because the oscillatory integrand decays in the average sense. The limits of the complex Fresnel function are:

$$\lim_{v\to+\infty} Z(v) = \frac{1}{2} + i\frac{1}{2}, \qquad \lim_{v\to-\infty} Z(v) = -\frac{1}{2} - i\frac{1}{2}. \tag{66}$$

Geometrically, this means the spiral starts at the lower-left focal point $(-1/2, -i/2)$ when $v = -\infty$, spirals outwards through the origin $(0, 0)$ at $v = 0$, and finally converges to the upper-right focal point $(1/2, i/2)$ as $v \to +\infty$.

Property 2: Arc length The most elegant property of the Cornu spiral is that the parameter v exactly equals the arc length measured from the origin. Indeed, differentiating the complex function $Z(v)$ gives:

$$dZ = dC + i\,dS = \left(\cos\frac{\pi v^2}{2} + i\sin\frac{\pi v^2}{2}\right) dv = e^{i\pi v^2/2}\,dv. \tag{67}$$

The infinitesimal arc length ds along the curve is the modulus of this complex differential:

$$ds = \sqrt{dC^2 + dS^2} = \left|dC + i\,dS\right| = \left|e^{i\pi v^2/2}\right| |dv| = |dv|. \tag{68}$$

Therefore, the arc length from the origin $(0,0)$ to the point $(C(v), S(v))$ for $v > 0$ is:

$$s(v) = \int_0^v ds = \int_0^v |dv| = v. \tag{69}$$

Hence, v is not merely a mathematical parameter; it has a direct geometrical interpretation as the physical distance traveled along the spiral.

Property 3: Diffraction intensity and geometric chord Intensity $I \propto |u|^2$. And I_0 is the intensity for the fully open case. For an aperture spanning from v_1 to v_2, the intensity is:

$$I = \frac{1}{2} I_0 \left[\big(C(v_2) - C(v_1)\big)^2 + \big(S(v_2) - S(v_1)\big)^2\right]. \tag{70}$$

Therefore, the diffraction intensity is directly proportional to the square of the chord distance between the two corresponding points on the spiral. For the symmetric case where $v = v_2 = -v_1$, we have:

$$I = 2I_0 \left[C^2(v) + S^2(v)\right]. \tag{71}$$

4 Diffraction in Optical Instruments

4.1 Young's Double Slit (Equal Widths)

For a double slit with equal slit width a and separation d (centre-to-centre), the intensity pattern is

$$I = I_0 \cos^2\left[\frac{kd\sin\theta}{2}\right]\mathrm{sinc}^2\left[\frac{ka\sin\theta}{2}\right]. \tag{72}$$

Two limiting cases:

- $a \to 0$: two coherent point sources, $I = I_0\cos^2\left(\frac{kd\sin\theta}{2}\right)$.
- $d \to 0$: single 1D aperture, $I = I_0\mathrm{sinc}^2\left(\frac{ka\sin\theta}{2}\right)$.

This result can be obtained as a special case ($w_1 = w_2 = a$) of the more general unequal-width double slit discussed in the next subsection.

4.2 Double Slit Interference with Unequal Slit Widths (General Case)

Let the two slits have widths w_1 and w_2, separated by a distance d (centre-to-centre). The field integral is

$$\begin{aligned}
u_P &\propto \int e^{ikr'}\,\mathrm{d}y \\
&\propto e^{ikr}\int e^{-iky\sin\theta}\,\mathrm{d}y \\
&\propto \frac{e^{ikr}}{-ik\sin\theta}\left[\int_{-\frac{d}{2}-\frac{w_2}{2}}^{-\frac{d}{2}+\frac{w_2}{2}}\mathrm{d}y\; e^{-iky\sin\theta} + \int_{\frac{d}{2}-\frac{w_1}{2}}^{\frac{d}{2}+\frac{w_1}{2}}\mathrm{d}y\; e^{-iky\sin\theta}\right] \\
&\propto e^{ikr}\left[e^{i\frac{kd\sin\theta}{2}}(-2i)\frac{\sin\frac{kw_2\sin\theta}{2}}{-ik\sin\theta} + e^{-i\frac{kd\sin\theta}{2}}(-2i)\frac{\sin\frac{kw_1\sin\theta}{2}}{-ik\sin\theta}\right].
\end{aligned} \tag{73}$$

Simplifying gives the complex amplitude

$$u_P \propto e^{ikr}\left[e^{i\frac{\delta}{2}}w_2\mathrm{sinc}\alpha_2 + e^{-i\frac{\delta}{2}}w_1\mathrm{sinc}\alpha_1\right], \tag{74}$$

with

$$\delta = kd\sin\theta, \qquad \alpha_1 = \frac{kw_1\sin\theta}{2}, \qquad \alpha_2 = \frac{kw_2\sin\theta}{2}. \tag{75}$$

The intensity $I_P \propto |u_P|^2$ expands to

$$\begin{aligned}
I_P &\propto \left(e^{i\frac{\delta}{2}}w_2\mathrm{sinc}\alpha_2 + e^{-i\frac{\delta}{2}}w_1\mathrm{sinc}\alpha_1\right)\left(e^{-i\frac{\delta}{2}}w_2\mathrm{sinc}\alpha_2 + e^{i\frac{\delta}{2}}w_1\mathrm{sinc}\alpha_1\right) \\
&\propto w_1^2\mathrm{sinc}^2\alpha_1 + w_2^2\mathrm{sinc}^2\alpha_2 + w_1w_2\mathrm{sinc}\alpha_1\mathrm{sinc}\alpha_2\left(e^{i\delta} + e^{-i\delta}\right) \\
&= w_1^2\mathrm{sinc}^2\alpha_1 + w_2^2\mathrm{sinc}^2\alpha_2 + 2w_1w_2\mathrm{sinc}\alpha_1\mathrm{sinc}\alpha_2\cos\delta.
\end{aligned} \tag{76}$$

Thus the final intensity formula is

$$I_P = I_1\mathrm{sinc}^2\alpha_1 + I_2\mathrm{sinc}^2\alpha_2 + 2\sqrt{I_1I_2}\,\mathrm{sinc}\alpha_1\,\mathrm{sinc}\alpha_2\cos\delta, \tag{77}$$

where

$$\delta = kd\sin\theta, \qquad \alpha_1 = \frac{kw_1\sin\theta}{2}, \qquad \alpha_2 = \frac{kw_2\sin\theta}{2}, \tag{78}$$

and $I_1 \propto w_1^2,\ I_2 \propto w_2^2$. Equivalently, with $\beta = \frac{\delta}{2} = \frac{kd\sin\theta}{2}$ and using $\cos\delta = 2\cos^2\beta - 1$, we have

$$I_P = I_1\mathrm{sinc}^2\alpha_1 + I_2\mathrm{sinc}^2\alpha_2 + 2\sqrt{I_1 I_2}\,\mathrm{sinc}\alpha_1\,\mathrm{sinc}\alpha_2\left(2\cos^2\beta - 1\right), \tag{79}$$

with

$$\beta = \frac{\pi d}{\lambda}\sin\theta, \qquad \alpha_1 = \frac{\pi w_1}{\lambda}\sin\theta, \qquad \alpha_2 = \frac{\pi w_2}{\lambda}\sin\theta. \tag{80}$$

When $w_1 = w_2 = a$, this reduces to the standard Young's double-slit formula given in §4.2.

4.3 Diffraction in Optical Instruments

4.3.1 Simple Imaging Lens

Lens equation:

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}. \tag{81}$$

4.3.2 Telescope

Angular resolution:

$$\Delta\theta = 1.22\frac{\lambda}{D}. \tag{82}$$

Resolving power:

$$R = \frac{1}{\Delta\theta} \approx \frac{D}{1.22\lambda}. \tag{83}$$

4.3.3 Microscope

Define optical path lengths:

$$S_P = nu + S_{\text{lens}} + v, \tag{84}$$

$$S_Q = n\sqrt{u^2 + d^2} + S_{\text{lens}} + \sqrt{v^2 + d'^2}, \tag{85}$$

$$S_{P,M} = n\sqrt{u^2 + \left(\frac{D}{2}\right)^2} + \sqrt{v^2 + \left(\frac{D}{2}\right)^2}, \tag{86}$$

$$S_{Q,M} = n\sqrt{u^2 + \left(\frac{D}{2} + d\right)^2} + \sqrt{v^2 + \left(\frac{D}{2} - d'\right)^2}. \tag{87}$$

The imaging condition requires $S_P = S_{P,M}$ and $S_Q = S_{Q,M}$. In the far-field limit ($u, v \gg D \gg d', d$), we obtain

$$n\frac{Dd}{\sqrt{u^2 + (D/2)^2}} = \frac{Dd'}{\sqrt{v^2 + (D/2)^2}}. \tag{88}$$

Simplified to

$$nd\sin\theta = d'\sin\theta', \tag{89}$$

assuming the image space is air ($n' = 1$). With the geometric approximations

$$\frac{d'}{v} = \tan\alpha \approx \sin\alpha \approx 1.22\frac{\lambda}{D}, \qquad \sin\theta' \approx \frac{D}{2v}, \tag{90}$$

we obtain the resolution criteria.

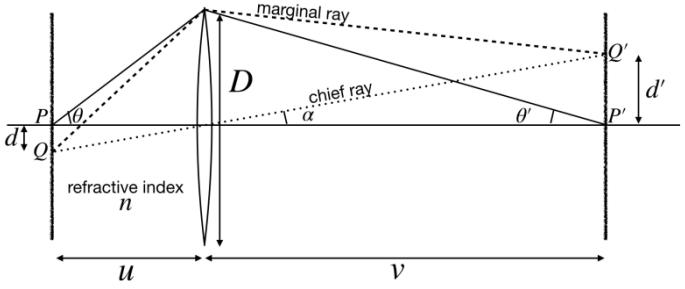


Figure 5: Schematic of a compound microscope: the objective lens forms a magnified real image, which is further magnified by the eyepiece for visual observation.

4.4 Microscope Resolution

Microscope resolution:

$$\text{Linear resolution: } d \approx 0.61\frac{\lambda}{n\sin\theta}, \tag{91}$$

$$\text{Resolving power: } R = \frac{1}{d} = \frac{\text{NA}}{0.61\lambda}, \tag{92}$$

$$\text{Numerical aperture: } \text{NA} = n\sin\theta. \tag{93}$$

4.5 Diffraction Grating

The intensity profile for a grating with N slits, slit width a, and period d is obtained by multiplying the interference factor from §2.5 (point sources) with the single-slit diffraction envelope:

$$I(\theta) = I_0\left[\frac{\sin(N\beta)}{\sin\beta}\right]^2 \text{sinc}^2\alpha, \tag{94}$$

where

$$\beta = \frac{kd\sin\theta}{2} = \frac{\pi d}{\lambda}\sin\theta, \qquad \alpha = \frac{ka\sin\theta}{2} = \frac{\pi a}{\lambda}\sin\theta. \tag{95}$$

For a transmission grating with $a = d/2$, the maxima condition is

$$\frac{kd\sin\theta}{2} = n\pi \quad \Longrightarrow \quad n\lambda = d\sin\theta. \tag{96}$$

Angular dispersion:

$$\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = \frac{n}{d\cos\theta}. \tag{97}$$

Free spectral range:

$$\Delta\lambda_{\text{FSR}} = \lambda_{\text{red}} - \lambda_{\text{blue}}, \qquad n\lambda_{\text{red}} = (n+1)\lambda_{\text{blue}} \quad \Longrightarrow \quad \Delta\lambda_{\text{FSR}} = \frac{\lambda_{\text{blue}}}{n}. \tag{98}$$

Rayleigh's criterion: the diffraction maximum of one wavelength coincides with the first minimum of the adjacent wavelength.

$$\text{Maxima: } \frac{kd\sin\theta}{2} = n\pi \quad \Longrightarrow \quad d\sin\theta = n\lambda, \tag{99}$$

$$\text{Minima: } \frac{Nkd\sin\theta'}{2} = m\pi \quad \Longrightarrow \quad d\sin\theta' = \frac{m\lambda}{N}. \tag{100}$$

Set $m = nN + 1$. Then

$$\Delta\sin\theta = \sin\theta' - \sin\theta = \frac{\lambda}{d}\left(\frac{m}{N} - n\right) = \frac{\lambda}{Nd}, \tag{101}$$

$$\Delta\lambda = \Delta\left(\frac{d\sin\theta}{n}\right) = \frac{d}{n}\cos\theta\,\Delta\theta = \frac{d}{n}\Delta\sin\theta. \tag{102}$$

Thus the resolving power is

$$R = \frac{\lambda}{\Delta\lambda} = \frac{d\sin\theta}{n}\cdot\frac{1}{\frac{d}{Nd}\Delta\sin\theta} \quad \Longrightarrow \quad R = Nn. \tag{103}$$

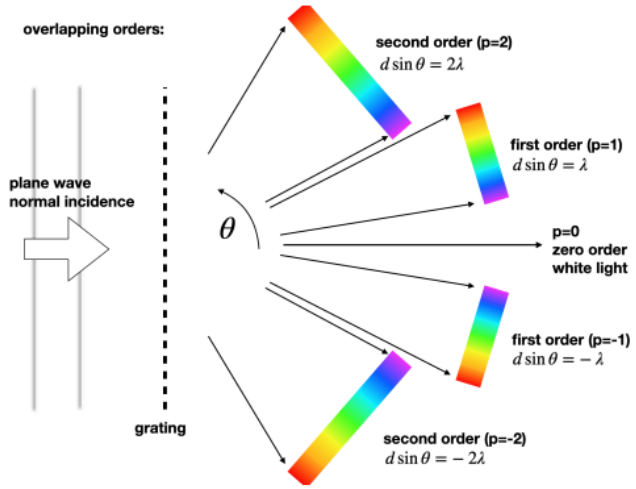


Figure 6: Diffraction grating principle: a periodic array of slits produces sharp interference maxima; the angular position of each order is determined by the grating equation $d\sin\theta = n\lambda$.

5 Interferometry

5.1 Interference Fringes

5.1.1 Equal inclination fringes

The optical path difference is

$$\Delta L = 2n_2 d\cos\theta_r. \tag{104}$$

With phase shift $\delta = k_0\Delta L - \pi$, where $k_0 = \frac{2\pi}{\lambda_2}$, the intensity is

$$I = I_0\cos^2\left(\frac{\delta}{2}\right). \tag{105}$$

Maxima occur when $\frac{\delta}{2} = m\pi$, i.e.,

$$d\cos\theta_r = \frac{(2m+1)\lambda_2}{4}. \tag{106}$$

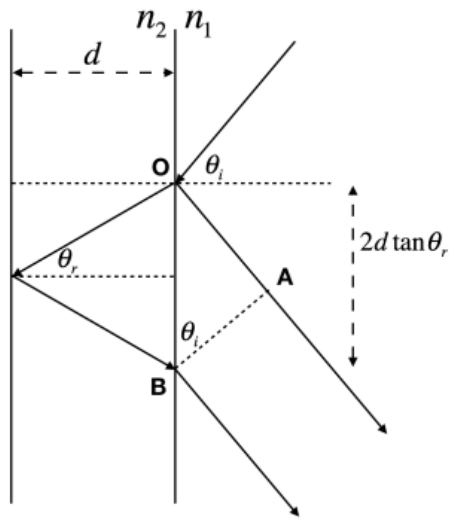


Figure 7: Equal-inclination fringes (Haidinger fringes): circular fringes observed when a parallel plate is illuminated by an extended monochromatic source.

5.1.2 Equal thickness fringes

For normal incidence, $\theta_i = 0 \Rightarrow \theta_r = 0$. If $d = \alpha x$, then

$$d\cos\theta_r = \frac{(2m+1)\lambda_2}{4} \quad \Longrightarrow \quad x = \frac{(2m+1)\lambda_2}{4\alpha}. \tag{107}$$

5.2 The Michelson Interferometer

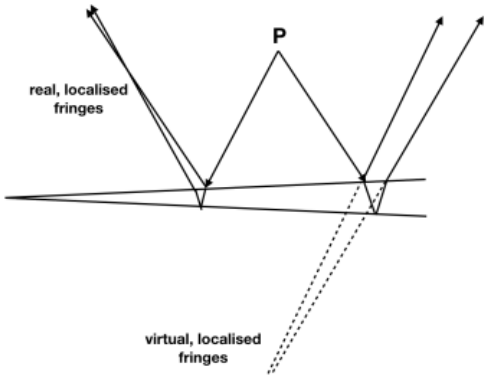


Figure 8: Equal-thickness fringes (Fizeau fringes): straight or curved fringes corresponding to contours of constant film thickness in a wedge-shaped thin film.

Fringe geometry The optical path difference is $\Delta L = 2x\cos\theta$, and the phase difference is

$$\delta = k\Delta L = 2kx\cos\theta. \tag{108}$$

Maxima conditions:

$$2x = m_0\lambda, \qquad 2x\cos\theta_p = (m_0 - p)\lambda \quad (p\text{-th ring}). \tag{109}$$

For small angles $\theta_p \ll \pi/2$, $\cos\theta_p \approx 1 - \theta_p^2/2$, we get

$$2x(1-\cos\theta_p) = p\lambda \quad \Longrightarrow \quad \theta_p^2 = p\,\frac{\lambda}{x}. \tag{110}$$

With $\theta = r/f$, the ring spacing is

$$\theta_{p+1}^2 - \theta_p^2 = \frac{\lambda}{x}, \qquad r_{p+1}^2 - r_p^2 = \frac{f^2\lambda}{x}. \tag{111}$$

Spectral resolution On axis ($\theta = 0$):

$$u(x) \propto \left(1 + e^{i2kx}\right) \quad \Longrightarrow \quad I(x) \propto \left(1 + \cos(2kx)\right). \tag{112}$$

Thus

$$I(x) = \frac{1}{2} I_{\max}\left(1 + \cos(2kx)\right). \tag{113}$$

For two spectral lines λ_1, λ_2 with k_1, k_2:

$$\begin{aligned} I(x) &= \frac{1}{2} I_{\max}\left[1 + \cos(2k_1x) + 1 + \cos(2k_2x)\right] \\ &= I_{\max}\left\{1 + \cos\left[(k_1+k_2)x\right]\cos\left[(k_1-k_2)x\right]\right\}. \end{aligned} \tag{114}$$

The minimum resolvable wavevector difference is

$$\Delta k = k_1 - k_2 = \frac{\pi}{x_{\max}} = \frac{2\pi}{L_{\max}}, \tag{115}$$

where $L_{\max} = 2x_{\max}$ is the maximum optical path difference. The resolving power is

$$R = \frac{\lambda}{\Delta\lambda} = \frac{k}{\Delta k} = \frac{L_{\max}}{\lambda}. \tag{116}$$

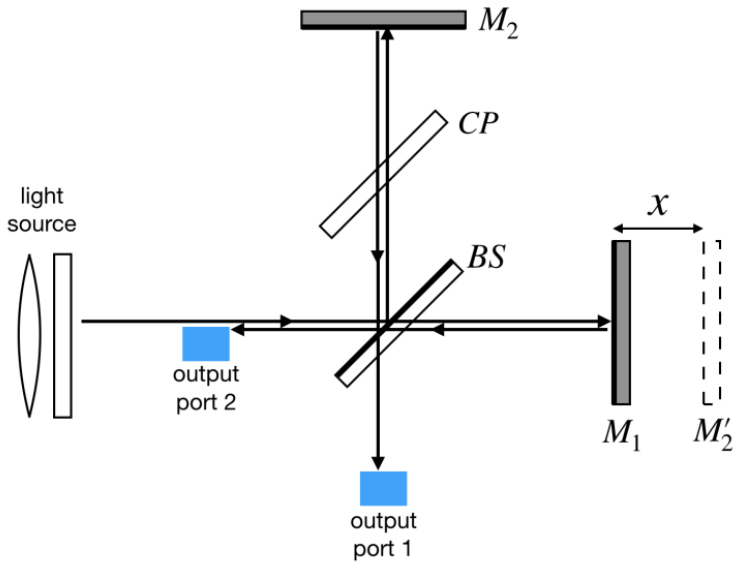


Figure 9: Michelson interferometer: a beam splitter divides the incident light into two paths; the recombined beams produce interference fringes whose visibility depends on the optical path difference.

5.3 The Fabry–Perot Etalon and Spectrometer

In contrast to the two-beam interference of the Michelson interferometer, the Fabry–Perot etalon exploits multiple-beam interference between two parallel partially reflecting surfaces. This yields much sharper transmission peaks and consequently a significantly higher resolving power.

Let the reflection and transmission coefficients at the interfaces be r, t (from n_1 to n_2) and r', t' (from n_2 to n_1). The reflected and transmitted intensities are $I_R = |u_R|^2$, $I_T = |u_T|^2$. The optical path difference is

$$\Delta L = 2n_2 d\cos\theta_t, \tag{117}$$

and phase difference $\delta = k\Delta L = 2kn_2 d\cos\theta_t$. Reciprocity relations: $r = -r'$, $r^2 + tt' = 1$; reflectance and transmittance are $R = r^2$, $T = tt'$.

The reflected field is

$$u_R = ur + utt'r'e^{i\delta}\left(1 + r'^2 e^{i\delta} + r'^4 e^{i2\delta} + \cdots\right) = ur\,\frac{1 - e^{i\delta}}{1 - r^2 e^{i\delta}}. \tag{118}$$

The transmitted field is

$$u_T = utt' e^{ikd/\cos\theta_t}\left(1 + r^2 e^{i\delta} + r^4 e^{i2\delta} + \cdots\right) = ue^{ikd/\cos\theta_t}\,\frac{T}{1 - Re^{i\delta}}. \tag{119}$$

The intensity formulas:

$$I_R = I \cdot \frac{4R\sin^2(\delta/2)}{(1-R)^2 + 4R\sin^2(\delta/2)}, \tag{120}$$

$$I_T = I \cdot \frac{1}{1 + \mathcal{F}\sin^2(\delta/2)}, \tag{121}$$

where the finesse coefficient is

$$\mathcal{F} = \frac{4R}{(1-R)^2}. \tag{122}$$

Transmission maxima occur when $\delta = 2\pi m$, i.e.,

$$2kn_2 d\cos\theta_t = 2\pi m \quad \Longrightarrow \quad \cos\theta_t = \frac{m\lambda}{2n_2 d}. \tag{123}$$

For small angles $0 < \theta_t \ll \pi/2$, the angular separation of fringes is

$$\theta_m^2 - \theta_{m+1}^2 = \frac{\lambda}{n_2 d}. \tag{124}$$

The finesse and resolving power are

$$F = \frac{\pi\sqrt{R}}{1-R}, \qquad R = \frac{\lambda}{\Delta\lambda} = mF. \tag{125}$$

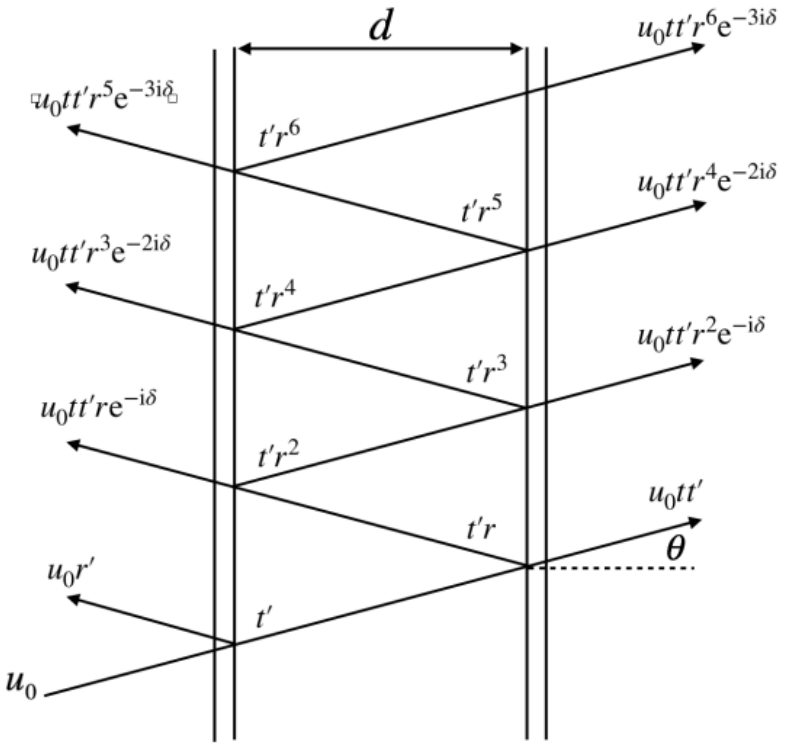


Figure 10: Fabry–Perot etalon: multiple-beam interference between two parallel partially reflective surfaces produces sharp transmission peaks, with resolving power proportional to the finesse and the order m.

6 Fundamentals of Lasers

6.1 Interaction of Atoms

The radiative transition rates per unit volume are

$$\gamma_{21} = A_{21}n_2 + B_{21}u(\omega)n_2 \quad \text{(spontaneous + stimulated emission)}, \tag{126}$$
$$\gamma_{12} = B_{12}u(\omega)n_1 \quad \text{(stimulated absorption)}. \tag{127}$$

The photon energy is $\hbar\omega = E_2 - E_1$. In thermal equilibrium, the population ratio is

$$\frac{n_2}{n_1} = \frac{g_2}{g_1}\exp\left(-\frac{\hbar\omega}{k_BT}\right). \tag{128}$$

The blackbody spectral energy density is

$$u(\omega) = \frac{\hbar\omega^3}{\pi^2c^3}\frac{1}{e^{\hbar\omega/k_BT}-1}. \tag{129}$$

Detailed balance in equilibrium ($\gamma_{21} = \gamma_{12}$, $A_{21} = A$) gives

$$A = \left(B_{12}\frac{n_1}{n_2} - B_{21}\right)u(\omega). \tag{130}$$

Substituting the Boltzmann distribution:

$$A = \left[B_{12}\frac{g_1}{g_2}\exp\left(\frac{\hbar\omega}{k_BT}\right) - B_{21}\right]\frac{\hbar\omega^3}{\pi^2c^3}\frac{1}{e^{\hbar\omega/k_BT}-1}. \tag{131}$$

Since A, B_{12}, B_{21} are temperature-independent, we require $B_{12}g_1 = B_{21}g_2$. Hence the Einstein relations:

$$A_{21} = B_{12}\frac{g_1}{g_2}\frac{\hbar\omega^3}{\pi^2c^3} = B_{21}\frac{\hbar\omega^3}{\pi^2c^3}. \tag{132}$$

Final form:

$$\frac{A_{21}}{B_{21}} = \frac{\hbar\omega^3}{\pi^2c^3}, \qquad B_{12}g_1 = B_{21}g_2. \tag{133}$$

6.2 Conditions for Laser Gain

Net stimulated emission requires

$$n_2B_{21}u(\omega) > n_1B_{12}u(\omega) \quad \Longrightarrow \quad n_2 > n_1. \tag{134}$$

If $g_1 = g_2$, then $B_{21} = B_{12}$. The gain threshold is $B_{21}u(\omega) > A_{21}$.

Pumping Let τ_1, τ_2 be the excited-state lifetimes, and R_1, R_2 the pump rates into the lower and upper levels. The rate equations are

$$\frac{\mathrm{d}n_2}{\mathrm{d}t} = R_2 - \frac{n_2}{\tau_2}, \tag{135}$$
$$\frac{\mathrm{d}n_1}{\mathrm{d}t} = R_1 + n_2A_{21} - \frac{n_1}{\tau_1}. \tag{136}$$

In the steady state ($\mathrm{d}n_1/\mathrm{d}t = \mathrm{d}n_2/\mathrm{d}t = 0$):

$$n_2 = R_2\tau_2, \tag{137}$$

$$n_1 = R_1\tau_1 + R_2\tau_2 A_{21}\tau_1. \tag{138}$$

The population inversion condition $n_2 > n_1$ yields

$$\frac{R_2\tau_2}{R_1\tau_1}\left(1 - A_{21}\tau_1\right) > 1. \tag{139}$$

Thus two inequalities are required:

$$R_2\tau_2 > R_1\tau_1, \qquad A_{21}\tau_1 < 1. \tag{140}$$

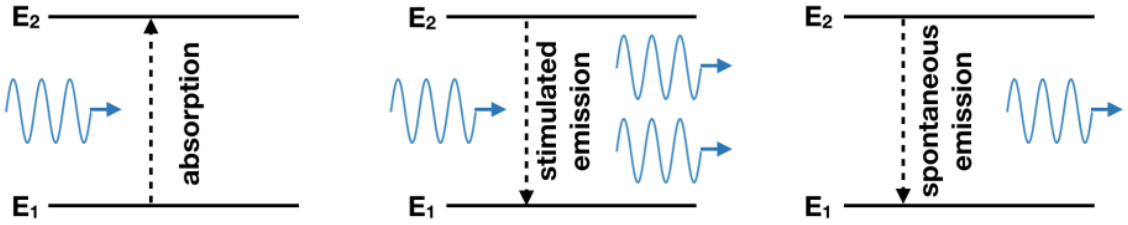


Figure 11: Laser operation principle: population inversion between two energy levels enables stimulated emission to dominate over absorption, producing coherent optical amplification.

6.3 Laser Longitudinal Modes

6.3.1 Resonance Condition

$$2L = m\lambda, \tag{141}$$

where L is the cavity length and m is the mode number.

6.3.2 Frequency Separation of Consecutive Longitudinal Modes

$$\nu = \frac{c}{\lambda} = \frac{c}{2L}m. \tag{142}$$

The separation is

$$\Delta\nu = \nu(m+1) - \nu(m) = \frac{c}{2L}. \tag{143}$$

6.3.3 Wavelength Linewidth

$$\Delta\nu_{\text{line}} = \frac{c}{\lambda^2}\Delta\lambda_{\text{line}}, \tag{144}$$

where λ is the laser wavelength and $\Delta\lambda_{\text{line}}$ is the spontaneous emission linewidth.

6.3.4 Number of Lasing Longitudinal Modes

$$N \approx \frac{\Delta\nu_{\text{line}}}{\Delta\nu}. \tag{145}$$

7 Coherence Theory

7.1 Coherence

Classification:

- Temporal coherence: phase correlation over time.
- Spatial (transverse) coherence: phase correlation at different spatial positions.

7.1.1 Temporal Coherence

The spectral bandwidth limits temporal coherence. The coherence time and length are

$$\tau_c = \frac{1}{\Delta\nu}, \qquad L_c = c\tau_c = \frac{\lambda^2}{\Delta\lambda}. \tag{146}$$

In a Michelson interferometer, the fringe visibility is

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}. \tag{147}$$

Limits:

- $V = 1$: perfect coherence,
- $V = 0$: complete incoherence,
- $0 < V < 1$: partial coherence.

With the complex degree of coherence $\gamma = \frac{\langle u_A u_B^* \rangle}{\sqrt{I_A I_B}}$, the intensity is

$$I = I_A + I_B + 2\sqrt{I_A I_B}\ \mathrm{Re}[\gamma]. \tag{148}$$

The visibility is related to γ by

$$V = \frac{2\sqrt{I_A I_B}}{I_A + I_B}\,|\gamma|. \tag{149}$$

7.1.2 Spatial (Transverse) Coherence

1) Intensity from two symmetric coherent sources:

$$I(\theta) = I_0\left[1 + \cos(ka\sin\theta)\cos(ka\sin\theta_s)\right]. \tag{150}$$

Alternative forms:

$$I(\theta) = I_0\cos^2\left(\frac{ka(\sin\theta \mp \sin\theta_s)}{2}\right) = \frac{I_0}{2}\left(1 + \cos\left[ka(\sin\theta \mp \sin\theta_s)\right]\right). \tag{151}$$

The total intensity is $I(\theta) = I(\theta)_+ + I(\theta)_-$.

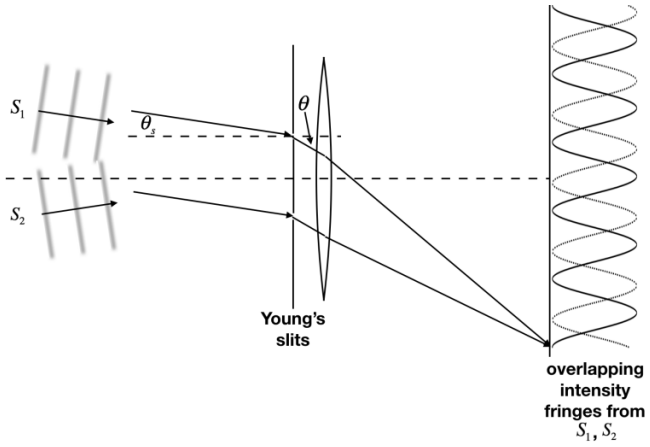


Figure 12: Transverse coherence: the degree of coherence between two points in the observation plane is determined by the angular size of the source, as described by the Van Cittert–Zernike theorem.

7.2 Extended Source Spatial Coherence

For two point sources, the maximum and minimum intensities are

$$I_{\max} = I_0 \left(1 + \cos(ka \sin \theta_s)\right), \qquad I_{\min} = I_0 \left(1 - \cos(ka \sin \theta_s)\right). \tag{152}$$

Thus

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \cos(ka \sin \theta_s). \tag{153}$$

In the point-source limit $\theta_s \to 0$, we get $V \to 1$, $I_{\max} \to 2I_0$, $I_{\min} \to 0$, sharp fringes.

2) Extended uniform source $-\Theta_s < \theta_s < \Theta_s$. Let $k' = k \sin \theta_s$. The intensity is proportional to

$$I(\theta) \propto \int_0^{k \sin \Theta_s} \left[1 + \cos(ka \sin \theta) \cos(k'a)\right] \mathrm{d}k', \tag{154}$$

where we have used symmetry to integrate only half the range. Evaluation gives

$$I(\theta) \propto k \sin \Theta_s + \cos(ka \sin \theta) \cdot \frac{\sin\left(ka \sin \Theta_s\right)}{a}. \tag{155}$$

Compact form:

$$I(\theta) = I_0 \left[1 + \cos(ka \sin \theta) \cdot \mathrm{sinc}\left(ka \sin \Theta_s\right)\right]. \tag{156}$$

The extrema are

$$I_{\max} = I_0 \left[1 + \mathrm{sinc}(ka \sin \Theta_s)\right], \qquad I_{\min} = I_0 \left[1 - \mathrm{sinc}(ka \sin \Theta_s)\right]. \tag{157}$$

Hence the visibility is

$$V = \mathrm{sinc}\left(ka \sin \Theta_s\right). \tag{158}$$

7.3 Van Cittert–Zernike Theorem

The theorem states that the complex degree of coherence between two points in the interference plane equals the Fourier transform of the source intensity distribution.

7.3.1 Astronomical Imaging Interference

For a single telescope of diameter D:

$$\theta_{\min} = 1.22\frac{\lambda}{D}. \tag{159}$$

For an interferometer with baseline B:

$$\theta_{\min} \approx \frac{\lambda}{B}. \tag{160}$$

8 Polarization of Light

8.1 General Formulation

The orthogonal field components are

$$E_x = E_{x0}\cos(\omega t - kz + \phi_x), \qquad E_y = E_{y0}\cos(\omega t - kz + \phi_y). \tag{161}$$

Define $\delta = \phi_y - \phi_x$. The polarization ellipse equation is

$$\left(\frac{E_x}{E_{x0}}\right)^2 + \left(\frac{E_y}{E_{y0}}\right)^2 - 2\frac{E_x E_y}{E_{x0}E_{y0}}\cos\delta = \sin^2\delta. \tag{162}$$

8.2 Polarization States

1. Linear polarization: $\delta = 0$ or $\delta = \pi$.
2. Circular polarization: $\delta = \pm\frac{\pi}{2}$, $E_{x0} = E_{y0}$.
   - $\delta > 0$: right-handed circular,
   - $\delta < 0$: left-handed circular.
3. Elliptical polarization: all other phase/amplitude combinations.

8.3 Jones Vectors

The general Jones vector is

$$\mathbf{E} = \begin{pmatrix} E_{x0} \\ E_{y0} \end{pmatrix} = E_x \begin{pmatrix} 1 \\ |E_y/E_{x0}|e^{i\delta} \end{pmatrix}. \tag{163}$$

Standard polarization vectors:

$$\text{Linear } x\text{-axis: } \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \tag{164}$$

$$\text{Linear } y\text{-axis: } \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \tag{165}$$

$$\text{Linear } 45^\circ: \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \tag{166}$$

$$\text{Right circular: } \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}, \tag{167}$$

$$\text{Left circular: } \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}. \tag{168}$$

8.4 Jones Matrices

The field transformation is $\mathbf{E}_{\text{out}} = J\,\mathbf{E}_{\text{in}}$. Standard matrices:

$$J_{\text{LP},x} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad J_{\text{LP},y} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad J_{\text{LP},45^\circ} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \tag{169}$$

$$J_{\text{HWP}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J_{\text{QWP}} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad J_{\text{WP}} = \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\delta} \end{pmatrix}. \tag{170}$$

The rotation matrix (counterclockwise rotation by θ) is

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}. \tag{171}$$

8.5 Stokes Parameters

$$I = E_{x0}^2 + E_{y0}^2 \quad \text{(total intensity)}, \tag{172}$$
$$Q = E_{x0}^2 - E_{y0}^2 \quad \text{(horizontal/vertical linear)}, \tag{173}$$
$$U = 2E_{x0}E_{y0}\cos\delta \quad (\pm 45^\circ \text{ linear}), \tag{174}$$
$$V = 2E_{x0}E_{y0}\sin\delta \quad \text{(circular)}. \tag{175}$$

The degree of polarization is

$$P = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{\sqrt{Q^2 + U^2 + V^2}}{I}. \tag{176}$$

8.6 Malus's Law, Brewster Angle, Faraday Effect

8.6.1 Malus's Law

$$I_{\text{out}} = I_{\text{in}} \cos^2\theta, \tag{177}$$

where θ is the angle between the incident linear polarization axis and the polarizer transmission axis.

8.6.2 Brewster's Angle

The Brewster angle θ_B produces purely s-polarized reflected light (perpendicular to the plane of incidence). It satisfies

$$\tan\theta_B = \frac{n_2}{n_1} \iff r_p = \frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)} = 0 \iff \theta_i + \theta_t = \frac{\pi}{2}. \tag{178}$$

8.6.3 Faraday Effect

The rotation angle of linear polarized light is

$$\beta = VBd, \tag{179}$$

where V is the Verdet constant, B the magnetic field strength parallel to propagation, and d the propagation distance.

8.7 Polarization Optical Components and Birefringence

8.7.1 Polarization Optical Components

Linear polarizer (LP):

- Polarized input: $I_{\text{out}} = I_{\text{in}} \cos^2\theta$,
- Unpolarized input: $I_{\text{out}} = I_{\text{in}}/2$.

Wave plates (phase retarders):

1. Half-wave plate (HWP): $\delta = \pi$, $d|n_o - n_e| = \frac{\lambda}{2}$,
2. Quarter-wave plate (QWP): $\delta = \frac{\pi}{2}$, $d|n_o - n_e| = \frac{\lambda}{4}$.

8.7.2 Birefringence

- Ordinary ray (o-ray): obeys Snell's law; $\mathbf{E} \perp$ incident plane (s-polarization); $n_o = \frac{c}{v_o} = \frac{\sin\theta_i}{\sin\theta_o}$ (constant).
- Extraordinary ray (e-ray): does not obey Snell's law; $\mathbf{E} \parallel$ incident plane (p-polarization); $n_e = \frac{c}{v_e} = \frac{\sin\theta_i}{\sin\theta_e}$ (angle-dependent).

Gavin's Lectures on Physics - Theoretical Mechanics



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1 The Principle of Least Action (Hamilton's Principle)

1.1 Action Functional and Variation

The action for a mechanical system with generalized coordinates $q_\alpha(t)$ is defined as

$$S[q(t)] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, dt,$$

where $L = T - V$ is the Lagrangian. The principle of least action states that the physical trajectory is an extremum of S with respect to variations $\delta q_\alpha(t)$ that vanish at the endpoints:

$$\delta q_\alpha(t_1) = \delta q_\alpha(t_2) = 0.$$

Requiring $\delta S = 0$ yields

$$\delta S = \int_{t_1}^{t_2} \sum_\alpha \left(\frac{\partial L}{\partial q_\alpha} \delta q_\alpha + \frac{\partial L}{\partial \dot{q}_\alpha} \delta \dot{q}_\alpha \right) dt = 0.$$

Integrating the second term by parts and using endpoint conditions, we obtain the Euler–Lagrange equations:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_\alpha} \right) - \frac{\partial L}{\partial q_\alpha} = 0. \tag{1}$$

1.2 Relation to Newton's Second Law

For a single particle in Cartesian coordinates, $L = \frac{1}{2} m \dot{\mathbf{r}}^2 - V(\mathbf{r})$. The Euler–Lagrange equation gives

$$m\ddot{\mathbf{r}} = -\nabla V = \mathbf{F},$$

which is Newton's second law. Thus the variational principle contains Newtonian mechanics as a special case.

2 Lagrangian Mechanics and Applications

2.1 Constraints and Generalized Coordinates

A system of N particles has $3N$ coordinates. In the presence of constraints, the number of independent degrees of freedom is reduced. Constraints can be classified in several ways, and their treatment often involves the introduction of generalized coordinates and, when constraint forces are of interest, the method of Lagrange multipliers.

2.1.1 Classification by Form of Constraint Equation

- **Holonomic constraints**: These are constraints that can be expressed as algebraic equations relating the coordinates and time:

  $$f(\mathbf{r}_1, \ldots, \mathbf{r}_N, t) = 0.$$

  They are integrable and reduce the number of degrees of freedom. Example: a rigid rod connecting two particles imposes $|\mathbf{r}_1 - \mathbf{r}_2| = l$.

- **Non-holonomic constraints**: These cannot be written as algebraic relations among coordinates alone. They often appear as inequalities (e.g., a particle confined to a table, $z \geq 0$) or as non-integrable differential relations. Example: a disk rolling without slipping on a plane involves velocity constraints that cannot be integrated to position constraints.

- **Differential constraints**: A special class of non-holonomic constraints are those given by differential forms:

  $$\sum_{\alpha} a_\alpha(q, t)\, dq_\alpha + a_0(q, t)\, dt = 0.$$

  If this form is integrable (i.e., there exists an integrating factor making it an exact differential), then the constraint is actually holonomic; otherwise, it is non-holonomic.

2.1.2 Classification by Time Dependence

- **Scleronomic (stable) constraints**: Constraints that do not depend explicitly on time:

  $$f(\mathbf{r}_1, \ldots, \mathbf{r}_N) = 0.$$

  The configuration space is fixed. Example: a bead on a rigid circular wire.

- **Rheonomic (unstable) constraints**: Constraints that depend explicitly on time:

  $$f(\mathbf{r}_1, \ldots, \mathbf{r}_N, t) = 0.$$

  The constraints move or change with time. Example: a bead on a circular wire whose radius changes with time.

2.1.3 Geometric vs. Kinematic Constraints

- **Geometric constraints**: Restrict the positions of particles (e.g., a particle on a sphere).
- **Kinematic constraints**: Restrict velocities (e.g., rolling condition $\dot{x} - R\dot{\theta} = 0$). If kinematic constraints are integrable, they reduce to geometric (holonomic) constraints.

2.1.4 Generalized Coordinates

If there are k independent holonomic constraints, the number of independent degrees of freedom is $s = 3N - k$. We introduce generalized coordinates $q_1, \ldots, q_s$ such that

$$\mathbf{r}_i = \mathbf{r}_i(q_1, \ldots, q_s, t).$$

Equivalently, for holonomic constraints $f_\alpha(x_A, t) = 0$ with $\alpha = 1, \ldots, 3N - n$, we can write

$$x_A = x_A\left(q_1, \ldots, q_n\right).$$

Virtual displacements $\delta\mathbf{r}_i$ are defined as infinitesimal changes consistent with the constraints at fixed time:

$$\delta\mathbf{r}_i = \sum_\alpha \frac{\partial \mathbf{r}_i}{\partial q_\alpha} \delta q_\alpha.$$

2.1.5 Lagrange Multipliers for Holonomic Constraints

To find the forces of constraint, we treat holonomic constraints $f_\alpha(x_A, t) = 0$ by introducing multipliers λ_α. The modified Lagrangian is

$$L' = L\left(x^A, \dot{x}^A\right) + \lambda_\alpha f_\alpha\left(x^A, t\right).$$

The constraint equations are obtained by variation with respect to λ_α:

$$\frac{\partial L'}{\partial \lambda_\alpha} = f_\alpha\left(x^A, t\right) = 0,$$

and the equations of motion become

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}^A}\right) - \frac{\partial L}{\partial x^A} = \lambda_\alpha \frac{\partial f_\alpha}{\partial x^A}.$$

In terms of generalized coordinates, we can choose

$$x_A \to \begin{cases} q_i, & (i = 1, \ldots, n), \\ f_\alpha, & (\alpha = 1, \ldots, 3N - n), \end{cases}$$

so that the equations reduce to

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = \lambda_\alpha \frac{\partial f_\alpha}{\partial q_i}. \tag{2}$$

The multipliers λ_α are directly related to the **generalized constraint forces**. Specifically, the generalized force of constraint is given by

$$Q_i^{\text{constraint}} = \sum_\alpha \lambda_\alpha \frac{\partial f_\alpha}{\partial q_i}.$$

For example, in the case of a simple pendulum with constraint $f(r - l) = 0$, this formalism yields the tension in the string.

2.2 Remarks on the Lagrangian Formalism

For a system described by generalized coordinates q_i and velocities $\dot{q}_i$, the Lagrangian is $L = L(q_i, \dot{q}_i, t)$. The Euler–Lagrange equations and the definition of canonical momenta are summarized as

$$\begin{cases} \dfrac{d}{dt}\left(\dfrac{\partial L}{\partial \dot{q}_i}\right) - \dfrac{\partial L}{\partial q_i} = 0, \\ p_i = \dfrac{\partial L}{\partial \dot{q}_i}, \end{cases} \qquad \Rightarrow \quad \dot{p}_i = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i}.$$

The Lagrangian is not unique: it can be multiplied by a constant (if the action is scaled) or augmented by a total time derivative of any function of coordinates and time. That is, the following transformations leave the equations of motion invariant:

$$L' = \alpha L, \quad (\alpha \in \mathbb{R}), \qquad L' = L + \frac{df}{dt}.$$

2.3 D'Alembert's Principle and Virtual Work

D'Alembert's principle extends the principle of virtual work to dynamics:

$$\sum_i (\mathbf{F}_i - \dot{\mathbf{p}}_i) \cdot \delta \mathbf{r}_i = 0,$$

where $\mathbf{p}_i = m_i \dot{\mathbf{r}}_i$. For ideal constraints (no work done by constraint forces), this reduces to

$$\sum_i \mathbf{F}_i^{\text{applied}} \cdot \delta \mathbf{r}_i = \sum_\alpha Q_\alpha \delta q_\alpha,$$

with generalized forces

$$Q_\alpha = \sum_i \mathbf{F}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_\alpha}.$$

For static equilibrium, $\dot{\mathbf{p}}_i = 0$, so the virtual work vanishes: $\sum_\alpha Q_\alpha \delta q_\alpha = 0$, hence $Q_\alpha = 0$ for all α.

2.4 Lagrange's Equations from D'Alembert's Principle

D'Alembert's principle for a system of N particles with ideal constraints is

$$\sum_i (\mathbf{F}_i - m_i \ddot{\mathbf{r}}_i) \cdot \delta \mathbf{r}_i = 0,$$

where $\delta \mathbf{r}_i = \sum_\alpha \frac{\partial \mathbf{r}_i}{\partial q_\alpha} \delta q_\alpha$ are virtual displacements. Define generalized forces

$$Q_\alpha = \sum_i \mathbf{F}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_\alpha}.$$

Using the kinetic energy $T = \frac{1}{2} \sum_i m_i \dot{\mathbf{r}}_i^2$, one can show (by expressing $\ddot{\mathbf{r}}_i \cdot \delta \mathbf{r}_i$ in terms of T and its derivatives) that

$$\sum_i m_i \ddot{\mathbf{r}}_i \cdot \delta \mathbf{r}_i = \sum_\alpha \left[\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_\alpha}\right) - \frac{\partial T}{\partial q_\alpha} \right] \delta q_\alpha.$$

Substituting into D'Alembert's principle yields

$$\sum_{\alpha} \left[Q_\alpha - \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_\alpha}\right) + \frac{\partial T}{\partial q_\alpha} \right] \delta q_\alpha = 0.$$

Since the δq_α are independent, we obtain Lagrange's equations of the first kind:

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_\alpha}\right) - \frac{\partial T}{\partial q_\alpha} = Q_\alpha.$$

For Conservative Forces $Q_\alpha = -\partial V/\partial q_\alpha$, and with $L = T - V$, this reduces to the standard Euler–Lagrange equations (1).

For Non-Conservative Forces When non-conservative forces (e.g., friction) are present, we write

$$Q_\alpha = -\frac{\partial V}{\partial q_\alpha} + Q_\alpha^{\text{nc}}.$$

If the dissipative forces are linear in velocities, we introduce the Rayleigh dissipation function

$$\mathcal{F} = \frac{1}{2}\sum_{i,j} c_{ij}\dot{q}_i\dot{q}_j,$$

where the coefficient matrix c_{ij} is taken to be symmetric. Then the generalized non-conservative forces are

$$Q_\alpha^{\text{nc}} = -\frac{\partial \mathcal{F}}{\partial \dot{q}_\alpha} = \sum_i c_{\alpha i}\dot{q}_i.$$

The modified Lagrange equations become

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_\alpha}\right) - \frac{\partial L}{\partial q_\alpha} = Q_\alpha^{\text{nc}} = -\frac{\partial \mathcal{F}}{\partial \dot{q}_\alpha}. \tag{3}$$

2.5 Small Oscillations (Normal Modes)

Near stable equilibrium q_α^0, let $\eta_\alpha = q_\alpha - q_\alpha^0$. Expand the potential and kinetic energy to quadratic order:

$$V = V_0 + \frac{1}{2}\sum_{\alpha,\beta} K_{\alpha\beta}\eta_\alpha\eta_\beta, \qquad T = \frac{1}{2}\sum_{\alpha,\beta} M_{\alpha\beta}\dot{\eta}_\alpha\dot{\eta}_\beta,$$

where

$$K_{\alpha\beta} = \left.\frac{\partial^2 V}{\partial q_\alpha \partial q_\beta}\right|_0$$

and $M_{\alpha\beta}$ is the mass matrix. The Lagrangian is

$$L = \frac{1}{2}\sum_{\alpha,\beta}\left(M_{\alpha\beta}\dot{\eta}_\alpha\dot{\eta}_\beta - K_{\alpha\beta}\eta_\alpha\eta_\beta\right).$$

The equations of motion yield the generalized eigenvalue problem

$$\sum_\beta \left(K_{\alpha\beta} - \omega^2 M_{\alpha\beta}\right) A_\beta = 0. \tag{4}$$

The s eigenfrequencies ω_k determine the normal modes. In terms of normal coordinates ξ_k, the Lagrangian decouples:

$$L = \frac{1}{2}\sum_k \left(\dot{\xi}_k^2 - \omega_k^2 \xi_k^2\right).$$

2.6 Rigid Body Rotation

For a rigid body, the kinetic energy separates into translational and rotational parts via König's theorem:

$$T = \frac{1}{2}MV_{\rm cm}^2 + \frac{1}{2}\boldsymbol{\omega}\cdot\boldsymbol{I}\cdot\boldsymbol{\omega}. \tag{5}$$

Inertia tensor and angular momentum For rigid rotation, $\dot{\boldsymbol{r}}_i' = \boldsymbol{\omega}\times\boldsymbol{r}_i'$ The angular momentum is

$$\begin{aligned}\boldsymbol{L} &= \sum_i \boldsymbol{r}_i' \times m_i\dot{\boldsymbol{r}}_i' \\ &= \sum_i m_i \boldsymbol{r}_i' \times (\boldsymbol{\omega}\times\boldsymbol{r}_i') \\ &= \sum_i m_i \left[\boldsymbol{\omega}\, r_i'^2 - \boldsymbol{r}_i'(\boldsymbol{r}_i'\cdot\boldsymbol{\omega})\right] \\ &= \boldsymbol{I}\cdot\boldsymbol{\omega}.\end{aligned}$$

In component notation:

$$L_\alpha = \sum_\beta I_{\alpha\beta}\omega_\beta.$$

The inertia tensor components are explicitly:

$$I_{xx} = \int (y^2+z^2)\,dm, \qquad I_{yy} = \int (z^2+x^2)\,dm, \qquad I_{zz} = \int (x^2+y^2)\,dm,$$

$$I_{xy} = I_{yx} = -\int xy\,dm, \qquad I_{yz} = I_{zy} = -\int yz\,dm, \qquad I_{zx} = I_{xz} = -\int xz\,dm.$$

In the principal axis frame, the tensor is diagonal with moments I_1, I_2, I_3, and

$$T_{\rm rot} = \frac{1}{2}\left(I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_3^2\right).$$

Euler's equations for torque-free motion In the inertial frame, $d\boldsymbol{L}/dt\big|_{\rm inertial} = \boldsymbol{M}_{\rm ext} = 0$. The time derivative rule for a rotating frame gives

$$\left.\frac{d\boldsymbol{L}}{dt}\right|_{\rm inertial} = \left.\frac{d\boldsymbol{L}}{dt}\right|_{\rm rot} + \boldsymbol{\omega}\times\boldsymbol{L} = 0.$$

In principal axes, $\boldsymbol{L} = (I_1\omega_1, I_2\omega_2, I_3\omega_3)$. The cross product is

$$\boldsymbol{\omega}\times\boldsymbol{L} = \big((I_3-I_2)\omega_2\omega_3,\ (I_1-I_3)\omega_3\omega_1,\ (I_2-I_1)\omega_1\omega_2\big).$$

Separating components yields Euler's equations:

$$\begin{aligned} I_1\dot{\omega}_1 + (I_3 - I_2)\omega_2\omega_3 &= 0, \\ I_2\dot{\omega}_2 + (I_1 - I_3)\omega_3\omega_1 &= 0, \\ I_3\dot{\omega}_3 + (I_2 - I_1)\omega_1\omega_2 &= 0. \end{aligned} \tag{6}$$

The Euler angles (ϕ, θ, ψ) describe the orientation; their time derivatives relate to $\boldsymbol{\omega}$ via standard formulas.

2.7 Central Force and the Runge–Lenz Vector

Effective radial potential For a particle in a central potential $V(r)$, the angular momentum is conserved. In polar coordinates,

$$T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2).$$

Since θ is cyclic, $p_\theta = mr^2\dot{\theta} = L$ is conserved, so $\dot{\theta} = L/(mr^2)$. The total energy is

$$E = \frac{1}{2}m\dot{r}^2 + \frac{L^2}{2mr^2} + V(r).$$

Thus the effective radial motion is governed by the effective potential:

$$V_{\text{eff}}(r) = V(r) + \frac{L^2}{2mr^2}. \tag{7}$$

Binet's orbit equation To eliminate time and directly solve the orbital shape $r(\theta)$, we use the radial force equation:

$$m(\ddot{r} - r\dot{\theta}^2) = F(r).$$

Change variable $u = 1/r$. We have

$$\dot{r} = \frac{dr}{d\theta}\dot{\theta} = -\frac{1}{u^2}\frac{du}{d\theta}\cdot\frac{Lu^2}{m} = -\frac{L}{m}\frac{du}{d\theta},$$

$$\ddot{r} = \frac{d\dot{r}}{d\theta}\dot{\theta} = -\frac{L^2u^2}{m^2}\frac{d^2u}{d\theta^2}, \qquad r\dot{\theta}^2 = \frac{L^2u^3}{m^2}.$$

Substituting into the radial equation gives Binet's equation:

$$\frac{d^2u}{d\theta^2} + u = -\frac{m}{L^2u^2}F\left(\frac{1}{u}\right). \tag{8}$$

Runge–Lenz vector for the Kepler problem For the Kepler potential $V(r) = -\alpha/r$, the Runge–Lenz vector is defined as

$$\boldsymbol{A} = \boldsymbol{p} \times \boldsymbol{L} - m\alpha\frac{\boldsymbol{r}}{r}. \tag{9}$$

To prove its conservation, differentiate with respect to time:

$$\frac{d\boldsymbol{A}}{dt} = \dot{\boldsymbol{p}} \times \boldsymbol{L} + \boldsymbol{p} \times \dot{\boldsymbol{L}} - m\alpha\frac{d}{dt}\left(\frac{\boldsymbol{r}}{r}\right).$$

For a central force, $\dot{\boldsymbol{L}} = 0$, so the second term vanishes. Newton's law gives

$$\dot{\boldsymbol{p}} = \boldsymbol{F} = -\frac{\alpha}{r^3}\boldsymbol{r}.$$

Thus

$$\dot{\boldsymbol{p}} \times \boldsymbol{L} = -\frac{\alpha}{r^3}\boldsymbol{r} \times (\boldsymbol{r} \times \boldsymbol{p}) = \alpha\left(\frac{\boldsymbol{p}}{r} - \frac{\boldsymbol{r}(\boldsymbol{r}\cdot\boldsymbol{p})}{r^3}\right).$$

The derivative of the unit vector is

$$\frac{d}{dt}\left(\frac{\boldsymbol{r}}{r}\right) = \frac{\boldsymbol{p}}{mr} - \frac{\boldsymbol{r}(\boldsymbol{r}\cdot\boldsymbol{p})}{mr^3}.$$

Therefore

$$-m\alpha\frac{d}{dt}\left(\frac{\boldsymbol{r}}{r}\right) = \alpha\left(-\frac{\boldsymbol{p}}{r} + \frac{\boldsymbol{r}(\boldsymbol{r}\cdot\boldsymbol{p})}{r^3}\right).$$

Summing all terms yields complete cancellation, hence

$$\frac{d\boldsymbol{A}}{dt} = 0.$$

The vector $\boldsymbol{A}$ points from the focus to the perihelion, and its magnitude is proportional to the orbital eccentricity.

3 Noether's Theorem and Symmetries

Noether's theorem establishes a profound connection between symmetries of a physical system and its conserved quantities. For a Lagrangian system, any continuous symmetry of the action (or of the Lagrangian) implies the existence of a corresponding constant of motion. Below we formalize this idea and illustrate its most important applications.

A function $F(q_i, \dot{q}_i, t)$ is a constant of motion (conserved quantity) if

$$0 = \frac{dF}{dt} = \sum_{j=1}^{n} \left(\frac{\partial F}{\partial q_j} \dot{q}_j + \frac{\partial F}{\partial \dot{q}_j} \ddot{q}_j \right) + \frac{\partial F}{\partial t}.$$

The Hamiltonian is defined as

$$H = \sum_j \dot{q}_j \frac{\partial L}{\partial \dot{q}_j} - L.$$

If $\partial L/\partial t = 0$, then

$$\frac{dH}{dt} = \sum_j \left[\ddot{q}_j \frac{\partial L}{\partial \dot{q}_j} + \dot{q}_j \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} \dot{q}_j - \frac{\partial L}{\partial \dot{q}_j} \ddot{q}_j \right] = 0.$$

3.1 Ignorable / Cyclic Coordinate: q_j

If a coordinate q_j does not appear explicitly in the Lagrangian,

$$\begin{cases} \dfrac{\partial L}{\partial q_j} = 0, \\ p_j = \dfrac{\partial L}{\partial \dot{q}_j}, \end{cases} \qquad \Rightarrow \quad \frac{d}{dt} p_j = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) = \frac{\partial L}{\partial q_j} = 0.$$

Thus, a coordinate that does not appear explicitly in the Lagrangian (a cyclic coordinate) gives rise to a conserved canonical momentum.

3.2 Noether's Theorem

Consider a one-parameter family of transformations:

$$\begin{cases} q_i(t) \to Q_i(s, t), \quad s \in \mathbb{R}, \\ Q_i(s = 0, t) = q_i(t). \end{cases}$$

If the transformation leaves the Lagrangian invariant (up to a total time derivative), i.e.,

$$\frac{\partial}{\partial s} L\Big(Q_i(s, t), \dot{Q}_i(s, t), t \Big) = 0,$$

then this transformation is a continuous symmetry of the Lagrangian L, and there exists a conserved quantity F with $dF/dt = 0$.

Proof: Evaluating the derivative at $s = 0$:

$$\begin{aligned}
0 &= \left.\frac{\partial L}{\partial s}\right|_{s=0} \\
&= \left[\frac{\partial L}{\partial Q_i}\frac{\partial Q_i}{\partial s} + \frac{\partial L}{\partial \dot{Q}_i}\frac{\partial \dot{Q}_i}{\partial s}\right]_{s=0} \\
&= \left.\frac{\partial L}{\partial q_i}\frac{\partial Q_i}{\partial s}\right|_{s=0} + \left.\frac{\partial L}{\partial \dot{q}_i}\frac{\partial \dot{Q}_i}{\partial s}\right|_{s=0} \\
&= \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right)\left.\frac{\partial Q_i}{\partial s}\right|_{s=0} + \frac{\partial L}{\partial \dot{q}_i}\frac{d}{dt}\left(\left.\frac{\partial Q_i}{\partial s}\right|_{s=0}\right) \\
&= \frac{d}{dt}\left(\left.\frac{\partial L}{\partial \dot{q}_i}\frac{\partial Q_i}{\partial s}\right|_{s=0}\right).
\end{aligned}$$

Hence the conserved quantity is

$$F = \left.\frac{\partial L}{\partial \dot{q}_i}\frac{\partial Q_i}{\partial s}\right|_{s=0}, \qquad \frac{dF}{dt} = 0. \tag{10}$$

3.2.1 Homogeneity of Space ⇒ Conservation of Momentum

For a free particle,

$$L = \frac{1}{2}m\dot{\mathbf{r}}^2 - V(\mathbf{r}).$$

Under a spatial translation $\mathbf{r} \to \mathbf{r}' = \mathbf{r} + s\hat{n}$, the Lagrangian is invariant. Then

$$0 = \frac{\partial L}{\partial s} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\mathbf{r}}}\cdot\hat{n}\right) = \frac{d}{dt}\left(m\dot{\mathbf{r}}\cdot\hat{n}\right) = \frac{d}{dt}\left(\mathbf{p}\cdot\hat{n}\right).$$

Since $\hat{n}$ is arbitrary,

$$\Rightarrow \frac{d\mathbf{p}}{dt} = 0.$$

3.2.2 Isotropy of Space ⇒ Conservation of Angular Momentum

Under an infinitesimal rotation $\mathbf{r} \to \mathbf{r}' = \mathbf{r} + \alpha\left(\hat{n}\times\mathbf{r}\right)$ (with α the parameter), invariance of the Lagrangian gives

$$\begin{aligned}
0 &= \frac{\partial L}{\partial \alpha} \\
&= \frac{d}{dt}\left[\frac{\partial L}{\partial \dot{\mathbf{r}}}\cdot\left(\hat{n}\times\mathbf{r}\right)\right] \\
&= \frac{d}{dt}\left[\mathbf{p}\cdot\left(\hat{n}\times\mathbf{r}\right)\right] \\
&= \frac{d}{dt}\left[\hat{n}\cdot\left(\mathbf{r}\times\mathbf{p}\right)\right] \\
&= \frac{d}{dt}\left(\hat{n}\cdot\mathbf{L}\right).
\end{aligned}$$

Since $\hat{n}$ is arbitrary,

$$\Rightarrow \frac{d\mathbf{L}}{dt} = 0.$$

3.2.3 Homogeneity of Time ⇒ Conservation of Energy

If the Lagrangian does not depend explicitly on time, i.e., $\partial L/\partial t = 0$, then time translation $t \to t' = t + s$ is a symmetry. This implies

$$\frac{\partial L}{\partial t} = 0 \quad \Rightarrow \quad \frac{dH}{dt} = \frac{d}{dt}\left[\sum_i \dot{q}_i \left(\frac{\partial L}{\partial \dot{q}_i}\right) - L\right] = 0,$$

so the Hamiltonian (energy) is conserved.

4 Hamiltonian Mechanics and Applications

4.1 Hamilton's Canonical Equations

Given the Lagrangian $L(q, \dot{q}, t)$, define canonical momenta

$$p_\alpha = \frac{\partial L}{\partial \dot{q}_\alpha}.$$

The Hamiltonian is obtained by Legendre transformation:

$$H(q, p, t) = \sum_\alpha p_\alpha \dot{q}_\alpha - L.$$

Then the equations of motion take the symmetric form

$$\dot{q}_\alpha = \frac{\partial H}{\partial p_\alpha}, \qquad \dot{p}_\alpha = -\frac{\partial H}{\partial q_\alpha}. \tag{11}$$

A coordinate q_α that does not appear explicitly in H is cyclic; its conjugate momentum p_α is conserved.

4.2 Poisson Brackets and Poisson's Theorem

For any two functions $f(q, p, t)$ and $g(q, p, t)$, their Poisson bracket is defined as

$$\{f, g\} = \sum_{\alpha=1}^{s} \left(\frac{\partial f}{\partial q_\alpha}\frac{\partial g}{\partial p_\alpha} - \frac{\partial f}{\partial p_\alpha}\frac{\partial g}{\partial q_\alpha} \right).$$

The time evolution of any function f is given by

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \{f, H\}. \tag{12}$$

Poisson's theorem states that if f and g are constants of motion (i.e., their time derivatives vanish) **and** they do not depend explicitly on time ($\partial f/\partial t = \partial g/\partial t = 0$), then their Poisson bracket $\{f, g\}$ is also a constant of motion. This property is essential for integrability.

4.3 Liouville's Theorem

The volume of a region in phase space is invariant under Hamiltonian flow:

$$V(R) = \int_R dq_1 \cdots dq_s \, dp_1 \cdots dp_s.$$

Proof: The phase-space flow velocity is

$$\boldsymbol{v} = (\dot{q}_1, \ldots, \dot{q}_s, \dot{p}_1, \ldots, \dot{p}_s).$$

Its divergence is

$$\nabla \cdot \boldsymbol{v} = \sum_{\alpha=1}^{s} \left(\frac{\partial \dot{q}_\alpha}{\partial q_\alpha} + \frac{\partial \dot{p}_\alpha}{\partial p_\alpha} \right).$$

Using Hamilton's canonical equations,

$$\dot{q}_\alpha = \frac{\partial H}{\partial p_\alpha}, \qquad \dot{p}_\alpha = -\frac{\partial H}{\partial q_\alpha},$$

we have

$$\frac{\partial \dot{q}_\alpha}{\partial q_\alpha} = \frac{\partial^2 H}{\partial q_\alpha \partial p_\alpha}, \qquad \frac{\partial \dot{p}_\alpha}{\partial p_\alpha} = -\frac{\partial^2 H}{\partial p_\alpha \partial q_\alpha}.$$

Since mixed partial derivatives commute, each pair cancels:

$$\frac{\partial \dot{q}_\alpha}{\partial q_\alpha} + \frac{\partial \dot{p}_\alpha}{\partial p_\alpha} = 0 \quad \Rightarrow \quad \nabla \cdot \boldsymbol{v} = 0.$$

By Reynolds transport theorem,

$$\frac{dV}{dt} = \int_R \nabla \cdot \boldsymbol{v} \, d\Gamma = 0.$$

Thus the Hamiltonian flow is incompressible in phase space, and the density of representative points is constant along trajectories.

4.4 Canonical Transformations

A transformation $(q, p) \to (Q, P)$ is canonical if it preserves the form of Hamilton's equations. The fundamental condition for a canonical transformation is that there exists a generating function F such that

$$\sum_\alpha p_\alpha dq_\alpha - H dt = \sum_\alpha P_\alpha dQ_\alpha - K dt + dF. \tag{13}$$

Type-1 generating function $F_1(q, Q, t)$ For a generating function depending on the old coordinates q and new coordinates Q, the total differential is

$$dF_1 = \sum_\alpha \frac{\partial F_1}{\partial q_\alpha} dq_\alpha + \sum_\alpha \frac{\partial F_1}{\partial Q_\alpha} dQ_\alpha + \frac{\partial F_1}{\partial t} dt.$$

Substituting into the fundamental condition and grouping independent differentials dq_α, dQ_α, and dt, their coefficients must vanish individually. This yields the transformation rules:

$$p_\alpha = \frac{\partial F_1}{\partial q_\alpha}, \qquad P_\alpha = -\frac{\partial F_1}{\partial Q_\alpha}, \qquad K = H + \frac{\partial F_1}{\partial t}.$$

Type-2 generating function $F_2(q, P, t)$ A Legendre transform from F_1 gives $F_2 = F_1 + \sum P_\alpha Q_\alpha$. The transformation rules become

$$p_\alpha = \frac{\partial F_2}{\partial q_\alpha}, \qquad Q_\alpha = \frac{\partial F_2}{\partial P_\alpha}, \qquad K = H + \frac{\partial F_2}{\partial t}.$$

This form is particularly useful for Hamilton-Jacobi theory.

Type-3 generating function $F_3(p, Q, t)$ A Legendre transform from F_1 gives $F_3 = F_1 - \sum p_\alpha q_\alpha$. The total differential is

$$dF_3 = -\sum_\alpha q_\alpha dp_\alpha - \sum_\alpha P_\alpha dQ_\alpha + (K - H)dt = \sum_\alpha \frac{\partial F_3}{\partial p_\alpha} dp_\alpha + \sum_\alpha \frac{\partial F_3}{\partial Q_\alpha} dQ_\alpha + \frac{\partial F_3}{\partial t} dt.$$

Thus the transformation rules are

$$q_\alpha = -\frac{\partial F_3}{\partial p_\alpha}, \qquad P_\alpha = -\frac{\partial F_3}{\partial Q_\alpha}, \qquad K = H + \frac{\partial F_3}{\partial t}.$$

Type-4 generating function $F_4(p, P, t)$ Similarly, $F_4 = F_1 - \sum p_\alpha q_\alpha + \sum P_\alpha Q_\alpha$ gives

$$dF_4 = -\sum_\alpha q_\alpha dp_\alpha + \sum_\alpha Q_\alpha dP_\alpha + (K - H)dt = \sum_\alpha \frac{\partial F_4}{\partial p_\alpha} dp_\alpha + \sum_\alpha \frac{\partial F_4}{\partial P_\alpha} dP_\alpha + \frac{\partial F_4}{\partial t} dt,$$

yielding

$$q_\alpha = -\frac{\partial F_4}{\partial p_\alpha}, \qquad Q_\alpha = \frac{\partial F_4}{\partial P_\alpha}, \qquad K = H + \frac{\partial F_4}{\partial t}.$$

4.5 Hamilton-Jacobi Theory

If we find a canonical transformation such that the new Hamiltonian $K \equiv 0$, then the new coordinates and momenta are constants of motion. The type-2 generating function $S(q, P, t)$ satisfies the Hamilton-Jacobi equation:

$$\frac{\partial S}{\partial t} + H\left(q, \frac{\partial S}{\partial q}, t\right) = 0. \tag{14}$$

Derivation and physical interpretation We seek a canonical transformation with new Hamiltonian $K = 0$. Using a type-2 generating function $S(q, \alpha, t)$, where $P_\alpha = \alpha_\alpha$ are constants, the canonical transformation rule gives $K = H + \partial S/\partial t$. Setting $K = 0$ and substituting $p_\alpha = \partial S/\partial q_\alpha$ into H directly produces the Hamilton-Jacobi equation.

To verify that S is the classical action, compute its total time derivative:

$$\frac{dS}{dt} = \sum \frac{\partial S}{\partial q_\alpha} \dot{q}_\alpha + \frac{\partial S}{\partial t} = \sum p_\alpha \dot{q}_\alpha - H = L.$$

Integrating gives $S = \int L\, dt$, confirming that S is indeed the action functional.

A complete solution $S(q, \alpha, t)$ with s constants α_α (where $\alpha_\alpha = P_\alpha$) gives the trajectories through

$$p_\alpha = \frac{\partial S}{\partial q_\alpha}, \qquad \beta_\alpha = \frac{\partial S}{\partial \alpha_\alpha},$$

where β_α are new constants (the new coordinates Q_α).

Time-independent Hamilton-Jacobi equation For time-independent Hamiltonians ($\partial H/\partial t = 0$), we separate variables:

$$S = W(q, \alpha) - Et.$$

Substituting into the Hamilton-Jacobi equation yields the time-independent equation:

$$H\left(q, \frac{\partial W}{\partial q}\right) = E,$$

where W is Hamilton's characteristic function.

4.6 Action-Angle Variables

4.6.1 The Simple Harmonic Oscillator

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2 \tag{1}$$

$$\Rightarrow \begin{cases} \dot{q} = \dfrac{\partial H}{\partial p} = \dfrac{p}{m}, \\ \dot{p} = -\dfrac{\partial H}{\partial q} = -m\omega^2 q \end{cases} \tag{2}$$

Change variable $(q, p) \to (\theta, I)$:

$$\begin{cases} q = \sqrt{\dfrac{2I}{m\omega}}\ \sin\theta, \\ p = \sqrt{2Im\omega}\ \cos\theta \end{cases} \tag{3}$$

$$H = \omega I \tag{4}$$

$$\begin{cases} \dot{\theta} = \dfrac{\partial H}{\partial I} = \omega, \\ \dot{I} = -\dfrac{\partial H}{\partial \theta} = 0 \end{cases} \tag{5}$$

4.6.2 Liouville's Theorem on Integrable systems

If we can find n mutually Poisson commuting constants of motion $I_1, I_2, \cdots, I_n$, then this implies the existence of angle-action variables (θ_i, I_i), and the system is integrable.

$$\{I_i, I_j\} = 0 \tag{6}$$

4.6.3 Action-Angle Variables for 1d Systems

$$H = \frac{p^2}{2m} + V(q), \qquad H = H(I) = E \tag{7}$$

Choose I as the phase integral:

$$I = \frac{1}{2\pi}\oint p\, dq \tag{8}$$

$$\begin{cases} \dfrac{dI}{dE} = \dfrac{1}{2\pi}\dfrac{d}{dE}\oint p\, dq, \\ E = \dfrac{p^2}{2m} + V(q), \\ p = m\dfrac{dq}{dt} \end{cases} \tag{9}$$

$$\Rightarrow \ \frac{dI}{dE} = \frac{1}{2\pi}\oint dt = \frac{1}{2\pi}\cdot T = \frac{1}{\omega} \tag{10}$$

$$\Rightarrow \begin{cases} \dot{\theta} = \dfrac{\partial H}{\partial I} = \left(\dfrac{dI}{dE}\right)^{-1} = \omega, \\ \dot{I} = -\dfrac{\partial H}{\partial \theta} = 0 \end{cases} \tag{11}$$

$$t = \frac{d}{dE}\int p\,dq \tag{12}$$

$$\theta = \omega t = \frac{dE}{dI}\cdot\frac{d}{dE}\int p\,dq = \frac{d}{dI}\int p\,dq \tag{13}$$

$$\Rightarrow \begin{cases} \theta = \dfrac{d}{dI}\int p\,dq, \\ I = \dfrac{1}{2\pi}\oint p\,dq \end{cases} \tag{14}$$

For higher dimensions, assuming separability, one usually defines each action as the line integral over the corresponding closed orbit:

$$I_i = \frac{1}{2\pi}\oint_{\Gamma_i} p_i\,dq_i \tag{15}$$

where Γ_i are the periods of the invariant tori. In more invariant form:

$$\begin{cases} \theta_i = \dfrac{\partial}{\partial I_i}\int_{\Gamma_i}\sum_j p_j\,dq_j, \\ I_i = \dfrac{1}{2\pi}\oint_{\Gamma_i}\sum_j p_j\,dq_j \end{cases} \tag{16}$$

with Γ_i being the periods of the invariant tori.

4.6.4 Action-Angle Variables for the Kepler Problem

$$\begin{cases} H = E = \dfrac{1}{2}m\dot{r}^2 + \dfrac{1}{2}mr^2\dot{\phi}^2 - \dfrac{G\,Mm}{r}, \\ k = G\,Mm, \\ p_r = m\dot{r}, \qquad p_\phi = L_\phi = mr^2\dot{\phi} \end{cases} \tag{17}$$

$$\Rightarrow\ H = E = \frac{p_r^2}{2m} + \frac{p_\phi^2}{2mr^2} - \frac{k}{r} \tag{18}$$

For ϕ

$$\begin{cases} \dot{\phi} = \dfrac{\partial H}{\partial p_\phi} = \dfrac{p_\phi}{mr^2}, \\ \dot{p}_\phi = -\dfrac{\partial H}{\partial \phi} = 0 \end{cases} \tag{19}$$

$$I_\phi = \frac{1}{2\pi}\oint p_\phi\,d\phi = p_\phi\cdot\frac{1}{2\pi}\oint d\phi = p_\phi \tag{20}$$

$$\begin{cases} \dot{\phi} = \dfrac{I_\phi}{mr^2}, \\ \dot{I}_\phi = 0 \end{cases} \tag{21}$$

For r

$$\begin{cases} \dot{r} = \dfrac{\partial H}{\partial p_r} = \dfrac{p_r}{m}, \\ \dot{p}_r = -\dfrac{\partial H}{\partial r} = -\dfrac{p_\phi^2}{mr^3} + \dfrac{k}{r^2} \end{cases} \tag{22}$$

The action for the radial motion is

$$I_r = \frac{1}{2\pi} \oint p_r \, dr = \frac{1}{\pi} \int_{r_{min}}^{r_{max}} p_r \, dr \tag{23}$$

with

$$E = \frac{p_r^2}{2m} + \frac{I_\phi^2}{2mr^2} - \frac{k}{r} = -|E| \tag{24}$$

Thus

$$p_r = \frac{\sqrt{2m|E|}}{r} \sqrt{-r^2 + \frac{k}{|E|} r - \frac{I_\phi^2}{2m|E|}} \tag{25}$$

and consequently

$$I_r = \frac{1}{\pi} \int_{r_{min}}^{r_{max}} \frac{\sqrt{2m|E|}}{r} \cdot \sqrt{-r^2 + \frac{k}{|E|} r - \frac{I_\phi^2}{2m|E|}} \, dr \tag{26}$$

The turning points satisfy

$$\begin{cases} r_{max} \cdot r_{min} = \dfrac{I_\phi^2}{2m|E|}, \\ r_{min} + r_{max} = \dfrac{k}{|E|} \end{cases} \tag{27}$$

The integral evaluates to

$$I_r = \frac{\sqrt{2m|E|}}{\pi} \int_{r_{min}}^{r_{max}} \frac{1}{r} \sqrt{(r_{max} - r)(r - r_{min})} \, dr \tag{28}$$

$$= \frac{\sqrt{2m|E|}}{\pi} \left[\frac{\pi}{2} \big(r_{min} + r_{max} \big) - \pi \sqrt{r_{min} r_{max}} \right] \tag{29}$$

$$= \frac{k}{2} \sqrt{\frac{2m}{|E|}} - I_\phi \tag{30}$$

Solving for the energy:

$$H = E = -|E| = -\frac{mk^2}{2(I_r + I_\phi)^2} \tag{31}$$

The frequencies are given by

$$\begin{cases} \dot{\theta}_r = \dfrac{\partial H}{\partial I_r}, \\ \dot{I}_r = -\dfrac{\partial H}{\partial \theta_r} = 0 \end{cases} \tag{32}$$

and since

$$H(I_r, I_\phi) = -\frac{mk^2}{2(I_r + I_\phi)^2} \tag{33}$$

is symmetric in I_r and I_ϕ, the motion is degenerate.

4.7 Adiabatic Invariants

Consider a system whose Hamiltonian depends on a slowly varying parameter $\lambda(t)$:

$$H = H(q, p, \lambda(t)).$$

Let τ be the timescale of variation of λ and T the period of the system. The adiabatic condition is $\tau \gg T$. Under this condition, the action variable

$$I = \frac{1}{2\pi} \oint p\, dq$$

is approximately conserved, i.e., it is an adiabatic invariant.

To show this, define

$$I = I(E, \lambda) = \frac{1}{2\pi} \oint p\, dq.$$

Then

$$\dot{I} = \left.\frac{\partial I}{\partial E}\right|_{\lambda} \dot{E} + \left.\frac{\partial I}{\partial \lambda}\right|_{E} \dot{\lambda}.$$

Using

$$\left.\frac{\partial I}{\partial E}\right|_{\lambda} = \frac{T}{2\pi}, \qquad \left.\frac{\partial I}{\partial \lambda}\right|_{E} = -\frac{1}{2\pi} \int_0^T \left.\frac{\partial H}{\partial \lambda}\right|_{E} dt',$$

and the equations of motion, one obtains

$$\dot{I} = \frac{\dot{\lambda}}{2\pi} \left[T \left.\frac{\partial H}{\partial \lambda}\right|_{E} - \int_0^T \left.\frac{\partial H}{\partial \lambda}\right|_{E} dt' \right].$$

Averaging over one fast period gives

$$\langle \dot{I} \rangle = 0,$$

so the action is conserved on average. Thus the action variable is an adiabatic invariant.

The Bohr-Sommerfeld quantization condition states that for integrable systems,

$$I = \frac{1}{2\pi} \oint p\, dq = n\hbar, \qquad n \in \mathbb{Z}.$$

This quantization rule played a crucial role in the old quantum theory and remains a cornerstone of semiclassical physics.

Gavin's Lectures on Physics - Electrodynamics



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1 Maxwell Equations

Maxwell equations can be expressed in 3 equivalent formulations.

1.1 Maxwell Equations

$$\begin{cases} \nabla \cdot \mathbf{D} = \rho \\ \nabla \times \mathbf{E} = -\dfrac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{H} = \mathbf{J} + \dfrac{\partial \mathbf{D}}{\partial t} \end{cases} \tag{1}$$

In linear, homogeneous, isotropic media

$$\begin{cases} \mathbf{J} = \sigma \mathbf{E} \\ \mathbf{D} = \varepsilon \mathbf{E} = \varepsilon_0 \varepsilon_r \mathbf{E} = \varepsilon_0 \mathbf{E} + \mathbf{P} = \varepsilon_0 (1 + \chi_e) \mathbf{E} \\ \mathbf{B} = \mu \mathbf{H} = \mu_0 \mu_r \mathbf{H} = \mu_0 (\mathbf{H} + \mathbf{M}) = \mu_0 (1 + \chi_m) \mathbf{H} \end{cases} \tag{2}$$

Boundary Conditions

$$\begin{cases} \hat{n} \cdot (\mathbf{D_2} - \mathbf{D_1}) = \sigma_f \\ \hat{n} \cdot (\mathbf{B_2} - \mathbf{B_1}) = 0 \\ \hat{n} \times (\mathbf{E_2} - \mathbf{E_1}) = 0 \\ \hat{n} \times (\mathbf{H_2} - \mathbf{H_1}) = \mathbf{k_f} \end{cases} \tag{3}$$

$\sigma_f = \dfrac{dQ_f}{ds}$, free charge surface density.
$k_f = \dfrac{dI_f}{dl}$, free current linear density.
$\hat{n}$ points from medium 1 to medium 2.

1.2 d'Alembert Equation

Relation of potentials and electromagnetic field:

$$\begin{cases} \mathbf{E} = -\nabla \phi - \dfrac{\partial \mathbf{A}}{\partial t} \\ \mathbf{B} = \nabla \times \mathbf{A} \end{cases} \tag{4}$$

d'Alembert Equation

$$\Box A^\mu = -\mu_0 J^\mu \tag{5}$$

$$\begin{cases} \Box = -\dfrac{1}{c^2}\dfrac{\partial^2}{\partial t^2} + \nabla^2 = \partial_\mu \partial^\mu \\ A^\mu = (\phi/c,\ \mathbf{A}) \\ J^\mu = (\rho c,\ \mathbf{J}) \\ \eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{cases}$$

$$\begin{cases} \partial_\mu = \left(\frac{1}{c}\frac{\partial}{\partial t},\ \nabla \right) \\ \partial^\mu = \left(-\frac{1}{c}\frac{\partial}{\partial t},\ \nabla \right) \end{cases}$$

Continuity Equation:

$$\partial_\mu J^\mu = \frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0 \tag{6}$$

Lorenz gauge:

$$\partial_\mu A^\mu = \frac{1}{c^2}\frac{\partial \phi}{\partial t} + \nabla \cdot \mathbf{A} = 0 \tag{7}$$

Retarded Potential

$$\begin{cases} t_{\text{ret}} = t - |\mathbf{r} - \mathbf{r}'|/c \\ A^\mu(\mathbf{r}, t) = \frac{\mu_0}{4\pi}\int \frac{J^\mu(\mathbf{r}', t_{\text{ret}})}{|\mathbf{r} - \mathbf{r}'|} d^3 r' \end{cases} \tag{8}$$

1.3 Covariant Maxwell Equations

Electromagnetic field tensor:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{9}$$

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & B_z & -B_y \\ E_y/c & -B_z & 0 & B_x \\ E_z/c & B_y & -B_x & 0 \end{pmatrix} \tag{10}$$

$F^{\mu\nu}: F^{0i} = F_{i0},\ F^{i0} = F_{0i},\ F^{ij} = F_{ij}$

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu \tag{11}$$

Covariant Maxwell Equations

$$\begin{cases} \partial_\mu F^{\mu\nu} = \Box A^\nu = -\mu_0 J^\nu \\ \partial_{[\mu} F_{\nu\rho]} = \partial_\mu F_{\nu\rho} + \partial_\rho F_{\mu\nu} + \partial_\nu F_{\rho\mu} = 0 \end{cases} \tag{12}$$

2 Dielectric and Magnetic Media

2.1 Electric Polarization

Molecular electric dipole moment:

$$\mathbf{p} = q\mathbf{l} \tag{13}$$

Electric polarization intensity vector:

$$\mathbf{P} = \frac{\sum_i \mathbf{p}_i}{\Delta V} = \frac{N_{\text{total}}\, q\mathbf{l}}{\Delta V} = Nq\mathbf{l} \tag{14}$$

Bound charge: charge distribution arising from polarization

$$\begin{aligned} Q_p &= -\oint Nq\mathbf{l}\cdot d\mathbf{S} = -\oint \mathbf{P}\cdot d\mathbf{S} \\ &= -\int \nabla\cdot\mathbf{P}\, dV \end{aligned}$$

Polarized volume bound charge density:

$$\rho_p = \frac{dQ_p}{dV} = -\nabla\cdot\mathbf{P} \tag{15}$$

Polarized surface bound charge surface density: between medium 2 and medium 1, $\hat{n}$: from $1 \to 2$

$$\sigma_p = \frac{dQ_{p2} - dQ_{p1}}{dS} = -\frac{(\mathbf{P}_2 - \mathbf{P}_1)\cdot d\mathbf{S}}{dS} = -\hat{n}\cdot(\mathbf{P}_2 - \mathbf{P}_1) \tag{16}$$

Free charge density:

$$\begin{aligned} \rho_f &= \varepsilon_0 \nabla\cdot\mathbf{E} - \rho_p \\ &= \varepsilon_0 \nabla\cdot\mathbf{E} + \nabla\cdot\mathbf{P} \\ &= \nabla\cdot(\varepsilon_0\mathbf{E} + \mathbf{P}) = \nabla\cdot\mathbf{D} \end{aligned}$$

2.2 Magnetization

i: small loop current; $\mathbf{a}$: loop area vector.

Magnetic moment:

$$\mathbf{m} = i\mathbf{a} \tag{17}$$

Magnetization intensity:

$$\mathbf{M} = \frac{\sum_i \mathbf{m}_i}{\Delta V} = \frac{N_{\text{total}}\, i\mathbf{a}}{\Delta V} = Ni\mathbf{a} \tag{18}$$

Total magnetization current:

$$\begin{aligned} I_m &= \oint Ni\mathbf{a}\cdot d\mathbf{l} = \oint \mathbf{M}\cdot d\mathbf{l} \\ &= \int (\nabla\times\mathbf{M})\cdot d\mathbf{S} \end{aligned}$$

Magnetization current density:

$$\mathbf{J}_m = \nabla \times \mathbf{M} \tag{19}$$

Magnetization surface current density:

$$\alpha_m = \hat{n} \times (\mathbf{M}_2 - \mathbf{M}_1) \tag{20}$$

Polarization current density:

$$\mathbf{J}_p = \frac{\sum_i q_i \mathbf{v}_i}{\Delta V} = \frac{\partial \mathbf{P}}{\partial t} \tag{21}$$

Free current density:

$$\begin{aligned}
\mathbf{J}_f &= \frac{1}{\mu_0}\nabla \times \mathbf{B} - \mathbf{J}_m - \mathbf{J}_p - \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \\
&= \frac{1}{\mu_0}\nabla \times \mathbf{B} - \nabla \times \mathbf{M} - \frac{\partial \mathbf{P}}{\partial t} - \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \\
&= \nabla \times \left(\frac{1}{\mu_0}\mathbf{B} - \mathbf{M} \right) - \frac{\partial}{\partial t}\left(\varepsilon_0 \mathbf{E} + \mathbf{P}\right) \\
&= \nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t}
\end{aligned}$$

3 Action Principle

3.1 Action of a Charged Particle in EM Field

Action

$$S = \int L dt = \int \left(-mc^2 d\tau + qA_\mu dx^\mu\right) = \int \left(-\frac{mc^2}{\gamma} + qA_\mu \dot{x}^\mu\right) dt, \quad (\dot{x}^\mu = (c, \mathbf{v})) \tag{22}$$

Lagrangian

$$\begin{aligned} L &= -\frac{mc^2}{\gamma} + qA_\mu \dot{x}^\mu \\ &= -mc^2\sqrt{1 - \frac{v^2}{c^2}} + q\mathbf{v}\cdot\mathbf{A} - q\phi \end{aligned} \tag{23}$$

Euler-Lagrange equation

$$\frac{d}{dt}\left(\frac{\partial L}{\partial v_i}\right) = \frac{\partial L}{\partial x_i} \tag{24}$$

$$\iff \begin{cases} \dfrac{d\mathbf{p}}{dt} = (q\mathbf{E} + q\mathbf{v}\times\mathbf{B}) \\ \mathbf{p} = \dfrac{m\mathbf{v}}{\sqrt{1-\frac{v^2}{c^2}}} = \gamma m\mathbf{v} \\ \gamma = \dfrac{1}{\sqrt{1-\frac{v^2}{c^2}}} = \dfrac{dt}{d\tau} \end{cases} \tag{25}$$

Canonical Momentum:

$$\mathbf{P} = \frac{\partial L}{\partial \mathbf{v}} = \mathbf{p} + q\mathbf{A} \tag{26}$$

Hamiltonian

$$\begin{aligned} H(\mathbf{x}, \mathbf{P}) &= \mathbf{P}\cdot\mathbf{v} - L \\ &= \gamma mc^2 + q\phi \\ &= \sqrt{(\mathbf{P} - q\mathbf{A})^2 c^2 + m^2c^4} + q\phi \end{aligned} \tag{27}$$

Hamilton canonical equations

$$\begin{cases} \mathbf{v} = \dfrac{\partial H}{\partial \mathbf{P}} \\ \dot{\mathbf{P}} = -\dfrac{\partial H}{\partial \mathbf{x}} = -\nabla H \end{cases} \tag{28}$$

$$\iff \frac{d\mathbf{p}}{dt} = q\mathbf{E} + q\mathbf{v}\times\mathbf{B} \tag{29}$$

3.2 Action of the Electromagnetic Field

Lagrangian density

$$\begin{aligned} \mathcal{L} &= -\frac{1}{4\mu_0} F_{\mu\nu}F^{\mu\nu} + A_\mu J^\mu \\ &= \frac{1}{2}\left(\varepsilon_0 E^2 - \frac{1}{\mu_0}B^2\right) - \rho\phi + \mathbf{J}\cdot\mathbf{A}. \end{aligned} \tag{30}$$

Euler-Lagrange equation

$$\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} - \frac{\partial \mathcal{L}}{\partial A_\nu} = 0 \tag{31}$$

$$\iff \begin{cases} \partial_\mu F^{\mu\nu} = -\mu_0 J^\nu \\ \partial_{[\rho} F_{\mu\nu]} = \partial_\rho F_{\mu\nu} + \partial_\mu F_{\nu\rho} + \partial_\nu F_{\rho\mu} = 0. \end{cases} \tag{32}$$

Canonical Momentum Density

$$\Pi^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_0 A_\mu)} = -\frac{1}{\mu_0} F^{0\mu} = \left(0,\ -\frac{1}{\mu_0 c}\mathbf{E}\right) \tag{33}$$

Hamiltonian Density

$$\begin{aligned} \mathcal{H}(A_\mu,\ \Pi_\mu) &= \Pi^\mu(\partial_0 A_\mu) - \mathcal{L} \\ &= \Pi^i(\partial_0 A_i) - \mathcal{L} \end{aligned}$$

$$\begin{cases} \mathbf{E} = -\nabla\phi - \dfrac{\partial \mathbf{A}}{\partial t} \\ \nabla \cdot \mathbf{E} = \rho/\varepsilon_0 \end{cases}$$

$$\Rightarrow \mathcal{H} = \frac{1}{2}\left(\varepsilon_0 E^2 + \frac{1}{\mu_0} B^2\right) - \mathbf{J}\cdot\mathbf{A} + \nabla\cdot(\varepsilon_0\phi\mathbf{E}) \quad \langle \text{Precise} \rangle \tag{34}$$

$$\because \int \nabla\cdot(\varepsilon_0\phi\mathbf{E}) dV = \oint \varepsilon_0\phi\mathbf{E}\cdot d\mathbf{S} \to 0,\ \text{omit.}$$

$$\therefore \mathcal{H} = \frac{1}{2}\left(\varepsilon_0 E^2 + \frac{1}{\mu_0} B^2\right) - \mathbf{J}\cdot\mathbf{A} \tag{35}$$

$$\begin{aligned} \therefore d\mathcal{H} &= \varepsilon_0 \mathbf{E}\cdot d\mathbf{E} + \frac{1}{\mu_0}\mathbf{B}\cdot d\mathbf{B} - \mathbf{J}\cdot d\mathbf{A} + \varepsilon_0 \nabla\phi\cdot d\mathbf{E} \\ &= \varepsilon_0(\mathbf{E}+\nabla\phi)\cdot d\mathbf{E} - \mathbf{J}\cdot d\mathbf{A} + \frac{1}{\mu_0}\mathbf{B}\cdot d(\nabla\times\mathbf{A}) \\ &= -\frac{1}{c}(\mathbf{E}+\nabla\phi)\cdot d\ - \mathbf{J}\cdot d\mathbf{A} + \frac{1}{\mu_0}(\nabla\times\mathbf{B})\cdot d\mathbf{A} \\ &\quad - \frac{1}{\mu_0}\nabla\cdot(\mathbf{B}\cdot d\mathbf{A}) \quad \text{(omit)} \end{aligned}$$

Hamiltonian canonical equations

$$\begin{cases} \partial_0 A_i = \dfrac{\partial \mathcal{H}}{\partial \Pi^i} \\ \partial_0 \Pi^i = -\dfrac{\partial \mathcal{H}}{\partial A_i} \end{cases} \tag{36}$$

$$\Rightarrow \begin{cases} \mathbf{E} = -\nabla\phi - \dfrac{\partial \mathbf{A}}{\partial t} \\ \nabla\times\mathbf{H} = \mathbf{J} + \dfrac{\partial \mathbf{D}}{\partial t} \end{cases} \tag{37}$$

$$(\mathbf{B} = \mu_0\mathbf{H},\ \mathbf{D} = \varepsilon_0\mathbf{E})$$

4 Lorentz Transformation of EM Field

In this section we study how the electromagnetic field tensor $F^{\mu\nu}$ transforms under a Lorentz boost. Consider an inertial frame S' moving with velocity v along the x-axis relative to S. The Lorentz transformation matrix is

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \beta = \frac{v}{c}, \quad \gamma = \frac{1}{\sqrt{1-\beta^2}}. \tag{38}$$

The electromagnetic field tensor and its components are defined as

$$F^{\mu\nu} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{pmatrix}, \tag{39}$$

with the identifications

$$\frac{E_i}{c} = F^{0i}, \qquad B_i = \frac{1}{2}\varepsilon_{ijk}F^{jk}. \tag{40}$$

Under a Lorentz transformation, the tensor transforms as

$$F^{\mu\nu'} = \Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}F^{\alpha\beta}, \qquad x^{\mu'} = \Lambda^{\mu}{}_{\alpha}x^{\alpha}. \tag{41}$$

In matrix notation this reads

$$F' = \Lambda F \Lambda^T = \Lambda F \Lambda, \tag{42}$$

since $\Lambda^T = \Lambda$ for this boost. Carrying out the matrix multiplication gives the transformed tensor explicitly:

$$F' = \begin{pmatrix} 0 & \dfrac{E_x}{c} & \dfrac{\gamma(E_y - vB_z)}{c} & \dfrac{\gamma(E_z + vB_y)}{c} \\ -\dfrac{E_x}{c} & 0 & \gamma\left(B_z - \dfrac{v}{c^2}E_y\right) & -\gamma\left(B_y + \dfrac{v}{c^2}E_z\right) \\ -\dfrac{\gamma(E_y - vB_z)}{c} & -\gamma\left(B_z - \dfrac{v}{c^2}E_y\right) & 0 & B_x \\ -\dfrac{\gamma(E_z + vB_y)}{c} & \gamma\left(B_y + \dfrac{v}{c^2}E_z\right) & -B_x & 0 \end{pmatrix}. \tag{43}$$

Comparing the components of F' with the definition of $F^{\mu\nu}$, we obtain the transformation rules for the electric and magnetic fields:

$$\begin{cases} E'_x = E_x \\ E'_y = \gamma(E_y - vB_z) \\ E'_z = \gamma(E_z + vB_y) \end{cases} \qquad \begin{cases} B'_x = B_x \\ B'_y = \gamma\left(B_y + \dfrac{v}{c^2}E_z\right) \\ B'_z = \gamma\left(B_z - \dfrac{v}{c^2}E_y\right) \end{cases} \tag{44}$$

For a boost along the x-axis, these relations can also be written in the following compact matrix forms:

$$\begin{pmatrix} E'_x/c \\ B'_x \end{pmatrix} = \begin{pmatrix} E_x/c \\ B_x \end{pmatrix},$$

$$\begin{pmatrix} E'_y/c \\ B'_z \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} E_y/c \\ B_z \end{pmatrix},$$

$$\begin{pmatrix} E'_z/c \\ B'_y \end{pmatrix} = \begin{pmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} E_z/c \\ B_y \end{pmatrix}.$$

These equations show that electric and magnetic fields mix under Lorentz transformations, reflecting the unified nature of the electromagnetic field.

5 Static Fields

$r \gg r'$ far field:

$$\frac{1}{|\mathbf{r}-\mathbf{r}'|} \approx \frac{1}{r} - \mathbf{r}' \cdot \nabla \frac{1}{r} + \frac{1}{2}(\mathbf{r}' \cdot \nabla)^2 \frac{1}{r} \tag{45}$$

Potential in static field:

$$\begin{cases} \phi(\mathbf{r}) = \dfrac{1}{4\pi\varepsilon_0} \displaystyle\int \frac{\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\, d^3r' \\ \mathbf{A}(\mathbf{r}) = \dfrac{\mu_0}{4\pi} \displaystyle\int \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\, d^3r' \end{cases} \tag{46}$$

5.1 Static Electric Field

Multipole moments:

Charge:

$$Q = \int \rho(\mathbf{r}')\, d^3r' \tag{47}$$

Electric dipole moment:

$$\mathbf{p} = \int \rho(\mathbf{r}')\, \mathbf{r}'\, d^3r' \tag{48}$$

Electric quadrupole moment:

$$D_{ij} = \int 3\, r_i' r_j'\, \rho(\mathbf{r}')\, d^3r' \tag{49}$$

Scalar Potential:

$$\begin{cases} \phi(\mathbf{r}) \approx \phi^{(0)}(\mathbf{r}) + \phi^{(1)}(\mathbf{r}) + \phi^{(2)}(\mathbf{r}) \\ \phi^{(0)}(\mathbf{r}) = \dfrac{Q}{4\pi\varepsilon_0 r} \\ \phi^{(1)}(\mathbf{r}) = \dfrac{\mathbf{p}\cdot\hat{r}}{4\pi\varepsilon_0 r^2} \\ \phi^{(2)}(\mathbf{r}) = \dfrac{1}{24\pi\varepsilon_0} \displaystyle\sum_{i,j=1}^{3} D_{ij} \frac{\partial^2}{\partial r_i \partial r_j}\left(\frac{1}{r}\right) \end{cases} \tag{50}$$

Electric Field of Dipole:

$$\begin{aligned} \mathbf{E}^{(1)}(\mathbf{r}) &= -\nabla\phi^{(1)}(\mathbf{r}) \\ &= \frac{1}{4\pi\varepsilon_0} \frac{3(\mathbf{p}\cdot\hat{r})\hat{r} - \mathbf{p}}{r^3} - \frac{1}{3\varepsilon_0}\, \mathbf{p}\, \delta^3(\mathbf{r}) \end{aligned} \tag{51}$$

5.2 Static Magnetic Field

Magnetic Dipole:

$$\mathbf{m} = I\mathbf{S} = \frac{1}{2} I \oint \mathbf{r}' \times d\mathbf{l} \tag{52}$$

Vector Potential:

$$\begin{aligned} \mathbf{A}(\mathbf{r}) &\approx \mathbf{A}^{(0)}(\mathbf{r}) + \mathbf{A}^{(1)}(\mathbf{r}) \\ \mathbf{A}^{(0)}(\mathbf{r}) &= \frac{\mu_0 I}{4\pi r} \oint d\mathbf{l} = 0 \\ \mathbf{A}^{(1)}(\mathbf{r}) &= \frac{\mu_0}{4\pi r^2}(\mathbf{m} \times \hat{r}) \end{aligned} \tag{53}$$

Magnetic Field of Dipole:

$$\begin{aligned} \mathbf{B}^{(1)}(\mathbf{r}) &= \nabla \times \mathbf{A}^{(1)} \\ &= \frac{\mu_0}{4\pi} \frac{3(\mathbf{m} \cdot \hat{r})\hat{r} - \mathbf{m}}{r^3} + \frac{2}{3}\mu_0 \, \mathbf{m} \, \delta^3(\mathbf{r}) \end{aligned} \tag{54}$$

5.3 Ferromagnetism (铁磁性)

5.3.1 Magnetic Media (磁介质)

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mu_0(1 + \chi_m)\mathbf{H} = \mu_0\mu_r\mathbf{H} \tag{55}$$

$$\begin{cases} \chi_m < 0: \text{ diamagnet (抗磁体)} \\ 0 < \chi_m \ll 1: \text{ paramagnet (顺磁体)} \\ \chi_m \gg 1: \text{ ferromagnet (铁磁体).} \end{cases} \tag{56}$$

5.3.2 Ferromagnetism (磁畴)

Magnetic domain: A tiny microscopic region inside ferromagnetic materials. Within one single domain, all atomic magnetic moments are aligned uniformly in the same direction.

Under external **B**, domain walls move and domains reorient, leading to magnetization.

$T > T_c$ (Curie temperature), domains disappear, ferromagnet turns into paramagnet.

$$T_c = \begin{cases} 770°C, \text{ Fe} \\ 1131°C, \text{ Co} \\ 358°C, \text{ Ni.} \end{cases} \tag{57}$$

Key property

$$\begin{cases} \text{large } \chi_m \\ \text{Nonlinear response} \\ \text{hysteresis (磁滞)} \end{cases} \tag{58}$$

5.3.3 Hysteresis Loop (磁滞回线)

M_s: Saturation (饱和) Magnetization
M_r: Remanent (剩余) Magnetization
H_c: Coercive (矫顽) Field
$M = 0$ at $H = -H_c$

$W = \oint HdB$: Hysteresis Loss.

5.3.4 Magnetic Circuit

Ampère's Law : $\oint \mathbf{H} \cdot d\mathbf{l} = NI$ Magnetic Flux : $\Phi_B = \int \mathbf{B} \cdot d\mathbf{A}$

$$\Rightarrow \begin{cases} \text{Magnetic circuit law : } \ NI = \Phi_B R \\ \text{Magnetic Reluctance : } \ R = \dfrac{L}{\mu A} \end{cases} \tag{59}$$

6 Electromagnetic Wave

6.1 Wave Equation and Basic Properties

In vacuum, Maxwell equations lead to the wave equations for the electric and magnetic fields:

$$\Rightarrow \begin{cases} \left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2\right)\mathbf{E} = 0 \\ \left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2\right)\mathbf{B} = 0 \end{cases} \qquad \Rightarrow \begin{cases} \Box\mathbf{E} = 0 \\ \Box\mathbf{B} = 0 \end{cases} \tag{60}$$

The plane-wave solutions are

$$\Rightarrow \begin{cases} \mathbf{E} = E_0 \cos\left(\mathbf{k}\cdot\mathbf{r} - \omega t + \delta\right)\hat{n} \\ \mathbf{B} = \frac{1}{c}\left(\hat{k}\times\mathbf{E}\right) \end{cases} \tag{61}$$

Phase velocity:

$$v_p = \frac{\omega}{k}$$

Group velocity:

$$v_g = \frac{d\omega}{dk}$$

6.2 Telegrapher Equation

For a uniform transmission line, let R, L, G, and C be the resistance, inductance, conductance, and capacitance per unit length. The voltage $V(z,t)$ and current $I(z,t)$ satisfy the telegrapher equations:

$$\begin{cases} \frac{\partial V}{\partial z} = -L\frac{\partial I}{\partial t} - RI \\ \frac{\partial I}{\partial z} = -C\frac{\partial V}{\partial t} - GV \end{cases} \tag{62}$$

Eliminating I yields the wave equation for V:

$$\frac{\partial^2 V}{\partial z^2} = LC\frac{\partial^2 V}{\partial t^2} + (RC + GL)\frac{\partial V}{\partial t} + RG\,V.$$

In the lossless case ($R = G = 0$), this reduces to the ideal wave equation:

$$\frac{\partial^2 V}{\partial z^2} = LC\frac{\partial^2 V}{\partial t^2}, \tag{63}$$

with phase velocity $v_p = 1/\sqrt{LC}$, which is the transmission-line analogue of $v_p = 1/\sqrt{\mu\varepsilon}$ for plane EM waves.

For time-harmonic fields $V(z,t) = \mathrm{Re}\{V(z)e^{i\omega t}\}$, the telegrapher equation becomes

$$\frac{d^2 V}{dz^2} = (R + i\omega L)(G + i\omega C)\,V = \gamma^2 V, \tag{64}$$

where $\gamma = \alpha + i\beta$ is the complex propagation constant. The attenuation constant α and phase constant β are given by

$$\gamma = \sqrt{(R + i\omega L)(G + i\omega C)} \quad \Longrightarrow \quad \begin{cases} \alpha = \mathrm{Re}(\gamma) \\ \beta = \mathrm{Im}(\gamma) \end{cases} \tag{65}$$

The characteristic impedance is

$$Z_0 = \sqrt{\frac{R + i\omega L}{G + i\omega C}}. \tag{66}$$

This formulation is directly analogous to the propagation of EM waves in conducting media, where the conductivity introduces an effective complex permittivity, linking naturally to the energy dissipation term $-\mathbf{E} \cdot \mathbf{J}$ in the Poynting theorem that follows.

6.3 Energy and Energy Flux (Poynting Theorem)

Poynting theorem:

$$\frac{\partial w}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{E} \cdot \mathbf{J} \tag{67}$$

with

$$\begin{cases} w = \dfrac{1}{2}\left(\varepsilon E^2 + \dfrac{1}{\mu} B^2\right) \\ \mathbf{S} = \dfrac{1}{\mu} \mathbf{E} \times \mathbf{B} \end{cases} \tag{68}$$

For non-magnetic media ($\mu = \mu_0$) with refractive index n, the time-averaged intensity is

$$I = \langle S \rangle = \frac{1}{2} n \varepsilon_0 c E_0^2 \tag{69}$$

Power and total energy:

$$P = \int \mathbf{S} \cdot d\mathbf{A}, \qquad E_{\text{total}} = \int w \, dV$$

6.4 Interface Phenomena: Refraction, Reflection

Fermat's Principle

$$0 = \delta \int_S dt = \frac{1}{c} \delta \int_S n \, ds \quad \Longrightarrow \quad \text{Snell's Law} \tag{70}$$

Snell's Law

$$\begin{cases} n_1 \sin\theta_i = n_2 \sin\theta_t \\ \theta_i = \theta_r \end{cases} \tag{71}$$

Fresnel Equations From boundary conditions with $\sigma_f = k_f = 0$. Define polarisation relative to the incident plane:

- S polarization: $\mathbf{E} \perp$ incident plane
- P polarization: $\mathbf{E} \parallel$ incident plane

S Polarization:

$$\begin{cases} r_s = \dfrac{E_r}{E_i} = -\dfrac{\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)} \\ t_s = \dfrac{E_t}{E_i} = \dfrac{2\cos\theta_i \sin\theta_t}{\sin(\theta_i + \theta_t)} \end{cases} \tag{72}$$

P Polarization:

$$\begin{cases} r_p = \dfrac{E_r}{E_i} = \dfrac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)} \\ t_p = \dfrac{E_t}{E_i} = \dfrac{2\cos\theta_i \sin\theta_t}{\sin(\theta_i + \theta_t)\cos(\theta_i - \theta_t)} \end{cases} \tag{73}$$

Reflectance and transmittance:

$$R = r^2, \qquad T = 1 - R$$

Normal Incidence At normal incidence ($\theta_i \to 0$), the S/P distinction vanishes. Using Snell's law $n_1 \sin\theta_i = n_2 \sin\theta_t$, we obtain

$$\boxed{r = \frac{n_1 - n_2}{n_1 + n_2}}, \qquad \boxed{t = \frac{2n_1}{n_1 + n_2}}$$

Brewster Angle The Brewster angle is the incident angle for which $r_p = 0$. This occurs when the reflected and transmitted rays are orthogonal:

$$\theta_i + \theta_t = \frac{\pi}{2}.$$

Substituting into Snell's law gives

$$n_1 \sin\theta_i = n_2 \cos\theta_i \quad \Longrightarrow \quad \tan\theta_i = \frac{n_2}{n_1}.$$

Thus

$$\theta_B = \arctan\left(\frac{n_2}{n_1}\right).$$

6.5 Interference and Diffraction

Interference For two waves at a point P:

$$\begin{cases} u_P = u_A + u_B \\ u_A = u_{A0} e^{i(\omega_A t - k_A r_A)} \\ u_B = u_{B0} e^{i(\omega_B t - k_B r_B)} \end{cases} \tag{74}$$

Intensity is proportional to $\langle u u^* \rangle$.

If A and B are incoherent ($k_A \neq k_B,\ \omega_A \neq \omega_B$):

$$I_P = I_A + I_B.$$

If they are coherent ($k_A = k_B,\ \omega_A = \omega_B$):

$$I_P = I_A + I_B + 2\sqrt{I_A I_B}\,\cos(\phi_A - \phi_B).$$

Helmholtz Equation From the wave equation, assuming harmonic time dependence $\Psi(\mathbf{r},t) \propto e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$, we get

$$(\nabla^2 + k^2)\Psi = 0, \tag{75}$$

where Ψ stands for any component of $\mathbf{E}$ or $\mathbf{B}$.

The Green function satisfies

$$(\nabla^2 + k^2)G(\mathbf{R}) = -\delta(\mathbf{R}). \tag{76}$$

With spherical symmetry $\nabla^2 \to \frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial}{\partial R}\right)$, the outgoing solution is

$$G(R) = \frac{e^{ikR}}{4\pi R}, \qquad \mathbf{R} = \mathbf{r}_s - \mathbf{r}_0. \tag{77}$$

Using Green's identity:

$$\begin{aligned}\Psi(\mathbf{r}_0) &= \int \Psi(\mathbf{r}_s)\,\delta(\mathbf{r}_s - \mathbf{r}_0)\,dV \\ &= \int \left(G\nabla^2\Psi - \Psi\nabla^2 G\right) dV.\end{aligned}$$

Fresnel–Kirchhoff Diffraction Formula

$$\Psi(\mathbf{r}_0) = \frac{1}{i\lambda}\oint \Psi(\mathbf{r})\frac{e^{ikr}}{r}\frac{\cos\theta_s + \cos\theta}{2}\,ds. \tag{78}$$

6.6 Polarization

For a monochromatic plane wave propagating along z:

$$\begin{cases} E_x = E_{x0}\cos(\omega t - kz + \phi_x) \\ E_y = E_{y0}\cos(\omega t - kz + \phi_y) \\ \delta = \phi_y - \phi_x \end{cases} \tag{79}$$

The general polarisation equation is

$$\left(\frac{E_x}{E_{x0}}\right)^2 + \left(\frac{E_y}{E_{y0}}\right)^2 - \frac{2E_xE_y}{E_{x0}E_{y0}}\cos\delta = \sin^2\delta. \tag{80}$$

Polarisation states:

$$\begin{cases} \text{Linear}: \ \delta = 0 \text{ or } \pi \\ \text{Circular}: \ \delta = \pm\dfrac{\pi}{2} \text{ and } E_{x0} = E_{y0} \\ \text{Elliptical}: \ \text{otherwise} \end{cases} \tag{81}$$

Stokes parameters:

$$\begin{cases} I = E_{x0}^2 + E_{y0}^2 \\ Q = E_{x0}^2 - E_{y0}^2 \\ U = 2E_{x0}E_{y0}\cos\delta \\ V = 2E_{x0}E_{y0}\sin\delta \end{cases} \tag{82}$$

Degree of polarisation:

$$P = \frac{\sqrt{Q^2 + U^2 + V^2}}{I}. \tag{83}$$

7 Electromagnetic Radiation

Retarded Potentials

$$\begin{cases} \Box A^\mu = -\mu_0 J^\mu \\ A^\mu = \left(\dfrac{\phi}{c}, \mathbf{A}\right) \\ J^\mu = (\rho c, \mathbf{J}) \\ \Box = -\dfrac{1}{c^2}\dfrac{\partial^2}{\partial t^2} + \nabla^2 \end{cases} \tag{84}$$

$$\Rightarrow \begin{cases} t' = t - \dfrac{|\mathbf{r} - \mathbf{r}'|}{c} \\ A^\mu(\mathbf{r}, t) = \dfrac{\mu_0}{4\pi}\int \dfrac{J^\mu(\mathbf{r}', t')}{|\mathbf{r} - \mathbf{r}'|} d^3 r' \end{cases} \tag{85}$$

7.1 Point Charge Radiation

For field : $r \gg r'$

Point particle :

$$\begin{cases} \rho = q \cdot \delta(\mathbf{r}') \\ \mathbf{J} = \rho \mathbf{v} = q\mathbf{v} \cdot \delta(\mathbf{r}') \end{cases} \tag{86}$$

7.1.1 Liénard–Wiechert Potentials

In rest frame : $\mathbf{v} = 0$

$$\begin{cases} \tilde{\phi}(t) = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{\tilde{r}} \\ \tilde{\mathbf{A}}(t) = 0 \end{cases} \tag{87}$$

In moving frame : $\mathbf{v} \neq 0$

Lorentz transformation :

$$\begin{pmatrix} \phi/c \\ A^1 \\ A^2 \\ A^3 \end{pmatrix} = \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tilde{\phi}/c \\ \tilde{A}^1 \\ \tilde{A}^2 \\ \tilde{A}^3 \end{pmatrix} \tag{88}$$

$$\beta = \frac{v}{c}, \quad \gamma = \frac{1}{\sqrt{1 - \dfrac{v^2}{c^2}}}$$

$$\begin{cases} \tilde{r} = c(\tilde{t} - \tilde{t}') \\ \tilde{t} = \gamma\left(t - \dfrac{\beta \cdot \mathbf{x}}{c}\right) \\ \tilde{t}' = \gamma\left(t' - \dfrac{\beta \cdot \mathbf{x}'}{c}\right) \\ \mathbf{r} = \mathbf{x} - \mathbf{x}' \end{cases}$$

$$\Rightarrow \begin{cases} \dfrac{\partial t'}{\partial t} = \dfrac{1}{1 - \dfrac{\mathbf{v}\cdot\hat{r}}{c}} \\ \nabla t' = -\dfrac{\hat{r}}{c\left(1 - \dfrac{\mathbf{v}\cdot\hat{r}}{c}\right)} \end{cases} \tag{89}$$

$$\Rightarrow \begin{cases} \phi = \dfrac{q}{4\pi\varepsilon_0\left(r - \dfrac{\mathbf{v}\cdot\mathbf{r}}{c}\right)} \\ \mathbf{A} = \dfrac{q\mathbf{v}}{4\pi\varepsilon_0 c^2\left(r - \dfrac{\mathbf{v}\cdot\mathbf{r}}{c}\right)} \end{cases} \quad \text{(Liénard–Wiechert Potential)} \tag{90}$$

7.1.2 Non-Relativistic Radiation Fields

In the non-relativistic limit $v \ll c$, the Liénard–Wiechert potentials reduce to the simpler forms

$$\begin{cases} \phi(\mathbf{r},t) \approx \dfrac{q}{4\pi\varepsilon_0 r}\left[1 + \dfrac{\mathbf{v}(t')\cdot\hat{r}}{c}\right] \\ \mathbf{A}(\mathbf{r},t) \approx \dfrac{q\mathbf{v}(t')}{4\pi\varepsilon_0 c^2 r} \end{cases} \tag{91}$$

where $r = |\mathbf{r} - \mathbf{r}'|$ and all quantities on the right are evaluated at the retarded time $t' = t - r/c$. Using $\nabla t' = -\hat{r}/c$ and $\partial t'/\partial t = 1$, we derive the electric field from

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}. \tag{92}$$

The gradient of the scalar potential gives the static Coulomb term:

$$\begin{aligned} -\nabla\phi &= -(\nabla\phi)_{t'} - \nabla t'\frac{\partial\phi}{\partial t'} \\ &= -\frac{q}{4\pi\varepsilon_0}\nabla\frac{1}{r} - \left(-\frac{\hat{r}}{c}\right)\cdot\frac{q}{4\pi\varepsilon_0 r}\cdot\frac{\mathbf{a}\cdot\hat{r}}{c} \\ &= \frac{q\hat{r}}{4\pi\varepsilon_0 r^2} + \frac{q}{4\pi\varepsilon_0 c^2 r}\,\hat{r}\,(\mathbf{a}\cdot\hat{r}) \end{aligned}$$

For the vector potential,

$$-\frac{\partial\mathbf{A}}{\partial t} = -\frac{q}{4\pi\varepsilon_0 c^2}\frac{d}{dt'}\left(\frac{\mathbf{v}}{r}\right) = -\frac{q}{4\pi\varepsilon_0 c^2}\left(\frac{\mathbf{a}}{r} + \mathbf{v}\frac{\partial}{\partial t}\frac{1}{r}\right),$$

with $\partial_t(1/r) = -(\hat{r}\cdot\mathbf{v})/r^2$. Keeping only the radiation term that falls off as $1/r$ and dropping the higher-order velocity corrections, we obtain

$$\begin{aligned} \mathbf{E} &= -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t} \\ &= \frac{q\hat{r}}{4\pi\varepsilon_0 r^2} + \frac{q}{4\pi\varepsilon_0 c^2 r}\left[\hat{r}(\mathbf{a}\cdot\hat{r}) - \mathbf{a}\right] \\ &= \frac{q\hat{r}}{4\pi\varepsilon_0 r^2} + \frac{q}{4\pi\varepsilon_0 c^2 r}\,\hat{r}\times(\hat{r}\times\mathbf{a}) \end{aligned} \tag{93}$$

so the full non-relativistic electric field is

$$\boxed{\mathbf{E} = \frac{q}{4\pi\varepsilon_0}\frac{\hat{r}}{r^2} + \frac{q}{4\pi\varepsilon_0 c^2 r}\,\hat{r}\times(\hat{r}\times\mathbf{a})}. \tag{94}$$

Take acceleration in z direction $\mathbf{a} = a\hat{z}$. We can rewrite as:

$$\begin{aligned}
\mathbf{E} &= \mathbf{E}_{\text{static}} + \mathbf{E}_{\text{rad}},\\
\mathbf{E}_{\text{static}} &= \frac{q}{4\pi\varepsilon_0 r^2}\hat{r},\\
\mathbf{E}_{\text{rad}} &= \frac{qa\sin\theta}{4\pi\varepsilon_0 c^2 r}\hat{\theta}
\end{aligned} \tag{95}$$

For the magnetic field, we compute $\mathbf{B} = \nabla\times\mathbf{A}$. The gradient acts on $1/r$ and on the retarded velocity:

$$\nabla\times\left(\frac{\mathbf{v}}{r}\right) = \nabla\left(\frac{1}{r}\right)\times\mathbf{v} + \frac{1}{r}(\nabla\times\mathbf{v}).$$

The first term gives the non-relativistic magnetostatic field (velocity field):

$$\mathbf{B}_{\text{static}} = \frac{q}{4\pi\varepsilon_0 c^2}\frac{\mathbf{v}\times\hat{r}}{r^2}, \tag{96}$$

which decays as $1/r^2$ and corresponds to the field of a uniformly moving charge. The second term yields the radiation field (acceleration field). Since

$$\nabla\times\mathbf{v} = \nabla t'\times\dot{\mathbf{v}} = -\frac{\hat{r}}{c}\times\dot{\mathbf{v}},$$

we find the radiation magnetic field

$$\mathbf{B}_{\text{rad}} = \frac{q}{4\pi\varepsilon_0 c^3 r}\,\dot{\mathbf{v}}\times\hat{r}. \tag{97}$$

Using the vector identity, this is exactly

$$\mathbf{B}_{\text{rad}} = \frac{1}{c}\,\hat{r}\times\mathbf{E}_{\text{rad}}. \tag{98}$$

Thus the complete non-relativistic magnetic field is

$$\boxed{\begin{aligned}
\mathbf{B} &= \frac{q}{4\pi\varepsilon_0 c^2}\frac{\mathbf{v}\times\hat{r}}{r^2} + \frac{q}{4\pi\varepsilon_0 c^3 r}\,\dot{\mathbf{v}}\times\hat{r}\\
&= \frac{1}{c^2}\mathbf{v}\times\mathbf{E}_{\text{static}} + \frac{1}{c}\hat{r}\times\mathbf{E}_{\text{rad}}
\end{aligned}} \tag{99}$$

Take acceleration in z direction $\mathbf{a} = a\hat{z}$. We can rewrite as:

$$\begin{aligned}
\mathbf{B} &= \mathbf{B}_{\text{static}} + \mathbf{B}_{\text{rad}},\\
\mathbf{B}_{\text{static}} &= \frac{q}{4\pi\varepsilon_0 c^2 r^2}\,(\mathbf{v}\times\hat{r}),\\
\mathbf{B}_{\text{rad}} &= \frac{qa\sin\theta}{4\pi\varepsilon_0 c^3 r}\,\hat{\phi}
\end{aligned} \tag{100}$$

The far-zone Poynting vector is therefore

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} = \frac{1}{\mu_0 c} |\mathbf{E}|^2 \, \hat{r}. \tag{101}$$

With $|\mathbf{E}_{\rm rad}| = \dfrac{q\dot{v}}{4\pi\varepsilon_0 c^2 r} \sin\theta$, where θ is the angle between $\hat{r}$ and $\dot{\mathbf{v}}$, we have

$$\mathbf{S} = \frac{q^2 \dot{v}^2}{16\pi^2 \varepsilon_0 c^3 r^2} \sin^2\theta \, \hat{r}. \tag{102}$$

Integrating over a sphere,

$$P = \int \mathbf{S} \cdot d\mathbf{A} = \int_0^{2\pi} d\phi \int_0^{\pi} d\theta \, \sin\theta \, r^2 (\mathbf{S} \cdot \hat{r}) = \frac{q^2 \dot{v}^2}{6\pi\varepsilon_0 c^3}, \tag{103}$$

which is the classical Larmor formula:

$$\boxed{P = \frac{q^2 \dot{v}^2}{6\pi\varepsilon_0 c^3}}. \tag{104}$$

Radiation Reaction Force To obtain the radiation reaction force, consider the energy balance over one period of motion. The work done by the reaction force must equal the negative of the energy radiated:

$$\int_0^T \mathbf{F}_{\rm rad} \cdot \mathbf{v} \, dt = -\int_0^T P \, dt = -\frac{q^2}{6\pi\varepsilon_0 c^3} \int_0^T \dot{\mathbf{v}}^{\,2} \, dt.$$

We now perform integration by parts on the right-hand side. Using the identity

$$\dot{\mathbf{v}}^{\,2} = \dot{\mathbf{v}} \cdot \dot{\mathbf{v}} = \frac{d}{dt}(\dot{\mathbf{v}} \cdot \mathbf{v}) - \mathbf{v} \cdot \ddot{\mathbf{v}},$$

and assuming periodic motion (or that the boundary term vanishes), we have

$$\int_0^T \dot{\mathbf{v}}^{\,2} \, dt = -\int_0^T \mathbf{v} \cdot \ddot{\mathbf{v}} \, dt.$$

Substituting this back gives

$$\int_0^T \mathbf{F}_{\rm rad} \cdot \mathbf{v} \, dt = \frac{q^2}{6\pi\varepsilon_0 c^3} \int_0^T \mathbf{v} \cdot \ddot{\mathbf{v}} \, dt = \int_0^T \left(\frac{q^2}{6\pi\varepsilon_0 c^3} \, \ddot{\mathbf{v}} \right) \cdot \mathbf{v} \, dt.$$

Since this equality holds for arbitrary velocity variations, we can read off the non-relativistic radiation reaction force (the Abraham–Lorentz force) as

$$\boxed{\mathbf{F}_{\rm rad} = \frac{q^2}{6\pi\varepsilon_0 c^3} \, \ddot{\mathbf{v}} = \frac{q^2}{6\pi\varepsilon_0 c^3} \, \dot{\mathbf{a}}}. \tag{105}$$

7.1.3 Relativistic Radiation Fields: Poynting Vector and Power

For relativistic motion, define the Lorentz factor

$$\boxed{\gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{1-v^2/c^2}}}, \qquad \beta = \frac{\mathbf{v}}{c}. \tag{106}$$

For a charge moving with constant velocity (uniform rectilinear motion, $\dot{\beta} = 0$), the electric and magnetic fields at the observation point $\mathbf{R}$ (measured from the instantaneous position of the charge) are given by the exact expressions

$$\mathbf{E} = \frac{q}{4\pi\varepsilon_0} \frac{1-\beta^2}{(1-\beta^2\sin^2\theta)^{3/2}} \frac{\mathbf{R}}{R^3}, \qquad \mathbf{B} = \frac{1}{c}\,\beta \times \mathbf{E}, \tag{107}$$

where θ is the angle between $\mathbf{R}$ and β, and $R = |\mathbf{R}|$. These fields contain no radiation (no $1/R$ far-zone energy flux) because the Poynting vector vanishes in the far zone.

When the charge accelerates ($\dot{\beta} \neq 0$), the radiation (acceleration) fields are obtained from the full Liénard–Wiechert fields by keeping only the terms that fall off as $1/r$:

$$\mathbf{E}_{\rm rad}(\mathbf{r},t) = \frac{q}{4\pi\varepsilon_0 c} \frac{\hat{r} \times [(\hat{r}-\beta)\times\dot{\beta}]}{r(1-\beta\cdot\hat{r})^3}, \qquad \mathbf{B}_{\rm rad} = \frac{1}{c}\hat{r} \times \mathbf{E}_{\rm rad}, \tag{108}$$

where $\dot{\beta} = d\beta/dt$, and all quantities are evaluated at the retarded time.

The far-zone Poynting vector is therefore

$$\mathbf{S} = \frac{1}{\mu_0 c}|\mathbf{E}_{\rm rad}|^2\,\hat{r} = \frac{q^2}{16\pi^2\varepsilon_0 c} \frac{|\hat{r}\times[(\hat{r}-\beta)\times\dot{\beta}]|^2}{r^2(1-\beta\cdot\hat{r})^6}\,\hat{r}. \tag{109}$$

The angular distribution of the radiated power is

$$\frac{dP}{d\Omega} = r^2\mathbf{S}\cdot\hat{r} = \frac{q^2}{16\pi^2\varepsilon_0 c} \frac{|\hat{r}\times[(\hat{r}-\beta)\times\dot{\beta}]|^2}{(1-\beta\cdot\hat{r})^6}. \tag{110}$$

Integrating over all solid angles yields the relativistic Larmor formula (Liénard's formula)

$$P = \int \frac{dP}{d\Omega}\,d\Omega = \frac{q^2}{6\pi\varepsilon_0 c}\,\gamma^6\left[\dot{\beta}^2 - (\beta\times\dot{\beta})^2\right], \tag{111}$$

or equivalently, in terms of acceleration $\mathbf{a} = \dot{\mathbf{v}}$,

$$\boxed{P = \frac{q^2}{6\pi\varepsilon_0 c^3}\,\gamma^6\left[a^2 - \frac{(\mathbf{v}\times\mathbf{a})^2}{c^2}\right]}. \tag{112}$$

In the low-velocity limit $v \ll c$, $\gamma \simeq 1$ and the term $(\mathbf{v}\times\mathbf{a})^2/c^2$ is negligible, so this reduces to the classical Larmor formula.

7.1.4 Complete Liénard–Wiechert Fields

The full electric and magnetic fields of a point charge in arbitrary relativistic motion are

$$\mathbf{E}(\mathbf{r},t) = \frac{q}{4\pi\varepsilon_0}\,\frac{1}{(1-\beta\cdot\hat{r})^3}\left[\frac{\hat{r}-\beta}{\gamma^2 r^2} + \frac{\hat{r}\times[(\hat{r}-\beta)\times\dot{\beta}]}{cr}\right], \tag{113}$$

$$\mathbf{B}(\mathbf{r},t) = \frac{1}{c}\,\hat{r} \times \mathbf{E}(\mathbf{r},t), \tag{114}$$

where again all quantities on the right-hand side are evaluated at the retarded time $t' = t - r/c$, with $r = |\mathbf{r} - \mathbf{r}'|$ and $\hat{r} = (\mathbf{r} - \mathbf{r}')/r$. The first term in the brackets is the velocity field (generalized Coulomb field), which decays as $1/r^2$; the second term is the acceleration field (radiation field), which decays as $1/r$ and dominates in the far zone. In the non-relativistic limit $\beta \ll 1$, the above expressions reduce to the familiar forms

$$\mathbf{E} \simeq \frac{q}{4\pi\varepsilon_0}\frac{\hat{r}}{r^2} + \frac{q}{4\pi\varepsilon_0 c^2 r}\,\hat{r} \times (\hat{r} \times \dot{\mathbf{v}}), \qquad \mathbf{B} \simeq \frac{1}{c}\hat{r} \times \mathbf{E}, \tag{115}$$

recovering the static Coulomb field plus the radiation field given in Sec. 3.

7.2 Electric Dipole Radiation

Electric Dipole is also called as Hertzian Dipole.

Charge motion Consider a charge q oscillating along the z-axis with amplitude l:

$$\begin{cases} z(t') = l\sin(\omega t'), \quad l = \text{const} \\ v(t') = \dfrac{dz}{dt'} = \omega l\cos(\omega t') \\ a(t') = \dfrac{dv(t')}{dt'} = -\omega^2 l\sin(\omega t') \end{cases} \tag{116}$$

Fields and power (Larmor formula) For an observation point at distance r and polar angle θ (with respect to the dipole axis), the transverse electric field is

$$\mathbf{E}_{\text{rad}} = \frac{qa\sin\theta}{4\pi\varepsilon_0}\frac{\hat{\theta}}{c^2 r} \tag{117}$$

and the magnetic field $B = E/c$:

$$\mathbf{B}_{\text{rad}} = \frac{q}{4\pi\varepsilon_0 c^3 r}\,\mathbf{a} \times \hat{r}. \tag{118}$$

The instantaneous radiated power is given by the Larmor formula:

$$P(t) = \frac{q^2 a^2(t')}{6\pi\varepsilon_0 c^3}, \qquad t' = t - \frac{r}{c}, \qquad k = \frac{\omega}{c}. \tag{119}$$

Substituting $a(t') = -\omega^2 l\sin(\omega t')$, we obtain

$$\begin{aligned} E_{\text{rad}} &= \frac{ql\omega^2\sin\theta}{4\pi\epsilon c^2 r}\sin(kr - \omega t), \\ B_{\text{rad}} &= E_{\text{rad}}/c \\ P(t) &= \frac{q^2 l^2\omega^4}{6\pi\varepsilon_0 c^3}\sin^2(kr - \omega t). \end{aligned}$$

The time-averaged power is

$$\langle P\rangle = \frac{q^2 l^2\omega^4}{12\pi\varepsilon_0 c^3}. \tag{120}$$

Dipole picture Define the dipole moment amplitude $p_0 = ql$. Then

$$\begin{aligned} E_{\text{rad}} &= \frac{p_0 \omega^2 \sin\theta}{4\pi\epsilon c^2 r} \sin\left(kr - \omega t\right), \\ B_{\text{rad}} &= E_{\text{rad}}/c \\ P(t) &= \frac{p_0^2 \omega^4}{6\pi\varepsilon_0 c^3} \sin^2(kr - \omega t). \end{aligned} \tag{121}$$

The time-averaged power is

$$\langle P \rangle = \frac{p_0^2 \omega^4}{12\pi\varepsilon_0 c^3}. \tag{122}$$

Current picture The current is the time derivative of the charge $q(t') = q\sin(\omega t')$:

$$I(t') = \frac{d}{dt'}\left[q\sin(\omega t')\right] = q\omega\cos(\omega t') = q\omega\cos(kr - \omega t). \tag{123}$$

Define the current amplitude

$$I_0 = q\omega. \tag{124}$$

Then

$$\begin{aligned} E_{\text{rad}} &= \frac{I_0 l \omega \sin\theta}{4\pi\epsilon c^2 r} \sin\left(kr - \omega t\right), \\ B_{\text{rad}} &= E_{\text{rad}}/c \\ P(t) &= \frac{I_0^2 l^2 \omega^2}{6\pi\varepsilon_0 c^3} \sin^2(kr - \omega t). \end{aligned}$$

and the average power becomes

$$\langle P \rangle = \frac{I_0^2 l^2 \omega^2}{12\pi\varepsilon_0 c^3}. \tag{125}$$

Define the radiation resistance R_{rad} such that

$$P(t) = I_0^2 R_{\text{rad}} \sin^2(kr - \omega t), \tag{126}$$

$$\langle P \rangle = \frac{1}{2} I_0^2 R_{\text{rad}}. \tag{127}$$

Comparing with the expression above,

$$R_{\text{rad}} = \frac{l^2\omega^2}{6\pi\varepsilon_0 c^3} = \frac{l^2}{6\pi\varepsilon_0 c^3}\left(\frac{2\pi c}{\lambda}\right)^2 = \frac{2}{3}\pi\mu_0 c\left(\frac{l}{\lambda}\right)^2 \approx 80\pi^2\,\Omega\,\left(\frac{l}{\lambda}\right)^2. \tag{128}$$

8 Radiation in Matter

Group and phase velocity For EM wave:

Group velocity is often under c:

$$v_g = \frac{d\omega}{dk} \tag{129}$$

More precisely, $v_g < c$ is typical for ordinary transparent media; in anomalous dispersion v_g can exceed c (or become negative), while the signal/front velocity remains bounded by c.

Phase velocity can be larger than c:

$$v_p = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\epsilon}} \tag{130}$$

Refractive index:

$$n = \frac{c}{v_p} \tag{131}$$

For most media $\mu \approx \mu_0$, so,

$$\frac{\varepsilon}{\varepsilon_0} \approx \frac{c^2}{v_p^2} = \frac{c^2 k^2}{\omega^2} \tag{132}$$

Wave equation with current density If there is no net charge $\rho = 0$, but charge particles can move (**J** exist), from Maxwell Equations, we can get wave equation with current density.

$$-\frac{1}{v_p^2}\frac{\partial^2 \mathbf{E}}{\partial t^2} + \nabla^2 \mathbf{E} = \mu \frac{\partial \mathbf{J}}{\partial t} \tag{133}$$

Electric Field of EM wave:

$$\mathbf{E}(\mathbf{x}, t) = \mathbf{E}_0 e^{i(\mathbf{k}\cdot\mathbf{x} - \omega t)} \tag{134}$$

Take the direction of electric field as z axis:

$$\mathbf{E}_0 = E_0 \hat{\mathbf{z}} \tag{135}$$

We get:

$$(\frac{\omega^2}{v_p^2} - k^2)\mathbf{E} = \mu \frac{\partial \mathbf{J}}{\partial t} \tag{136}$$

Dispersion relation depends on relation between **J** and **E**

Full dynamical equation of electron

$$m_e \ddot{\mathbf{x}} + m_e \gamma_c \dot{\mathbf{x}} + m_e \omega_0^2 \mathbf{x} = -e(\mathbf{E} + \dot{\mathbf{x}} \times \mathbf{B}) + \mathbf{F}_{\text{rad}} \tag{137}$$

Here,

$m_e \ddot{\mathbf{x}}$: accelaration term.

$m_e \gamma_c \dot{\mathbf{x}}$: collision term.

$m_e \omega_0^2 \mathbf{x}$: harmonic oscillation term for bound electron.

$-e\mathbf{E}$: electric force term.

$-e\dot{\mathbf{x}} \times \mathbf{B}$: Lorentz force from magnetic field.

$\mathbf{F}_{\text{rad}} = q^2 \dot{\mathbf{a}}/6\pi\varepsilon_0 c^3$:radiation force reaction term.

Classical electron radius

$$r_{\mathrm{e}} = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} \tag{138}$$

Incident intensity of EM wave

$$I_0 = \frac{1}{2}\varepsilon_0 c E_0^2 \tag{139}$$

Radiation of electron accelerating in z axis

$$\mathbf{E}_{\mathrm{rad}} = \frac{qa\sin\theta}{4\pi\varepsilon_0 c^2 r}\hat{\theta} \tag{140}$$

Time-averaged Poynting vector

$$\langle \mathbf{S} \rangle = \langle S \rangle \hat{\mathbf{r}} = \varepsilon_0 c \langle E_{\mathrm{rad}}^2 \rangle \hat{\mathbf{r}} \tag{141}$$

Time-averaged total scattered power

$$\langle P \rangle = \int \langle \mathbf{S} \rangle \cdot d\mathbf{A} = \int r^2 \langle S \rangle d\Omega \tag{142}$$

Scattering cross section

$$\sigma(\omega) = \frac{\langle P \rangle}{I_0} \tag{143}$$

Differential scattering cross section

$$\frac{d\sigma}{d\Omega} = \frac{1}{I_0}\frac{d\langle P \rangle}{d\Omega} = \frac{r^2 \langle S \rangle}{I_0} \tag{144}$$

Mean free path

$$\lambda_{\mathrm{mfp}} = \frac{1}{N\sigma} \tag{145}$$

N is the number density of electrons.

8.1 Wave in Plasma: Free Electron

Now we consider ionospheric (电离层) reflection

Wave equation with current density in vacuum becomes:

$$(\frac{\omega^2}{c^2} - k^2)\mathbf{E} = \mu_0 \frac{\partial \mathbf{J}}{\partial t} \tag{146}$$

Define N as electron number density. Current Density:

$$\mathbf{J} = -Ne\mathbf{v} \tag{147}$$

Electrons are influenced by EM field of EM wave, omit force due to radiation from electron acceleration. Non-relativistic case, dynamical equation of free electron:

$$m_e\ddot{\mathbf{x}} + m_e\gamma_c\dot{\mathbf{x}} = -e\mathbf{E} \tag{148}$$

Then,

$$\mathbf{x} = \frac{e\mathbf{E}}{m(\omega^2 + i\omega\gamma_c)}$$

$$\ddot{\mathbf{x}} = -\omega^2\mathbf{x} = -\frac{e\mathbf{E}}{m}\frac{\omega^2}{\omega^2 + i\omega\gamma_c}$$

8.1.1 Dispersion relation of plasma

:

Def:

$$\omega_p^2 = \frac{Ne^2}{m_e\varepsilon_0} \tag{149}$$

Dispersion relation:

$$c^2k^2 = \omega^2 - \frac{\omega^2}{\omega^2 + i\omega\gamma_c}\omega_p^2 \tag{150}$$

If $\omega \gg \gamma_c$, dispersion relation is simplified:

$$c^2k^2 = \omega^2 - \omega_p^2 \tag{151}$$

When the EM wave travels from vacuum($n_1 = 1$) to refractive index n media:

$$n^2 = (\frac{c}{v_p})^2 = \frac{\varepsilon(\omega)}{\varepsilon_0} = 1 - \frac{\omega_p^2}{\omega^2} \tag{152}$$

Propagation conditions in plasma

$$\begin{cases} \omega > \omega_p: & \text{EM wave propagate in Plasma.} \\ \omega < \omega_p: & \text{EM wave total reflection.} \end{cases} \tag{153}$$

8.1.2 Free-electron scattering

We now examine the radiation emitted by a single free electron driven by an incident EM wave, including the same damping term γ_c as in the plasma dispersion relation.

Acceleration of free electron due to EM wave:

$$\ddot{\mathbf{x}} = -\omega^2\mathbf{x} = -\frac{e\mathbf{E}}{m}\frac{\omega^2}{\omega^2 + i\omega\gamma_c}$$

Radiation of accelerating electron:

$$\mathbf{E}_{\text{rad}} = \frac{qa\sin\theta}{4\pi\varepsilon_0 c^2 r}\hat{\theta} \tag{154}$$

Def: the classical electron radius

$$r_{\text{e}} = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} \tag{155}$$

We get the far-zone radiation electric field :

$$\mathbf{E}_{\rm rad}(r,t) = E_0 \cos(kr - \omega t)\frac{r_{\rm e}}{r}\frac{\omega^2}{\omega^2 + i\omega\gamma_c}\sin\theta\,\hat{\boldsymbol{\theta}} \tag{156}$$

Incident intensity

$$I_0 = \frac{1}{2}\varepsilon_0 c E_0^2 \tag{157}$$

The time-averaged Poynting vector is:

$$\langle \mathbf{S} \rangle = \varepsilon_0 c \langle E_{\rm rad}^2 \rangle \hat{\mathbf{r}} = I_0 \frac{r_{\rm e}^2}{r^2}\frac{\omega^2}{\omega^2 + \gamma_c^2}\sin^2\theta\,\hat{\mathbf{r}} \tag{158}$$

and integrating over a sphere gives the total scattered power

$$\langle P \rangle = \int \langle \mathbf{S} \rangle \cdot d\mathbf{A} = \frac{8\pi}{3} I_0 \frac{r_{\rm e}^2 \omega^2}{\omega^2 + \gamma_c^2} \tag{159}$$

Thus the scattering cross section is

$$\boxed{\sigma(\omega) = \frac{\langle P \rangle}{I_0} = \frac{8\pi}{3}\frac{r_{\rm e}^2\omega^2}{\omega^2 + \gamma_c^2}} \tag{160}$$

The differential scattering cross section is,

$$\frac{d\sigma}{d\Omega} = \frac{r^2\langle S \rangle}{I_0} = \frac{\omega^2}{\omega^2 + \gamma_c^2} r_e^2 \sin^2\theta \tag{161}$$

The frequency-dependent mean free path is therefore

$$\lambda_{\rm mfp}(\omega) = \frac{1}{N\sigma(\omega)} = \frac{3(\omega^2 + \gamma_c^2)}{8\pi N r_{\rm e}^2 \omega^2}. \tag{162}$$

Limiting cases:

For $\omega \gg \gamma_c$ (Thomson limit), the cross section becomes constant and the mean free path reduces to

$$\frac{d\sigma}{d\Omega} \to r_e^2 \sin^2\theta, \qquad \sigma \to \frac{8\pi}{3} r_{\rm e}^2, \qquad \lambda_{\rm mfp} \to \frac{3}{8\pi N r_{\rm e}^2}. \tag{163}$$

For $\omega \ll \gamma_c$ (strong damping), scattering is suppressed and the medium becomes effectively transparent:

$$\frac{d\sigma}{d\Omega} \to 0, \qquad \sigma \to 0, \qquad \lambda_{\rm mfp} \to \infty. \tag{164}$$

8.2 Wave in Dielectrics（电介质）: Bound Electron

Now we consider dielectrics deflection

Wave equation with current density in vacuum becomes:

$$(\frac{\omega^2}{c^2} - k^2)\mathbf{E} = \mu_0 \frac{\partial \mathbf{J}}{\partial t} \tag{165}$$

Define N as electron number density. Current Density:

$$\mathbf{J} = -Ne\mathbf{v} \tag{166}$$

Electrons are influenced by EM field of EM wave, omit force due to radiation from electron acceleration. Non-relativistic case, dynamical equation of bound electron in harmonic oscillation:

$$m_e\ddot{\mathbf{x}} + m_e\gamma_c\dot{\mathbf{x}} + m_e\omega_0^2\mathbf{x} = -e\mathbf{E} \tag{167}$$

Then,

$$\mathbf{x} = \frac{e\mathbf{E}}{m(\omega^2 - \omega_0^2 + i\omega\gamma_c)}$$

8.2.1 Dispersion relation of dielectrics

:

Def:

$$\omega_p^2 = \frac{Ne^2}{m_e\varepsilon_0} \tag{168}$$

Dispersion relation:

$$c^2k^2 = \omega^2 - \frac{\omega^2}{\omega^2 - \omega_0^2 + i\omega\gamma_c}\omega_p^2 \tag{169}$$

When the EM wave travels from vacuum($n_1 = 1$) to refractive index n media:

$$n^2 = (\frac{c}{v_p})^2 = \frac{\varepsilon(\omega)}{\varepsilon_0} = 1 - \frac{\omega_p^2}{\omega^2 - \omega_0^2 + i\omega\gamma_c} \tag{170}$$

8.2.2 Bound-electron scattering

We now examine the radiation emitted by a single bound electron (treated as a damped harmonic oscillator) driven by an incident EM wave, including the same damping term γ_c and a natural frequency ω_0.

Acceleration of bound electron due to EM wave:

$$\ddot{\mathbf{x}} = -\frac{e\mathbf{E}}{m_e}\frac{\omega^2}{\omega^2 - \omega_0^2 + i\omega\gamma_c}$$

Radiation of accelerating electron:

$$\mathbf{E}_{\text{rad}} = \frac{qa\sin\theta}{4\pi\varepsilon_0 c^2 r}\hat{\theta} \tag{171}$$

Def: the classical electron radius

$$r_{\text{e}} = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} \tag{172}$$

We get the far-zone radiation electric field :

$$\mathbf{E}_{\text{rad}}(r,t) = E_0\cos(kr - \omega t)\frac{r_{\text{e}}}{r}\frac{\omega^2}{\omega^2 - \omega_0^2 + i\omega\gamma_c}\sin\theta\,\hat{\boldsymbol{\theta}} \tag{173}$$

Incident intensity

$$I_0 = \frac{1}{2}\varepsilon_0 c E_0^2 \tag{174}$$

The time-averaged Poynting vector is:

$$\langle \mathbf{S} \rangle = I_0 \frac{r_\mathrm{e}^2}{r^2} \frac{\omega^4}{(\omega^2 - \omega_0^2)^2 + \omega^2 \gamma_c^2} \sin^2\theta \, \hat{\mathbf{r}} \tag{175}$$

and integrating over a sphere gives the total scattered power

$$\langle P \rangle = \frac{8\pi}{3} I_0 \frac{r_\mathrm{e}^2 \omega^4}{(\omega^2 - \omega_0^2)^2 + \omega^2 \gamma_c^2} \tag{176}$$

Thus the scattering cross section is

$$\boxed{\sigma(\omega) = \frac{\langle P \rangle}{I_0} = \frac{8\pi}{3} \frac{r_\mathrm{e}^2 \omega^4}{(\omega^2 - \omega_0^2)^2 + \omega^2 \gamma_c^2}}. \tag{177}$$

The differential scattering cross section is,

$$\frac{d\sigma}{d\Omega} = \frac{r^2 \langle S \rangle}{I_0} = \frac{\omega^4}{(\omega^2 - \omega_0^2)^2 + \omega^2 \gamma_c^2} \, r_e^2 \sin^2\theta \tag{178}$$

The frequency-dependent mean free path is therefore

$$\lambda_\mathrm{mfp}(\omega) = \frac{1}{N\sigma(\omega)} = \frac{3}{8\pi N r_\mathrm{e}^2} \frac{(\omega^2 - \omega_0^2)^2 + \omega^2 \gamma_c^2}{\omega^4}. \tag{179}$$

Limiting cases:

For $\omega \gg \omega_0$ and $\omega \gg \gamma_c$ (Thomson limit), the cross section becomes constant and the mean free path reduces to

$$\frac{d\sigma}{d\Omega} \to r_e^2 \sin^2\theta, \qquad \sigma \to \frac{8\pi}{3} r_\mathrm{e}^2, \qquad \lambda_\mathrm{mfp} \to \frac{3}{8\pi N r_\mathrm{e}^2}. \tag{180}$$

For $\omega \to \omega_0$ (resonant scattering peak), the cross section reaches its maximum and the mean free path becomes

$$\frac{d\sigma}{d\Omega} \to \frac{\omega_0^2}{\gamma_c^2} \, r_e^2 \sin^2\theta, \qquad \sigma \to \frac{8\pi}{3} r_\mathrm{e}^2 \frac{\omega_0^2}{\gamma_c^2}, \qquad \lambda_\mathrm{mfp} \to \frac{3\gamma_c^2}{8\pi N r_\mathrm{e}^2 \omega_0^2}. \tag{181}$$

For $\omega \ll \omega_0$ (Rayleigh scattering regime), the cross section scales as ω^4 and the mean free path scales as ω^{-4}:

$$\frac{d\sigma}{d\Omega} \to \frac{\omega^4}{\omega_0^4} \, r_e^2 \sin^2\theta, \qquad \sigma \to \frac{8\pi}{3} r_\mathrm{e}^2 \frac{\omega^4}{\omega_0^4}, \qquad \lambda_\mathrm{mfp} \to \frac{3\omega_0^4}{8\pi N r_\mathrm{e}^2 \omega^4}. \tag{182}$$

8.3 Wave in Conductors

Wave equation with current density in media becomes:

$$(\frac{\omega^2}{v_p^2} - k^2)\mathbf{E} = \mu \frac{\partial \mathbf{J}}{\partial t} \tag{183}$$

In linear, homogeneous, isotropic media, current density:

$$\mathbf{J} = \sigma \mathbf{E} \tag{184}$$

Dispersion relation of conducting media

$$k^2 = \mu\omega^2\left(\varepsilon + i\frac{\sigma}{\omega}\right) \tag{185}$$

Good or Poor Conductor

$$\begin{cases} \sigma \gg \varepsilon\omega, \text{ Good conductor}, & k \approx (1+i)\sqrt{\dfrac{\mu\omega\sigma}{2}} \\ \sigma \ll \varepsilon\omega, \text{ Poor conductor}, & k \approx \sqrt{\mu\varepsilon}\cdot\omega + i\frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}} \end{cases} \tag{186}$$

General Skin Depth Distance where wave amplitude decays to $1/e$ of its initial value.

$$e^{ikx} \to e^{i\mathrm{Re}(k)\cdot x}\cdot e^{-\mathrm{Im}(k)\cdot x}$$

From $\exp\left[-\mathrm{Im}(k)\cdot\delta\right] = 1/e$, we define general skin depth,

$$\delta = \frac{1}{\mathrm{Im}(k)} = \begin{cases} \sqrt{\dfrac{2}{\mu\omega\sigma}}, & \text{good conductor} \\ \dfrac{2}{\sigma}\sqrt{\dfrac{\varepsilon}{\mu}}, & \text{poor conductor} \end{cases} \tag{187}$$

9 Kramers–Kronig Relation

9.1 Cauchy Integral Formula

When α outside contour C,

$$I = \oint_C (z-\alpha)^n dz = 0 \tag{188}$$

When α inside contour C,

$$z = \alpha + Re^{i\varphi}$$

$$\begin{aligned} I &= \oint_C (z-\alpha)^n \, dz \\ &= \oint_C R^n e^{in\varphi} \, d\big(\alpha + Re^{i\varphi}\big) \\ &= iR^{n+1} \oint_C e^{i(n+1)\varphi} \, d\varphi \\ &= iR^{n+1} \int_0^{2\pi} e^{i(n+1)\varphi} \, d\varphi \end{aligned}$$

$$I = \begin{cases} 0, & n \neq -1 \\ 2\pi i, & n = -1 \end{cases} \tag{189}$$

Therefore:

$$\frac{1}{2\pi i} \oint_C (z-\alpha)^n \, dz = 0, \qquad n \neq -1 \tag{190}$$

$$\frac{1}{2\pi i} \oint_C \frac{dz}{z-\alpha} = \begin{cases} 0, & \alpha \text{ outside } C \\ 1, & \alpha \text{ inside } C \end{cases} \tag{191}$$

Bounding the integral on the small circular arc: Let C_ε be a small circle of radius ε centered at α, with $z = \alpha + \varepsilon e^{i\varphi}$.

$$\begin{aligned} \left| \oint_{C_\varepsilon} \frac{f(z)-f(\alpha)}{z-\alpha} \, dz \right| &\leq \frac{\max_{z \in C_\varepsilon} |f(z)-f(\alpha)|}{\varepsilon} \cdot \left| \oint_{C_\varepsilon} dz \right| \\ &\leq \frac{\max_{z \in C_\varepsilon} |f(z)-f(\alpha)|}{\varepsilon} \cdot \left| \int_0^{2\pi} i\varepsilon \, e^{i\varphi} \, d\varphi \right| \\ &\leq \frac{\max_{z \in C_\varepsilon} |f(z)-f(\alpha)|}{\varepsilon} \cdot \varepsilon \cdot 2\pi \\ &= 2\pi \cdot \max_{z \in C_\varepsilon} |f(z)-f(\alpha)| \end{aligned}$$

As $\varepsilon \to 0$, $\max_{z \in C_\varepsilon} |f(z)-f(\alpha)| \to 0$
Thus,

$$\lim_{\varepsilon \to 0} \oint_{C_\varepsilon} \frac{f(z)-f(\alpha)}{z-\alpha} \, dz = 0.$$

Since the integrand is analytic in the region between the original contour C and the small circle C_ε, Cauchy's theorem gives

$$\oint_C \frac{f(z)-f(\alpha)}{z-\alpha}\,dz = \oint_{C_\varepsilon} \frac{f(z)-f(\alpha)}{z-\alpha}\,dz = 0$$

Cauchy Integral Formula:

$$f(\alpha) = \frac{1}{2\pi i}\oint_C \frac{f(\alpha)}{z-\alpha}\,dz = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z-\alpha}\,dz \tag{192}$$

9.2 Sokhotski–Plemelj Equation

$$\frac{1}{x-i\varepsilon} = \frac{x+i\varepsilon}{x^2+\varepsilon^2} = \frac{x}{x^2+\varepsilon^2} + i\,\frac{\varepsilon}{x^2+\varepsilon^2}$$

$$\int_{-\infty}^{+\infty} \frac{\varepsilon}{x^2+\varepsilon^2}\,dx = \int_{-\infty}^{+\infty} \frac{1}{\left(\frac{x}{\varepsilon}\right)^2+1}\,d\Big(\frac{x}{\varepsilon}\Big) = \arctan\Big(\frac{x}{\varepsilon}\Big)\Big|_{-\infty}^{+\infty} = \pi$$

As $\varepsilon \to 0^+$:

$$\frac{\varepsilon}{x^2+\varepsilon^2} = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases}$$

So,

$$\lim_{\varepsilon\to 0^+} \frac{\varepsilon}{x^2+\varepsilon^2} = \pi\,\delta(x) \tag{193}$$

Cauchy principal value integral:

$$\mathcal{P}\int_a^b \frac{f(x)}{x-x_0}dx = \lim_{\varepsilon\to 0}\left(\int_a^{x_0-\varepsilon} \frac{f(x)}{x-x_0}dx + \int_{x_0+\varepsilon}^b \frac{f(x)}{x-x_0}dx\right) \tag{194}$$

Sokhotski–Plemelj Equation:

$$\lim_{\varepsilon\to 0^+} \frac{1}{x-i\varepsilon} = \mathcal{P}\left(\frac{1}{x}\right) + i\pi\,\delta(x) \tag{195}$$

where $\mathcal{P}$ denotes the Cauchy principal value, and $\delta(x)$ is the Dirac delta function.

9.3 Kramers–Kronig Relation

As α inside contour C,

$$z \to w', \qquad \alpha \to w + i\varepsilon$$

Electric susceptibility as a complex number, must satisfy Cauchy Integral Formula:

$$\chi(w) = \frac{1}{2\pi i}\oint_C \frac{\chi(w')}{w'-w-i\varepsilon}\,dw' \tag{196}$$

Apply Sokhotski–Plemelj identity:

$$\begin{aligned}\chi(w) &= \frac{1}{2\pi i}\int_{-\infty}^{+\infty} \chi(w')\left[\mathcal{P}\left(\frac{1}{w'-w}\right) + i\pi\,\delta(w'-w)\right]dw' \\ &= \frac{1}{2\pi i}\int_{-\infty}^{+\infty} \chi(w')\,\mathcal{P}\left(\frac{1}{w'-w}\right)dw' + \frac{1}{2}\chi(w)\end{aligned}$$

Rearrange terms:

$$\chi(w) = \frac{1}{\pi i}\,\mathcal{P}\int_{-\infty}^{+\infty} \frac{\chi(w')}{w'-w}\,dw' \tag{197}$$

Decompose susceptibility into real and imaginary parts:

$$\chi(w) = \chi_1(w) + i\,\chi_2(w)$$

Causality / Hermiticity condition:

$$\chi(-w) = \chi^*(w)$$

This gives symmetry relations:

$$\begin{cases} \chi_1(-w) = \chi_1(w) \\ \chi_2(-w) = -\chi_2(w) \end{cases}$$

χ_1 is even; χ_2 is odd function of frequency.

Real part of susceptibility:

$$\begin{aligned} \chi_1(w) &= \frac{1}{\pi}\,\mathcal{P}\int_{-\infty}^{+\infty} \frac{\chi_2(w')}{w'-w}\,dw' \\ &= \frac{1}{\pi}\,\mathcal{P}\left[\int_0^{\infty} \chi_2(w')\left(\frac{1}{w'-w} + \frac{1}{w'+w}\right)dw'\right] \\ &= \frac{2}{\pi}\,\mathcal{P}\int_0^{\infty} \frac{w'\,\chi_2(w')}{w'^2-w^2}\,dw' \end{aligned}$$

Imaginary part of susceptibility:

$$\begin{aligned} \chi_2(w) &= -\frac{1}{\pi}\,\mathcal{P}\int_{-\infty}^{+\infty} \frac{\chi_1(w')}{w'-w}\,dw' \\ &= -\frac{1}{\pi}\,\mathcal{P}\left[\int_0^{\infty} \chi_1(w')\left(\frac{1}{w'-w} - \frac{1}{w'+w}\right)dw'\right] \\ &= -\frac{2w}{\pi}\,\mathcal{P}\int_0^{\infty} \frac{\chi_1(w')}{w'^2-w^2}\,dw' \end{aligned}$$

Relation to complex permittivity:

$$\chi(w) = \frac{\varepsilon(w)}{\varepsilon_0} - 1$$

Substitute $\chi_1 = \mathrm{Re}(\varepsilon/\varepsilon_0) - 1$, $\chi_2 = \mathrm{Im}(\varepsilon/\varepsilon_0)$:

$$\mathrm{Re}\left(\frac{\varepsilon(w)}{\varepsilon_0}\right) = 1 + \frac{2}{\pi}\,\mathcal{P}\int_0^{\infty} \frac{w'\,\mathrm{Im}\left(\dfrac{\varepsilon(w')}{\varepsilon_0}\right)}{w'^2-w^2}\,dw' \tag{198}$$

$$\mathrm{Im}\left(\frac{\varepsilon(w)}{\varepsilon_0}\right) = -\frac{2w}{\pi}\,\mathcal{P}\int_0^{\infty} \frac{\mathrm{Re}\left(\dfrac{\varepsilon(w')}{\varepsilon_0}\right) - 1}{w'^2-w^2}\,dw' \tag{199}$$

These are the standard Kramers–Kronig dispersion relations, which connect the real dispersive part and imaginary absorptive part of complex relative permittivity.

Let $\tilde{n}(\omega) = n(\omega) + i\kappa(\omega)$ be the complex refractive index, where n is the ordinary refractive index and κ is the extinction coefficient. Since

$$\frac{\varepsilon(\omega)}{\varepsilon_0} = \big(\tilde{n}(\omega)\big)^2 = (n + i\kappa)^2 = n^2 - \kappa^2 + i\,2n\kappa,$$

we have

$$\begin{cases} \mathrm{Re}\left(\dfrac{\varepsilon(\omega)}{\varepsilon_0}\right) = n^2(\omega) - \kappa^2(\omega), \text{ related to dispersion and extinction} \\ \mathrm{Im}\left(\dfrac{\varepsilon(\omega)}{\varepsilon_0}\right) = 2n(\omega)\kappa(\omega), \text{ related to EM wave absorption.} \end{cases} \tag{200}$$

In the lossless limit ($\kappa \to 0$), the real part reduces to $n^2(\omega)$.

Gavin's Lectures on Physics - Quantum Mechanics



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1 Foundational Postulates & Early Quantum Tools

1.1 Fundamental Postulates

1.1.1 State Postulate (Wave Function)

The state of a physical system is completely described by a wave function $\Psi(\mathbf{r}, t)$.

1.1.2 Born Interpretation

$$P(\mathbf{r}, t) = |\Psi(\mathbf{r}, t)|^2, \qquad \int_{-\infty}^{\infty} |\Psi(\mathbf{r}, t)|^2 \, d^3r = 1. \tag{1}$$

1.1.3 Correspondence Postulate (Operators)

Every measurable physical quantity corresponds to a linear, Hermitian operator.

$$\hat{x} = x, \tag{2}$$

$$\hat{p} = -i\hbar\nabla, \tag{3}$$

$$\hat{T} = \frac{\hat{p}^2}{2m} = -\frac{\hbar^2}{2m}\nabla^2, \tag{4}$$

$$\hat{H} = \hat{T} + V(\mathbf{x}) = -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{x}). \tag{5}$$

1.1.4 Measurement Postulate (Eigenvalue Equation)

$$\hat{A}\Psi = a\Psi, \tag{6}$$

where Ψ is an eigenfunction of $\hat{A}$ with eigenvalue a.

1.1.5 Collapse Postulate

Immediately after a measurement yielding eigenvalue a_n, the state collapses instantaneously to the corresponding eigenfunction Ψ_n.

1.1.6 Evolution Postulate (Schrödinger Equation)

$$i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r}, t) = \hat{H}\Psi(\mathbf{r}, t). \tag{7}$$

The wave function of a closed system evolves deterministically according to the time-dependent Schrödinger equation.

1.1.7 Spin-Statistics Postulate

- **Fermions**: half-integer spin, anti-symmetric total wave function, obey Pauli exclusion principle.
- **Bosons**: integer spin, symmetric total wave function.

1.2 Essential Formulas

1.2.1 Origin of Operators

For a matter wave $\Psi = \Psi_0 e^{i(kx-\omega t)}$.

de Broglie wave tells momentum: $p = \hbar k = \frac{h}{\lambda}$:

$$-i\hbar\frac{\partial\Psi}{\partial x} = -i\hbar(ik)\Psi = \hbar k\Psi = p\Psi, \tag{8}$$

hence

$$\hat{p} = -i\hbar\frac{\partial}{\partial x} \xrightarrow{3D} \hat{p} = -i\hbar\nabla. \tag{9}$$

Einstein photon tells energy: $E = \hbar\omega$

$$i\hbar\frac{\partial\Psi}{\partial t} = E\Psi = \hat{H}\Psi, \tag{10}$$

hence

$$\hat{H} = i\hbar\frac{\partial}{\partial t} \tag{11}$$

1.2.2 The Schrödinger Equation

Considering conservation of energy:

$$\hat{H} = \frac{\hat{p}^2}{2m} + V \tag{12}$$

We got The Schrödinger Equation:

Time-dependent:

$$i\hbar\frac{\partial\Psi}{\partial t} = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r},t)\right]\Psi. \tag{13}$$

Time-independent:

$$\Psi(\mathbf{r},t) = \psi(\mathbf{r})e^{-iEt/\hbar}, \qquad E\psi(\mathbf{r}) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r})\right]\psi(\mathbf{r}). \tag{14}$$

1.2.3 Commutation Relations

$$[\hat{x},\hat{p}_x] = i\hbar, \qquad [\hat{L}_i,\hat{L}_j] = i\hbar\varepsilon_{ijk}\hat{L}_k. \tag{15}$$

1.2.4 Heisenberg Uncertainty Principle

$$\Delta x\,\Delta p_x \geq \frac{\hbar}{2}, \qquad \Delta E\,\Delta t \geq \frac{\hbar}{2}. \tag{16}$$

1.2.5 Representation Transformation

$$|\Psi\rangle = \sum_i a_i |u_i\rangle = \sum_j b_j |v_j\rangle , \tag{17}$$

$$b_j = \sum_i S_{ji} a_i, \qquad S_{ji} = \langle v_j | u_i \rangle. \tag{18}$$

Momentum and position representations:

$$\langle p|\Psi\rangle = \Psi(p), \qquad \langle x|\Psi\rangle = \Psi(x), \tag{19}$$

$$\Psi(p) = \int \langle p|x\rangle \Psi(x)\ \mathrm{d}x, \qquad \langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} = \psi_p(x). \tag{20}$$

1.3 Ehrenfest Theorem

Ehrenfest Theorem is the connection between quantum mechanics and classical mechanics.

For any operator $\hat{A}$ with no explicit time dependence:

$$\frac{d}{dt}\langle \hat{A} \rangle = \frac{1}{i\hbar} \langle [\hat{A}, \hat{H}] \rangle. \tag{21}$$

In particular:

$$\frac{d}{dt}\langle x \rangle = \frac{\langle p \rangle}{m}, \qquad \frac{d}{dt}\langle p \rangle = -\left\langle \frac{\partial V}{\partial x} \right\rangle = \langle F \rangle. \tag{22}$$

This corresponds to classical Poisson brackets: $\frac{\partial A}{\partial t} + [A, H] = 0$.

2 1D Motion

2.1 The 1D Infinite Square Well

Potential:

$$V(x) = \begin{cases} 0, & 0 < x < L, \\ \infty, & \text{otherwise.} \end{cases} \tag{23}$$

Energy eigenvalues:

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}, \qquad n = 1, 2, 3, \ldots \tag{24}$$

Normalised eigenfunctions:

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \qquad 0 < x < L. \tag{25}$$

2.2 The 1D Harmonic Oscillator

Potential:

$$V(x) = \frac{1}{2}kx^2, \qquad \omega = \sqrt{\frac{k}{m}}, \tag{26}$$

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}m\omega^2x^2. \tag{27}$$

Define

$$\xi = \sqrt{\frac{m\omega}{\hbar}}\, x, \tag{28}$$

$$\hat{a} = \frac{1}{\sqrt{2}}\left(\xi + \frac{\partial}{\partial\xi}\right), \tag{29}$$

$$\hat{a}^\dagger = \frac{1}{\sqrt{2}}\left(\xi - \frac{\partial}{\partial\xi}\right), \tag{30}$$

with

$$[\hat{a}, \hat{a}^\dagger] = 1. \tag{31}$$

Then

$$E\Psi(x) = \hat{H}\Psi(x) = \frac{1}{2}\hbar\omega\left(\xi^2 - \frac{\partial^2}{\partial\xi^2}\right)\Psi(x) = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)\Psi(x). \tag{32}$$

Ground state:

$$\hat{a}\,|E_0\rangle = 0 \implies E_0 = \frac{1}{2}\hbar\omega, \tag{33}$$

and

$$\hat{H}\hat{a}^\dagger\,|E_n\rangle = (E_n + \hbar\omega)\hat{a}^\dagger\,|E_n\rangle\,, \qquad \hat{H}\hat{a}\,|E_n\rangle = (E_n - \hbar\omega)\hat{a}\,|E_n\rangle\,. \tag{34}$$

Thus

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \qquad |E_n\rangle = \frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^n\,|E_0\rangle\,, \tag{35}$$

$$\hat{a}\,|n\rangle = \sqrt{n}\;|n-1\rangle\,, \qquad \hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\;|n+1\rangle\,. \tag{36}$$

The ground state wavefunction satisfies

$$\hat{H}\psi_0(x) = E_0\psi_0(x) = \frac{1}{2}\hbar\omega\,\psi_0(x), \tag{37}$$

and

$$0 = \hat{a}\psi_0 = \frac{1}{\sqrt{2}}\left(\xi + \frac{\partial}{\partial\xi}\right)\psi_0, \tag{38}$$

with normalisation $\int_{-\infty}^{\infty}|\psi_0|^2dx = 1$, giving

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\xi^2/2}. \tag{39}$$

Excited states:

$$\begin{aligned}\psi_n(x) &= \frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^n\psi_0 = \frac{1}{\sqrt{n!}}\left[\frac{1}{\sqrt{2}}\left(\xi - \frac{\partial}{\partial\xi}\right)\right]^n\left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\xi^2/2} \\ &= \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\frac{1}{\sqrt{2^n n!}}\left(\xi - \frac{\partial}{\partial\xi}\right)^n e^{-\xi^2/2} = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\frac{1}{\sqrt{2^n n!}}H_n(\xi)e^{-\xi^2/2}.\end{aligned} \tag{40}$$

Here $H_n(\xi)$ are Hermite polynomials:

$$H_0(\xi) = 1, \quad H_1(\xi) = 2\xi, \quad H_2(\xi) = 4\xi^2 - 2, \quad \ldots \tag{41}$$

Final result:

$$\boxed{E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \qquad \psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\frac{1}{\sqrt{2^n n!}}H_n(\xi)e^{-\xi^2/2}.} \tag{42}$$

3 3D Motion, Angular Momentum & Spin

3.1 Two-Body Problem

Centre-of-mass and relative coordinates:

$$\mathbf{R} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}, \tag{43}$$

$$\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1, \tag{44}$$

$$M = m_1 + m_2, \qquad \mu = \frac{m_1 m_2}{m_1 + m_2}, \tag{45}$$

$$\mathbf{r}_1 = \mathbf{R} - \frac{m_2}{m_1 + m_2}\mathbf{r}, \qquad \mathbf{r}_2 = \mathbf{R} + \frac{m_1}{m_1 + m_2}\mathbf{r}. \tag{46}$$

Momenta:

$$\mathbf{P} = M\dot{\mathbf{R}}, \qquad \mathbf{p} = \mu\dot{\mathbf{r}}, \qquad \mathbf{p}_1 = m_1\dot{\mathbf{r}}_1, \qquad \mathbf{p}_2 = m_2\dot{\mathbf{r}}_2. \tag{47}$$

Total energy:

$$E = \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} + V(r) = \frac{P^2}{2M} + \frac{p^2}{2\mu} + V(r). \tag{48}$$

Quantisation:

$$\hat{H}_C\psi_C(\mathbf{R}) = \frac{\hat{P}^2}{2M}\psi_C(\mathbf{R}) = E_C\psi_C(\mathbf{R}), \tag{49}$$

$$\hat{H}_r\psi_r(\mathbf{r}) = \frac{\hat{p}^2}{2\mu}\psi_r(\mathbf{r}) + V(r)\psi_r(\mathbf{r}) = E_r\psi_r(\mathbf{r}), \tag{50}$$

$$\Psi = \psi_C(\mathbf{R})\psi_r(\mathbf{r}), \tag{51}$$

$$\hat{P} = -i\hbar\nabla_R, \qquad \hat{p} = -i\hbar\nabla_r. \tag{52}$$

3.2 Angular Momentum and Spin

$$\hat{\mathbf{L}} = \mathbf{r} \times \hat{\mathbf{p}}, \tag{53}$$

so

$$\hat{L}_x = y\hat{p}_z - z\hat{p}_y, \quad \hat{L}_y = z\hat{p}_x - x\hat{p}_z, \quad \hat{L}_z = x\hat{p}_y - y\hat{p}_x. \tag{54}$$

In spherical coordinates ($x = r\sin\theta\cos\phi,\ y = r\sin\theta\sin\phi,\ z = r\cos\theta$):

$$\hat{L}_x = i\hbar\left(\sin\phi\frac{\partial}{\partial\theta} + \cot\theta\cos\phi\frac{\partial}{\partial\phi}\right), \tag{55}$$

$$\hat{L}_y = i\hbar\left(-\cos\phi\frac{\partial}{\partial\theta} + \cot\theta\sin\phi\frac{\partial}{\partial\phi}\right), \tag{56}$$

$$\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}, \tag{57}$$

$$\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]. \tag{58}$$

Define

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2, \qquad \hat{L}_\pm = \hat{L}_x \pm i\hat{L}_y, \tag{59}$$

and spin operators $\hat{\mathbf{S}} = \frac{\hbar}{2}\hat{\sigma}$, with Pauli matrices

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{60}$$

satisfying

$$\hat{\sigma}_x^2 = \hat{\sigma}_y^2 = \hat{\sigma}_z^2 = \hat{I}, \qquad [\hat{\sigma}_i, \hat{\sigma}_j] = 2i\varepsilon_{ijk}\hat{\sigma}_k. \tag{61}$$

Commutation relations for $\hat{L}$:

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z, \tag{62}$$
$$[\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x, \tag{63}$$
$$[\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y, \tag{64}$$
$$[\hat{L}^2, \hat{L}_z] = 0. \tag{65}$$

Eigenvalues:

$$\hat{L}^2 \left|lm\right\rangle = l(l+1)\hbar^2 \left|lm\right\rangle, \tag{66}$$
$$\hat{L}_z \left|lm\right\rangle = m\hbar \left|lm\right\rangle, \tag{67}$$
$$\hat{L}_\pm \left|lm\right\rangle = \hbar\sqrt{l(l+1) - m(m\pm 1)}\ \left|l, m\pm 1\right\rangle, \tag{68}$$

with

$$l = 0, 1, 2, \ldots, \qquad m = -l, -l+1, \ldots, l. \tag{69}$$

3.3 Stern–Gerlach Experiment

A beam of spin-1/2 particles in a non-uniform magnetic field splits into two discrete paths, demonstrating quantisation of spin.

3.4 Molecular Rotational Energy Levels

For a diatomic molecule with reduced mass μ, moment of inertia $I = \mu r^2$:

$$\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]. \tag{70}$$

The full Hamiltonian:

$$\hat{H}_r = -\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{\hat{L}^2}{2\mu r^2} + V(r). \tag{71}$$

If the potential is harmonic $V(r) = \frac{1}{2}m\omega^2 r^2$, then for vibrational and rotational modes:

$$\begin{aligned}\hat{H}_r\psi_{nl} &= \left[-\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{\hat{L}^2}{2\mu r^2} + V(r)\right]\psi_{nl} \\ &= \left[\left(n+\frac{1}{2}\right)\hbar\omega + \frac{l(l+1)\hbar^2}{2I}\right]\psi_{nl}.\end{aligned} \tag{72}$$

Thus

$$E_{n+1} - E_n = \hbar\omega = h\nu, \qquad E_{l+1} - E_l = \frac{\hbar^2}{I}. \tag{73}$$

3.5 Clebsch–Gordan Coefficients

$$|j_1, j_2; j, m\rangle = \sum_{m_1+m_2=m} C^{jm}_{j_1 m_1 j_2 m_2} \, |j_1, m_1; j_2, m_2\rangle \,, \tag{74}$$

$$C^{jm}_{j_1 m_1 j_2 m_2} = (-1)^{j_1 - j_2 + m} \sqrt{2j+1} \begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & -m \end{pmatrix} = \langle j_1, m_1; j_2, m_2 | j_1, j_2, j, m \rangle. \tag{75}$$

Allowed values:

$$\begin{cases} j = |j_1 - j_2|,\ |j_1 - j_2| + 1, \ldots, j_1 + j_2 - 1, j_1 + j_2, \\ m = -j, -j+1, \ldots, j-1, j, \\ m_1 = -j_1, \ldots, j_1, \quad m_2 = -j_2, \ldots, j_2. \end{cases} \tag{76}$$

The 3-j symbol:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = \delta_{m_1+m_2+m_3,0} \, \Delta(j_1, j_2, j_3) \sqrt{(j_1+m_1)!(j_1-m_1)!(j_2+m_2)!(j_2-m_2)!(j_3+m_3)!(j_3-m} \\ \times \sum_k \frac{(-1)^k}{k!(j_1+j_2-j_3-k)!(j_1-m_1-k)!(j_2+m_2-k)!(j_3-j_2+m_1+k)!(j_3-j_1-} \tag{77}$$

with

$$\Delta(j_1, j_2, j_3) = \sqrt{\frac{(j_1+j_2-j_3)!(j_1-j_2+j_3)!(-j_1+j_2+j_3)!}{(j_1+j_2+j_3+1)!}}. \tag{78}$$

In practice, coefficients are often looked up in tables.

4 Hydrogen Atom & Stationary Perturbation Theory

4.1 Spherically Symmetric Potential and the Hydrogen Atom

4.1.1 General radial equation

For $V = V(r)$:

$$\hat{H}_r = -\frac{\hbar^2}{2\mu}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{\hat{L}^2}{\hbar^2 r^2}\right] + V(r), \tag{79}$$

$$\Psi_E(r,\theta,\phi) = R(r)Y_l(\theta,\phi). \tag{80}$$

Then

$$\hat{L}^2 Y_l = l(l+1)\hbar^2 Y_l, \tag{81}$$

$$\left[-\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d}{dr}\right) + V(r) + \frac{l(l+1)\hbar^2}{2\mu r^2}\right] R(r) = ER(r). \tag{82}$$

4.1.2 Angular solutions

$$Y_l(\theta,\phi) = \Theta(\theta)\Phi(\phi), \tag{83}$$

$$\hat{L}_z^2\Phi = -\hbar^2\frac{d^2\Phi}{d\phi^2} = m^2\hbar^2\Phi \implies \Phi(\phi) \propto e^{im\phi}, \quad m = -l,\ldots,l, \tag{84}$$

$$\left[-\frac{\hbar^2}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d}{d\theta}\right) + \frac{m^2\hbar^2}{\sin^2\theta}\right]\Theta = l(l+1)\hbar^2\Theta, \tag{85}$$

so $\Theta(\theta) = P_l^m(\cos\theta)$ (associated Legendre polynomials).

Normalised spherical harmonics:

$$Y_l^m(\theta,\phi) = (-1)^{(m+|m|)/2}\sqrt{\frac{(2l+1)(l-|m|)!}{4\pi(l+|m|)!}}\,P_l^m(\cos\theta)\,e^{im\phi}, \tag{86}$$

with

$$Y_l^{-m}(\theta,\phi) = (-1)^m\left[Y_l^m(\theta,\phi)\right]^*. \tag{87}$$

4.1.3 Hydrogen atom –zeroth order

Coulomb potential:

$$V(r) = -\frac{e^2}{4\pi\varepsilon_0 r} = -\frac{\alpha\hbar c}{r}, \qquad \alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \approx \frac{1}{137}. \tag{88}$$

The radial equation becomes

$$0 = \left[-\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d}{dr}\right) - \frac{\alpha\hbar c}{r} + \frac{l(l+1)\hbar^2}{2\mu r^2} - E\right]R, \tag{89}$$

or

$$\frac{d^2R}{dr^2} + \frac{2}{r}\frac{dR}{dr} + \left[\frac{2\mu E}{\hbar^2} + \frac{2\mu\alpha}{\hbar r} - \frac{l(l+1)}{r^2}\right]R = 0. \tag{90}$$

Asymptotic behaviour:

- $r \to \infty$: $R(r) \propto e^{-\sqrt{2\mu|E|}\, r/\hbar}$ (bound states, $E<0$).
- $r \to 0$: $R(r) \propto r^{l}$.

Define

$$A = \sqrt{\frac{2\mu|E|}{\hbar^2}}, \qquad B = \frac{2\mu c\alpha}{\hbar}, \qquad u(r) = R(r)r. \tag{91}$$

Then

$$u(r) = e^{-Ar} r^{l+1} F(r), \qquad \frac{u''}{u} + \left[-A^2 + \frac{B}{r} - \frac{l(l+1)}{r^2}\right] = 0. \tag{92}$$

With $F(r) = \sum_{i=0}^{\infty} a_i r^i$, one obtains the recurrence

$$\frac{a_{i+1}}{a_i} = \frac{2A\left[i + l + 1 - \frac{B}{2A}\right]}{(i+1)(i+2l+2)}. \tag{93}$$

Quantisation requires $n = \frac{B}{2A} = \alpha c\sqrt{\frac{\mu}{2|E|}}$ with $l + 1 - n \le 0$. Hence

$$E_n = -\frac{\mu c^2 \alpha^2}{2n^2}, \qquad A = \frac{\mu c\alpha}{n\hbar}, \qquad n = 1, 2, \ldots \tag{94}$$

Bohr radius:

$$a_0 = \frac{\hbar}{\mu c\alpha} = \frac{4\pi\varepsilon_0\hbar^2}{\mu e^2} \approx 0.529\,\text{Å}. \tag{95}$$

Radial wavefunctions:

$$R_{nl}(r) \propto e^{-r/na_0} \left(\frac{2r}{na_0}\right)^{l} L_{n+l}^{2l+1}\left(\frac{2r}{na_0}\right), \tag{96}$$

where L_{n+l}^{2l+1} are associated Laguerre polynomials. Normalised:

$$R_{nl}(r) = -\sqrt{\left(\frac{2}{na_0}\right)^3 \frac{(n-l-1)!}{2n[(n+l)!]^3}}\, e^{-r/na_0} \left(\frac{2r}{na_0}\right)^{l} L_{n+l}^{2l+1}\left(\frac{2r}{na_0}\right). \tag{97}$$

Full hydrogen wavefunctions:

$$\Psi_{nlm} = R_{nl}(r) Y_l^m(\theta, \phi). \tag{98}$$

Explicit lowest states:

$$\Psi_{100} = \frac{1}{\sqrt{\pi}} \left(\frac{1}{a_0}\right)^{3/2} e^{-r/a_0}, \tag{99}$$

$$\Psi_{200} = \frac{1}{4\sqrt{2\pi}} \left(\frac{1}{a_0}\right)^{3/2} \left(2 - \frac{r}{a_0}\right) e^{-r/2a_0}, \tag{100}$$

$$\Psi_{210} = \frac{1}{4\sqrt{2\pi}} \left(\frac{1}{a_0}\right)^{3/2} \frac{r}{a_0} e^{-r/2a_0} \cos\theta, \tag{101}$$

$$\Psi_{21,\pm1} = \mp\frac{1}{8\sqrt{\pi}} \left(\frac{1}{a_0}\right)^{3/2} \frac{r}{a_0} e^{-r/2a_0} \sin\theta\, e^{\pm i\phi}. \tag{102}$$

Most probable radius (for circular orbits $l = n-1$) is $r_{\max} \approx n^2 a_0$.

Energy levels:

$$E_n^{(0)} = -\frac{m_e c^2 \alpha^2}{2n^2} = -\frac{m_e e^4}{8\varepsilon_0^2 \hbar^2}\frac{1}{n^2} = -\frac{13.6\text{ eV}}{n^2}. \tag{103}$$

Degeneracy including spin:

$$\text{degeneracy} = 2\sum_{l=0}^{n-1}(2l+1) = 2n^2. \tag{104}$$

Spin-orbit coupling (to be added in fine structure):

$$\hat{H}_{so} = \xi(r)\hat{\mathbf{L}}\cdot\hat{\mathbf{S}}, \qquad \xi(r) = \frac{1}{2m^2c^2}\frac{e^2}{4\pi\varepsilon_0}\frac{1}{r^3}. \tag{105}$$

4.2 Time-Independent Perturbation Theory

4.2.1 Non-degenerate case

Given $\hat{H} = \hat{H}_0 + \lambda\hat{H}'$, with known $E_n^{(0)}, |n^{(0)}\rangle$:

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \cdots, \tag{106}$$

$$|n\rangle = |n^{(0)}\rangle + \lambda|n^{(1)}\rangle + \lambda^2|n^{(2)}\rangle + \cdots, \tag{107}$$

and $\langle n^{(l)}|n^{(k)}\rangle = \delta_{lk}$.

Order-by-order equations:

$$0\text{th}: \quad (\hat{H}_0 - E_n^{(0)})|n^{(0)}\rangle = 0, \tag{108}$$

$$1\text{st}: \quad (\hat{H}_0 - E_n^{(0)})|n^{(1)}\rangle + (\hat{H}' - E_n^{(1)})|n^{(0)}\rangle = 0, \tag{109}$$

$$2\text{nd}: \quad (\hat{H}_0 - E_n^{(0)})|n^{(2)}\rangle + (\hat{H}' - E_n^{(1)})|n^{(1)}\rangle + E_n^{(2)}|n^{(0)}\rangle = 0, \tag{110}$$

and generally for the l-th order:

$$(\hat{H}_0 - E_n^{(0)})|n^{(l)}\rangle + (\hat{H}' - E_n^{(1)})|n^{(l-1)}\rangle + E_n^{(2)}|n^{(l-2)}\rangle + \cdots + E_n^{(l)}|n^{(0)}\rangle = 0. \tag{111}$$

Define $H'_{nm} = \langle n^{(0)}|\hat{H}'|m^{(0)}\rangle$. Then

$$E_n^{(1)} = H'_{nn}, \qquad E_n^{(2)} = \sum_{m\neq n}\frac{|H'_{nm}|^2}{E_n^{(0)} - E_m^{(0)}}. \tag{112}$$

Also

$$|n^{(1)}\rangle = \sum_{m\neq n}\frac{H'_{mn}}{E_n^{(0)} - E_m^{(0)}}|m^{(0)}\rangle. \tag{113}$$

4.2.2 Degenerate case

Let the unperturbed subspace have dimension g_n with states $|n^{(0)}, \nu\rangle,\ \nu = 1, \ldots, g_n$. To first order in λ:

$$\hat{H}(\lambda) = \hat{H}_0 + \lambda\hat{H}', \tag{114}$$

$$E_k = E_k^{(0)} + \lambda E_k^{(1)}, \tag{115}$$

$$|k\rangle = |k^{(0)}\rangle + \lambda|k^{(1)}\rangle. \tag{116}$$

Define

$$H'_{n\nu,m\mu} = \langle n^{(0)}, \nu | \hat{H}' | m^{(0)}, \mu \rangle. \tag{117}$$

The first-order correction is found by diagonalising $\hat{H}'$ within the degenerate subspace. The secular equation:

$$\det\left(H'_{k\nu,k\mu} - E_k^{(1)}\delta_{\nu\mu}\right) = 0, \tag{118}$$

which yields g_k eigenvalues $E_{k,\alpha}^{(1)}$.

4.3 Fine Structure of the Hydrogen Atom

Corrections of order $\alpha^4 m_e c^2$.

4.3.1 Relativistic correction

$$E = \sqrt{m_e^2c^4 + p^2c^2} \approx m_e c^2 + \frac{p^2}{2m_e} - \frac{1}{8}\frac{p^4}{m_e^3c^2}. \tag{119}$$

Thus

$$\hat{H}_0 = \frac{\hat{p}^2}{2m_e} - \frac{\alpha\hbar c}{r}, \qquad \hat{H}'_{rel} = -\frac{1}{8}\frac{\hat{p}^4}{m_e^3c^2} = -\frac{1}{2m_ec^2}\left(\hat{H}_0 + \frac{\alpha\hbar c}{r}\right)^2. \tag{120}$$

Useful expectation values:

$$\left\langle nlm \left| \frac{\alpha\hbar c}{r} \right| nlm \right\rangle = -2E_n^{(0)} \quad \text{(virial theorem)}, \tag{121}$$

$$\left\langle nlm \left| \left(\frac{\alpha\hbar c}{r}\right)^2 \right| nlm \right\rangle = \frac{4n}{l+\frac{1}{2}}\left[E_n^{(0)}\right]^2. \tag{122}$$

Then

$$E_{n,rel}^{(1)} = -\frac{1}{2m_ec^2}\left[E_n^{(0)}\right]^2\left(-3 + \frac{4n}{l+\frac{1}{2}}\right) = -\frac{1}{2}m_ec^2\alpha^4\left[-\frac{3}{4n^4} + \frac{1}{n^3(l+\frac{1}{2})}\right]. \tag{123}$$

4.3.2 Spin-orbit interaction(S-L)

e spin - e orbit, perturbation Hamiltonian scaling $\hat{H}'_{\mathrm{SO}} \sim \alpha^4 m_e c^2$

Classical derivation of orbital magnetic moment Definition of magnetic moment for current loop:

$$\mu_L = IA = I\pi r^2 \tag{124}$$

Current generated by orbiting electron:

$$I = \frac{e}{T} = \frac{ev}{2\pi r} = \frac{e}{2\pi m_e r^2}L \tag{125}$$

Combine the two equations to eliminate current, obtain the magnitude relation of orbital magnetic moment:

$$\mu_L = \frac{e}{2m_e}L \tag{126}$$

Electron spin magnetic moment, non-relativistic approximation for electron spin g-factor $g_e = 2$:

$$\mu_S = g_e\frac{e}{2m_e}\cdot\boldsymbol{S}, \quad g_e = 2 \tag{127}$$

Biot–Savart's Law Remark: $\boldsymbol{B}$ due to proton

$$B = \oint \frac{\mu_0}{4\pi} \frac{I d\boldsymbol{\ell} \times \hat{r}}{r^2} \tag{128}$$

$$= \frac{I}{4\pi\varepsilon_0 c^2 r^2} \oint d\ell \tag{129}$$

$$= \frac{I \cdot 2\pi r}{4\pi\varepsilon_0 c^2 r^2} \tag{130}$$

$$= \frac{I}{2\varepsilon_0 c^2 r} \tag{131}$$

Vector Operator System of coupled equations for spin magnetic moment operator and magnetic field operator:

$$\begin{cases} \hat{\boldsymbol{\mu}}_S = -g_e \dfrac{e}{2m_e} \hat{\boldsymbol{S}} \\ \hat{\boldsymbol{B}} = \dfrac{e}{4\pi\varepsilon_0 m_e c^2 r^3} \hat{\boldsymbol{L}} \end{cases} \tag{132}$$

Spin-orbit perturbation Hamiltonian $\hat{H}'_{\mathrm{SO}} = -\hat{\boldsymbol{\mu}}_S \cdot \hat{\boldsymbol{B}}$. Substitute the operators and correct the typo $m_e \to m_e^2$ in denominator:

$$\hat{H}'_{\mathrm{SO}} = -\hat{\boldsymbol{\mu}}_S \cdot \hat{\boldsymbol{B}} \tag{133}$$

$$= g_e \cdot \frac{e^2}{8\pi\varepsilon_0 m_e^2 c^2 r^3} \left(\hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{S}} \right) \tag{134}$$

Substitute the approximation $g_e = 2$.

Total angular momentum identity:

$$\hat{\boldsymbol{J}} = \hat{\boldsymbol{L}} + \hat{\boldsymbol{S}} \Rightarrow \hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{S}} = \frac{1}{2}\left(\hat{J}^2 - \hat{L}^2 - \hat{S}^2\right)$$

Substitute the dot product of angular momentum into the Hamiltonian:

$$\hat{H}'_{\mathrm{SO}} = \frac{e^2}{8\pi\varepsilon_0 m_e^2 c^2 r^3} \left(\hat{J}^2 - \hat{L}^2 - \hat{S}^2 \right) \tag{135}$$

Fine-structure constant and energy scaling estimation Definition of fine-structure constant, equivalent form of Bohr radius, scaling of squared angular momentum operators:

$$\because \alpha = \frac{e^2}{4\pi\varepsilon_0 \hbar c}, \quad \langle r \rangle \sim a_0 = \frac{\hbar}{\alpha \cdot m_e \cdot c}, \quad \hat{J}^2, \hat{L}^2, \hat{S}^2 \sim \hbar^2$$

Scaling of expectation value of spin-orbit interaction energy:

$$\langle \hat{H}'_{\mathrm{SO}} \rangle \sim \frac{1}{2}\alpha^4 \cdot m_e c^2 \sim \alpha^2 \left| E_n^{(0)} \right| \tag{136}$$

the first-order energy shift is

$$E^{(1)}_{nlj,so} = -\frac{Z^2\alpha^2}{2nl(l+1)(l+2)} E_n^{(0)} \begin{cases} l, & j = l + \frac{1}{2}, \\ -(l+1), & j = l - \frac{1}{2}. \end{cases} \tag{137}$$

4.3.3 Lamb shift

Dirac theory predicts degeneracy between $2S_{1/2}$ and $2P_{1/2}$. The Lamb shift (order $\alpha^5 m_e c^2$) lifts this:

- **Electron self-energy** ($\approx$97%):

$$\Delta E_{SE}(2S_{1/2}) = \frac{\alpha^5 m_e c^2}{6\pi n^3}\left(\ln\frac{1}{\alpha^2} - \frac{19}{30}\right)\frac{4(2\alpha)^4 m_e c^2}{32}, \tag{138}$$

$$\Delta E_{SE}(2P_{1/2}) \approx 0. \tag{139}$$

- **Vacuum polarisation** ($\approx$3%):

$$\Delta E_{VP}(2S_{1/2}) = \left(\frac{\alpha^5 m_e c^2}{15\pi n^3}\right)\frac{\alpha}{\pi}\frac{(2\alpha)^4}{n^3}F_{VP}\, m_e c^2, \tag{140}$$

$$\Delta E_{VP}(2P_{1/2}) \approx 0. \tag{141}$$

For $n = 2,\ Z = 1,\ \alpha = 1/137.036,\ m_e = 0.511$ MeV:

$$\Delta E_{\text{Lamb}} = E(2S_{1/2}) - E(2P_{1/2}) \approx 4.37 \times 10^{-6}\ \text{eV}, \tag{142}$$

$$\nu = \frac{\Delta E_{\text{Lamb}}}{h} \approx 1057.8\ \text{MHz}. \tag{143}$$

4.4 Hyperfine Structure of the Hydrogen Atom (I-S)

Interaction between electron spin and proton spin; order $\alpha^4 \frac{m_e}{m_p} m_e c^2$.

Electron magnetic moment:

$$\hat{\mu}_s = -g_e \frac{\mu_B}{\hbar}\hat{\mathbf{S}}, \qquad \mu_B = \frac{e\hbar}{2m_e}. \tag{144}$$

Proton magnetic moment:

$$\hat{\mu}_p = g_p \frac{\mu_N}{\hbar}\hat{\mathbf{I}}, \qquad \mu_N = \frac{e\hbar}{2m_p}. \tag{145}$$

Magnetic field due to proton spin at the origin:

$$\hat{\mathbf{B}}_p(r) \approx \frac{2}{3}\mu_0\, \delta^3(r)\, \hat{\mu}_p. \tag{146}$$

Hyperfine Hamiltonian:

$$\hat{H}'_{hfs} = -\hat{\mathbf{B}}_p \cdot \hat{\mu}_s = \frac{2}{3}\mu_0 \frac{g_e g_p \mu_N \mu_B}{\hbar^2}(\hat{\mathbf{I}}\cdot\hat{\mathbf{S}})\,\delta^3(r). \tag{147}$$

Total spin $\hat{\mathbf{F}} = \hat{\mathbf{S}} + \hat{\mathbf{I}}$, with

$$\hat{F}^2|FM_F\rangle = F(F+1)\hbar^2|FM_F\rangle, \qquad \hat{F}_z|FM_F\rangle = M_F\hbar|FM_F\rangle. \tag{148}$$

Then

$$\hat{\mathbf{I}}\cdot\hat{\mathbf{S}} = \frac{1}{2}(\hat{F}^2 - \hat{I}^2 - \hat{S}^2). \tag{149}$$

For the $1s$ state, $|\psi_{1s}(0)|^2 = \frac{1}{\pi a_0^3}$. Hence

$$E_{hfs,1s}(F) = \frac{1}{3}\mu_0 g_e g_p \mu_N \mu_B \left[F(F+1) - \frac{3}{2}\right] |\psi_{1s}(0)|^2, \tag{150}$$

since $S = I = 1/2$, so $F = 0, 1$.

The splitting between $F = 1$ and $F = 0$:

$$\Delta E_{hfs,1s} = E(F=1) - E(F=0) = \frac{2}{3\pi a_0^3}\mu_0 g_e g_p \mu_N \mu_B = \frac{hc}{\lambda_{hfs}}. \tag{151}$$

Using $a_0 = 5.3 \times 10^{-11}$ m and constants $\mu_N = 3.15 \times 10^{-8}$ eV/T, $\mu_B = 5.8 \times 10^{-5}$ eV/T, $g_e \approx 2$, $g_p \approx 5.6$, $\mu_0 = 4\pi \times 10^{-7}$ N/A^2, one obtains

$$\Delta E_{hfs,1s} \approx 6 \times 10^{-6} \text{ eV}, \qquad \lambda_{hfs} \approx 21 \text{ cm}. \tag{152}$$

5 Atoms in External Magnetic Fields

5.1 Electron Spin Resonance (B-S)

For a free electron in $\mathbf{B} = B\hat{z}$:

$$\hat{H}'_{ESR} = g_e \frac{\mu_B}{\hbar} \hat{S}_z B, \tag{153}$$

so

$$E_{ESR} = g_e \mu_B m_s B, \qquad m_s = \pm\frac{1}{2}, \tag{154}$$

and the transition frequency:

$$h\nu = g_e \mu_B B \implies \nu = \frac{g_e \mu_B B}{h}. \tag{155}$$

5.2 Nuclear Magnetic Resonance (B-I)

For a proton:

$$E_{NMR} = -g_p \mu_N m_I B, \qquad m_I = \pm\frac{1}{2}, \tag{156}$$

$$h\nu = g_p \mu_N B \implies \nu = \frac{g_p \mu_N B}{h}. \tag{157}$$

5.3 Normal Zeeman effect (B-L)

$$\hat{H}'_{NZ} = \frac{\mu_B}{\hbar} \hat{L}_z B, \qquad E_{NZ} = \mu_B m_l B. \tag{158}$$

For a transition $2P \to 1S$:

$$h\nu = \frac{3}{8} m_e c^2 \alpha^2 + \mu_B m_l B, \tag{159}$$

so with $\nu_0 = \frac{3 m_e c^2 \alpha^2}{8h}$:

$$\nu = \nu_0 + \frac{\mu_B B}{h} \ (m_l = 1), \quad \nu = \nu_0 \ (m_l = 0), \quad \nu = \nu_0 - \frac{\mu_B B}{h} \ (m_l = -1). \tag{160}$$

5.4 Anomalous Zeeman effect (weak B - L+S coupling)

$$\hat{\mu} = -\frac{\mu_B}{\hbar}(\hat{\mathbf{L}} + 2\hat{\mathbf{S}}) \implies \hat{\mu} = -g\frac{\mu_B}{\hbar}\hat{\mathbf{J}}, \tag{161}$$

with

$$g = 1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)}. \tag{162}$$

Then

$$E_{AZ} = g \mu_B m_j B. \tag{163}$$

For example:

- ${}^2S_{1/2}$: $g = 2$, $m_j = \pm\frac{1}{2}$.
- ${}^2P_{1/2}$: $g = \frac{2}{3}$, $m_j = \pm\frac{1}{2}$.
- ${}^2P_{3/2}$: $g = \frac{4}{3}$, $m_j = -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}$.

For the $2P_{1/2} \to 1S_{1/2}$ transition, four lines appear at

$$\nu = \nu_0 \pm \frac{2}{3}\frac{\mu_B B}{h}, \qquad \nu = \nu_0 \pm \frac{4}{3}\frac{\mu_B B}{h}. \tag{164}$$

5.5 Paschen–Back effect (strong B - L+S decoupled)

$$\hat{H}'_{PB} = \frac{\mu_B}{\hbar}(\hat{L}_z + 2\hat{S}_z)B, \tag{165}$$

so

$$E_{PB} = \mu_B B(m_l + 2m_s). \tag{166}$$

For a $2P \to 1S$ transition, selection rules $\Delta m_l = 0, \pm 1$, $\Delta m_s = 0$ give three lines:

$$\nu = \nu_0 - \frac{\mu_B B}{h}, \quad \nu_0, \quad \nu_0 + \frac{\mu_B B}{h}. \tag{167}$$

6 Many-Body Systems & Identical Particles

6.1 Identical Particles

- Bosons (integer spin): symmetric wavefunction.
- Fermions (half-integer spin): anti-symmetric wavefunction.

Spin states for two spin-1/2 particles:

- Symmetric triplet:

$$|11\rangle = |{+}{+}\rangle, \tag{168}$$

$$|10\rangle = \frac{1}{\sqrt{2}}(|{+}{-}\rangle + |{-}{+}\rangle), \tag{169}$$

$$|1\ -1\rangle = |{-}{-}\rangle. \tag{170}$$

- Anti-symmetric singlet:

$$|00\rangle = \frac{1}{\sqrt{2}}(|{+}{-}\rangle - |{-}{+}\rangle). \tag{171}$$

For fermions, the total wavefunction must be anti-symmetric:

$$|\Psi^A\rangle = |\Psi^A_{spatial}\rangle|\Psi^S_{spin}\rangle = |\Psi^S_{spatial}\rangle|\Psi^A_{spin}\rangle. \tag{172}$$

Interacting identical particles:

- Direct integral:

$$J_{nm} = \int \phi_n^*(\mathbf{x}_1)\phi_m^*(\mathbf{x}_2)V_{int}(|\mathbf{x}_2 - \mathbf{x}_1|)\phi_n(\mathbf{x}_1)\phi_m(\mathbf{x}_2)\, d\mathbf{x}_1 d\mathbf{x}_2. \tag{173}$$

- Exchange integral:

$$K_{nm} = \int \phi_n^*(\mathbf{x}_1)\phi_m^*(\mathbf{x}_2)V_{int}(|\mathbf{x}_2 - \mathbf{x}_1|)\phi_m(\mathbf{x}_1)\phi_n(\mathbf{x}_2)\, d\mathbf{x}_1 d\mathbf{x}_2. \tag{174}$$

- Bosons (spin 0): $\Delta E_{12} = J_{12} + K_{12}$.
- Fermions: symmetric spatial $\rightarrow J_{12} + K_{12}$; anti-symmetric spatial $\rightarrow J_{12} - K_{12}$.

6.2 Helium Atom

Hamiltonian:

$$\hat{H} = -\frac{\hbar^2}{2m_e}\nabla_1^2 - \frac{\hbar^2}{2m_e}\nabla_2^2 - \frac{2e^2}{4\pi\varepsilon_0 r_1} - \frac{2e^2}{4\pi\varepsilon_0 r_2} + \frac{e^2}{4\pi\varepsilon_0 r_{12}}. \tag{175}$$

Ground state: $|\Psi^S_{spatial}\rangle = |1s, 1s\rangle$, so total wavefunction is

$$|\Psi^A_{He}\rangle = |\Psi^S_{spatial}\rangle|\Psi^A_{spin}\rangle. \tag{176}$$

Excited states: spatial part must be anti-symmetric:

$$|\Psi^A_{spatial}\rangle = \frac{1}{\sqrt{2}}(|1s,2s\rangle - |2s,1s\rangle), \quad \frac{1}{\sqrt{2}}(|1s,2p\rangle - |2p,1s\rangle). \tag{177}$$

For a configuration like $1s\,3d$: $L = 2$, $S = 0$ or 1, so

$$J = \begin{cases} 2 & (S=0), \\ 1,2,3 & (S=1). \end{cases} \tag{178}$$

Allowed transitions require $\Delta S = 0$.

Pauli exclusion principle: no two fermions can occupy the same quantum state.

6.3 Bose–Einstein Condensation (BEC)

Any number of bosons can occupy the same state. For hydrogen atoms, total spin $F = I + S$ with $I = S = 1/2$ gives $F = 0, 1$, so H atoms are bosons.

For a gas at temperature T:

$$\frac{1}{2}m_H v^2 = \frac{3}{2}kT \implies p = m_H v = \sqrt{3m_H kT}, \tag{179}$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{3m_H kT}}. \tag{180}$$

When the de Broglie wavelength becomes comparable to inter-atom spacing, condensation occurs at $T_c \sim 10^{-6}$ K.

7 TIME-DEPENDENT PERTURBATION & TRANSITIONS

The logic of this section is as follows. We first write the time-dependent perturbation problem in the interaction picture and obtain the transition amplitude from the Dyson series. We then discuss constant perturbations, Fermi's golden rule, second-order matrix elements, harmonic perturbations, Rabi oscillations, and finally apply electric-dipole transitions to the hydrogen atom and state the selection rules.

7.1 Interaction Picture

For a total Hamiltonian $H_0 + V(t)$, define the interaction-picture state and perturbation operator by

$$|\Psi(t)\rangle_I = e^{iH_0 t}|\Psi(t)\rangle, \qquad V_I(t) = e^{iH_0 t}V(t)e^{-iH_0 t}. \tag{181}$$

The interaction-picture state then satisfies

$$i\frac{\partial}{\partial t}|\Psi(t)\rangle_I = V_I(t)|\Psi(t)\rangle_I, \qquad |\Psi(t)\rangle_I = U_I(t,t_0)|\Psi(t_0)\rangle_I = \sum_n c_n(t)|n\rangle. \tag{182}$$

7.2 Dyson Series

The interaction-picture time-evolution operator satisfies

$$\begin{cases} i\dfrac{\partial U_I(t,t_0)}{\partial t} = V_I(t)U_I(t,t_0), \\ U_I(t_0,t_0) = 1. \end{cases}$$

Iterating this equation gives the Dyson series

$$\begin{aligned} U_I(t,t_0) &= 1 - i\int_{t_0}^{t} dt_1\, V_I(t_1)U_I(t_1,t_0) \\ &= 1 - i\int_{t_0}^{t} dt_1\, V_I(t_1)\left[1 - i\int_{t_0}^{t_1} dt_2\, V_I(t_2)U_I(t_2,t_0)\right] \\ &= \cdots \\ &= \sum_{n=0}^{\infty}(-i)^n \int_{t_0}^{t} dt_1 \int_{t_0}^{t_1} dt_2 \cdots \int_{t_0}^{t_{n-1}} dt_n\, V_I(t_1)V_I(t_2)\cdots V_I(t_n). \end{aligned} \tag{183}$$

Equivalently, it can be written as the time-ordered exponential

$$U_I(t,t_0) = \mathcal{T}\exp\left[-i\int_{t_0}^{t} dt'\, V_I(t')\right].$$

7.3 Transition Probability

Take the initial state to be $|i\rangle$. Then

$$|\Psi(t_0)\rangle_I = |i\rangle, \qquad |\Psi(t)\rangle_I = U_I(t,t_0)|i\rangle = \sum_m c_m(t)|m\rangle.$$

The transition amplitude is therefore

$$c_n(t) = \langle n|U_I(t,t_0)|i\rangle. \tag{184}$$

Define the Bohr frequency and the perturbation matrix element by

$$\omega_{nm} = E_n - E_m, \qquad \langle n|V_I(t)|m\rangle = e^{i\omega_{nm}t}V_{nm}, \qquad V_{nm} = \langle n|V|m\rangle.$$

To second order, the time-evolution operator is approximated by

$$\begin{aligned} U_I(t,t_0) \approx 1 &- i\int_{t_0}^{t} dt_1\, V_I(t_1) \\ &+ (-i)^2 \int_{t_0}^{t} dt_1 \int_{t_0}^{t_1} dt_2\, V_I(t_1)V_I(t_2). \end{aligned}$$

Thus

$$\begin{aligned} c_n(t) &= \langle n|U_I(t,t_0)|i\rangle \\ &\approx \delta_{ni} - i\int_{t_0}^{t} dt_1\, e^{i\omega_{ni}t_1}V_{ni}(t_1) \\ &\quad + (-i)^2 \int_{t_0}^{t} dt_1 \int_{t_0}^{t_1} dt_2 \sum_m e^{i\omega_{nm}t_1}e^{i\omega_{mi}t_2}V_{nm}(t_1)V_{mi}(t_2). \end{aligned}$$

The transition probability is

$$P(i \to n) = |c_n(t)|^2.$$

7.4 Constant Perturbation

Consider

$$V(t) = \begin{cases} 0, & t < 0, \\ V, & t \geq 0, \end{cases} \qquad t_0 = 0.$$

The zeroth- and first-order amplitudes are

$$\begin{cases} c_n^{(0)} = \delta_{ni}, \\ c_n^{(1)} = -iV_{ni}\displaystyle\int_0^t dt_1\, e^{i\omega_{ni}t_1} = \dfrac{V_{ni}}{\omega_{ni}}\left(1 - e^{i\omega_{ni}t}\right). \end{cases} \tag{185}$$

Its squared modulus is

$$\begin{aligned} \left|c_n^{(1)}\right|^2 &= \frac{|V_{ni}|^2}{|\omega_{ni}|^2} \cdot 4\sin^2\frac{\omega_{ni}t}{2} \\ &= \frac{4|V_{ni}|^2}{|E_n - E_i|^2}\sin^2\left[\frac{(E_n - E_i)t}{2}\right] \\ &= |V_{ni}|^2\, t^2\, \mathrm{sinc}^2\left[\frac{(E_n - E_i)t}{2}\right], \end{aligned} \tag{186}$$

where $\mathrm{sinc}\, x = \sin x/x$.

7.5 Fermi's Golden Rule

For continuum final states, take the long-time limit. Using

$$\int_{-\infty}^{\infty} \mathrm{sinc}^2 y \, dy = \int_{-\infty}^{\infty} \left(\frac{\sin y}{y}\right)^2 dy = \pi,$$

and

$$\lim_{\alpha\to\infty} \frac{\alpha}{\pi} \mathrm{sinc}^2(\alpha x) = \delta(x),$$

one obtains

$$\lim_{t\to\infty} \frac{t}{2\pi} \mathrm{sinc}^2 \left[\frac{(E_n - E_i)t}{2}\right] = \delta(E_n - E_i).$$

Therefore

$$\lim_{t\to\infty} \left|c_n^{(1)}(t)\right|^2 = 2\pi \, \overline{|V_{ni}|^2} \, t \, \delta(E_n - E_i).$$

Let $\rho(E)$ be the density of final states, i.e. the number of states in the energy interval $(E, E + dE)$. In the long-time limit,

$$\begin{aligned} P(i \to n) &= \sum_{\substack{n \\ E_n \approx E_i}} \left|c_n^{(1)}\right|^2_{t\to\infty} = \int dE_n \, \rho(E_n) \left|c_n^{(1)}(t)\right|^2_{t\to\infty} \\ &= \int dE_n \, \rho(E_n) \, 2\pi \overline{|V_{ni}|^2} \, t \, \delta(E_n - E_i) \\ &= 2\pi \overline{|V_{ni}|^2} \rho(E_i) \, t. \end{aligned}$$

Writing the transition matrix element as M_{fi}, the transition rate per unit time is

$$\Gamma_{fi} = \frac{d}{dt} P(i \to f) = 2\pi |M_{fi}|^2 \rho(E_f), \qquad E_f \approx E_i. \tag{187}$$

Here the first-order transition matrix element is

$$M_{fi} = \langle f|V|i\rangle = V_{fi}.$$

The second-order correction gives

$$M_{fi} = V_{fi} + \sum_m \frac{V_{fm} V_{mi}}{E_i - E_m}, \qquad \Gamma_{fi} = 2\pi |M_{fi}|^2 \rho(E_f), \quad E_f \approx E_i. \tag{188}$$

7.6 Harmonic Perturbation

Take

$$V(t) = V e^{i\omega t} + V^\dagger e^{-i\omega t}, \qquad t_0 = 0.$$

The first-order amplitude is

$$\begin{aligned} c_n^{(1)}(t) &= -i \int_0^t dt_1 \, e^{i\omega_{ni} t_1} V_{ni}(t_1) \\ &= -i \int_0^t dt_1 \, e^{i\omega_{ni} t_1} \left(V_{ni} e^{i\omega t_1} + V_{ni}^\dagger e^{-i\omega t_1}\right) \\ &= V_{ni} \frac{1 - e^{i(\omega_{ni} + \omega)t}}{\omega_{ni} + \omega} + V_{ni}^\dagger \frac{1 - e^{i(\omega_{ni} - \omega)t}}{\omega_{ni} - \omega}. \end{aligned}$$

As $t \to \infty$, $\left|c_n^{(1)}\right|^2$ is appreciable only if

$$\omega_{ni} + \omega \approx 0, \qquad \omega_{ni} - \omega \approx 0, \qquad \omega_{ni} = E_n - E_i.$$

Equivalently,

$$E_n \approx E_i - \omega, \qquad E_n \approx E_i + \omega.$$

The corresponding transition rate is

$$\Gamma_{ni} = \frac{dP(i \to n)}{dt} = \begin{cases} 2\pi \overline{|V_{ni}|^2} \rho(E_n), & E_n \approx E_i - \omega, \\ 2\pi \overline{|V_{ni}^\dagger|^2} \rho(E_n), & E_n \approx E_i + \omega. \end{cases} \tag{189}$$

7.7 Rabi Oscillation

Consider a two-level system driven by a sinusoidal perturbation. In the Schrödinger picture,

$$|\Psi(t)\rangle = C_1(t) e^{-iE_1 t}|1\rangle + C_2(t) e^{-iE_2 t}|2\rangle,$$
$$H_0 = E_1|1\rangle\langle 1| + E_2|2\rangle\langle 2|, \qquad V(t) = re^{i\omega t}|1\rangle\langle 2| + re^{-i\omega t}|2\rangle\langle 1|.$$

In the interaction picture, $|\Psi\rangle_I = e^{iH_0 t}|\Psi\rangle$ and $V_I(t) = e^{iH_0 t} V(t) e^{-iH_0 t}$, so that

$$i\frac{d}{dt}\begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} 0 & re^{i(\omega-\omega_{21})t} \\ re^{-i(\omega-\omega_{21})t} & 0 \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}, \qquad \omega_{21} = E_2 - E_1.$$

For the initial condition $C_2(0) = 0$, $|C_1(0)| = 1$, the exact solution is

$$\begin{aligned} |C_2(t)|^2 &= \frac{r^2}{\Omega^2} \sin^2(\Omega t), \\ |C_1(t)|^2 &= 1 - |C_2(t)|^2, \\ \Omega &= \sqrt{\frac{(\omega - \omega_{21})^2}{4} + r^2}. \end{aligned} \tag{190}$$

At resonance, $\omega = \omega_{21}$, the Rabi frequency reduces to

$$\Omega = r,$$

and the population oscillates completely between $|1\rangle$ and $|2\rangle$:

$$|C_2(t)|^2 = \sin^2(rt), \qquad |C_1(t)|^2 = \cos^2(rt).$$

In contrast to Fermi's golden rule, which predicts a constant transition rate in the weak-coupling limit, the Rabi solution shows that for a discrete two-level system under strong coherent driving the dynamics are fully reversible and periodic.

7.8 Electron transition for H atom

For z-polarized light,

$$\mathbf{E}(t) = E_0 \mathbf{z} \cos(\omega t) = \frac{1}{2} E_0 \mathbf{z} \left(e^{i\omega t} + e^{-i\omega t}\right).$$

The electron-field interaction is

$$V(t) = -e\mathbf{E}(t)\cdot\mathbf{r} = -\frac{1}{2}eE_0 r\cos\theta\left(e^{i\omega t}+e^{-i\omega t}\right).$$

Thus it can be written in the harmonic-perturbation form

$$V(t) = V\left(e^{i\omega t}+e^{-i\omega t}\right), \qquad V = -\frac{1}{2}eE_0 r\cos\theta.$$

The transition matrix element is

$$\begin{aligned} V_{fi} &= \langle f|V|i\rangle = -\frac{1}{2}eE_0\langle f|r\cos\theta|i\rangle \\ &= -\frac{1}{2}eE_0\int \psi_f^*\psi_i\,(r\cos\theta)\,dV \\ &= -\frac{1}{2}eE_0\int R_f^*Y_f^*\,R_iY_i\,r\cos\theta\,r^2\sin\theta\,dr\,d\theta\,d\phi \\ &= -\frac{1}{2}eE_0\left[\int_0^\infty R_f^*(r)R_i(r)\,r^3dr\right] \\ &\quad\times\left[\int Y_f^*(\theta,\phi)Y_i(\theta,\phi)\sin\theta\cos\theta\,d\theta\,d\phi\right]. \end{aligned} \tag{191}$$

7.9 Selection Rules for Electric Dipole (E1) Transitions

From the matrix element derived above, the angular integral

$$\int Y_f^*(\theta,\phi)Y_i(\theta,\phi)\sin\theta\cos\theta\,d\theta\,d\phi$$

determines whether V_{fi} vanishes. Using

$$\cos\theta \propto Y_{1,0}(\theta,\phi)$$

and the Wigner-Eckart theorem, one obtains the electric-dipole selection rules:

$$\begin{cases} \text{Parity: } \pi_f = -\pi_i, \\ \text{Orbital angular momentum: } \Delta\ell = \pm 1, \\ \text{Magnetic quantum number: } \Delta m = 0,\ \pm 1. \end{cases} \tag{192}$$

For multi-electron atoms, these generalize to

$$\begin{cases} \text{Parity: } \pi_f = -\pi_i, \\ \text{Total angular momentum: } \Delta J = 0,\ \pm 1 \quad (J_i = 0 \to J_f = 0 \text{ forbidden}), \\ \Delta m_J = 0,\ \pm 1, \\ \text{Spin: } \Delta S = 0, \\ \text{Orbital: } \Delta L = 0,\ \pm 1 \quad (L_i = 0 \to L_f = 0 \text{ forbidden}). \end{cases} \tag{193}$$

These rules follow directly from Fermi's golden rule and the electric-dipole interaction Hamiltonian

$$H_{\text{int}} = -e\mathbf{E}\cdot\mathbf{r},$$

where $e > 0$ is the elementary charge.

Gavin's Lectures on Physics - Statistical Mechanics



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1 Linkage : Thermodynamics Statistical Mechanics

1.1 Boltzmann Entropy Formula

The entropy is expressed as the logarithm of the number of microstates:

$$S = k_B \ln \Omega \tag{1}$$

1.2 Thermodynamic Identity

The combined first and second laws for a system with variable particle number are:

$$TdS = dQ = dU + PdV - \mu dN \tag{2}$$

The temperature, pressure and chemical potential are defined as partial derivatives of the entropy:

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{V,N}, \quad \frac{P}{T} = \left(\frac{\partial S}{\partial V}\right)_{U,N}, \quad \frac{\mu}{T} = -\left(\frac{\partial S}{\partial N}\right)_{U,V} \tag{3}$$

2 Laws of Thermodynamics

2.1 Zeroth Law of Thermodynamics

Transitivity of thermal equilibrium: if two systems are each in thermal equilibrium with a third system, they are in thermal equilibrium with each other.

$$T_1 = T_2,\ T_3 = T_2 \implies T_1 = T_3$$

2.2 First Law of Thermodynamics

Law of conservation of energy. In this document, dW denotes the work done *on* the system (so that for a gas, $dW = -PdV$).

$$dU = dW + dQ$$

2.3 Second Law of Thermodynamics

Entropy increase principle: the entropy of an isolated system never decreases, and any spontaneous process increases the total entropy.

$$dS \geq \frac{\delta Q}{T}, \quad \Delta S_{\text{isolated}} \geq 0$$

2.4 Third Law of Thermodynamics

Nernst's theorem: as the temperature approaches absolute zero, the entropy change of any isothermal process approaches zero.

$$\lim_{T \to 0} (\Delta S)_T = 0$$

3 Basic Statistical Distributions

Define:

$$\alpha = -\frac{\mu}{k_B T}, \quad \beta = \frac{1}{k_B T} \tag{4}$$

a_l: particle number in level l
w_l: degeneracy in level l
ε_l: energy of a single particle in level l
Then

$$N = \sum_l a_l, \qquad U = \sum_l a_l \varepsilon_l.$$

Using Stirling's approximation $\ln(m!) \approx m \ln m - m$ and maximizing $\ln \Omega$ subject to constraints gives

$$\Phi = \ln \Omega - \alpha \left(\sum_l a_l - N \right) - \beta \left(\sum_l a_l \varepsilon_l - E \right)$$

Number of micro states satisfies:

$$0 = \frac{\partial \Phi}{\partial a_l} = \frac{\partial \ln \Omega}{\partial a_l} - \alpha - \beta \varepsilon_l \tag{5}$$

3.1 Maxwell-Boltzmann Distribution (Classical)

For the classical case where $a_l \ll w_l$, the number of microstates is

$$\Omega_{\mathrm{MB}} = N! \prod_l \frac{w_l^{a_l}}{a_l!}, \quad (a_l \ll w_l).$$

Taking the logarithm and applying Stirling's approximation gives

$$\ln \Omega_{\mathrm{MB}} \approx \ln(N!) + \sum_l \left(a_l \ln w_l - a_l \ln a_l + a_l \right).$$

Differentiation with respect to a_l yields

$$\alpha + \beta \varepsilon_l = \frac{\partial \ln \Omega_{\mathrm{MB}}}{\partial a_l} \approx \ln \frac{w_l}{a_l}.$$

Thus the Maxwell–Boltzmann distribution is

$$a_l = w_l e^{-\alpha - \beta \varepsilon_l} = w_l e^{-(\varepsilon_l - \mu)/k_B T} \tag{6}$$

3.2 Bose-Einstein Distribution (Bosons)

For bosons, the number of ways to distribute a_l identical particles among w_l degenerate states is given by the combinatorics of dividers:

$$\Omega_{\mathrm{BE}} = \prod_l \frac{(w_l + a_l - 1)!}{a_l! \, (w_l - 1)!}.$$

Using Stirling's approximation, the logarithm becomes

$$\ln \Omega_{\mathrm{BE}} \approx (w_l + a_l - 1)\ln(w_l + a_l - 1) - a_l \ln a_l - (w_l - 1)\ln(w_l - 1) \\ - (w_l + a_l - 1) + a_l + (w_l - 1).$$

Differentiation gives

$$\alpha + \beta\varepsilon_l = \frac{\partial \ln \Omega_{\mathrm{BE}}}{\partial a_l} \approx \ln\left(\frac{w_l}{a_l} + 1\right).$$

Hence the Bose–Einstein distribution is

$$a_l = \frac{w_l}{e^{\alpha+\beta\varepsilon_l} - 1} = \frac{w_l}{e^{(\varepsilon_l - \mu)/k_B T} - 1} \tag{7}$$

3.3 Fermi-Dirac Distribution (Fermions)

For fermions, each state can be occupied by at most one particle, so the multiplicity is

$$\Omega_{\mathrm{FD}} = \prod_l C_{w_l}^{a_l} = \prod_l \frac{w_l!}{a_l!\,(w_l - a_l)!}.$$

Using Stirling's approximation,

$$\ln \Omega_{\mathrm{FD}} \approx w_l \ln w_l - w_l - a_l \ln a_l + a_l \\ - (w_l - a_l)\ln(w_l - a_l) + (w_l - a_l).$$

Differentiation yields

$$\alpha + \beta\varepsilon_l = \frac{\partial \ln \Omega_{\mathrm{FD}}}{\partial a_l} = \ln\left(\frac{w_l}{a_l} - 1\right).$$

Thus the Fermi–Dirac distribution is

$$a_l = \frac{w_l}{e^{\alpha+\beta\varepsilon_l} + 1} = \frac{w_l}{e^{(\varepsilon_l - \mu)/k_B T} + 1} \tag{8}$$

In the classical limit $a_l \ll w_l$, both Bose and Fermi distributions reduce to the Maxwell–Boltzmann form, as seen from

$$\Omega_{\mathrm{BE}},\ \Omega_{\mathrm{FD}} \xrightarrow{a_l \ll w_l} \frac{\Omega_{\mathrm{MB}}}{N!}.$$

The maximisation of $\Omega(a_l)$ determines the final occupation numbers.

4 Statistics of Distributions

4.1 Boltzmann Statistics

4.1.1 Conservation of energy

The first law in differential form:

$$\begin{aligned} dU &= dW + dQ \\ &= Ydy + TdS, \end{aligned}$$

and also

$$\sum_l d(a_l\varepsilon_l) = \sum_l a_l d\varepsilon_l + \sum_l \varepsilon_l da_l.$$

The connection between macroscopic and microscopic world:

$$\begin{aligned} dU &= \sum_l d(a_l\varepsilon_l) \\ dW &= Ydy = \sum_l a_l d\varepsilon_l \\ dQ &= TdS = \sum_l \varepsilon_l da_l \end{aligned} \tag{9}$$

4.1.2 Thermodynamic quantity

For Boltzmann statistics, the occupation numbers and the entropy logarithm are:

$$\begin{cases} a_l = w_l e^{-\alpha-\beta\varepsilon_l} \\ \ln\Omega_{\mathrm{MB}} \approx N\ln N + \alpha N + \beta U \end{cases}$$

We define the single-particle partition function

$$Z_1 = \sum_l w_l e^{-\beta\varepsilon_l} \tag{10}$$

Then the total particle number and energy are expressed as

$$\begin{cases} N = \sum_l a_l = e^{-\alpha} Z_1 \\ U = \sum_l a_l\varepsilon_l = -N\frac{\partial \ln Z_1}{\partial\beta} \end{cases} \tag{11}$$

As ω_l and β are independent of y, only ε_l is a function of y. Then,

$$Z_1 = Z_1(y,\beta)$$

The generalised force conjugate to an external parameter y is

$$Y = \sum_l \frac{\partial\varepsilon_l}{\partial y} a_l = -\frac{N}{\beta}\frac{\partial \ln Z_1}{\partial y},$$

so the work done on the system is

$$dW = Ydy = -\frac{N}{\beta}\left(d\ln Z_1 - \frac{\partial \ln Z_1}{\partial\beta} d\beta\right).$$

From the first law, we obtain

$$\begin{aligned} TdS = dQ &= dU - dW \\ &= -Nd\frac{\partial \ln Z_1}{\partial \beta} + \frac{N}{\beta}\left(d\ln Z_1 - \frac{\partial \ln Z_1}{\partial \beta}d\beta\right) \\ &= \frac{N}{\beta}d\left[\ln Z_1 - \frac{\partial \ln Z_1}{\partial \beta}\beta\right]. \end{aligned}$$

Setting $\beta = 1/(k_B T)$, we get

$$\begin{aligned} S &= k_B N\left(\ln Z_1 - \frac{\partial \ln Z_1}{\partial \beta}\beta\right) \\ &= k_B N\left(\ln N + \alpha + \beta\frac{U}{N}\right) \\ &= k_B\left(N\ln N + \alpha N + \beta U\right). \end{aligned}$$

Thus the entropy is given by

$$S = k_B \ln \Omega_{\mathrm{MB}} \tag{12}$$

From the relations U and S as above:

$$\begin{cases} U = -N\frac{\partial \ln Z_1}{\partial \beta} \\ S = k_B N\left(\ln Z_1 - \frac{\partial \ln Z_1}{\partial \beta}\beta\right) \\ \beta = \frac{1}{k_B T} \end{cases}$$

The Helmholtz free energy:

$$F = U - TS = -Nk_B T \ln Z_1 \tag{13}$$

From the derivative of $\ln Z_1$ with respect to volume:

$$\begin{cases} Y = -\frac{N}{\beta}\frac{\partial \ln Z_1}{\partial y} \\ dW = Ydy = -PdV \end{cases}$$

The pressure:

$$P = \frac{N}{\beta}\frac{\partial \ln Z_1}{\partial V} \tag{14}$$

4.1.3 Ideal gas equation of state

For a single particle in a box, the energy and phase-space element are

$$\begin{cases} \varepsilon = \left(p_x^2 + p_y^2 + p_z^2\right)/(2m) \\ dw = dxdydz\ dp_x dp_y dp_z/h^3 \end{cases}$$

Then

$$Z_1 = \sum_l w_l e^{-\beta\varepsilon_l} = \int e^{-\beta\varepsilon} dw = V\left(\frac{2\pi m}{h^2\beta}\right)^{\frac{3}{2}}.$$

Using the pressure formula above, we find

$$P = \frac{N}{\beta}\frac{\partial \ln Z_1(\beta, V)}{\partial V} = \frac{N}{\beta V} = \frac{N k_B T}{V},$$

which yields the ideal gas law:

$$PV = N k_B T \tag{15}$$

The condition for the classical (ideal) gas is

$$e^{\alpha} = \frac{Z_1}{N} = \frac{V}{N}\left(\frac{2\pi m}{h^2 \beta}\right)^{\frac{3}{2}} \gg 1, \quad E_p = 0,\ \varepsilon = \frac{1}{2}mv^2.$$

4.1.4 Maxwell velocity distribution Law

Starting from the phase-space element

$$dw = \frac{dxdydz\ dp_x dp_y dp_z}{h^3},$$

we have

$$\begin{aligned} Z_1 &= \int e^{-\beta\varepsilon} dw = \frac{V}{h^3}\int e^{-\frac{mv^2}{2k_B T}} m^3 dv_x dv_y dv_z \\ &= \frac{m^3 V}{h^3}\int_0^{\infty} e^{-\frac{mv^2}{2k_B T}} \cdot 4\pi v^2 dv. \end{aligned}$$

Using derived partition function,

$$Z_1 = V\left(\frac{2\pi m k_B T}{h^2}\right)^{3/2}$$

We get the Maxwell speed distribution:

$$f(v) = 4\pi\left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 e^{-mv^2/(2k_B T)} \tag{16}$$

With normalization:

$$\int_0^{\infty} f(v)dv = 1$$

4.1.5 Velocities

The Gamma function is

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt,$$

with properties $\Gamma(z+1) = z\,\Gamma(z), \quad \Gamma(n) = (n-1)!, \quad \Gamma(\frac{1}{2}) = \sqrt{\pi}$.

The most probable, mean, and root-mean-square speeds are

$$\begin{cases} f'(v_p) = 0 \implies v_p = \sqrt{\frac{2k_B T}{m}} \\ \langle v \rangle = \int_0^{\infty} v f(v) dv = \sqrt{\frac{8k_B T}{\pi m}} \\ \sqrt{\langle v^2 \rangle} = \left[\int_0^{\infty} v^2 f(v) dv\right]^{\frac{1}{2}} = \sqrt{\frac{3k_B T}{m}}. \end{cases}$$

4.1.6 The Equipartition Theorem

For a single-atom molecule with 3 degrees of freedom.

Consider partition,

$$Z_1 = V\left(\frac{2\pi m k_B T}{h^2}\right)^{\frac{3}{2}} = V\left(\frac{2\pi m}{\beta h^2}\right)^{\frac{3}{2}}$$

Total internal energy,

$$U = -N\frac{\partial \ln Z_1}{\partial \beta} = \frac{3}{2}\frac{N}{\beta} = \frac{3}{2}Nk_BT$$

Then the average energy for the x-direction is

$$\langle \varepsilon_x \rangle = \frac{1}{3}\frac{U}{N} = \frac{1}{2}k_BT.$$

Each degree of freedom contributes $\frac{1}{2}k_BT$ to the average energy.

In general, the total internal energy is

$$U = \frac{s}{2}Nk_BT, \tag{17}$$

where s is the number of degrees of freedom.

4.1.7 Heat Capacity

The heat capacities,

$$\begin{cases} C_V = \frac{\partial U}{\partial T}\big|_V \\ C_P = \frac{\partial H}{\partial T}\big|_P \end{cases}$$

From

$$\begin{cases} PV = Nk_BT \\ H = U + PV \end{cases}$$

We get the relation of capacities,

$$C_P - C_V = Nk_B \tag{18}$$

Define the capacity ratio:

$$\gamma = \frac{C_P}{C_V}$$

For a monatomic gas (He, Ne, Ar):
$s = 3,\ U = \frac{3}{2}Nk_BT,\ C_V = \frac{3}{2}Nk_B,\ C_P = \frac{5}{2}Nk_B,\ \gamma = \frac{5}{3}$.
For a diatomic gas ($H_2, O_2, \ldots$):
$s = 5,\ U = \frac{5}{2}Nk_BT,\ C_V = \frac{5}{2}Nk_B,\ C_P = \frac{7}{2}Nk_B,\ \gamma = \frac{7}{5}$.

4.2 Bose Statistics

For bosons, the occupation number is

$$a_l = \frac{w_l}{e^{\alpha+\beta\varepsilon_l} - 1}.$$

The grand partition function,

$$\mathcal{Z} = \prod_l \left(1 - e^{-\alpha-\beta\varepsilon_l}\right)^{-w_l} \tag{19}$$

So,

$$\ln \mathcal{Z} = -\sum_l w_l \ln\left(1 - e^{-\alpha-\beta\varepsilon_l}\right).$$

The total particle number and energy,

$$\begin{cases} N = \sum_l a_l = -\frac{\partial \ln \mathcal{Z}}{\partial \alpha} \\ U = \sum_l a_l \varepsilon_l = -\frac{\partial \ln \mathcal{Z}}{\partial \beta}. \end{cases} \tag{20}$$

With $dW = Ydy = -PdV$, the generalised force is

$$Y = \sum_l \frac{\partial \varepsilon_l}{\partial y} a_l = -\frac{1}{\beta}\frac{\partial \ln \mathcal{Z}}{\partial y},$$

and the pressure is

$$P = \frac{1}{\beta}\frac{\partial \ln \mathcal{Z}}{\partial V},$$

4.3 Fermi Statistics

For fermions, the occupation number is

$$a_l = \frac{w_l}{e^{\alpha+\beta\varepsilon_l} + 1}.$$

The grand partition function,

$$\mathcal{Z} = \prod_l \left(1 + e^{-\alpha-\beta\varepsilon_l}\right)^{w_l} \tag{21}$$

So,

$$\ln \mathcal{Z} = \sum_l w_l \ln\left(1 + e^{-\alpha-\beta\varepsilon_l}\right).$$

The total particle number and energy,

$$\begin{cases} N = \sum_l a_l = -\frac{\partial \ln \mathcal{Z}}{\partial \alpha} \\ U = \sum_l a_l \varepsilon_l = -\frac{\partial \ln \mathcal{Z}}{\partial \beta} \end{cases}$$

With $dW = Ydy = -PdV$, the generalised force is

$$Y = \sum_l \frac{\partial \varepsilon_l}{\partial y} a_l = -\frac{1}{\beta}\frac{\partial \ln \mathcal{Z}}{\partial y},$$

and the pressure is

$$P = \frac{1}{\beta}\frac{\partial \ln \mathcal{Z}}{\partial V},$$

5 Ensembles' Theory

5.1 Definition of Ensemble

An ensemble is a collection of identical copies of a system, each representing a possible microstate of the real system.

As $N \to \infty$, all ensembles are equivalent for large systems, differing only in their fixed macroscopic variables.

5.2 The Three Ensembles

5.2.1 Microcanonical Ensemble

Fixed: N, V, E.

Potential: $S = k_B \ln \Omega$.

Example: Einstein Solid Multiplicity. For N independent quantum harmonic oscillators and q identical energy quanta, the multiplicity is

$$\Omega(N, q) = \frac{(q + N - 1)!}{q!\,(N - 1)!}.$$

5.2.2 Canonical Ensemble

Fixed: N, V, T.

Potential: $F = U - TS = -k_B T \ln Z$.

The canonical partition function is the sum over microstates with the Boltzmann factor:

$$Z = \sum_S e^{-\beta E_S}, \qquad \beta = \frac{1}{k_B T}. \tag{22}$$

The internal energy is obtained from the derivative of $\ln Z$ with respect to β:

$$U = \overline{E} = -\frac{\partial \ln Z}{\partial \beta}. \tag{23}$$

The entropy in terms of the partition function is

$$S = -\left(\frac{\partial F}{\partial T}\right)_{V,N} = k_B \left[\ln Z + T \frac{\partial \ln Z}{\partial T}\right]. \tag{24}$$

The chemical potential follows from the dependence of F on N:

$$\mu = \left(\frac{\partial F}{\partial N}\right)_{T,V} = -k_B T \left(\frac{\partial \ln Z}{\partial N}\right)_{T,V}. \tag{25}$$

The heat capacity is related to energy fluctuations:

$$C_V = \left(\frac{\partial \overline{E}}{\partial T}\right)_V = \frac{1}{k_B T^2} \sigma_E^2, \qquad \sigma_E^2 = \overline{E^2} - (\overline{E})^2.$$

5.2.3 Grand Canonical Ensemble

Fixed: μ, V, T.

Potential: $\Omega_G = F - \mu N = -k_B T \ln \mathcal{Z}$.

The grand partition function sums over all particle numbers and microstates:

$$\mathcal{Z} = \sum_{N=0}^{\infty} \sum_{S} e^{-\beta(E_S - \mu N_S)}. \tag{26}$$

The average particle number is obtained from the grand potential:

$$\overline{N} = -\left(\frac{\partial \Omega_G}{\partial \mu}\right)_{T,V}. \tag{27}$$

The internal energy from the grand partition function is

$$\overline{E} = -\left(\frac{\partial \ln \mathcal{Z}}{\partial \beta}\right)_{\mu,V} + \mu \overline{N}. \tag{28}$$

The entropy is given by the temperature derivative of the grand potential:

$$S = -\left(\frac{\partial \Omega_G}{\partial T}\right)_{\mu,V}. \tag{29}$$

5.3 Thermodynamic Potentials

The thermodynamic potentials are defined as:

- Internal energy: U
- Enthalpy: $H = U + PV$
- Helmholtz free energy: $F = U - TS$
- Gibbs free energy: $G = H - TS = F + PV$
- Grand potential: $\Omega_G = F - \mu N$

Their differential forms are:

$$\begin{cases} dU = T\,dS - P\,dV + \mu\,dN, \\ dH = T\,dS + V\,dP + \mu\,dN, \\ dF = -S\,dT - P\,dV + \mu\,dN, \\ dG = -S\,dT + V\,dP + \mu\,dN, \\ d\Omega_G = -S\,dT - P\,dV - N\,d\mu. \end{cases} \tag{30}$$

5.4 Maxwell Relations

From the equality of mixed second derivatives, we obtain the Maxwell relations:

$$\begin{cases} \left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V, \\ \left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P, \\ \left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V, \\ \left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P. \end{cases} \tag{31}$$

6 Applications

6.1 Blackbody Radiation

Ensemble : Grand Canonical with $\mu_\gamma = 0$. Distribution : Bose-Einstein gives the Planck distribution $\overline{n}_\gamma = \frac{1}{e^{\varepsilon/k_B T}-1}$.

6.1.1 Photon Density of States

For photons in a box, with $k_i = n_i\pi/L$, volume $V = L^3$, two polarisations and positive octant (since $n_i > 0$), the number of states in $[k, k+dk]$ is

$$g(k)\,dk = 2 \cdot \frac{1}{8} \cdot \frac{4\pi k^2 dk}{(\pi/L)^3} = \frac{V}{\pi^2} k^2 dk.$$

Using the photon dispersion relation $\varepsilon = h\nu = hck/(2\pi)$, we convert to energy space:

$$g(\varepsilon)\,d\varepsilon = g(k)\frac{dk}{d\varepsilon}\,d\varepsilon = \frac{8\pi V}{(hc)^3}\varepsilon^2 d\varepsilon.$$

The photon density of states per unit energy is therefore:

$$g(\varepsilon) = \frac{8\pi V}{(hc)^3}\varepsilon^2. \tag{32}$$

The two polarisations correspond to $S_\gamma = 1$ with $m_\gamma = \pm 1$ (the longitudinal mode $m_\gamma = 0$ is forbidden), giving linear x/y or right/left circular polarisations.

6.1.2 Planck Energy Spectrum

The spectral energy density is obtained by multiplying three factors:

- the single-photon energy ε,
- the density of states $g(\varepsilon) = 8\pi V\varepsilon^2/(hc)^3$,
- the average occupation number of each energy state $\overline{n}_\gamma = 1/(e^{\varepsilon/k_B T} - 1)$, from Bose-Einstein statistics with zero chemical potential ($\mu_\gamma = 0$).

Thus the energy per unit volume in modes with energies between ε and $\varepsilon + d\varepsilon$ is

$$\begin{aligned} u(\varepsilon)\,d\varepsilon &= \frac{1}{V}\,\varepsilon\,\overline{n}_\gamma\,g(\varepsilon)\,d\varepsilon \\ &= \frac{8\pi\varepsilon^3}{(hc)^3}\frac{1}{e^{\varepsilon/k_B T}-1}\,d\varepsilon. \end{aligned}$$

Hence the Planck spectral energy density is

$$u(\varepsilon) = \frac{8\pi\varepsilon^3}{(hc)^3}\frac{1}{e^{\varepsilon/k_B T}-1}. \tag{33}$$

This spectrum correctly reproduces the Rayleigh-Jeans law in the low-frequency limit ($\varepsilon \ll k_B T$, where the exponential approximates to $1 + \varepsilon/k_B T$, giving $u(\varepsilon) \propto \varepsilon^2 k_B T$) and the Wien law in the high-frequency limit ($\varepsilon \gg k_B T$, where the exponential dominates). It resolves the ultraviolet catastrophe and was the foundation for the birth of quantum mechanics.

6.1.3 Stefan-Boltzmann Law

For a general electromagnetic field, the local energy density and the energy flux density are

$$w = \frac{1}{2}\left(\varepsilon E^2 + \frac{1}{\mu}B^2\right), \qquad \mathbf{S} = \frac{1}{\mu}\mathbf{E}\times\mathbf{B}.$$

The total energy in a volume and the power flowing through a surface are obtained by integration:

$$E_{\text{total}} = \int w\, dV, \qquad P = \int \mathbf{S}\cdot d\mathbf{A}.$$

For blackbody radiation inside a cavity, w and $\mathbf{S}$ fluctuate rapidly in time. We are interested in their statistical averages. The average energy density is

$$\langle w\rangle = \frac{U}{V}.$$

This quantity is obtained by integrating the Planck spectrum over all photon energies:

$$\begin{aligned}\frac{U}{V} &= \int_0^\infty u(\varepsilon)\, d\varepsilon \\ &= \frac{8\pi}{(hc)^3}\int_0^\infty \frac{\varepsilon^3}{e^{\varepsilon/k_BT}-1}\, d\varepsilon \\ &= \frac{8\pi(k_BT)^4}{(hc)^3}\int_0^\infty \frac{x^3}{e^x-1}\, dx.\end{aligned}$$

Using the standard integral $\int_0^\infty x^3/(e^x-1)\, dx = \pi^4/15$, (from the Riemann zeta function $\zeta(4) = \pi^4/90$), we obtain

$$\frac{U}{V} = \frac{8\pi(k_BT)^4}{(hc)^3}\cdot\frac{\pi^4}{15} = a_S T^4, \tag{34}$$

with the radiation constant

$$a_S = \frac{8\pi^5 k_B^4}{15(hc)^3}.$$

Next we determine the average power radiated through a surface. For isotropic radiation, the flux through a surface element is reduced by the factor $\cos\theta$, where θ is the angle between the propagation direction and the surface normal. Averaging over the forward hemisphere gives

$$\langle S_n\rangle = \frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi/2}\frac{U}{V}\, c\cos\theta\, \sin\theta\, d\theta\, d\phi = \frac{c}{4}\frac{U}{V}.$$

Thus the power per unit area escaping through the surface is $\frac{c}{4}\frac{U}{V}$.

Finally, the total radiated power from a blackbody surface of area A follows from $P = \int \mathbf{S}\cdot d\mathbf{A}$:

$$P_r = \langle S_n\rangle A = \frac{c}{4}\frac{U}{V}A.$$

Substituting $U/V = a_S T^4$ gives

$$P_r = \frac{c}{4}a_S T^4 A = \sigma T^4 A, \tag{35}$$

where the Stefan–Boltzmann constant is

$$\sigma = \frac{c}{4}a_S = \frac{2\pi^5 k_B^4}{15h^3c^2} = 5.67\times10^{-8}\ \text{W}/(\text{m}^2\cdot\text{K}^4).$$

6.1.4 Wien's Displacement Law

The peak of the Planck spectrum is found by maximising $u(\varepsilon)$ with respect to ε, i.e. solving $du/d\varepsilon = 0$. Let $x = \varepsilon/(k_B T)$. The condition $du/d\varepsilon = 0$ is equivalent to

$$\frac{d}{dx}\left(\frac{x^3}{e^x - 1}\right) = 0.$$

The positive solution of this transcendental equation is

$$x = 2.82.$$

Thus the peak energy is

$$\varepsilon_{\text{peak}} = 2.82\, k_B T.$$

Using the relation $\varepsilon = hc/\lambda$, we obtain Wien's displacement law:

$$\lambda_{\text{peak}} T = \frac{hcT}{\varepsilon_{\text{peak}}} = \frac{hc}{2.82\, k_B} = 2.898 \times 10^{-3}\ \text{m} \cdot \text{K}. \tag{36}$$

6.2 Black Hole Thermodynamics

6.2.1 Schwarzschild Radius

For a non-rotating black hole of mass M, the Schwarzschild radius (the event horizon radius) is

$$R_s = \frac{2GM}{c^2}.$$

The surface area of the event horizon is therefore

$$A = 4\pi R_s^2 = \frac{16\pi G^2 M^2}{c^4}.$$

6.2.2 Bekenstein–Hawking Entropy

We use a characteristic length scale $R_{\text{BH}} = GM/c^2$ (which is $R_s/2$) as an order-of-magnitude estimate for the typical wavelength of Hawking quanta. Note that this is not the general Kerr radius; it is a convenient scale for this heuristic estimate:

$$R_{\text{BH}} = \frac{GM}{c^2}.$$

The internal energy of the black hole is simply its rest mass energy:

$$U = Mc^2.$$

A Hawking photon has wavelength of order R_{BH}, so its energy is

$$\varepsilon = \frac{hc}{\lambda}, \qquad \lambda = R_{\text{BH}} = \frac{GM}{c^2}.$$

Thus $\varepsilon \sim hc^3/(GM)$.

The number of such photons that could account for the black hole's internal energy is

$$N = \frac{U}{\varepsilon} = \frac{Mc^2}{hc^3/(GM)} = \frac{GM^2}{hc}.$$

Assuming each photon contributes one bit of entropy, the entropy scales as

$$S_{\rm BH} \sim N k_B = \frac{GM^2 k_B}{hc}.$$

In terms of the horizon area $A = 16\pi G^2 M^2/c^4$, a full general-relativistic calculation gives the exact result of the Bekenstein–Hawking entropy is

$$S_{\rm BH} = \frac{8\pi^2 GM^2 k_B}{hc} = k_B \frac{\pi c^3}{2hG} A. \tag{37}$$

6.2.3 Hawking Temperature

The thermodynamic identity for the black hole is

$$T\, dS_{\rm BH} = dQ = dU,$$

where $U = Mc^2$. Using the exact entropy $S_{\rm BH} = 8\pi^2 GM^2 k_B/(hc)$. Thus

$$\frac{1}{T} = \left(\frac{\partial S_{\rm BH}}{\partial U}\right) = \frac{1}{c^2}\frac{dS_{\rm BH}}{dM}. = \frac{16\pi^2 GMk_B}{hc^3}.$$

Rearranging gives the Hawking temperature:

$$T_{\rm BH} = \frac{hc^3}{16\pi^2 GMk_B}. \tag{38}$$

This shows that $T_{\rm BH} \propto 1/M$: smaller black holes are hotter.

6.2.4 Hawking Radiation

A black hole radiates as a black body at temperature $T_{\rm BH}$. From Wien's law, the peak photon energy is $\varepsilon_{\rm peak} = 2.82\, k_B T_{\rm BH}$.

The horizon area scales as

$$A = \frac{16\pi G^2 M^2}{c^4} \propto M^2.$$

The radiated power is given by the Stefan–Boltzmann law:

$$P_r = \sigma A T_{\rm BH}^4.$$

Substituting the expressions for A and $T_{\rm BH}$,using $\sigma = 2\pi^5 k_B^4/(15h^3c^2)$:

$$\begin{aligned} P_r &= \frac{2\pi^5 k_B^4}{15h^3c^2} \cdot \frac{16\pi G^2 M^2}{c^4} \cdot \left(\frac{hc^3}{16\pi^2 GMk_B}\right)^4 \\ &= \frac{c^2 \mathcal{H}}{M^2}. \end{aligned}$$

where we have defined the Hawking constant

$$\mathcal{H} = \frac{hc^4}{30720\pi^2 G^2} = 4.0 \times 10^{15}\ \text{kg}^3/\text{s}. \tag{39}$$

Energy conservation requires that the mass loss rate equals the radiated power:

$$\frac{d(Mc^2)}{dt} = -P_r = -\frac{c^2 \mathcal{H}}{M^2},$$

or equivalently

$$\frac{dM}{dt} = -\frac{\mathcal{H}}{M^2}. \tag{40}$$

6.2.5 Evaporation Lifetime

From the mass loss rate

$$\frac{dM}{dt} = -\frac{\mathcal{H}}{M^2},$$

so the evaporation lifetime is

$$\tau = \int_0^\tau dt = -\frac{1}{\mathcal{H}} \int_{M_{\rm BH}}^0 M^2 \, dM = \frac{M_{\rm BH}^3}{3\mathcal{H}}. \tag{41}$$

6.3 Degenerate Fermi Gases

Ensemble : Grand Canonical ($T \ll T_F$, fully degenerate).

Distribution : Fermi-Dirac : $\overline{n}_{\rm FD} = \dfrac{1}{e^{(\varepsilon-\mu)/k_B T}+1}$.

This section uses the quantum mechanics of free electrons in a 3D infinite square well of side L. States are labelled by (n_x, n_y, n_z).

The longest wavelength is set by the box: for the ground state, $L = \lambda/2$, so $\lambda_{\rm max} = 2L$. The shortest wavelength is set by the highest occupied level $n_{\rm max}$, which is fixed by the electron number N.

Thus the Fermi energy depends on the density N/V, not on an external energy input.

6.3.1 Fermi Energy

At zero temperature, the Fermi-Dirac distribution becomes a step function:

$$\lim_{T\to 0} \overline{n}_{\rm FD} = \begin{cases} 1, & \varepsilon < \mu = \varepsilon_F, \\ 0, & \varepsilon > \mu = \varepsilon_F. \end{cases}$$

Thus all states with energy below the Fermi energy ε_F are occupied, and all states above are empty.

For electrons in a cubic box of volume $V = L^3$, the quantum numbers satisfy

$$n^2 = n_x^2 + n_y^2 + n_z^2 \le n_{\rm max}^2,$$

with n_x, n_y, n_z positive integers. The de Broglie wavelength and momentum are

$$\lambda_n = \frac{2L}{n}, \qquad p_n = \frac{h}{\lambda_n} = \frac{hn}{2L} = \frac{hn}{2V^{1/3}}.$$

Counting both spin states (factor of 2) and integrating over the positive octant (factor of 1/8), the total number of electrons is

$$N = 2 \times \frac{1}{8} \times \frac{4\pi}{3} n_{\rm max}^3 = \frac{\pi}{3} n_{\rm max}^3.$$

Non-relativistic case ($v \ll c$)**:** The non-relativistic single-particle energy is therefore

$$\varepsilon_n = \frac{p_n^2}{2m_e} = \frac{h^2 n^2}{8 m_e V^{2/3}}.$$

The Fermi energy: the energy at the Fermi surface.

$$\varepsilon_F = \varepsilon_{\max} = \frac{h^2 n_{\max}^2}{8 m_e V^{2/3}} = \frac{h^2}{8 m_e}\left(\frac{3N}{\pi V}\right)^{2/3} = k_B T_F, \tag{42}$$

where T_F is the Fermi temperature. For constant particle number N, we have $\varepsilon_F \propto V^{-2/3}$.

The total internal kinetic energy at zero temperature is obtained by summing over all occupied states:

$$\begin{aligned} U_k &= 2 \sum_{n_x, n_y, n_z \geq 1} \varepsilon(n_x, n_y, n_z) \\ &= 2 \times \frac{1}{8} \int_0^{n_{\max}} \varepsilon_n \cdot 4\pi n^2 \, dn \\ &= \frac{\pi h^2}{40 m_e V^{2/3}} n_{\max}^5 \\ &= \frac{\pi h^2}{40 m_e V^{2/3}} (\frac{3N}{\pi})^{5/3} \\ &= \frac{3}{5} N \varepsilon_F \end{aligned}$$

Thus the internal energy scales with volume as $U \propto \varepsilon_F \propto V^{-2/3}$.

Neglecting gravitational energy, so that $U = U_k$.

The degeneracy pressure is obtained from the thermodynamic relation $P_e = -(\partial U/\partial V)_{S,N}$.

$$P_e = -\left(\frac{\partial U}{\partial V}\right)_{S,N} = \frac{2}{3}\frac{U}{V} = \frac{2}{5}\frac{N}{V}\varepsilon_F. \tag{43}$$

This shows that $P_e \propto \varepsilon_F / V \propto V^{-5/3}$.

The bulk modulus (体模量) is

$$B = -V\left(\frac{\partial P_e}{\partial V}\right)_T = \frac{5}{3} P_e = \frac{10}{9}\frac{U}{V}. \tag{44}$$

Relativistic case ($v \sim c$)**:** The relativistic single-particle energy is therefore

$$\varepsilon_n \approx p_n c = \frac{hcn}{2V^{1/3}}.$$

The Fermi energy: the energy at the Fermi surface.

$$\varepsilon_F = \varepsilon_{\max} = \frac{hcn_{\max}}{2V^{1/3}} \tag{45}$$

For constant particle number N, we have $\varepsilon_F \propto V^{-1/3}$.

The total internal kinetic energy at zero temperature is obtained by summing over all occupied states:

$$\begin{aligned} U_k &= 2 \times \frac{1}{8} \int_0^{n_{\max}} \varepsilon_n \cdot 4\pi n^2 \, dn \\ &= \frac{\pi hc}{8V^{1/3}} n_{\max}^4 \\ &= \frac{\pi hc}{8V^{1/3}} (\frac{3N}{\pi})^{4/3} \\ &= \frac{3}{4} N \varepsilon_F \end{aligned}$$

Thus the internal energy scales with volume as $U \propto \varepsilon_F \propto V^{-1/3}$.

6.3.2 White Dwarf Stars

In the fully degenerate limit ($T \approx 0$), a white dwarf star is supported against gravitational collapse by electron degeneracy pressure.

For a spherical star,

$$V = \frac{4}{3}\pi R^3$$

Assuming uniform density, the mass inside radius r is

$$m(r) = \frac{4\pi}{3} r^3 \rho$$

The mass of the white dwarf star

$$M_{\rm WD} = \frac{4\pi}{3} R^3 \rho,$$

where ρ is the mass density.

For a uniform sphere, the gravitational potential energy is

$$U_p = -\int_0^R \frac{Gm(r)}{r}\, dm = -\frac{3}{5}\frac{GM_{\rm WD}^2}{R} = -\frac{\alpha_p}{R},$$

with

$$\alpha_p = \frac{3}{5} G M_{\rm WD}^2.$$

Assuming the white dwarf is composed of fully ionised helium (or carbon/oxygen), each electron is associated with two nucleons ($\mu_e = A/Z = 2$), so

$$M_{\rm WD} = \mu_e N m_p = 2 N m_p,$$

Non-relativistic case ($v \ll c$)**:** For non-relativistic electrons, the kinetic energy can be written as

$$U_k = \frac{\pi h^2}{40 m_e V^{2/3}} (\frac{3N}{\pi})^{5/3} = \frac{\beta_k}{R^2}.$$

where

$$\beta_k = \frac{3}{40}\left(\frac{9}{4\pi^2}\right)^{2/3} \frac{h^2}{m_e}\left(\frac{M_{\rm WD}}{2m_p}\right)^{5/3}.$$

The equilibrium condition is obtained by minimizing the total energy $U_k + U_p$ with respect to R:

$$\frac{d}{dR}(U_k + U_p) = 0.$$

Since $U_k = \beta_k R^{-2}$ and $U_p = -\alpha_p R^{-1}$, we have

$$-\frac{2\beta_k}{R^3} + \frac{\alpha_p}{R^2} = 0.$$

Thus the virial theorem gives

$$2U_k + U_p = 0.$$

Thus:

$$R = \frac{2\beta_k}{\alpha_p} = 0.029 \frac{h^2}{G m_e m_p^{5/3} M_{\rm WD}^{1/3}}. \tag{46}$$

Relativistic case ($v \sim c$)**:** The kinetic energy in the relativistic limit is

Using the same V and N relations as above, we obtain

$$\begin{aligned} U_k &= \frac{\pi hc}{8V^{1/3}}(\frac{3N}{\pi})^{4/3} \\ &= \frac{\pi}{8}\left(\frac{3}{2\pi}\right)^{4/3}\left(\frac{3}{4\pi}\right)^{1/3}\left(\frac{M_{\rm WD}}{m_p}\right)^{4/3}\frac{hc}{R} \\ &\approx 0.091\left(\frac{M_{\rm WD}}{m_p}\right)^{4/3}\frac{hc}{R}. \end{aligned}$$

The gravitational potential energy remains

$$U_p = -\frac{3}{5}\frac{GM_{\rm WD}^2}{R}.$$

The equilibrium condition is obtained by minimizing the total energy $U_k + U_p$ with respect to R:

$$\frac{d}{dR}(U_k + U_p) = 0.$$

We get the virial theorem in relativistic limit as $U_k + U_p = 0$.

The Chandrasekhar mass limit:

$$M_{\rm WD} \leq \left(\frac{5}{3}\times 0.091 \times \frac{hc}{G}\right)^{3/2}\frac{1}{m_p^2} \approx 1.7M_\odot.$$

The full general-relativistic calculation gives a slightly lower value:

$$M_{\rm WD} \leq 1.44M_\odot. \tag{47}$$

6.3.3 Neutron Stars

For neutron stars, the degeneracy pressure comes from neutrons rather than electrons. Neutrons have much larger mass, consider the case in the non-relativistic regime, we have

$$U_k = \frac{3}{5}N\varepsilon_F, \quad \varepsilon_F = \frac{h^2}{8m_n}\left(\frac{3N}{\pi V}\right)^{2/3}, \quad M_{\rm NS} = Nm_n.$$

Following the same procedure as for white dwarfs, the kinetic energy is

$$U_k = \frac{\beta_k'}{R^2},$$

with

$$\beta_k' = \frac{3}{40}\left(\frac{9}{4\pi^2}\right)^{2/3}\frac{h^2}{m_n}\left(\frac{M_{\rm NS}}{m_n}\right)^{5/3}.$$

The gravitational potential energy is

$$U_p = -\frac{\alpha_p'}{R}, \quad \alpha_p' = \frac{3}{5}GM_{\rm NS}^2.$$

The virial theorem $2U_k + U_p = 0$ gives the equilibrium radius:

$$R_{\rm NS} = \frac{2\beta'_k}{\alpha'_p} = 0.093 \frac{h^2}{G m_n^{8/3} M_{\rm NS}^{1/3}}. \tag{48}$$

In the relativistic limit, the Tolman–Oppenheimer–Volkoff (TOV) limit for non-rotating neutron stars is approximately,

$$M_{\rm NS} \leq 3 M_\odot. \tag{49}$$

6.4 Debye Theory of Solid

Phonons obey Bose-Einstein statistics with zero chemical potential ($\mu = 0$):

$$\overline{n}_p = \frac{1}{e^{\varepsilon/k_B T} - 1}.$$

6.4.1 Phonon Energy

Phonons(声子) are quantized lattice vibrations . The phonon heat capacity dominates at high temperatures.

For standing waves in a box of length L, the quantisation condition is

$$L = n \cdot \frac{\lambda}{2},$$

so the wavelength is $\lambda = 2L/n$.

The volume of the solid is

$$V = L^3.$$

The phonon energy is related to its frequency and wavelength by

$$\varepsilon_n = h\nu = \frac{hc_s}{\lambda} = \frac{hc_s n}{2L} = \frac{hc_s n}{2V^{1/3}} \tag{50}$$

where c_s is the speed of sound in the solid, and n is a positive integer.

6.4.2 Lattice Thermal Energy

For each polarisation, the number of vibrational modes with wavevector magnitude up to $k_{\rm max}$ is

$$N = \frac{1}{8} \cdot \frac{4\pi}{3} n_{\rm max}^3 = \frac{\pi}{6} n_{\rm max}^3.$$

Here N is the number of modes per polarisation; the total number of vibrational modes across all three polarisations is $3N$.

The Debye temperature is defined as the maximum phonon energy divided by k_B. Using $\varepsilon_{\rm max} = hc_s n_{\rm max}/(2V^{1/3})$ and $n_{\rm max}^3 = 6N/\pi$, we have:

$$T_D = \frac{\varepsilon_{\rm max}}{k_B} = \frac{hc_s}{2k_B} \left(\frac{6N}{\pi V}\right)^{1/3}. \tag{51}$$

The internal energy is obtained by summing over all phonon modes. There are three polarisations: two transverse and one longitudinal. Converting the sum to an integral over the positive octant:

$$\begin{aligned} U &= 3 \sum_{n_x \geq 1} \sum_{n_y \geq 1} \sum_{n_z \geq 1} \varepsilon_n \, \overline{n}_p(\varepsilon_n) \\ &= 3 \cdot \frac{1}{8} \int_0^{n_{\max}} \frac{\varepsilon_n}{e^{\varepsilon_n / k_B T} - 1} \cdot 4\pi n^2 \, dn. \end{aligned}$$

We introduce the dimensionless variable

$$x = \frac{\varepsilon_n}{k_B T} = \frac{T_D}{T} \cdot \frac{n}{n_{\max}}.$$

With this substitution, the Debye internal energy is:

$$U = 9 N k_B \frac{T^4}{T_D^3} \int_0^{T_D/T} \frac{x^3}{e^x - 1} \, dx. \tag{52}$$

6.4.3 Heat Capacity

The heat capacity is obtained by differentiating the internal energy with respect to temperature.

a) High-temperature limit ($T \gg T_D$): In this limit, $x \ll 1$, so we can approximate $e^x \approx 1 + x$. The integral reduces to

$$\int_0^{T_D/T} \frac{x^3}{e^x - 1} \, dx \approx \int_0^{T_D/T} x^2 \, dx = \frac{1}{3} \left(\frac{T_D}{T} \right)^3.$$

Substituting this into the expression for U:

$$U = 9 N k_B \frac{T^4}{T_D^3} \cdot \frac{1}{3} \left(\frac{T_D}{T} \right)^3 = 3 N k_B T.$$

This corresponds to 6 degrees of freedom per atom (3 kinetic + 3 potential). The heat capacity in the high-temperature limit:

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V = 3 N k_B \quad (T \gg T_D). \tag{53}$$

This is the Dulong–Petit law.

b) Low-temperature limit ($T \ll T_D$): In this limit, the upper limit of the integral extends to infinity ($T_D/T \to \infty$). The integral becomes

$$\int_0^{\infty} \frac{x^3}{e^x - 1} \, dx = \frac{\pi^4}{15}.$$

Thus

$$U = 9 N k_B \frac{T^4}{T_D^3} \cdot \frac{\pi^4}{15} = \frac{3\pi^4}{5} N k_B \frac{T^4}{T_D^3}.$$

Differentiating with respect to temperature:

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V = \frac{12 \pi^4}{5} N k_B \left(\frac{T}{T_D} \right)^3. \tag{54}$$

This is the Debye T^3 law, which holds for most insulating solids at low temperatures.

For metals, the conduction electrons contribute an additional linear term in temperature:

$$C_V = \frac{\pi^2 N k_B^2}{2\varepsilon_F} T + \frac{12\pi^4 N k_B}{5T_D^3} T^3 \quad \text{(Metals)}.$$

c) Intermediate temperatures: For intermediate temperatures, the full integral must be evaluated numerically:

$$C_V = 9Nk_B \left(\frac{T}{T_D}\right)^3 \int_0^{T_D/T} \frac{x^4 e^x}{(e^x - 1)^2}\, dx. \tag{55}$$

6.5 Bose-Einstein Condensation (BEC)

For a gas of non-interacting spin-0 bosons, the occupation number is given by the Bose-Einstein distribution:

$$\overline{n}_{\text{BE}} = \frac{1}{e^{(\varepsilon-\mu)/k_B T} - 1}.$$

6.5.1 Single Particle Energy

For a particle in a cubic box of volume $V = L^3$, the standing wave condition gives

$$L = \frac{n\lambda}{2},$$

so the wavevector is

$$k = \frac{2\pi}{\lambda} = \frac{n\pi}{L}.$$

The momentum and de Broglie wavelength are related by

$$p = \frac{h}{\lambda} = \frac{hk}{2\pi} = \frac{hn}{2L} = \frac{hn}{2V^{1/3}}$$

Thus the single-particle energy is

$$\varepsilon_n = \frac{p^2}{2m} = \frac{h^2 k^2}{8\pi^2 m} = \frac{h^2 n^2}{8mV^{2/3}}.$$

For spin-0 bosons such as ground-state ^{4}He, there is only one spin state. The density of states in k-space (positive octant only) is

$$g(k)\, dk = 1 \cdot \frac{1}{8} \cdot \frac{4\pi k^2 dk}{(\pi/L)^3} = \frac{Vk^2}{2\pi^2} dk.$$

Converting to energy space using $\varepsilon_n = h^2 k^2 / 8\pi^2 m$:

$$g(\varepsilon_n) = g(k)\frac{dk}{d\varepsilon_n} = \frac{2\pi V}{h^3}(2m)^{3/2}\sqrt{\varepsilon_n}. \tag{56}$$

6.5.2 Ground State Population

The quantum numbers satisfy

$$n^2 = n_x^2 + n_y^2 + n_z^2.$$

The ground state corresponds to $n_x = n_y = n_z = 1$, so its energy is

$$\varepsilon_0 = \frac{h^2}{8mV^{2/3}}(1^2 + 1^2 + 1^2) = \frac{3h^2}{8mL^2}.$$

The occupation number of the ground state is therefore

$$\overline{n}_{\mathrm{BE}_0} = \frac{1}{e^{(\varepsilon_0 - \mu)/k_B T} - 1}.$$

As $T \to 0$, the chemical potential approaches the ground state energy from below:

$$\lim_{T \to 0} \mu = \varepsilon_0.$$

Consequently, the denominator tends to zero, so the ground-state occupation diverges:

$$\overline{n}_{\mathrm{BE}_0} \to \infty \quad \text{as } T \to 0.$$

This indicates macroscopic occupation of the ground state, i.e. Bose-Einstein condensation.

6.5.3 Critical Condensation Temperature

At temperatures $T \geq T_c$, all particles are in excited states, so the ground-state occupation is negligible. At the critical point,

$$\lim_{T \to T_c} N_0 = 0, \qquad \lim_{T \to T_c} \mu = 0.$$

The total number of particles at $T = T_c$ is obtained by integrating the density of states times the Bose-Einstein distribution:

$$\begin{aligned} N &= \lim_{T \to T_c} \int_0^\infty g(\varepsilon) \frac{1}{e^{\varepsilon / k_B T_c} - 1} \, d\varepsilon \\ &= \frac{2\pi V}{h^3} (2m)^{3/2} \int_0^\infty \frac{\sqrt{\varepsilon}}{e^{\varepsilon / k_B T_c} - 1} \, d\varepsilon. \end{aligned}$$

Introducing the dimensionless variable $x = \varepsilon / (k_B T_c)$ gives

$$N = \left(\frac{2\pi m k_B T_c}{h^2} \right)^{3/2} V \left[\frac{2}{\sqrt{\pi}} \int_0^\infty \frac{\sqrt{x}}{e^x - 1} \, dx \right].$$

For Riemann zeta function,

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx.$$

Gamma function,

$$\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx$$

And $\Gamma(3/2) = \sqrt{\pi}/2$. Thus

$$\frac{2}{\sqrt{\pi}} \int_0^\infty \frac{\sqrt{x}}{e^x - 1}\, dx = \frac{1}{\Gamma(3/2)} \int_0^\infty \frac{\sqrt{x}}{e^x - 1}\, dx = \zeta(3/2) \approx 2.612.$$

Therefore,

$$N = 2.612 \left(\frac{2\pi m k_B T_c}{h^2} \right)^{3/2} V.$$

Thus the critical temperature T_c is:

$$k_B T_c = 0.527 \frac{h^2}{2\pi m} \left(\frac{N}{V} \right)^{2/3}. \tag{57}$$

For $T < T_c$, a macroscopic fraction of particles occupies the ground state. The number of particles remaining in excited states scales as

$$N_{\text{excited}} = N \left(\frac{T}{T_c} \right)^{3/2}, \quad (T < T_c).$$

The condensate fraction (the number of particles in the ground state) is therefore

$$N_0 = N - N_{\text{excited}} = N \left[1 - \left(\frac{T}{T_c} \right)^{3/2} \right], \quad (T < T_c). \tag{58}$$

Gavin's Lectures on Physics - Condensed Matter



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1 Foundations of Phase Transitions and Statistical Mechanics

1.1 Phase Transition

1.1.1 Discontinuous or First-order Phase Transition

Two distinct phases coexist at phase transition. Latent (潜在的) Heat associated with transition.

1.1.2 Continuous or second-Order Phase Transition

Two phases are indistinguishable at transition. No latent heat, but correlation length diverges at transition. system components are correlated in a universal way at transition criticality (临界点).

1.2 Ensemble

In canonical ensemble: N, V, T, fixed. Potential : $F = E - TS \quad (E = U)$ Minimized F:

- High T : S dominates, disordered system.
- Low T : E dominates, ordered system.

1.3 Thermodynamic Limit

$$N \to \infty, \quad V \to \infty, \quad n = \frac{N}{V} = \text{constant} \tag{1}$$

1.4 Symmetry

1.4.1 Basic symmetries

- Translation symmetry : momentum conservation
- Rotation symmetry : angular momentum conservation.
- Inversion symmetry
- Time-reversal symmetry
- Gauge symmetry.

1.4.2 Spontaneous (自发的) Symmetry Breaking (SSB)

Def: The Hamiltonian of a system processes a certain symmetry, but the equilibrium ground state spontaneously loses part of this symmetry.

1.4.3 Order Parameter

Def: A measurable physical quantity that quantifies the degree of symmetry breaking.

$$\text{Order Parameter} \begin{cases} = 0 & , \text{ high symmetry, disordered phases} \\ \neq 0 & , \text{ low symmetry, ordered phases.} \end{cases} \tag{2}$$

2 Disorder, Long-Range Order, and Universality

2.1 Disorder

2.1.1 Domain (畴)

1. Impurities (杂质) introduce local symmetry breaking fields, leading to magnetic/structural domains.

2. The formation of domains depends on the competition between domain wall energy cost and energy benefit from impurities.

3. Energy Scaling . (L: system size, d: dimension)

2.1.2 Existence of Long-Range Order (LRO) in the Thermodynamic Limit

Dimension	Discrete (e.g., Ising)	Continuous (e.g., XY)
1D	No LRO ($T > 0$)	No LRO (Mermin-Wagner)
2D	LRO exists ($T < T_c$, Onsager solution)	No LRO, but quasi-LRO ($T < T_{\mathrm{BKT}}$)
3D	LRO exists	LRO exists ($T < T_c$)

LRO: (Long-Range order). Correlations between microscopic degrees of freedom persist at macroscopic / infinitely large distance.

2.1.3 Mermin–Wagner Theorem

For systems with **continuous symmetry and short-range interactions**, no spontaneous symmetry breaking, and no long-range order exist at finite temperature in 1D and 2D systems. **Notably, the 2D XY model allows for quasi-long-range order (BKT phase), which is an exception to conventional LRO.**

2.2 Universality

2.2.1 Universality Class

A universality class is a group of distinct physical systems that share the same set of key global properties: identical critical exponents, scaling functions, critical behaviors at critical points.

2.2.2 Universality Principle

Near Critical Point T_C, microscopic details become irrelevant. Different systems exhibit identical critical behaviours and belong to the same universality class.

A universality class is determined only by system dimension, order parameter dimension, and symmetry.

2.2.3 Critical Exponents

Power-law behaviour near critical point T_C.

1. order parameter : $\Delta \sim |T - T_C|^{\beta}$
2. magnetic susceptibility (磁化率) : $\chi \sim |T - T_C|^{-\gamma}$
3. Correlation length : $\xi \sim |T - T_C|^{-\nu}$

3 Renormalization Group (RG) (重整化群)

3.1 Parameters

- Relevant parameters : Grow under RG transformation, dominate macroscopic long-wavelength behaviours.
- Irrelevant parameters : Decay under RG transformation, negligible for macroscopic properties.

3.2 Standard RG Procedures

1. Coarse-graining (粗粒化): Combine adjacent microscopic degrees of freedom into blocks.
2. Rescaling (尺度重标): shrink all spatial lengths by a scaling factor $b > 1$.
3. Renormalization (参数重整): Rescale physical parameters and order parameters to restore the original system form.

3.3 RG fixed point

Parameters keep invariant under RG transformation

- stable fixed point : corresponds to stable thermodynamic phase .
- Unstable fixed point : Corresponds to phase transition critical point.

3.4 RG flow

Evolution & trajectory of parameters during scale transformation .

4 Landau Theory of Phase Transitions

4.1 Theory

1. Phenomenological theory constrained by system symmetry.
2. Ignores thermal fluctuations. (MFT 平均场理论)
3. Expand free energy F, as power series of order parameter Δ.

$$\Delta \to \begin{cases} S \\ E \end{cases} \Rightarrow F. \tag{3}$$

4.2 Second-Order Phase Transition

4.2.1 Free energy formula

$$F(\Delta) = F_0 + r_0 \left(T - T_C\right) \Delta^2 + s_0 \Delta^4 \quad (r_0, s_0 > 0) \tag{4}$$

No odd-term due to symmetry constraint.

$$\begin{cases} T > T_C;\ F \text{ minimum at } \Delta = 0, \text{ disordered phase} \\ T < T_C:\ F \text{ minimum at } \Delta = \pm\sqrt{\frac{r_0(T_C - T)}{2s_0}}, \text{ ordered phase.} \end{cases} \tag{5}$$

4.2.2 Mean-field critical exponents

$$\begin{cases} \Delta \sim |T - T_C|^{\beta} \quad , \ \beta = \frac{1}{2} \\ \chi \sim |T - T_C|^{-\gamma} \quad , \ \gamma = 1 \\ \xi \sim |T - T_C|^{-\nu} \quad , \ \nu = \frac{1}{2} \end{cases} \tag{6}$$

4.3 First-order Phase Transition

4.3.1 Free energy formula

$$F(\Delta) = F_0 + a(T - T_0)\Delta^2 - b\Delta^4 + c\Delta^6 \quad (a, b, c > 0) \tag{7}$$

Order parameter jumps discontinuously.

4.3.2 Features

Latent heat, supercooling & superheating, discontinuous order parameter.

4.4 Ginzburg Criterion

MFT: mean field theory . (Landau theory $\in$ MFT)

$$G = \frac{\text{thermal fluctuation strength}}{\text{ordering strength}} \tag{8}$$

$$\begin{cases} G \ll 1 \quad : \text{ MFT valid} \\ G \gtrsim 1 \quad : \text{ MFT fails} \end{cases} \tag{9}$$

Dimension :

$$\begin{cases} d > 4 & ; \text{ MFT holds for all } T \\ d = 4 & ; \text{ Marginal} \\ d < 4 & ; \text{ MFT works far from } T_C, \text{ breaks down in critical region, requiring RG treatment.} \end{cases} \tag{10}$$

5 Ising Model

Discrete symmetry

5.1 Ising Hamiltonian

$$H = -J\sum_{\langle ij\rangle} s_i s_j - h\sum_i s_i \tag{11}$$

$$\begin{cases} s_i = \pm 1, \quad \text{Lattice (格点) spin} \\ J : \text{Exchange coupling.} \begin{cases} J > 0 & : \text{ ferromagnetism} \\ J < 0 & : \text{ anti-ferromagnetism} \end{cases} \\ h : \text{external magnetic field, } h = \mu B \\ \langle ij\rangle : \text{nearest-neighbour pairs} \end{cases} \tag{12}$$

5.2 Self-Consistent Equation

5.2.1 Mean-Field Decoupling

In mean-field approximation:

$$\sum_{\langle ij\rangle} s_i s_j \approx \sum_{\langle ij\rangle}[m(s_i+s_j)-m^2] = \sum_i qms_i - \frac{1}{2}Nqm^2 = Nqm^2 - \frac{1}{2}Nqm^2 = \frac{1}{2}Nqm^2 \tag{13}$$

$$H = -J\sum_{\langle ij\rangle} s_i s_j - h\sum_i s_i \tag{14}$$

$$= -\frac{1}{2}\sum_{ij} J_{ij}\, s_i s_j - h\sum_i s_i, \quad J_{ij} = \begin{cases} J & ,\langle i,j\rangle \text{ near} \\ 0 & ,\langle i,j\rangle \text{ far.} \end{cases} \tag{15}$$

$$\Rightarrow \frac{\partial H}{\partial s_i} = -\sum_{j,\text{near } i} J_{ij} s_j - h = -Jq\langle s\rangle - h \tag{16}$$

Def: $\bar{h} = -\frac{\partial H}{\partial s_i} = Jq\langle s\rangle + h \Rightarrow H = \frac{1}{2}NJqm^2 - \bar{h}\sum_i s_i$

5.2.2 Partition Function and Magnetization

$$Z = \sum_{s_1=\pm1}\sum_{s_2=\pm1}\cdots\sum_{s_N=\pm1} e^{-\beta H} \tag{17}$$

$$H \approx -\big(Jq\langle s\rangle + h\big)\sum_i s_i = -\bar{h}\sum_i s_i \tag{18}$$

$$\Rightarrow \begin{cases} Z = \prod_{i=1}^N \sum_{s_i=\pm1} e^{\beta\bar{h}s_i} = \prod_{i=1}^N Z_1 \\ Z_1 = e^{\beta\bar{h}} + e^{-\beta\bar{h}} \end{cases} \tag{19}$$

5.2.3 Self-Consistent Equation

$dQ = 0$

$$\Rightarrow dW = dU = d\left(-\bar{h}\sum_i s_i\right) = d\left(-N\bar{h}\langle s\rangle\right) = -N\langle s\rangle d\bar{h} \tag{20}$$

$$\because\ dW = Ydy, \quad Y = -\frac{N}{\beta}\frac{\partial \ln Z_1}{\partial y} \tag{21}$$

$$\therefore\ N\langle s\rangle = \frac{N}{\beta}\frac{\partial \ln Z_1}{\partial y} \tag{22}$$

$$\therefore \langle s\rangle = \frac{1}{\beta}\frac{\partial \ln Z_1}{\partial y} = \tanh(\beta\bar{h}) \tag{23}$$

Here,$\beta = \frac{1}{k_B T}$

$$m = \langle s\rangle \Rightarrow m = \tanh\left(\frac{Jqm + h}{k_B T}\right) \tag{24}$$

5.2.4 Critical Temperature

Without external field: $h = 0$:

$$\begin{cases} m = \tanh\left(\frac{Jqm}{k_B T}\right) \\ \text{Def}:\ T_C = \frac{Jq}{k_B} \end{cases} \Rightarrow m = \tanh\left(\frac{T_C}{T}\cdot m\right) \tag{25}$$

$$\begin{cases} T \ll T_C & : m = \tanh\left(\frac{T_C}{T}m\right) \to 1,\ \text{ordered system} \\ T \gg T_C & : m = \tanh\left(\frac{T_C}{T}m\right) \approx \frac{T_C}{T}m \Rightarrow m = 0,\ \text{disordered system} \end{cases} \tag{26}$$

5.2.5 Compare Mean-Field Theory with Exact Results

Dimension	Mean-Field Theory (MFT) Prediction	Exact Result
1D	$T_C = 2J/k_B$	$T_C = 0$
2D	$T_C = 4J/k_B$	$T_C \approx 2.269J/k_B$ (Onsager)
3D	$T_C = 6J/k_B$	$T_C \approx 4.511J/k_B$

5.3 Bragg-Williams Free Energy

5.3.1 Entropy from Combinatorics

$$\begin{cases} m = \frac{N_+ - N_-}{N} \\ N = N_+ + N_- \end{cases} \Rightarrow \begin{cases} N_+ = \frac{(1+m)N}{2} \\ N_- = \frac{(1-m)N}{2} \end{cases} \tag{27}$$

Stirling : $\ln(m!) \approx m\ln m - m$.

$$\therefore\ \Omega = C_N^{N_+} = \frac{N!}{N_+!\,N_-!} \tag{28}$$

$$\begin{aligned}\ln \Omega &= \ln \frac{N!}{N_+! \, N_-!} \\ &\approx N \ln 2 - \frac{N(1+m)}{2} \ln(1+m) - \frac{N(1-m)}{2} \ln(1-m)\end{aligned} \tag{29}$$

$$S = k_B \ln \Omega \tag{30}$$

5.3.2 Internal Energy

$$U = H = \frac{1}{2} N J q m^2 - \bar{h} \sum_i s_i = \frac{1}{2} N J q m^2 - \bar{h} N m \tag{31}$$

$$(\bar{h} = Jqm + h) \tag{32}$$

$$\Rightarrow U = -\frac{1}{2} N J q m^2 - N h m \tag{33}$$

5.3.3 Free Energy Formula

$$F = U - TS \tag{34}$$

$$= -\frac{1}{2} N J q m^2 - N h m - k_B T \ln \Omega \tag{35}$$

$$= N \left\{ -\frac{1}{2} J q m^2 - h m - k_B T \left[\ln 2 - \frac{1+m}{2} \ln(1+m) - \frac{1-m}{2} \ln(1-m) \right] \right\} \tag{36}$$

5.4 Verify Self-Consistent Equation

To minimize free energy:

$$\frac{dF}{dm} = 0 \tag{37}$$

We can derive Self-Consistent Equation:

$$\therefore \; m = \tanh \left(\frac{Jqm + h}{k_B T} \right) \text{ minimizes free energy } F(m) \tag{38}$$

6 XY Model

Continuous symmetry, 2D

6.1 XY Hamiltonian

$$H = -J \sum_{\langle i,j \rangle} \cos\left(\theta_i - \theta_j\right) \tag{39}$$

$$\begin{cases} \theta_i: \text{ spin azimuthal (方位的) angle} \\ J: \text{ exchange coupling} \\ \langle i,j \rangle: \text{ nearest-neighbour pairs} \end{cases} \tag{40}$$

Long range fluctuation : $|\theta_i - \theta_j| \ll 1$

$$\cos(\theta_i - \theta_j) \approx 1 - \frac{1}{2}\left(\theta_i - \theta_j\right)^2 \tag{41}$$

$$H \approx H_0 + \frac{1}{2}\left(\theta_i - \theta_j\right)^2 \tag{42}$$

$$\approx H_0 + \frac{J}{2}\int d^2x \left|\nabla\theta(\mathbf{x})\right|^2 = H_0 + H_V \tag{43}$$

6.2 Topological Defects (拓扑缺陷)

Topological Charge :

$$n = \frac{1}{2\pi}\oint_C \nabla\theta \cdot d\mathbf{r} \quad \in \mathbb{Z} \tag{44}$$

C: closed contour enclosing the defect core.

- Vortex (涡旋): spins rotate counterclockwise by 2π when circling the defect core , $n = +1$
- Anti-vortex (反涡旋): spins rotate clockwise by 2π when circling the defect core , $n = -1$.

6.3 Energy of single vortex

single vortex:

$$1 = n = \frac{1}{2\pi}\oint_C \nabla\theta \cdot d\mathbf{r} = |\nabla\theta| \cdot \frac{1}{2\pi}\oint_C dr = |\nabla\theta| \cdot \frac{1}{2\pi} \cdot 2\pi r \tag{45}$$

$$\Rightarrow |\nabla\theta| = \frac{1}{r} \tag{46}$$

$$\Rightarrow \int d^2x \left|\nabla\theta\right|^2 = \int_a^L \frac{1}{r^2} \cdot 2\pi r \, dr = 2\pi \ln\left(\frac{L}{a}\right) \tag{47}$$

$$\Rightarrow E_V = H_V = \frac{J}{2}\int d^2x \left|\nabla\theta\right|^2 = \pi J \ln\left(\frac{L}{a}\right) \tag{48}$$

Large system : $L \gg a$: $E_V \approx \pi J \ln L$ L: system radius. a: cell core radius

6.4 Vortex Entropy

$$\Omega \propto \frac{L^2}{a^2} \tag{49}$$

$$S = k_B \ln \Omega = 2k_B \ln \left(\frac{L}{a} \right) \tag{50}$$

Large system ($L \gg a$):

$$S \approx 2k_B \ln L \tag{51}$$

6.5 BKT Transition

6.5.1 Definition

BKT Transition : topological phase transition driven by the binding and unbinding of vortex–antivortex pairs. It occurs without symmetry breaking .

6.5.2 Free Energy ($L \gg a$)

$$\begin{cases} F_V = E_V - TS_V \\ E_V \approx \pi J \ln L \\ S_V \approx 2k_B \ln L \end{cases} \Rightarrow F_V = (\pi J - 2k_B T) \ln L \tag{52}$$

$$\begin{cases} F_V = (T_C - T)\, 2k_B \ln L \\ T_C = \frac{\pi J}{2k_B} \quad \text{critical temperature.} \end{cases} \tag{53}$$

$$\begin{cases} T < T_C \quad : \ F_V > 0, \ \text{Isolated vortices are unstable} \\ \qquad\quad \text{Vortex–antivortex pairs are tightly bound.} \\ T > T_C \quad : \ F_V < 0, \ \text{Isolated vortices are stable} \\ \qquad\quad \text{Vortex–antivortex pairs dissociate into free vortices.} \end{cases} \tag{54}$$

6.5.3 Correlation Function Behavior

- $T < T_C$: Quasi-long-range order (准长程序) exists.
- $T > T_C$: Only short-range order exists.

7 Hopfield Neural Network

7.1 Definition

A recurrent artificial neural network based on the energy function of Ising model.

7.2 Network Energy Function

$$H = -\frac{1}{2}\sum_{i \neq j} J_{ij}\, s_i s_j \tag{55}$$

$$\begin{cases} s_i = \pm 1 : \text{Neuron state. } (+1 : \text{Active, } -1 : \text{inactive}) \\ J_{ij} : \text{synaptic weight (突触权重)} \end{cases} \tag{56}$$

7.3 Hebbian Learning Rule

$J_{ii} = 0$ for all i.
Symmetric weight matrix :

$$J_{ij} = \frac{1}{N}\sum_{\mu=1}^{P} \xi_i^\mu \xi_j^\mu \tag{57}$$

$$\begin{cases} \xi_i^\mu : \text{ stored pattern, } i\text{-th component of the } \mu\text{-th pattern.} \\ \xi_i^\mu = \pm 1 \\ N : \text{ total number of neurons (神经元)} \\ P : \text{ number of stored patterns} \end{cases} \tag{58}$$

7.4 Neuron Update Rule

$$s_i(t+1) = \text{sign}\left(\sum_j J_{ij}\, s_j(t)\right) \tag{59}$$

Gavin's Lectures on Physics - Cosmology



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1 Introduction

This note draws primarily on "Modern Cosmology" (Second Edition) by Scott Dodelson.[2]

Einstein Field Equation:"Matter tells space-time how to curve."
Boltzmann Equation:"Space-time tells matter how to move."

Einstein Field Equation 0th Order: Friedmann Equation.
Boltazmann Equation 0th Order: Particle Evolution Equation.

Einstein Field Equation 1st Order: Poisson Equation $\leftrightarrow$ Newton's Gravitational Law.
Boltzmann Equation 1st Order: 1.Continuity Equation $\leftrightarrow$ Matter Conservation, 2.Euler Equation $\leftrightarrow$ Newton's 2nd Law

2 Expanding Universe

2.1 Structure in the Universe

Matter Power Spectrum:

$$\langle \delta(\mathbf{k_1})\delta(\mathbf{k_2})\rangle = (2\pi)^3 \delta_D^{(3)}(\mathbf{k_1}+\mathbf{k_2})P(k_1) \tag{1}$$

Correlation Function:

$$\xi(\mathbf{x}-\mathbf{x}') = \langle \delta(\mathbf{x_1})\delta(\mathbf{x_2})\rangle = \int \frac{d^3k}{(2\pi)^3} P(k) e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')} \tag{2}$$

2.2 Entropy Density

$$s = \frac{\rho + P}{T} \propto \frac{1}{a^3} \tag{3}$$

$$s = \frac{2\pi^2}{45}\Big(\sum_{i=bosons} g_i T_i^3 + \frac{7}{8}\sum_{i=fermions} g_i T_i^3\Big) \tag{4}$$

2.3 Cosmic Inventory

Photons: $\Omega_{\gamma 0}h^2 = 2.47 \times 10^{-5}$
Baryons: $\Omega_{b0}h^2 = 0.0225$
Matter: $\Omega_{m0}h^2 = 0.1431$
Neutrinos: $\Omega_{\nu 0}h^2 = \sum_i m_{\nu i}/94eV$

Distribution of Fermions and Bosons:

$$f_{FD}(E) = \frac{1}{e^{(E-\mu)/T}+1}, \quad f_{BE}(E) = \frac{1}{e^{(E-\mu)/T}-1} \tag{5}$$

Density:

$$\rho_s = \begin{cases} m_s n_s & ,\text{Matter} \\ g_s \frac{\pi^2}{30} T^4 & ,\text{Radiation Bosons} \\ \frac{7}{8} g_s \frac{\pi^2}{30} T^4 & ,\text{Radiation Fermions} \end{cases} \tag{6}$$

$$\Omega_{s0} = \frac{\rho_{s0}}{\rho_{cr0}} \tag{7}$$

Temperature:

$$T_s \propto \begin{cases} \frac{1}{a} & ,(m_s << T) \\ \frac{1}{a^2} & ,(m_s >> T) \end{cases} \tag{8}$$

2.4 Distances

Define a special function for different curvature K:

$$\sin_k(\chi) = \begin{cases} K^{-1/2}\sin(\sqrt{K}\chi), & K > 0 \\ \chi, & K = 0 \\ (-K)^{-1/2}\sinh(\sqrt{-K}\chi), & K < 0 \end{cases} \tag{9}$$

Comoving Distance:

$$\chi(t) = \int_t^{t_0} \frac{dt'}{a(t')} = \int_a^1 \frac{da_1}{a_1^2 H(a_1)} = \int_0^z \frac{dz_1}{H(z_1)} \tag{10}$$

Angular Diameter Distance:

$$d_A = \frac{l}{\theta} = a\sin_k(\chi) = \frac{\sin_k(\chi)}{1+z} \tag{11}$$

Luminosity Distance:

$$d_L = \frac{\sin_k(\chi)}{a} = (1+z)\sin_k(\chi) \tag{12}$$

Flux:

$$F = \frac{La^2}{4\pi\sin_k^2(\chi)(a)} = \frac{L}{4\pi d_L^2} \tag{13}$$

3 Einstein Field Equation

Einstein Field Equation:

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi G T_{\mu\nu} \tag{14}$$

Connection:

$$\Gamma^{\mu}_{\alpha\beta} = \frac{1}{2} g^{\mu\nu} \left(g_{\nu\alpha,\beta} + g_{\nu\beta,\alpha} - g_{\alpha\beta,\nu} \right) \tag{15}$$

Ricci Tensor:

$$R_{\mu\nu} = \Gamma^{\alpha}_{\mu\nu,\alpha} - \Gamma^{\alpha}_{\mu\alpha,\nu} + \Gamma^{\alpha}_{\beta\alpha}\Gamma^{\beta}_{\mu\nu} - \Gamma^{\alpha}_{\beta\nu}\Gamma^{\beta}_{\mu\alpha} \tag{16}$$

Geodesic Equation:

$$\frac{d^2 x^{\mu}}{d\lambda^2} + \Gamma^{\mu}_{\alpha\beta} \frac{dx^{\alpha}}{d\lambda} \frac{dx^{\beta}}{d\lambda} = 0 \tag{17}$$

4 Boltzmann Equation

Boltzmann Equation:

$$C[f] = \frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x^i}\frac{dx^i}{dt} + \frac{\partial f}{\partial p}\frac{dp}{dt} + \frac{\partial f}{\partial \hat{p}^i}\frac{d\hat{p}^i}{dt} \tag{18}$$

For reaction: $1 + 2 \longleftrightarrow 3 + 4$

Collision Term:

$$\begin{aligned} C[f_1(\mathbf{p})] = & \frac{1}{2E_1(p)} \int \frac{d^3q}{(2\pi)^3 2E_2(q)} \int \frac{d^3p'}{(2\pi)^3 2E_3\left(p'\right)} \int \frac{d^3q'}{(2\pi)^3 2E_4\left(q'\right)} \\ & \times |M|^2 (2\pi)^4 \delta_0^{(3)}\left(\mathbf{p} + \mathbf{q} - \mathbf{p}' - \mathbf{q}'\right) \delta_D^{(1)}\left(E_1 + E_2 - E_3 - E_4\right) \\ & \times \{f_3(\mathbf{p}') f_4(\mathbf{q}') \left[1 \pm f_1(\mathbf{p})\right] \left[1 \pm f_2(\mathbf{q})\right] - f_1(\mathbf{p}) f_2(\mathbf{q}) \left[1 \pm f_3(\mathbf{p}')\right] \left[1 \pm f_4(\mathbf{q}')\right]\} \end{aligned} \tag{19}$$

Here, + for bosons, - for fermions

5 Einstein Field Equation: 0th Order

5.1 Continuity Equation

From:

$$\nabla_\mu T^\mu_\nu = \frac{\partial T^\mu_\nu}{\partial x^\mu} + \Gamma^\mu_{\alpha\mu} T^\alpha_\nu - \Gamma^\alpha_{\mu\nu} T^\mu_\alpha = 0 \tag{20}$$

We get:

$$\frac{\partial \rho}{\partial t} + 3H(\rho + P) = 0 \tag{21}$$

Set:

$$\omega_s = \frac{P_s}{\rho_s} \tag{22}$$

Result in:

$$\rho_s \propto a^{-3(1+\omega_s)} \tag{23}$$

For different gradients:

$$\omega_m = 0, \omega_r = \frac{1}{3}, \omega_K = -\frac{1}{3}, \omega_\Lambda = -1 \tag{24}$$

5.2 FLRW metric

FLRW metric:

$$ds^2 = -dt^2 + a^2(t)[d\chi^2 + \sin_k^2(\chi)(d\theta^2 + \sin^2\theta d\phi^2)] \tag{25}$$

Energy-Momentum Tensor:

$$T^\mu_\nu = \begin{pmatrix} -\rho & & & \\ & P & & \\ & & P & \\ & & & P \end{pmatrix} \tag{26}$$

5.3 0th Order Einstein Field Equation (Friedmann Equation)

Friedmann equations:

$$\begin{aligned} (\frac{\dot{a}}{a})^2 + \frac{K}{a^2} &= \frac{8\pi G}{3}\rho \\ \frac{\ddot{a}}{a} &= -\frac{4\pi G}{3}(\rho + 3P) \end{aligned} \tag{27}$$

Friedmann Equations can also be expressed as:

$$\begin{aligned} (\frac{\dot{a}}{a})^2 = H^2(a) &= H_0^2(\frac{\Omega_{r0}}{a^4} + \frac{\Omega_{m0}}{a^3} + \frac{\Omega_{K0}}{a^2} + \Omega_{\Lambda 0}) \\ \frac{\ddot{a}}{a} &= -\frac{H_0^2}{2}(\frac{2\Omega_{r0}}{a^4} + \frac{\Omega_{m0}}{a^3} - 2\Omega_{\Lambda 0}) \end{aligned} \tag{28}$$

Critical density:

$$\rho_{cr0} = \frac{3H_0^2}{8\pi G} \tag{29}$$

Curvature density fraction:

$$K = -H_0^2 \Omega_{K0} \tag{30}$$

Curvature radius:

$$r_K = |K|^{-1/2} \tag{31}$$

Parameter values:

$$\begin{gathered} K = 0 \\ H_0 = 100h \cdot km \cdot s^{-1} \cdot Mpc^{-1} \\ a_{eq} = \frac{\Omega_{r0}}{\Omega_{m0}} = 2.9 \times 10^{-4} \end{gathered} \tag{32}$$

For dynamical dark energy:

$$\Omega_{\Lambda 0} \to \Omega_{DE} \exp[3 \int_a^1 (1 + \omega(a)) \frac{da}{a}] \tag{33}$$

6 Boltzmann Equation: 0th Order

6.1 0th Order Boltzmann Equation

$$C[f] = \frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{p}{E}\frac{\hat{p}^i}{a}\frac{\partial f}{\partial x^i} - Hp\frac{\partial f}{\partial p} \tag{34}$$

$$\frac{dn(t)}{dt} + 3Hn(t) = \int \frac{d^3p}{(2\pi)^3} C[f] \tag{35}$$

$$a^{-3}\frac{d(na^3)}{dt} = \int \frac{d^3p}{(2\pi)^3} C[f] \tag{36}$$

6.2 For Reaction at Low Temperature

Reaction $1 + 2 \longleftrightarrow 3 + 4$ when $T << E - \mu$:

$$a^{-3}\frac{d(n_1 a^3)}{dt} = n_1^{(0)} n_2^{(0)} \langle \sigma v \rangle \{ \frac{n_3 n_4}{n_3^{(0)} n_4^{(0)}} - \frac{n_1 n_2}{n_1^{(0)} n_2^{(0)}} \} \tag{37}$$

Number density:

$$n_s = n_s^{(0)} e^{\mu_s / T_s} = g_s \int \frac{d^3p}{(2\pi)^3} f(E) \tag{38}$$

Number density for $\mu_s = 0$:

$$n_s^{(0)} = \begin{cases} g_s (\frac{m_s T_s}{2\pi})^{\frac{3}{2}} e^{-m_s/T_s}, (m_s >> T) \\ g_s \frac{T_s^3}{\pi^2}, (m_s << T) \end{cases} \tag{39}$$

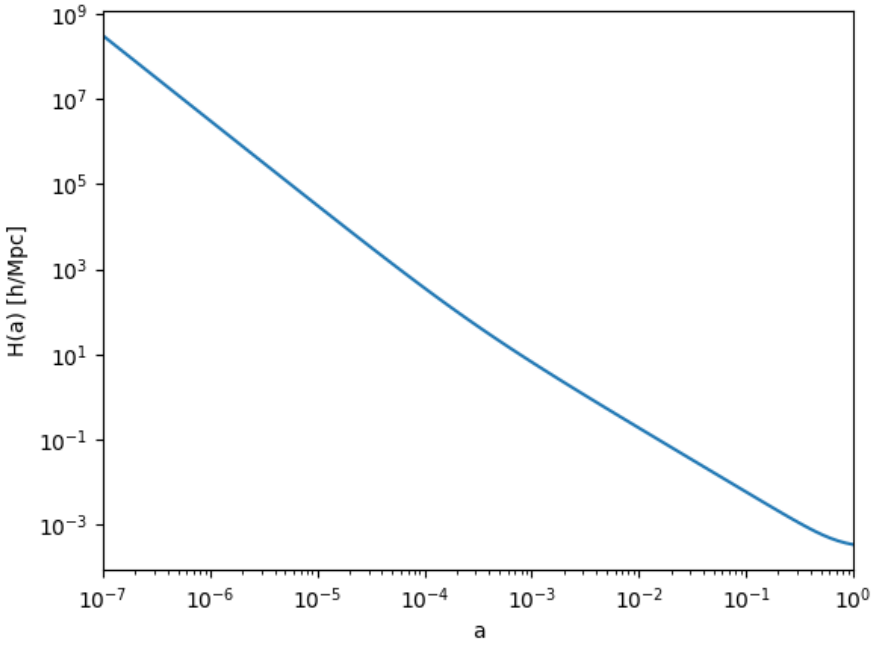


Figure 1: Hubble parameter H(a)

7 Homogeneous Results

7.1 Friedmann Equation Results

Age of the universe:

$$t = \int_0^1 \frac{da}{aH(a)} = 13.8\, billion\ years \tag{40}$$

Radius of visible universe:

$$r_{vi} = \int_0^1 \frac{da}{a^2 H(a)} = 46.3\, billion\ light\ years \tag{41}$$

Hubble parameter in Figure 1
Expansion rate $\dot{a} = H(z)/(1+z)$ in Figure 2 or Figure 3

7.2 Big Bang Nuclear Synthesis

Reaction between proton and neutron:

$$p + e^- \leftrightarrow n + \nu_e \tag{42}$$

Result in Figure 4

7.3 Recombination

Reaction at recombination:

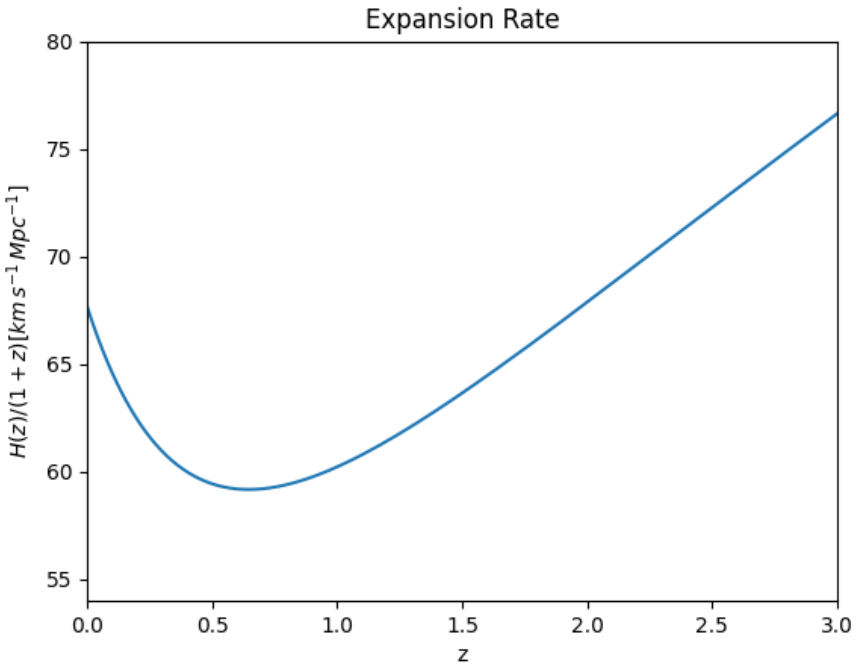


Figure 2: Expansion Rate

$$p + e^- \leftrightarrow H + \gamma \tag{43}$$

Fraction of free electrons:

$$X_e = \frac{n_e}{n_e + n_H} \tag{44}$$

X_e in Figure 5

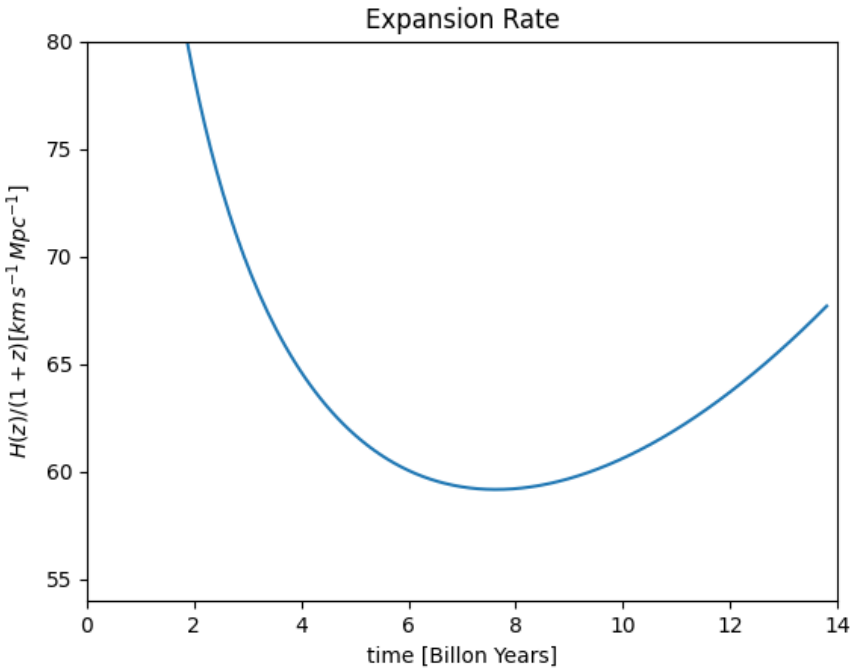


Figure 3: Expansion Rate

8 Einstein Field Equation: 1st Order

8.1 Metric

Scalar Perturbation:

$$\begin{cases} g_{00} = -1 - 2\Psi \\ g_{0i} = 0 \\ g_{ij} = a^2(t)\delta_{ij}(1 + 2\Phi) \end{cases} \tag{45}$$

Tensor Perturbation:

$$\begin{cases} \delta g_{ij} = a^2(t) h_{ij}^{TT} \\ h_{ij}^{TT} = \begin{pmatrix} h_+ & h_\times & 0 \\ h_\times & -h_+ & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{cases} \tag{46}$$

8.2 Energy-Momentum Tensor

$$\begin{aligned} T_\nu^\mu(\mathbf{x}, t) &= g \int \frac{d^3p}{(2\pi)^3} \frac{p^\mu p_\nu}{E(p)} f(\mathbf{x}, \mathbf{p}, t) \\ p^\mu &= [E(1 - \Psi), p^i(\frac{1 - \Phi}{a})] \\ -m^2 &= g_{\mu\nu} p^\mu p^\nu \end{aligned} \tag{47}$$

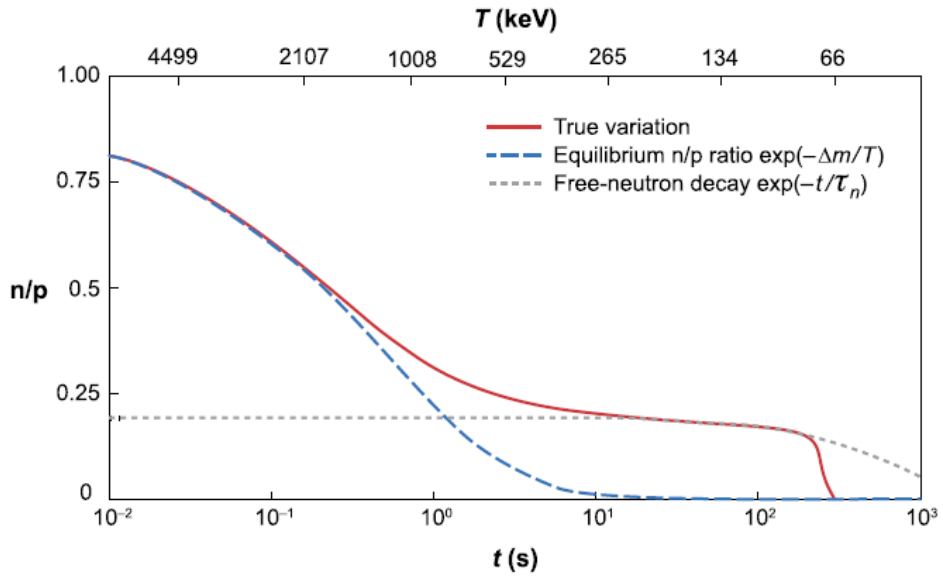


Figure 4: BBN

$$
\begin{aligned}
T^0{}_0(x,t) &= -g\int \frac{d^3p}{(2\pi)^3}E(p)f(\mathbf{x},\mathbf{p},t)\\
T^0{}_i(x,t) &= ga(1+\Phi-\Psi)\int \frac{d^3p}{(2\pi)^3}p_i f(\mathbf{x},\mathbf{p},t)\\
T^i{}_j(x,t) &= g\int \frac{d^3p}{(2\pi)^3}\frac{p^i p_j}{E(p)}f(\mathbf{x},\mathbf{p},t)
\end{aligned}
\tag{48}
$$

8.3 1st Order Einstein Field Equation

Scalar Perturbation(Poisson's Equation):

$$
\begin{aligned}
k^2\Phi + 3\frac{a'}{a}\left(\Phi' - \Psi\frac{a'}{a}\right) &= 4\pi G a^2[\rho_{\rm m}\delta_{\rm m} + 4\rho_{\rm r}\Theta_{\rm r,0}]\\
k^2(\Phi+\Psi) &= -32\pi G a^2\rho_{\rm r}\Theta_{\rm r,2}\\
k^2\Phi = 4\pi\, G a^2&\left[\rho_{\rm m}\delta_{\rm m} + 4\rho_{\rm r}\Theta_{\rm r,0} + \frac{3aH}{k}(i\rho_{\rm m}u_{\rm m} + 4\rho_{\rm r}\Theta_{\rm r,1})\right].
\end{aligned}
\tag{49}
$$

Tensor Perturbation:

$$
h'' + 2\frac{a'}{a}h_t' + k^2 h_t = 0 \quad ,(t=+,\times) \tag{50}
$$

Matter(m) contains CDM(c) and Baryons(b).
Radiation(r) contains Photons(γ) and Neutrinos(ν).

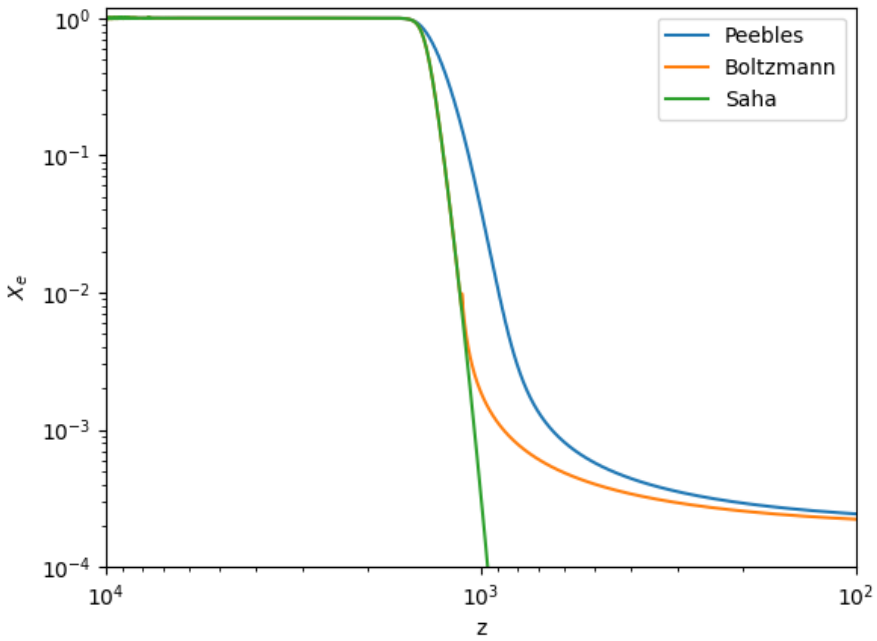


Figure 5: X_e

9 Boltzmann Equation: 1st Order

$$C[f] = \frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x^i}\frac{\hat{p}^i}{a}\frac{p}{E}(1 - \Phi + \Psi) - \frac{\partial f}{\partial p}\left\{\left[H + \dot{\Phi}\right]p + \frac{E}{a}\hat{p}^i\Psi_{,i}\right\} + \frac{\partial f}{\partial \hat{p}^i}\frac{E}{ap}\left[(\delta^{ik} - \hat{p}^i\hat{p}^k)(\frac{p^2}{E^2}\Phi - \Psi)_{,k}\right]. \tag{51}$$

9.1 Photons

9.1.1 Anisotropy

Temperature Perturbation:

$$\Theta = \delta T/T \tag{52}$$

Multipole:

$$\Theta_l = \frac{1}{(-i)^l}\int_{-1}^{1}\frac{d\mu}{2}\mathcal{P}_l(\mu)\Theta(\mu, k, \eta) \tag{53}$$

Optical depth:

$$\tau(\eta) = \int_{\eta}^{\eta_0} d\eta_1 n_e \sigma_T a \tag{54}$$

Visibility function:

$$g(\eta) = -\tau'(\eta)e^{-\tau(\eta)} \tag{55}$$

From visibility function, we get recombination at redshift about $z_{CMB} = 1080$ in Figure 6

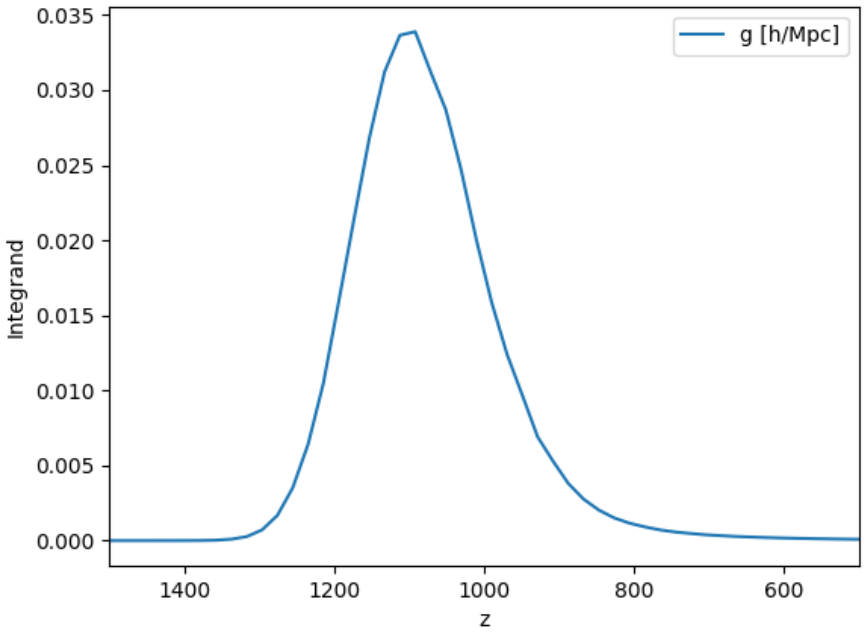


Figure 6: g(z)

Linear equation for photon anisotropy (CMB anisotropy):

$$\begin{aligned} \Theta' + ik\mu\Theta = -\Phi' - ik\mu\Psi - \tau'\left[\Theta_0 - \Theta + \mu u_{\rm b} - \frac{1}{2}\mathcal{P}_2(\mu)\Pi\right] \\ \Pi = \Theta_2 + \Theta_{P2} + \Theta_{P0} \end{aligned} \tag{56}$$

For each multipole:

$$\begin{aligned} l = 0: &\quad \Theta_0' + k\Theta_1 = -\Phi', \\ l = 1: &\quad \Theta_1' - \frac{1}{3}k(\Theta_0 - 2\Theta_2) = \frac{k\Psi}{3} + \tau'\left[\Theta_1 - \frac{iu_{\rm b}}{3}\right]. \\ l = 2: &\quad \Theta_2' - \frac{1}{5}k(2\Theta_1 - 3\Theta_3) = (\frac{9}{10}\Theta_2 - \frac{\Theta_{P0}}{10} - \frac{\Theta_{P2}}{10})\tau' \\ l > 2: &\quad \Theta_l' + \frac{k}{(-i)^{l+1}}\int_{-1}^{1}\frac{d\mu}{2}\mu\mathcal{P}_l(\mu)\Theta(\mu) = \tau'\Theta_l \end{aligned} \tag{57}$$

9.1.2 Polarization

Linear equation for photon polarization (CMB polarization, E mode):

$$\begin{aligned} \Theta_E' + ik\mu\Theta_E = \tau'\left[\Theta_E - \frac{3}{4}(1-\mu^2)\Pi\right] \\ \Pi = \Theta_2 + \Theta_{E2} + \Theta_{E0} \end{aligned} \tag{58}$$

9.2 Cold Dark Matter(CDM)

CDM perturbation:

$$\delta_c = \delta n_c/\bar{n} \tag{59}$$

Linear Equation for CDM:

$$\begin{aligned} \delta_c' + iku_c &= -3\Phi' \quad \text{(Continuity Equation)} \\ u_c' + \frac{a'}{a}u_c &= -ik\Psi \quad \text{(Euler's Equation)} \end{aligned} \tag{60}$$

9.3 Baryons

Baryon perturbation:

$$\delta_b = \delta\rho_e/\rho_e = \delta\rho_p/\rho_p \tag{61}$$

Linear Equation for Baryons:

$$\begin{aligned} \delta_b' + iku_b &= -3\Phi' \quad \text{(Continuity Equation)} \\ u_b' + \frac{a'}{a}u_b &= -ik\Psi + \frac{\tau'}{R_b}(u_b + 3i\Theta_1) \quad \text{(Euler's Equation)} \end{aligned} \tag{62}$$

The ratio of baryon to photon density:

$$R_b(\eta) = \frac{3\rho_b(\eta)}{4\rho_\gamma(\eta)} \tag{63}$$

9.4 Neutrinos

Neutrino perturbation:

$$\mathcal{N} = \delta T_\nu/T_\nu \tag{64}$$

Linear Equation for Neutrinos:

$$\mathcal{N}' + ik\mu\,\frac{p}{E_\nu(p)}\mathcal{N} - p\frac{a'}{a}\frac{\partial}{\partial p}\mathcal{N} = -\Phi' - ik\mu\frac{E_\nu(p)}{p}\Psi. \tag{65}$$

10 Initial Condition

10.1 Slow Roll Inflation Model

Scalar Inflation Field:

$$\begin{aligned} \rho &= \frac{1}{2}\dot{\phi}^2 + V(\phi) \\ \mathcal{P} &= \frac{1}{2}\dot{\phi}^2 - V(\phi) \end{aligned} \tag{66}$$

Slow Roll Variables:

$$\begin{aligned} \epsilon_{\rm sr} &\equiv \frac{d}{dt}\left(\frac{1}{H}\right) \\ \delta_{\rm sr} &\equiv \frac{1}{H}\frac{\ddot{\phi}}{\dot{\phi}} \end{aligned} \tag{67}$$

10.2 Initial Perturbations

$$\begin{aligned} \Psi = -\frac{2}{3}\mathcal{R}, \Phi = \frac{2}{3}\mathcal{R}, \\ \Theta_0 = \frac{1}{3}\mathcal{R}, \mathcal{N}_0 = \frac{1}{3}\mathcal{R} \\ \delta_b = \mathcal{R}, \delta_c = \mathcal{R} \\ \Theta_1 = -\frac{k}{9aH}\mathcal{R}, \mathcal{N}_1 = -\frac{k}{9aH}\mathcal{R} \\ iu_b = -\frac{k}{3aH}\mathcal{R}, iu_c = -\frac{k}{3aH}\mathcal{R} \end{aligned} \tag{68}$$

10.3 Initial Power Spectrum

Initial Power Spectrum:

$$\begin{aligned} P_{\mathcal{R}}(k) &= \frac{2\pi}{k^3}\frac{H^2}{m_{\rm Pl}^2\epsilon_{\rm Sr}}\bigg|_{aH=k} \equiv 2\pi^2\mathcal{A}_s k^{-3}\left(\frac{k}{k_{\rm P}}\right)^{n_s-1} \\ P_{\rm T}(k) &= 4P_h(k) = \frac{32\pi}{k^3}\frac{H^2}{m_{\rm Pl}^2}\bigg|_{aH=k} \equiv 2\pi^2 A_{\rm T} k^{-3}\left(\frac{k}{k_{\rm p}}\right)^{n_{\rm T}} \end{aligned} \tag{69}$$

Parameters:

$$\begin{aligned} n_s &= 1 - 4\epsilon_{\rm sr} - 2\delta_{\rm sr} = 0.9682 \\ n_{\rm T} &= -2\epsilon_{\rm sr} \approx 0 \\ k_p &= 0.05 Mpc^{-1} \\ ln(10^{10}A_s) &= 3.0480 \end{aligned} \tag{70}$$

Ratio of Initial Tensor and Scalar Power Spectrum:

$$r(k) \equiv \frac{P_{\rm T}(k)}{P_{\mathcal{R}}(k)} \overset{k=k_{\rm p}}{=} \frac{\mathcal{A}_{\rm T}}{\mathcal{A}_s} = 16\epsilon_{\rm sr}\Big|_{aH=k} \tag{71}$$

11 Linear Theory Summary

11.1 Linear Equations

Linear Equation for Scalar Perturbation:

$$\begin{aligned} k^2\Phi + 3\frac{a'}{a}\left(\Phi' - \Psi\frac{a'}{a}\right) &= 4\pi G a^2[\rho_{\rm m}\delta_{\rm m} + 4\rho_{\rm r}\Theta_{\rm r,0}](\text{Poisson's Equation}) \\ k^2(\Phi+\Psi) &= -32\pi G a^2\rho_{\rm r}\Theta_{\rm r,2} \end{aligned} \tag{72}$$

Linear Equation for Tensor Perturbation:

$$h'' + 2\frac{a'}{a}h_t' + k^2 h_t = 0 \quad ,(t=+,\times) \tag{73}$$

Linear Equation for Photon Anisotropy (CMB anisotropy):

$$\begin{aligned} \Theta' + ik\mu\Theta &= -\Phi' - ik\mu\Psi - \tau'\left[\Theta_0 - \Theta + \mu u_{\rm b} - \frac{1}{2}\mathcal{P}_2(\mu)\Pi\right] \\ \Pi &= \Theta_2 + \Theta_{P2} + \Theta_{P0}. \end{aligned} \tag{74}$$

Linear Equation for Photon Polarization (CMB polarization, E mode):

$$\begin{aligned} \Theta_E' + ik\mu\Theta_E &= \tau'\left[\Theta_E - \frac{3}{4}(1-\mu^2)\Pi\right] \\ \Pi &= \Theta_2 + \Theta_{E2} + \Theta_{E0}. \end{aligned} \tag{75}$$

Linear Equation for CDM:

$$\begin{aligned} \delta_c' + iku_c &= -3\Phi' \quad (\text{Continuity Equation}) \\ u_c' + \frac{a'}{a}u_c &= -ik\Psi \quad (\text{Euler's Equation}) \end{aligned} \tag{76}$$

Linear Equation for Baryons:

$$\begin{aligned} \delta_b' + iku_b &= -3\Phi' \quad (\text{Continuity Equation}) \\ u_b' + \frac{a'}{a}u_b &= -ik\Psi + \frac{\tau'}{R_b}(u_b + 3i\Theta_1) \quad (\text{Euler's Equation}) \end{aligned} \tag{77}$$

Linear Equation for Neutrinos:

$$\mathcal{N}' + ik\mu\,\frac{p}{E_\nu(p)}\mathcal{N} - p\frac{a'}{a}\frac{\partial}{\partial p}\mathcal{N} = -\Phi' - ik\mu\frac{E_\nu(p)}{p}\Psi. \tag{78}$$

11.2 Initial Condition

Initial Condition:

$$
\begin{aligned}
\Psi = -\frac{2}{3}\mathcal{R}, \Phi = \frac{2}{3}\mathcal{R}, \\
\Theta_0 = \frac{1}{3}\mathcal{R}, \mathcal{N}_0 = \frac{1}{3}\mathcal{R} \\
\delta_b = \mathcal{R}, \delta_c = \mathcal{R} \\
\Theta_1 = -\frac{k}{9aH}\mathcal{R}, \mathcal{N}_1 = -\frac{k}{9aH}\mathcal{R} \\
iu_b = -\frac{k}{3aH}\mathcal{R}, iu_c = -\frac{k}{3aH}\mathcal{R}
\end{aligned}
\tag{79}
$$

Initial Power Spectrum:

$$
\begin{aligned}
P_{\mathcal{R}}(k) &= \frac{2\pi}{k^3}\frac{H^2}{m_{\rm Pl}^2 \epsilon_{\rm Sr}}\bigg|_{aH=k} \equiv 2\pi^2 \mathcal{A}_s k^{-3}\left(\frac{k}{k_{\rm P}}\right)^{n_s-1} \\
P_{\rm T}(k) &= 4P_h(k) = \frac{32\pi}{k^3}\frac{H^2}{m_{\rm Pl}^2}\bigg|_{aH=k} \equiv 2\pi^2 A_{\rm T} k^{-3}\left(\frac{k}{k_{\rm p}}\right)^{n_{\rm T}}
\end{aligned}
\tag{80}
$$

12 Linear Results: Matter and Potential

Linear Matter Power Spectrum in Figure 7, Figure 8

$$P_L(k,a) = \left[\frac{2(k^2+3a^2H^2)}{5H_0^2\Omega_{m0}}T(k)D_+(a)\right]^2 P_{\mathcal{R}}(k) \tag{81}$$

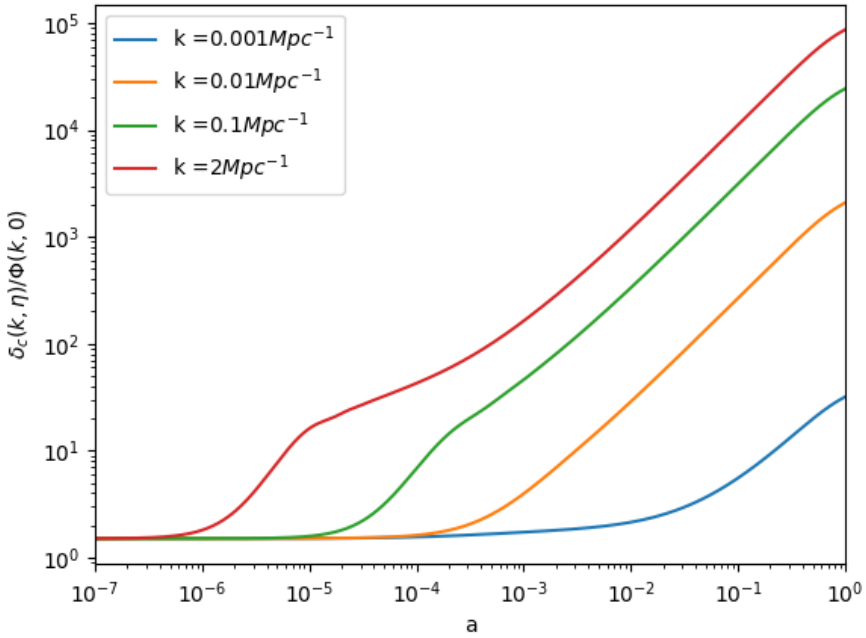


Figure 7: δ_c/Φ_0

Linear Potential Power Spectrum in Figure 9

$$P_\Phi(k,a) = \left[\frac{3}{5}T(k)\frac{D_+(a)}{a}\right]^2 P_{\mathcal{R}}(k) \tag{82}$$

Tensor perturbation in Figure 10. Smaller scale modes decay earlier.
Transfer function (BBKS):

$$\begin{aligned} T(k) = &\frac{ln(1+2.34\cdot q)}{2.34\cdot q} \\ &\cdot\left[1+(3.89\cdot q)+(16.1\cdot q)^2+(5.46\cdot q)^3+(6.71\cdot q)^4\right]^{-1/4} \end{aligned} \tag{83}$$

$$q(k) = \frac{k}{\Omega_{m0}h\cdot h/Mpc} \tag{84}$$

Growth Rate:

$$f(a) = \frac{dlnD_+(a)}{dlna} = \left[\frac{H_0^2}{H^2(a)}\frac{\Omega_{m0}}{a^3}\right]^{0.55} \tag{85}$$

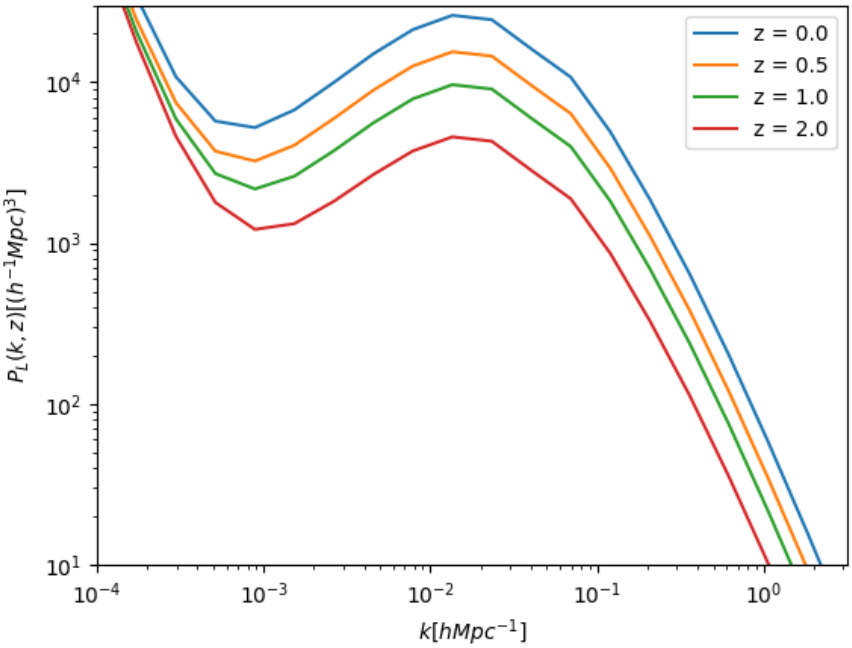


Figure 8: P_L

13 Linear Results: Photons (CMB)

13.1 Anisotropy

Cosmic Microwave Background(CMB) anisotropic power spectrum in Figure 11

$$C(l) = \frac{2}{\pi}\int_0^\infty dk k^2 P_{\mathcal{R}}(k)|T_l(k)|^2 \tag{86}$$

$$T_l(k) = \Theta_l(k,\eta_0)/\mathcal{R}(k) \tag{87}$$

The ratio of baryon to photon density:

$$R_b(\eta) = \frac{3\rho_b(\eta)}{4\rho_\gamma(\eta)} \tag{88}$$

Sound speed of baryon-photon fluid:

$$c_s(\eta) = \sqrt{\frac{1}{3[1+R_b(\eta)]}} \tag{89}$$

Sound horizon:

$$r_s(\eta) = \int_0^\eta d\tilde{\eta} c_s(\tilde{\eta}) \tag{90}$$

Multi-poles of Photons:

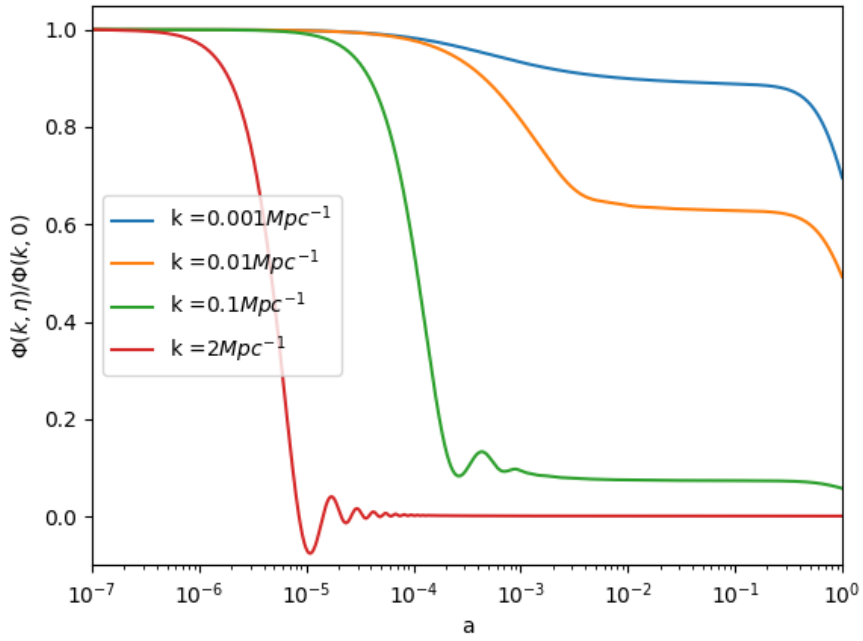


Figure 9: Φ/Φ_0

$$\begin{aligned}\Theta_l(k,\eta_0) \simeq & \left[\Theta_0(k,\eta_*)+\Psi(k,\eta_*)\right] j_l\left[k(\eta_0-\eta_*)\right] \\ & + 3\Theta_1(k,\eta_*)\left(j_{l-1}\left[k(\eta_0-\eta_*)\right]-(l+1)\frac{j_l\left[k(\eta_0-\eta_*)\right]}{k(\eta_0-\eta_*)}\right) \\ & + \int_0^{\eta_0} d\eta e^{-\tau}\left[\Psi'(k,\eta)-\Phi'(k,\eta)\right] j_l\left[k(\eta_0-\eta)\right]\end{aligned} \tag{91}$$

Monopole and Dipole:

$$\begin{aligned}\Theta_0(\mathbf{k},\eta) = & \{\left[\Theta_0(\mathbf{k},0)+\Phi(\mathbf{k},0)\right]\cos(kr_s)-\Phi(\mathbf{k},\eta) \\ & + \frac{k}{\sqrt{3}}\int_0^{\eta} d\tilde{\eta}[\Phi(\mathbf{k},\tilde{\eta})-\Psi(\mathbf{k},\tilde{\eta})]\sin[k(r_s(\eta)-r_s(\tilde{\eta}))]\}\cdot\exp(-\frac{k^2}{k_D^2})\end{aligned} \tag{92}$$

$$\begin{aligned}\Theta_1(\mathbf{k},\eta) = & \{\frac{1}{\sqrt{3}}[\Theta_0(\mathbf{k},0)+\Phi(\mathbf{k},0)]\sin(kr_s) \\ & - \frac{k}{3}\int_0^{\eta} d\tilde{\eta}[\Phi(\mathbf{k},\tilde{\eta})-\Psi(\mathbf{k},\tilde{\eta})]\cos[k(r_s(\eta)-r_s(\tilde{\eta}))]\}\cdot\exp(-\frac{k^2}{k_D^2})\end{aligned} \tag{93}$$

Monopole and Dipole in Wayne Hu's approximation (arxiv:9504057):

$$\frac{\Theta_0(k,\eta)}{\mathcal{R}(k)} = \left[-\frac{1}{5}(1+3\mathcal{R}_b)\cos(kr_s(\eta))+(\mathcal{R}_b+1)\frac{\Phi(k,\eta)}{\mathcal{R}(k)}\right]\cdot\exp(-\frac{k^2}{k_D^2}) \tag{94}$$

$$\frac{\Theta_1(k,\eta)}{\mathcal{R}(k)} = \left[-\frac{1}{5}(1+3\mathcal{R}_b)c_s\cdot\sin(kr_s(\eta))\right]\cdot\exp(-\frac{k^2}{k_D^2}) \tag{95}$$

Diffusion Damping:

$$\frac{1}{k_D^2(\eta)} \equiv \int_0^{\eta}\frac{d\tilde{\eta}}{6(1+R_b)n_e\sigma_{\mathrm{T}}a(\tilde{\eta})}\left[\frac{R_b^2}{1+R_b}+\frac{8}{9}\right] \tag{96}$$

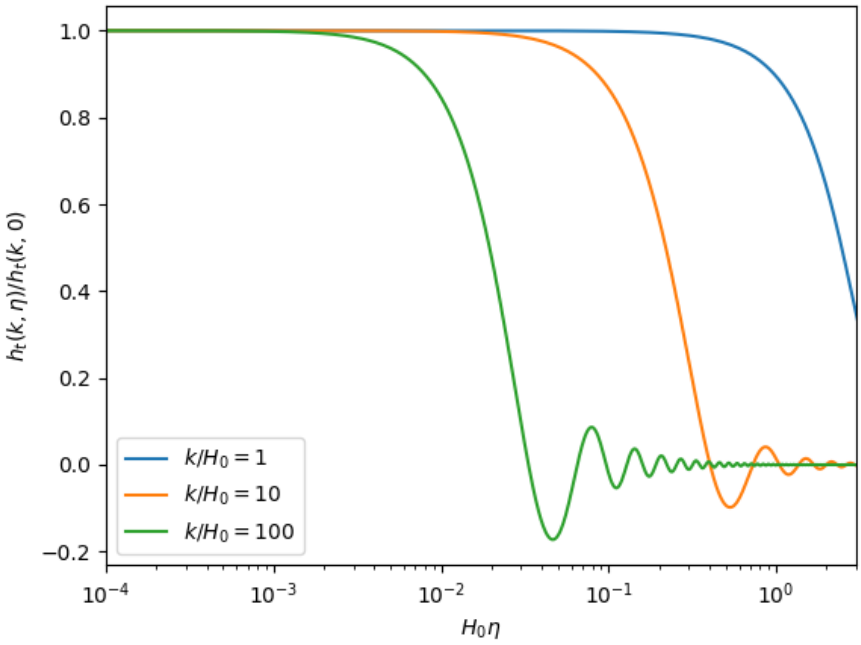


Figure 10: Tensor Perturbation

13.2 Polarization

(1) E and B modes
Polarization Tensor:

$$I_{ij} = \begin{pmatrix} I+Q & U \\ U & I-Q \end{pmatrix} \tag{97}$$

Meaning of polarization components Q,U in Figure 12
Change Q,U to rotation-invariant polarization components E,B in Figure 13

$$\begin{aligned} \nabla^2 E &= \partial_i \partial_j (I_{ij} - I\delta_{ij}) \\ \nabla^2 B &= \epsilon_{ik} \partial_j \partial_k (I_{ij} - I\delta_{ij}) \end{aligned} \tag{98}$$

Here,

$$\epsilon_{ij} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \tag{99}$$

In Fourier space:

$$\begin{aligned} E(\mathbf{l}) &= (\frac{l^i l^j}{l^2} - \frac{1}{2}\delta_{ij}) I_{ij} \\ B(\mathbf{l}) &= \epsilon_{ik} \frac{l^j l^k}{l^2} I_{ij} \end{aligned} \tag{100}$$

Set l as:

$$\mathbf{l} = l(\cos\phi_l, \sin\phi_l) \tag{101}$$

We get relation of Q,U and E,B:

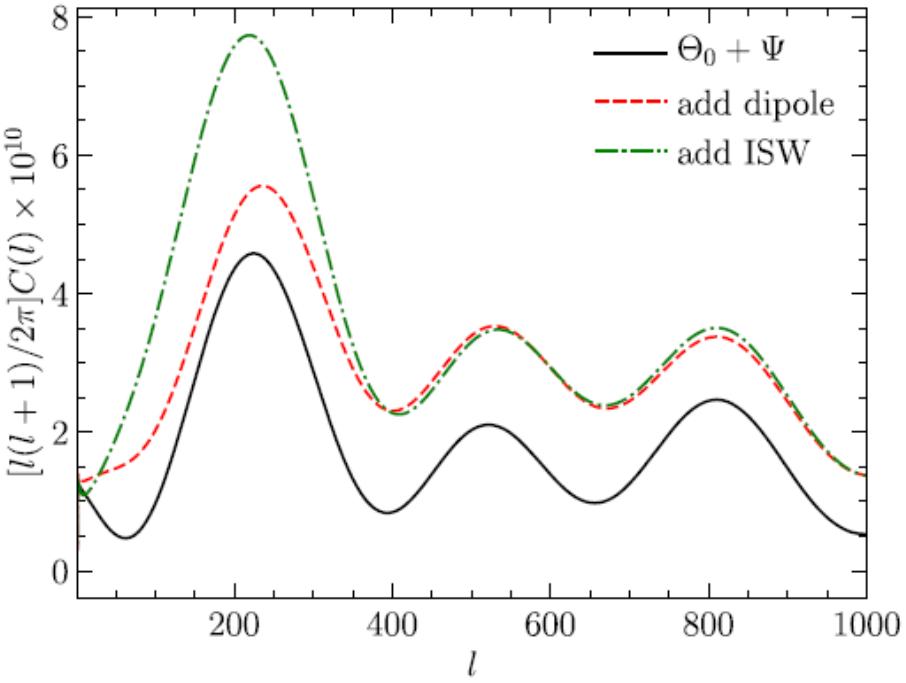


Figure 11: CMB Anisotropy

$$\begin{pmatrix} E(\mathbf{l}) \\ B(\mathbf{l}) \end{pmatrix} = \begin{pmatrix} \cos(2\phi_l) & \sin(2\phi_l) \\ -\sin(2\phi_l) & \cos(2\phi_l) \end{pmatrix} \begin{pmatrix} Q(\mathbf{l}) \\ U(\mathbf{l}) \end{pmatrix} \tag{102}$$

Reverse the matrix:

$$\begin{pmatrix} Q(\mathbf{l}) \\ U(\mathbf{l}) \end{pmatrix} = \begin{pmatrix} \cos(2\phi_l) & -\sin(2\phi_l) \\ \sin(2\phi_l) & \cos(2\phi_l) \end{pmatrix} \begin{pmatrix} E(\mathbf{l}) \\ B(\mathbf{l}) \end{pmatrix} \tag{103}$$

We can write polarization tensor I_{ij} in term of I, E and B:

$$I_{ij}(\mathbf{l}) = I(\mathbf{l})\delta_{ij} + \begin{pmatrix} \cos(2\phi_l) & \sin(2\phi_l) \\ \sin(2\phi_l) & -\cos(2\phi_l) \end{pmatrix} E(\mathbf{l}) + \begin{pmatrix} -\sin(2\phi_l) & \cos(2\phi_l) \\ \cos(2\phi_l) & \sin(2\phi_l) \end{pmatrix} B(\mathbf{l}) \tag{104}$$

(2) Power Spectrum
For different modes:

$$s = \{T, E, B\} \tag{105}$$

We have power spectrum in Figure 14

$$C_{s_1 s_2}(l) = \frac{2}{\pi} \int_0^\infty dk k^2 P_{\mathcal{R}}(k) |T_l^{s_1}(k) T_l^{s_2 *}(k)| \tag{106}$$

$$T_l^s(k) = \Theta_{s,l}(k, \eta_0)/\mathcal{R}(k) \tag{107}$$

E mode:

$$\Theta_{E,l}(k, \eta_0) = -\frac{5k}{6\tau'} \Theta_1(k, \eta_*) \frac{l^2}{(k\eta_0)^2} j_l(k\eta_0) \tag{108}$$

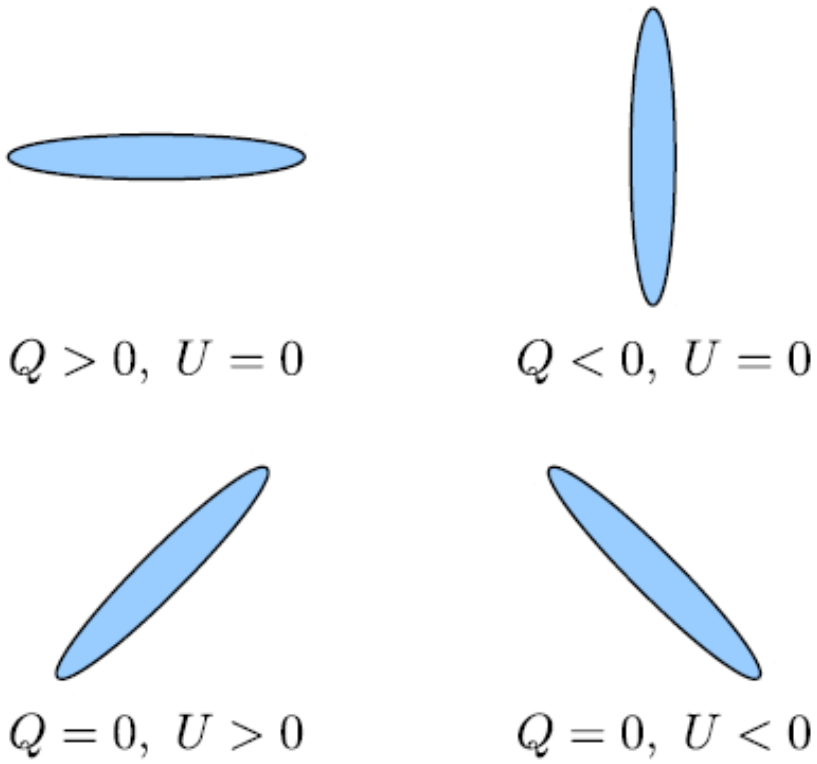


Figure 12: Polarization components:Q and U

B mode all comes from tensor perturbation:

$$B_{\times}^{T}(\mathbf{k}) \propto \frac{8\pi}{15} h_{\times}(\mathbf{k})(\hat{n} \cdot \hat{k}) \tag{109}$$

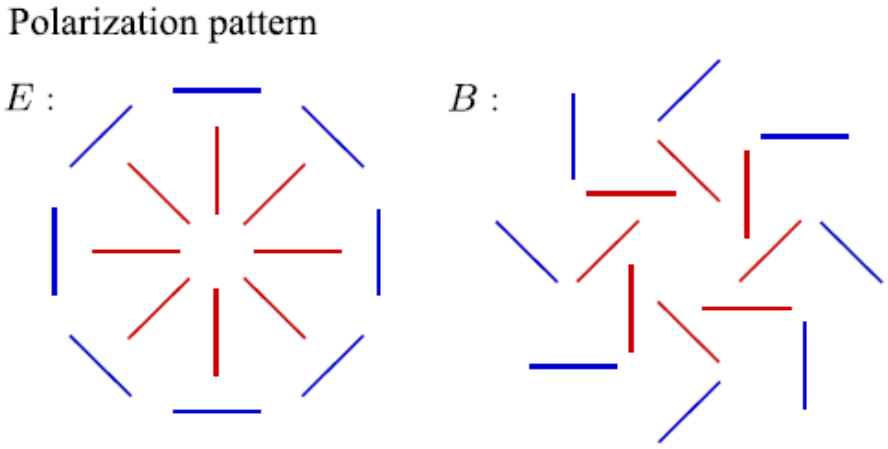


Figure 13: Rotation-invariant polarization components:E and B

14 Tracers

14.1 RSD + AP Effect

(1) Redshift space distortion (RSD) effect
RSD: Influence to power spectrum from peculiar velocity.

Galaxy bias:

$$\delta_g(\mathbf{x}, z) = b_1(z)\delta_m(\mathbf{x}, z) \tag{110}$$

Cosine of angle between the light of sight and the wave vector:

$$\mu_k = \widehat{\mathbf{e}}_z \cdot \widehat{\mathbf{k}} \tag{111}$$

(2) *Alcock − Paczyński* (AP) effect
AP effect: Influence to power spectrum from wrong cosmology.

Error of comoving distance and Hubble parameter between fiducial cosmology and real value:

$$\delta\chi = \chi_{fid} - \chi,\ \delta H = H_{fid} - H \tag{112}$$

Relation between true and observed $\mathbf{k}$:

$$\mathbf{k}[\mathbf{k}_{obs}] = ([1+\alpha_\perp]k^1_{\rm c,fid}, [1+\alpha_\perp]k^2_{\rm c,fid}, [1+\alpha_\parallel]k^3_{\rm c,fid}) \tag{113}$$

$$\alpha_\perp = \left.\frac{\delta\chi}{\chi_{fid}}\right|_{\bar{z}},\ \alpha_\parallel = -\left.\frac{\delta H}{H_{fid}}\right|_{\bar{z}} \tag{114}$$

(3) Observed galaxy power spectrum (RSD + AP effect):

$$P_{\rm g,obs}(\mathbf{k}_{\rm obs}, \bar{z}) = P_{\rm L}(k, \bar{z})\left[b_1 + f\mu_k^2\right]^2 + P_N \tag{115}$$

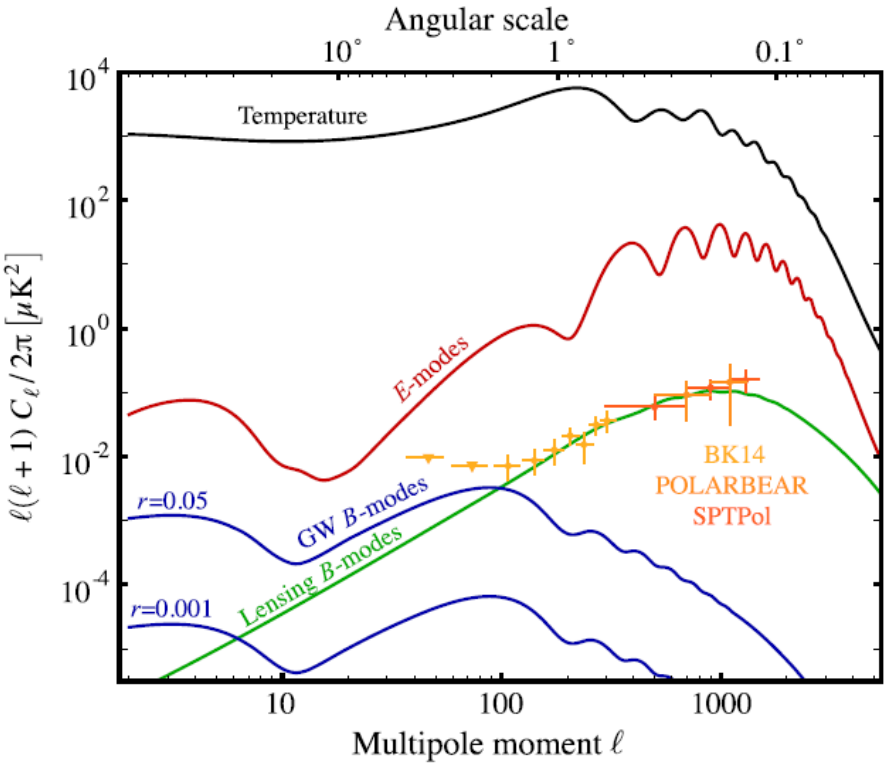


Figure 14: CMB Polarization

More accurate result, as shown in Figure 15:

$$P_{\mathrm{g,obs}}(\mathbf{k}_{\mathrm{obs}},\bar{z}) = P_{\mathrm{L}}(k,\bar{z})\left[b_1 + f\mu_k^2\right]^2 D^{FOG}(k\mu_k) + P_N \tag{116}$$

$$D^{FOG}(k_z = k\mu_k) = \left|\langle \exp(i\frac{k_z v_z}{H})\rangle\right|^2 \tag{117}$$

14.2 Angular Correlation

Distribution of galaxy distances:

$$W(\chi) = \frac{1}{N_g}\frac{dN_g}{d\chi} \tag{118}$$

Projected overdensity:

$$\Delta(\hat{\mathbf{n}}) = \int_0^\infty d\chi W(\chi)\delta_{g,obs}(\mathbf{x} = \hat{\mathbf{n}}\chi, \eta = \eta_0 - \chi) \tag{119}$$

Angular power spectrum:

$$\langle \Delta_{g,lm}\Delta^*_{g,l'm'}\rangle = \delta_{ll'}\delta_{mm'}C_g(l) \tag{120}$$

Angular power spectrum result (Limber approximation):

$$C_g(l) = \int \frac{d\chi}{\chi^2}W^2(\chi)P_{g,obs}\left(k = \frac{l+1/2}{\chi}, \mu_k = 0, \eta(\chi)\right) \tag{121}$$

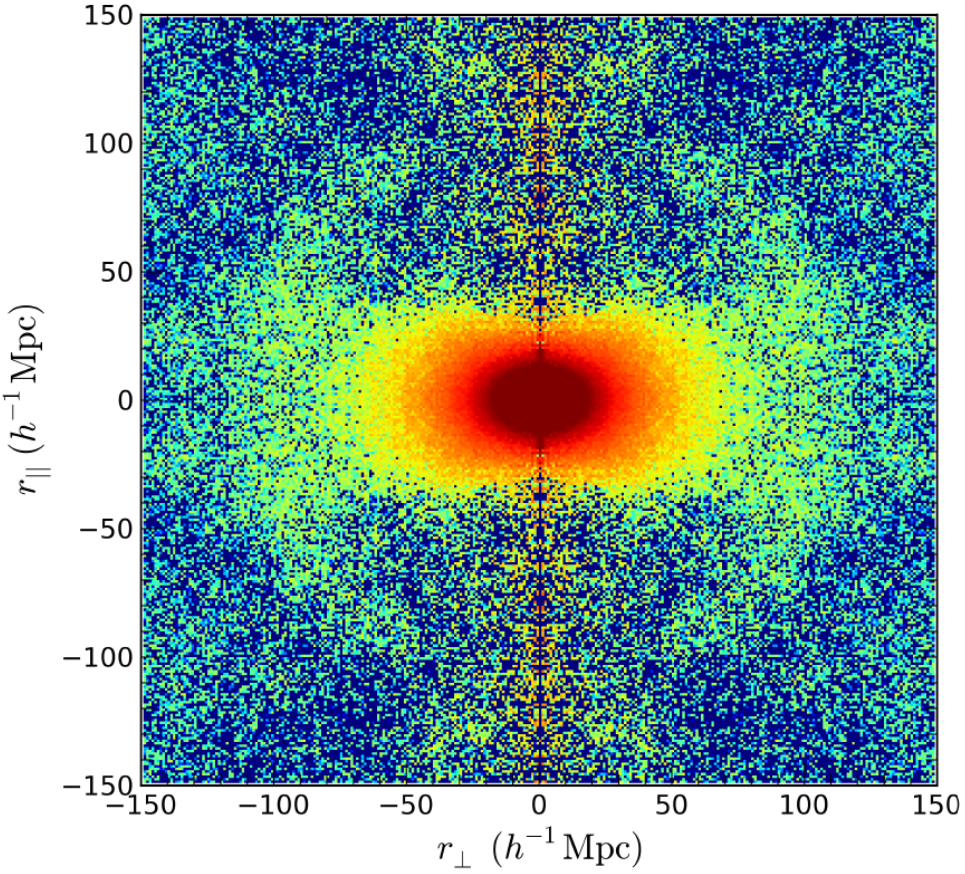


Figure 15: RSD

Angular correlation function:

$$w_g(\theta) = \int_0^\infty \frac{dl}{2\pi} l C_g(l) J_0(l\theta) \tag{122}$$

14.3 tSZ effect

Thermal Sunyaev-Zel'dovich (tSZ) effect: inverse-Compton scattering of high-temperature electrons in gas to CMB photons.

Gas pressure:

$$\mathcal{P}_{gas}(\mathbf{x}, \eta) = n_e(\mathbf{x}, \eta) T_e(\mathbf{x}, \eta) = \overline{n_e T_e}(\eta)[1 + \delta_{\mathcal{P}}(\mathbf{x}, \eta)] \tag{123}$$

tSZ term y is the integral over the kinetic energy density/pressure of the ionized gas along the line of sight:

$$y = \frac{\sigma_T}{m_e} \int n_e T_e a d\chi = \frac{\sigma_T}{m_e} \int \overline{n_e T_e}(\eta) a [1 + \delta_{\mathcal{P}}(\mathbf{x}, \eta)] d\chi \tag{124}$$

Distribution of CMB photons:

$$\begin{cases} f(x, y) \overset{y \ll 1}{=} [1 + y \frac{1}{x^2} \frac{\partial}{\partial x} (x^4 \frac{\partial}{\partial x})] \frac{1}{e^x - 1} \\ x = p/T \end{cases} \tag{125}$$

Radiation intensity at $T_0 = 2.73K$ in Figure 16

$$\begin{aligned} u_\nu(T, x, y) &= 4\pi I_\nu(T, x, y) \\ I_\nu(T, x, y) &= 4\pi\nu^3 f(x, y) \\ \nu &= p/(2\pi) = x \cdot T/(2\pi) \end{aligned} \tag{126}$$

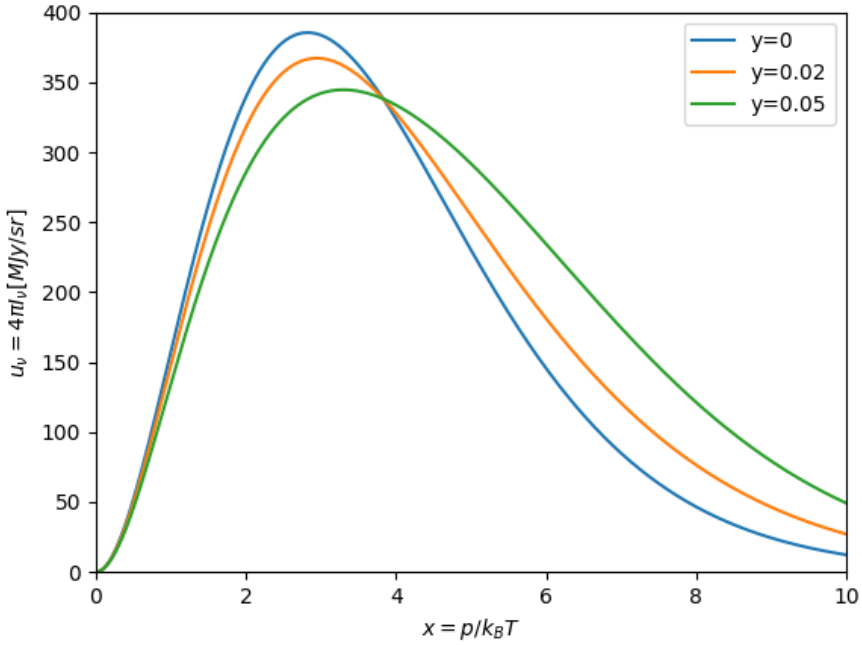


Figure 16: tSZ effect

Power spectrum of gas pressure perturbation:

$$\langle \delta_{\mathcal{P}}(\mathbf{k}_1)\delta_{\mathcal{P}}(\mathbf{k}_2)\rangle = (2\pi)^3 \delta_D^{(3)}(\mathbf{k}_1 + \mathbf{k}_2) P_{\mathcal{P}}(k_1) \tag{127}$$

Angular power spectrum of the tSZ y-distortion of the CMB:

$$C_y(l) = \left(\frac{\sigma_T}{m_e}\right)^2 \int_0^{\chi_*} \frac{d\chi}{\chi^2} [\overline{n_e T_e} a]^2 P_{\mathcal{P}}(k = \frac{l + 1/2}{\chi}, \eta(\chi)) \tag{128}$$

15 Perturbation Theory

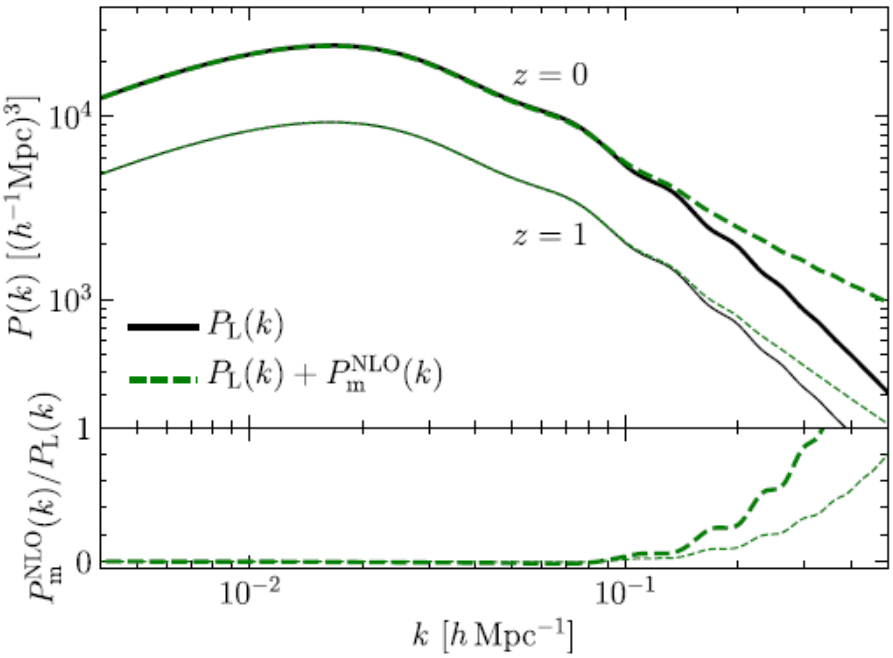


Figure 17: Power Spectrum in Perturbation Theory

Vlasov-Poisson System:

$$\begin{aligned} &\frac{\partial f_{\rm m}}{\partial t} + \frac{\partial f_{\rm m}}{\partial x^j}\frac{p^j}{ma} - \frac{\partial f_{\rm m}}{\partial p^j}\left[Hp^j + \frac{m}{a}\frac{\partial \Psi}{\partial x^j}\right] = 0, \\ &\nabla^2\Psi = 4\pi G a^2\left[\int \frac{d^3p}{(2\pi)^3} f_{\rm m}(x,p,t) - \rho_{\rm m}(t)\right]. \end{aligned} \tag{129}$$

Velocity divergence:

$$\theta_m = \partial_i u_m^i \tag{130}$$

From Vlasov-Poisson system, we get equations of perturbation theory:

$$\begin{aligned} {\delta_{\rm m}}' + \theta_{\rm m} &= -\delta_{\rm m}\theta_{\rm m} - u_{\rm m}^j\frac{\partial}{\partial x^j}\delta_{\rm m}, \\ {\theta_{\rm m}}' + aH\theta_{\rm m} + \nabla^2\Psi &= -u_{\rm m}^j\frac{\partial}{\partial x^j}\theta_{\rm m} - (\partial_i u_{\rm m}^j)(\partial_j u_{\rm m}^i), \\ \nabla^2\Psi &= \frac{3}{2}\Omega_{\rm m}(\eta)(aH)^2\delta_{\rm m}. \end{aligned} \tag{131}$$

Perturbation of matter density and velocity divergence:

$$\begin{aligned} \delta_{\rm m}(x,\eta) &= \delta^{(1)}(x,\eta) + \delta^{(2)}(x,\eta) + \cdots + \delta^{(n)}(x,\eta), \\ \theta_{\rm m}(x,\eta) &= \theta^{(1)}(x,\eta) + \theta^{(2)}(x,\eta) + \cdots + \theta^{(n)}(x,\eta), \end{aligned} \tag{132}$$

Solution of each order perturbation:

$$
\begin{aligned}
\delta^{(n)}(k,\eta) &= D_+^n(\eta)\left[\prod_{i=1}^n\int\frac{d^3k_i}{(2\pi)^3}\right](2\pi)^3\delta_{\rm D}^{(3)}\left(\mathbf{k}-\sum_{i=1}^n\mathbf{k}_i\right)\\
&\quad\times F_n(\mathbf{k}_1,\cdots,\mathbf{k}_n)\delta_0(\mathbf{k}_1)\cdots\delta_0(\mathbf{k}_n),\\
\theta^{(n)}(k,\eta) &= -aHfD_+^n(\eta)\left[\prod_{i=1}^n\int\frac{d^3k_i}{(2\pi)^3}\right](2\pi)^3\delta_{\rm D}^{(3)}\left(\mathbf{k}-\sum_{i=1}^n\mathbf{k}_i\right)\\
&\quad\times G_n(\mathbf{k}_1,\cdots,\mathbf{k}_n)\delta_0(\mathbf{k}_1)\cdots\delta_0(\mathbf{k}_n).
\end{aligned}
\tag{133}
$$

Where:

$$
\begin{aligned}
F_1 &= G_1 = 1\\
F_2(\mathbf{k}_1,\mathbf{k}_2) &= \frac{5}{7}+\frac{2}{7}\frac{(\mathbf{k}_1\cdot\mathbf{k}_2)^2}{k_1^2k_2^2}+\frac{1}{2}\widehat{\mathbf{k}}_1\cdot\widehat{\mathbf{k}}_2\left(\frac{k_1}{k_2}+\frac{k_2}{k_1}\right)\\
G_2(\mathbf{k}_1,\mathbf{k}_2) &= \frac{3}{7}+\frac{4}{7}\frac{(\mathbf{k}_1\cdot\mathbf{k}_2)^2}{k_1^2k_2^2}+\frac{1}{2}\widehat{\mathbf{k}}_1\cdot\widehat{\mathbf{k}}_2\left(\frac{k_1}{k_2}+\frac{k_2}{k_1}\right)\\
&\cdots
\end{aligned}
\tag{134}
$$

Power spectrum in Figure 17

$$
\begin{aligned}
&P(k,\eta) = P_{\rm L}(k,\eta)+P^{\rm NLO}(k,\eta)+\cdots,\\
&P^{\rm NLO}(k,\eta) = P^{(22)}(k,\eta)+2P^{(13)}(k,\eta),\\
&P^{(22)}(k,\eta) = 2\int\frac{d^3p}{(2\pi)^3}\Big[F_2(\mathbf{p},\mathbf{k}-\mathbf{p})\Big]^2P_{\rm L}(p,\eta)P_{\rm L}(|\mathbf{k}-\mathbf{p}|,\eta),\\
&P^{(13)}(k,\eta) = 3P_{\rm L}(k,\eta)\int\frac{d^3p}{(2\pi)^3}F_3(\mathbf{p},-\mathbf{p},\mathbf{k})P_{\rm L}(p,\eta).
\end{aligned}
\tag{135}
$$

Bispectrum:

$$
\begin{aligned}
\langle\delta_{\rm m}(\mathbf{k}_1,\eta)\delta_{\rm m}(\mathbf{k}_2,\eta)\delta_{\rm m}(\mathbf{k}_3,\eta)\rangle &= (2\pi)^3\delta_{\rm D}^{(3)}(\mathbf{k}_1+\mathbf{k}_2+\mathbf{k}_3)\\
&\quad\times\left[2F_2(\mathbf{k}_1,\mathbf{k}_2)P_{\rm L}(k_1,\eta)P_{\rm L}(k_2,\eta)+2\text{perm.}\right]
\end{aligned}
\tag{136}
$$

16 Spherical Collapse

16.1 Equations of spherical collapse

In Einstein-de Sitter Universe ($\Lambda = 0, k = 0$), and matter era. Omit pressure and baryon effects.

Newton's Gravitational Equation:

$$\ddot{r} = -\frac{GM}{r} \tag{137}$$

Energy conservation:

$$E = \frac{1}{2}\dot{r} - \frac{GM}{r} \tag{138}$$

Mass of perturbation M and energy E are constants.

16.2 Solution of spherical collapse

When $E < 0$, we get the solution:

$$\begin{aligned} r(\theta) &= \frac{GM}{2|E|}(1 - \cos\theta) \\ t(\theta) &= \frac{GM}{|2E|^{3/2}}(\theta - \sin\theta) \end{aligned} \tag{139}$$

16.3 Halo density perturbation

Halo density:

$$\rho_p(\theta) = \frac{M}{(4\pi/3)r^3} = \frac{6|E|^3}{\pi G^3 M^2}\frac{1}{(1-\cos\theta)^3} \tag{140}$$

Background density from Friedmann equation:

$$\bar{\rho}(\theta) = \frac{3H^2}{8\pi G} = \frac{1}{6\pi G t^2} = \frac{4|E|^3}{3\pi G^3 M^2}\frac{1}{(\theta - \sin\theta)^2} \tag{141}$$

Halo density perturbation:

$$1 + \delta_p(\theta) = \frac{\rho_p}{\bar{\rho}} = \frac{9}{2}\frac{(\theta - \sin\theta)^2}{(1-\cos\theta)^3} \propto \frac{t^2}{r^3} \tag{142}$$

Linear density perturbation ($\theta \ll 1$):

$$\delta_L(\theta) = \frac{3|E|}{10}\left(\frac{6}{GM}\right)^{2/3} t^{2/3} = \frac{3}{20}[6(\theta - \sin\theta)]^{2/3} \propto t^{2/3} \tag{143}$$

16.4 Perturbation on different stages

(1) Turn-around:

Time and radius:

$$\begin{aligned} \theta_{ta} &= \pi \\ t_{ta} = t(\theta_{ta}) &= \pi \cdot \frac{GM}{|2E|^{3/2}} \\ r_{ta} = r(\theta_{ta}) &= 2 \cdot \frac{GM}{2|E|} \end{aligned} \tag{144}$$

Perturbation:

$$\begin{aligned} \delta_{p,ta} &= \delta_p(\theta_{ta}) = 4.55 \\ \delta_{L,ta} &= \delta_L(\theta_{ta}) = 1.06 \end{aligned} \tag{145}$$

r(t) relation no longer holds at late collapse era, for pressure of baryons is large at late collapse era.

At late collapse era, compare with turn around perturbations and estimate perturbations from:

$$1 + \delta_p \propto \frac{t^2}{r^3}, \quad \delta_L \propto t^{2/3} \tag{146}$$

(2) Bounded:

The time when the halo is bounded.

Time and radius:

$$\begin{aligned} \theta_c &= \frac{3}{2}\pi \\ t_c = t(\theta_c) &\approx 2t_{ta} \\ r_c = r(\theta_c) &= \frac{1}{2}r_{ta} \end{aligned} \tag{147}$$

Perturbation:

$$\begin{aligned} 1 + \delta_{p,c} &= \left(\frac{t_c}{t_{ta}}\right)^2 \left(\frac{r_{ta}}{r_c}\right)^3 (1 + \delta_{p,ta}) = 177.6 \\ \delta_{cr} = \delta_{L,c} &= \left(\frac{t_c}{t_{ta}}\right)^{2/3} \delta_{L,ta} = 1.686 \end{aligned} \tag{148}$$

(3) Virialized:

The time when the halo is in virial equilibrium.

Time and radius:

$$\begin{aligned} \theta_{vir} &= 3\pi \\ t_{vir} = t(\theta_{vir}) &= 3t_{ta} \\ r_{vir} &= \frac{1}{2}r_{ta} \end{aligned} \tag{149}$$

Perturbation:

$$
\begin{aligned}
1 + \delta_{p,vir} &= \left(\frac{t_{vir}}{t_{ta}}\right)^2 \left(\frac{r_{ta}}{r_{vir}}\right)^3 (1 + \delta_{p,ta}) = 400 \\
\delta_{L,vir} &= \left(\frac{t_{vir}}{t_{ta}}\right)^{2/3} \delta_{L,ta} = 2.20
\end{aligned}
\tag{150}
$$

17 The Halo Model

17.1 Halo Density Profile

Halo density Navarro, Frenk and White profile (NFW profile) in Figure 18

$$\rho_{\rm h}(r) = \frac{\rho_s}{(r/r_{sh})(1+r/r_{sh})^2} \tag{151}$$

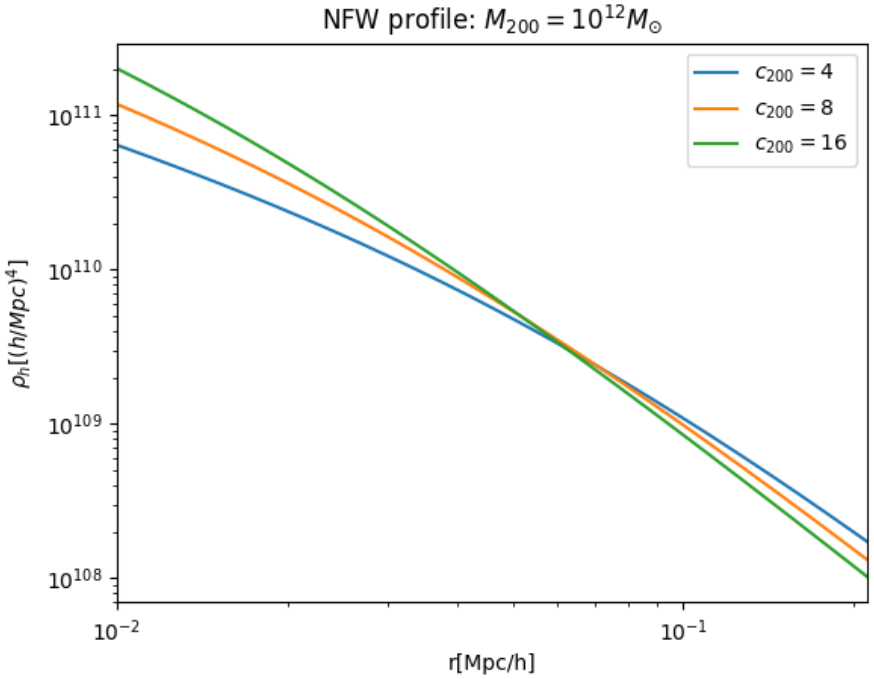


Figure 18: NFW Halo Density Profile $\rho_h(r)$

Concentration:

$$c_\Delta = R_\Delta / r_{sh} \tag{152}$$

Halo mass:

$$M_\Delta = M(< R_\Delta) = \frac{4\pi}{3} R_\Delta^3 \cdot \rho_{m0} \cdot \Delta \tag{153}$$

$$M_\Delta = \int_0^{R_\Delta} 4\pi r^2 \rho_h(r) dr \tag{154}$$

Usually set $\Delta = 200$.
For certain Δ,we can set $\rho_h(r)$ in terms of M_Δ and c_Δ:

$$
\begin{aligned}
\rho_{\rm h}(r) &= \frac{\rho_s}{(r/r_{sh})(1+r/r_{sh})^2} \\
R_\Delta &= \left(\frac{3M_\Delta}{4\pi\rho_{m0}\cdot\Delta}\right)^{1/3} \\
r_{sh} &= R_\Delta/c_\Delta \\
\rho_s &= \frac{M_\Delta}{4\pi r_{sh}^3}\left[\ln(c_\Delta+1)+\frac{1}{c_\Delta+1}-1\right]^{-1}
\end{aligned}
\tag{155}
$$

NFW halo density profile in k space:

$$
\begin{aligned}
\rho_h\left(k=\frac{x}{r_s},M\right) = \frac{M}{\mathcal{N}}[&\cos x\,[\mathrm{Ci}([c+1]x)-\mathrm{Ci}(x)] \\
&+\sin x\,[\mathrm{Si}([c+1]x)-\mathrm{Si}(x)]-\frac{\sin cx}{(c+1)x}]
\end{aligned}
\tag{156}
$$

$$
\begin{aligned}
\mathcal{N} &= \frac{1}{c+1}+\log(c+1)-1 \\
r_{sh} &= R_{200}/c, \quad c=8
\end{aligned}
\tag{157}
$$

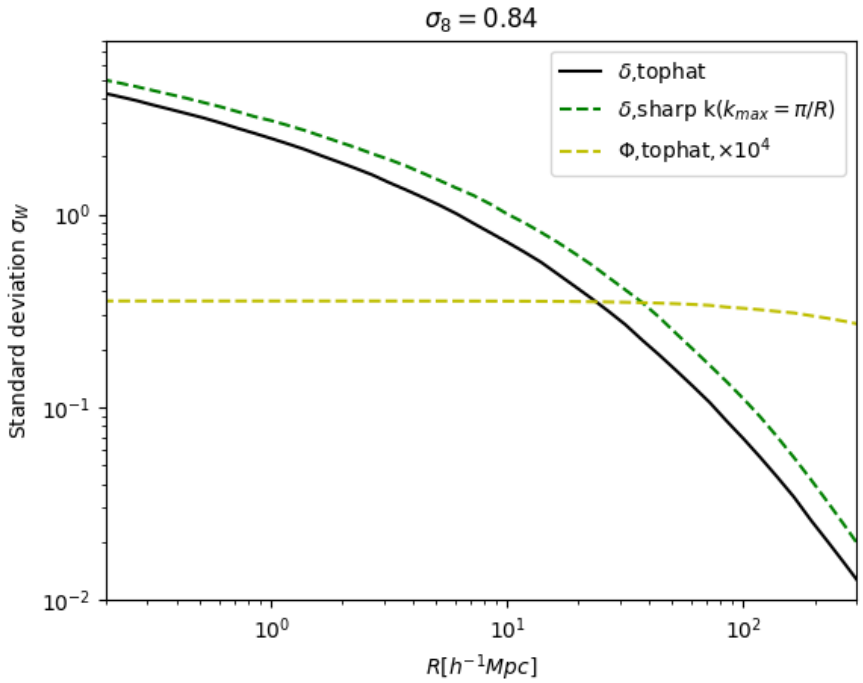


Figure 19: $\sigma_W(R)$

17.2 Halo Mass Function

Lagrangian radius (enclose mass of halo at the mean matter density):

$$
R_L(M) = 1.40h^{-1}\,\mathrm{Mpc}\left(\frac{M}{10^{12}h^{-1}M_\odot}\right)^{1/3} \tag{158}
$$

The top-hat window function describes a spherical perturbation with uniform density:

$$W_R(x) = \begin{cases} \frac{3}{4\pi R^3}, & 0 \le x < R \\ 0, & x \ge R \end{cases} \tag{159}$$

With FFT, we get top-hat window function in Fourier space:

$$W_R(k) = \frac{3}{(kR)^3}[\sin(kR) - (kR)\cos(kR)] \tag{160}$$

Correlation function in Figure 19

$$\begin{aligned} \sigma(M,z) &= \frac{D_+(z)}{D_+(z=0)}\sigma_W[R_L(M)] \\ \sigma_W^2(R) &= \frac{1}{2\pi^2}\int_0^\infty dk k^2 P_L(k)|W_R(k)|^2 \end{aligned} \tag{161}$$

And σ_8 :

$$\sigma_8 = \sigma_W(R = 8Mpc/h) \tag{162}$$

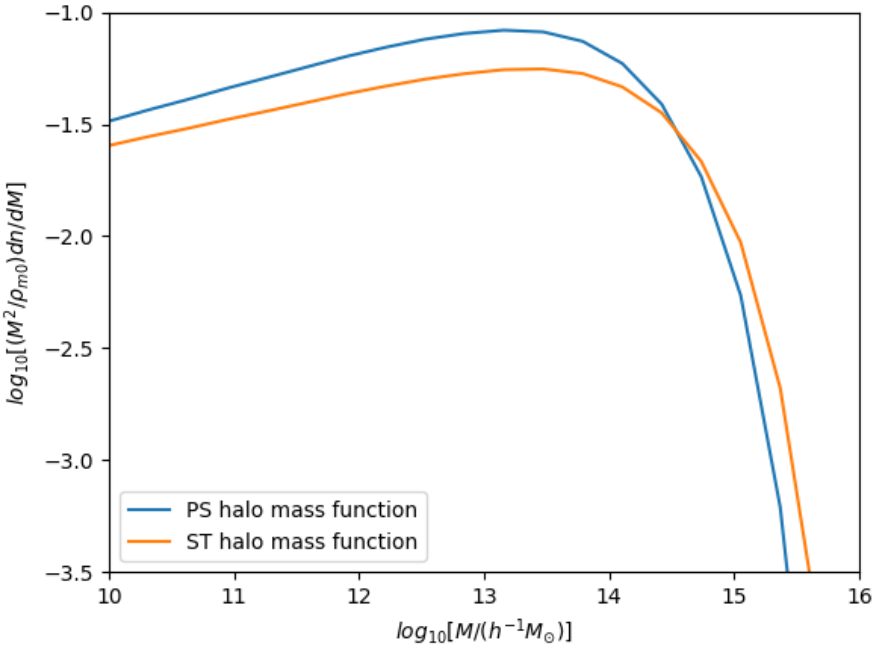


Figure 20: Halo Mass Function

Press-Schechter (PS) halo mass function:

$$\begin{aligned} \frac{dn(M,z)}{d\ln M} &= \frac{\rho_{\mathrm{m0}}}{M} f_{\mathrm{PS}}\left(\nu = \frac{\delta_{\mathrm{cr}}}{\sigma(M,z)}\right)\left|\frac{d\ln\sigma(M,z)}{d\ln M}\right| \\ f_{\mathrm{PS}}(\nu) &= \sqrt{\frac{2}{\pi}}\nu e^{-\nu^2/2}, \quad \delta_{cr} = 1.686 \end{aligned} \tag{163}$$

Sheth-Tormen (ST) halo mass function:

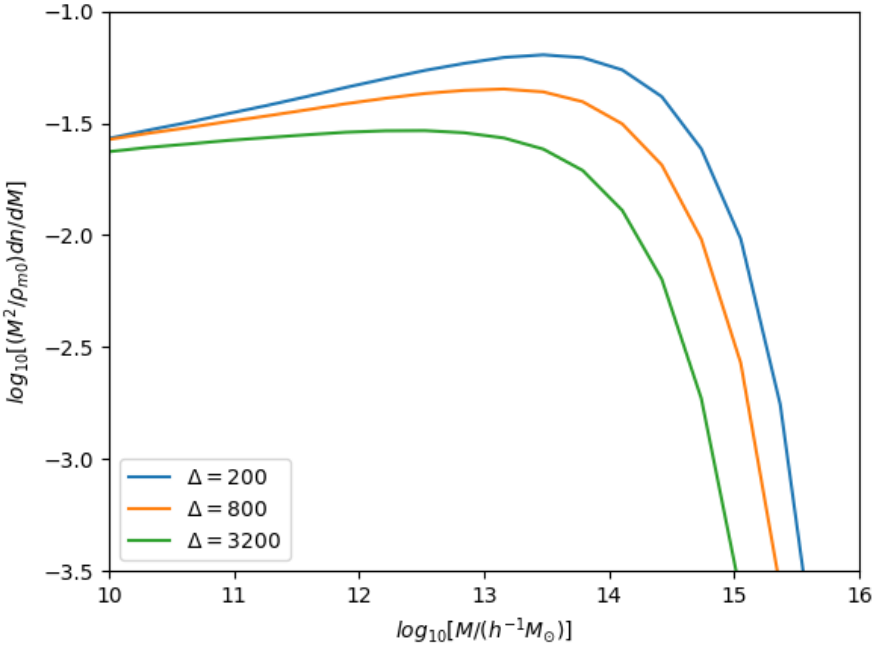


Figure 21: Tinker2008 Halo Mass Function

$$\begin{aligned}
\frac{dn(M,z)}{d\ln M} &= 0.322(1+\frac{1}{\nu'^{0.6}})\frac{\rho_{\rm m0}}{M}f_{\rm PS}\left(\nu'\right)\left|\frac{d\ln\sigma(M,z)}{d\ln M}\right| \\
f_{\rm PS}(\nu') &= \sqrt{\frac{2}{\pi}}\nu' e^{-\nu'^2/2}, \quad \delta_{cr} = 1.686 \\
\nu' &= \sqrt{0.707}\nu = \sqrt{0.707}\frac{\delta_{cr}}{\sigma(M,z)}
\end{aligned} \tag{164}$$

PS and ST halo mass function in Figure 20. Numerical halo mass function by Tinker2008 in Figure 21.

Number density of halos whose mass larger than M, in Figure 22

$$N(>M_1,z) = \int_{M_1}^{\infty} dn(M,z) = \int_{M_1}^{\infty}\frac{dn(M,z)}{d\ln M}d\ln M \tag{165}$$

17.3 Cluster of Halos

For PS halo mass function, halo - dark matter bias with normalization $P_{2h}(k) = P_L(k)$ for $k = 0.01h/Mpc$:

$$\begin{aligned}
\delta^{(1)}_{h,l}(\mathbf{x},t) &= b_1(M,z)\delta^{(1)}_l(\mathbf{x},t) \\
b_1^{PS}(M,z) &= 8.81\cdot\frac{\nu^2-1}{\delta_{cr}} \\
\nu &= \delta_{cr}/\sigma(M,z)
\end{aligned} \tag{166}$$

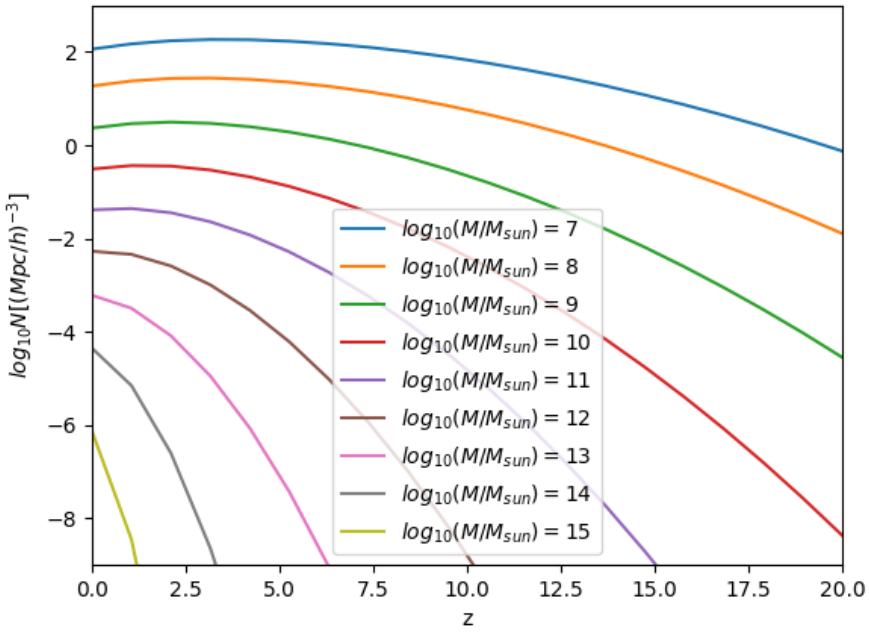


Figure 22: $N(>M, z)$

17.4 Power Spectrum in Halo Model

In observation, we get dark matter perturbation $\delta_{\rm m}^{\rm HM}$ from halo density profile $\rho_h(\mathbf{x}', M)$, halo mass function $dn/d\ln M$ and cluster of halos $\delta_h(\mathbf{x}', M)$.

Dark matter perturbation in halo model:

$$\delta_{\rm m}^{\rm HM}(x) = \int d\ln M \frac{1}{\rho_{\rm m0}} \frac{dn}{d\ln M} \int d^3x' \delta_h(\mathbf{x}', M) \rho_h(|\mathbf{x} - \mathbf{x}'|, M) \tag{167}$$

Power spectrum of halo model (NFW density profile + PS halo mass function + linear halo bias) in Figure 23

$$\begin{aligned} P^{\rm HM}(k, a) &= P_{\rm 2h}(k, a) + P_{\rm 1h}(k, a), \\ P_{\rm 2h}(k, a) &= \left[\frac{1}{\rho_{\rm m0}} \int d\ln M \frac{dn}{d\ln M} b_1(M, a) \rho_h(k, M)\right]^2 P_{\rm L}(k, a), \\ P_{\rm 1h}(k, a) &= \frac{1}{\rho_{\rm m0}^2} \int d\ln M \frac{dn}{d\ln M} [\rho_h(k, M)]^2. \end{aligned} \tag{168}$$

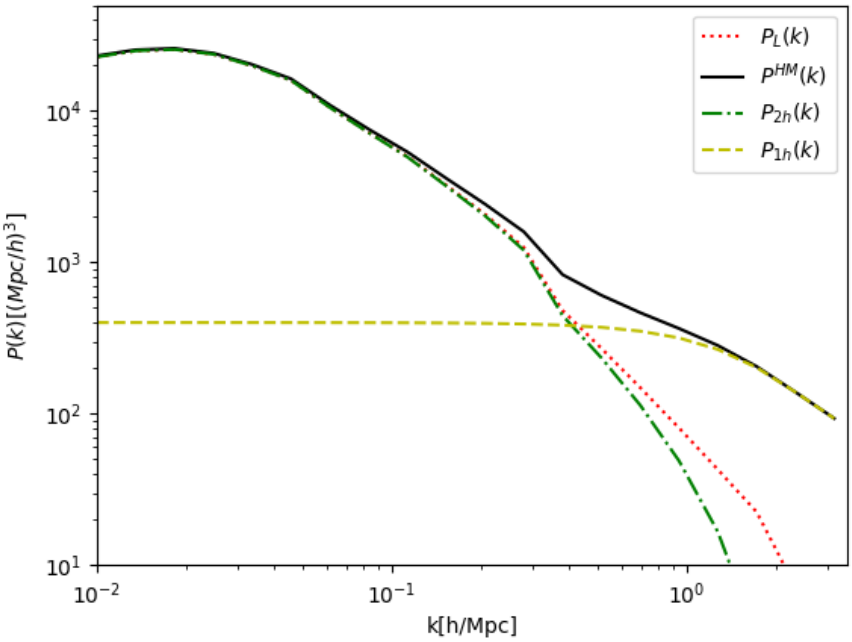


Figure 23: $P^{\mathrm{HM}}(k, a = 1)$

18 Galaxy Clusters

18.1 Mass-Temperature Relation

Kinetic energy:

$$E_k = \frac{1}{2} M \langle v^2 \rangle \tag{169}$$

Potential energy:

$$E_p = -\frac{3}{5} \frac{GM^2}{R_{vir}} \tag{170}$$

Halo in virial equilibrium:

$$E_k = -\frac{1}{2} E_p \tag{171}$$

Ideal gas in thermodynamic equilibrium:

$$\frac{1}{2} m_p \langle v^2 \rangle = \frac{3}{2} kT \tag{172}$$

Mass-Temperature relation of halo:

$$M = 1.38 \times 10^{14} h^{-1} M_\odot \left(\frac{T}{\mathrm{keV}} \right)^{3/2} \left(\frac{\Delta_{\mathrm{vir}}}{200} \right)^{-1/2} \tag{173}$$

Usually we set overdensity of halo $\triangle_{vir} = 200$

18.2 Power Spectrum with top-hat window function

Power spectrum of galaxy cluster / halo whose mass is M and density perturbation beyond δ_{cr} with top-hat window function at $z = 0$:

$$\begin{aligned} P_{\mathrm{g,thr}}(k, M) &= (b_1^{\mathrm{thr}})^2 P_{\mathrm{L}}(k) W_R^2(k) \\ &+ \frac{1}{2}(b_2^{\mathrm{thr}})^2 \int \frac{d^3p}{(2\pi)^3} P_{\mathrm{L}}(p) W_R^2(p) P_{\mathrm{L}}(|\mathbf{k}-\mathbf{p}|) W_R^2(|\mathbf{k}-\mathbf{p}|) \\ &+ \cdots \end{aligned} \tag{174}$$

Where:

$$\begin{aligned} b_1^{\mathrm{thr}} &= \sqrt{\frac{2}{\pi}} \frac{e^{-\nu^2/2}}{\mathrm{erfc}[\nu/\sqrt{2}]\sigma(M)} \overset{\nu \gg 1}{\approx} \frac{\nu}{\sigma(M)} \\ b_2^{\mathrm{thr}} &= \sqrt{\frac{2}{\pi}} \frac{e^{-\nu^2/2} \cdot \nu}{\mathrm{erfc}[\nu/\sqrt{2}]\sigma^2(M)} \overset{\nu \gg 1}{\approx} \frac{\nu^2}{\sigma^2(M)} \\ \nu &= \delta_{cr}/\sigma(M) \\ R &= R_L(M) \end{aligned} \tag{175}$$

19 Gravitational Lensing

19.1 Angle Deflection

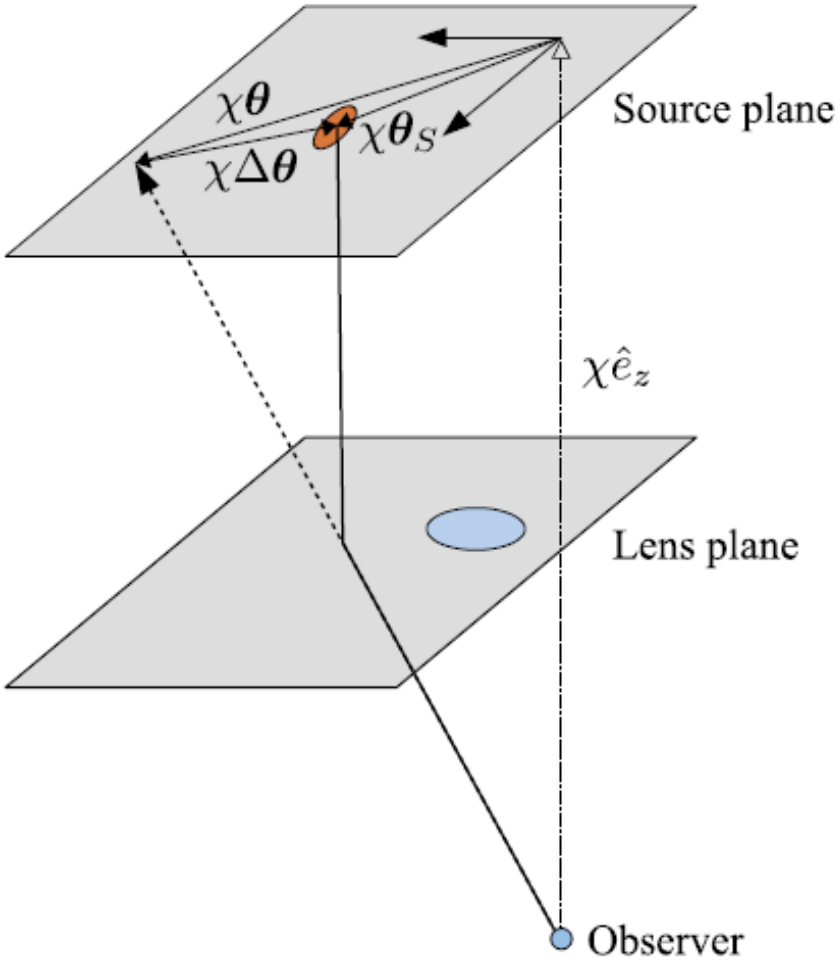


Figure 24: Gravitational Lensing

Gravitational lensing in Figure 24
Photon geodesics:

$$\frac{d\hat{p}}{d\chi} = -2(\delta^{ik} - \hat{p}^i\hat{p}^k)\Phi_{,k} \tag{176}$$

Lensing potential:

$$\phi_L(\theta) = 2\int_0^\chi \frac{d\chi'}{\chi'}\Phi(\theta,\chi')(1-\frac{\chi'}{\chi}) \tag{177}$$

Deflection angle:

$$\Delta\theta^i = \theta_s^i - \theta^i = \frac{\partial}{\partial\theta^i}\phi_L(\theta) \tag{178}$$

Distortion tensor:

$$-\Psi_{ij} = -\frac{\partial^2}{\partial\theta^i\partial\theta^j}\phi_L(\theta) = \kappa\delta_{ij} + \begin{pmatrix}\gamma_1 & \gamma_2\\ \gamma_2 & -\gamma_1\end{pmatrix} \tag{179}$$

Convergence:κ. Shear:γ_1, γ_2.
Angle deflection matrix:

$$A_{ij} = \frac{\partial \theta_s^i}{\partial \theta^i} = \delta_{ij} + \Psi_{ij} \tag{180}$$

19.2 Lensing of Point Mass

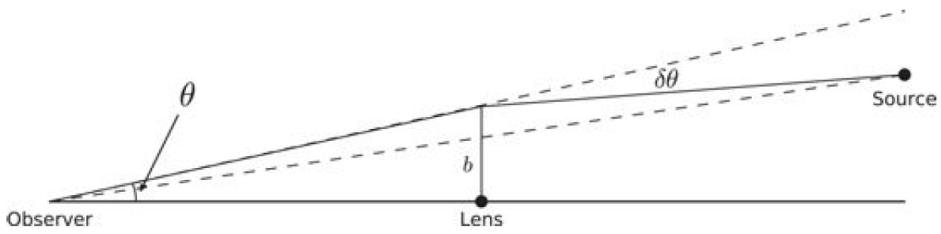


Figure 25: Point mass lensing to a very distant source

Deflection angle of lensing:

$$\Delta\theta^i = \theta_s^i - \theta^i = \frac{\partial}{\partial \theta^i}\phi_L(\theta) = 2\int_0^\chi \frac{d\chi'}{\chi'}\Phi_{,i}(\theta, \chi')(1 - \frac{\chi'}{\chi}) \tag{181}$$

Point mass lensing to a very distant source, in Figure 25 from "Gravitational Lensing" of Dodelson. [1]
For point mass at $(0, \chi_{lens})$,from Newton's Law of Gravity,

$$\Phi(y = \chi\theta, \chi) = \frac{GM}{\sqrt{y^2 + (\chi - \chi_{lens})^2}} \tag{182}$$

Source is very distant, $\chi_S \to \infty$, and $b \ll \chi_{lens}$,

$$\Delta\theta^i = 2\int_0^\infty \frac{d\chi'}{\chi'}\frac{\partial\Phi}{\partial\theta^i}\bigg|_{\chi\theta=b} = 2\int_0^\infty d\chi'\frac{\partial\Phi}{\partial y}\bigg|_{y=b} \tag{183}$$

We get:

$$\Delta\theta = -\frac{4GM}{b} \tag{184}$$

This result is verified in Eddington's eclipse expedition.

19.3 Galaxy Shape

Galaxy shape tensor:

$$q_{ij} = \frac{1}{2}q\delta_{ij} + \frac{1}{2}q\begin{pmatrix}\epsilon_1 & \epsilon_2 \\ \epsilon_2 & -\epsilon_1\end{pmatrix} \tag{185}$$

$$q_{ij} = \langle\theta_i\theta_j\rangle_{I_{obs}} = \frac{1}{F}\int d^2\theta I_{obs}(\theta)\theta_i\theta_j \tag{186}$$

Total flux:

$$F = \int d^2\theta I_{obs}(\theta) \tag{187}$$

Polarization tensor:

$$I_{ij} = I\delta_{ij} + \begin{pmatrix} Q & U \\ U & -Q \end{pmatrix} \tag{188}$$

From conservation of intensity:

$$I_{obs}(\theta) = I_{true}(\theta_s) \tag{189}$$

We get the relation of observed ellipticity of galaxy with true ellipticity , when $\kappa, \gamma_1, \gamma_2 \ll 1$:

$$\begin{aligned} \epsilon_1 &= (1 - 2[\epsilon_1^{true}\gamma_1 + \epsilon_2^{true}\gamma_2])\epsilon_1^{true} + 2\gamma_1 \\ \epsilon_2 &= (1 - 2[\epsilon_1^{true}\gamma_1 + \epsilon_2^{true}\gamma_2])\epsilon_2^{true} + 2\gamma_2 \end{aligned} \tag{190}$$

If true ellipticity is small $\epsilon_1^{true}, \epsilon_2^{true} \ll 1$:

$$\begin{aligned} \epsilon_1 &= \epsilon_1^{true} + 2\gamma_1 \\ \epsilon_2 &= \epsilon_2^{true} + 2\gamma_2 \end{aligned} \tag{191}$$

Intrinsic alignment:

$$q_{ij}^{IA} = -c_1(\eta)\chi^2\Phi_{,ij}(\mathbf{x}, \eta) \tag{192}$$

19.4 Lensing/Shear Angular Power Spectrum

In Fourier space:

$$\mathbf{l} = l(\cos\theta_l, \sin\theta_l) \tag{193}$$

Change shear γ_1, γ_2 to rotation-invariant E,B:

$$\begin{pmatrix} \gamma_1(\mathbf{l}) \\ \gamma_2(\mathbf{l}) \end{pmatrix} = \begin{pmatrix} \cos(2\phi_l) & -\sin(2\phi_l) \\ \sin(2\phi_l) & \cos(2\phi_l) \end{pmatrix} \begin{pmatrix} E(\mathbf{l}) \\ B(\mathbf{l}) \end{pmatrix} \tag{194}$$

In Fourier space:

$$\begin{aligned} E(\mathbf{l}) &= (\frac{l^i l^j}{l^2} - \frac{1}{2}\delta_{ij})(-\Psi_{ij}) \\ B(\mathbf{l}) &= \epsilon_{ik}\frac{l^j l^k}{l^2}(-\Psi_{ij}) \end{aligned} \tag{195}$$

We get:

$$\begin{aligned} E(\mathbf{l}) = \kappa(\mathbf{l}) &= \frac{1}{2}l^2\phi_L(\mathbf{l}) \\ B(\mathbf{l}) &= 0 \end{aligned} \tag{196}$$

Lensing/Shear angular power spectrum of galaxies in Figure 26

$$
\begin{aligned}
C_{EE}(l) = C_{\kappa\kappa}(l) &= \frac{1}{4} l^4 C_{\phi_L \phi_L}(l) \\
&= (\frac{3}{2}\Omega_{m0} H_0^2)^2 \int_0^\infty d\chi a^{-2}(\chi) g_L^2(\chi) P(k = \frac{l}{\chi}, \eta = \eta_0 - \chi) \\
g_L(\chi) &= \int_\chi^\infty d\chi' (1 - \frac{\chi}{\chi'}) W(\chi') \\
W(\chi) &= \frac{1}{N_g} \frac{dN_g}{d\chi}
\end{aligned}
\tag{197}
$$

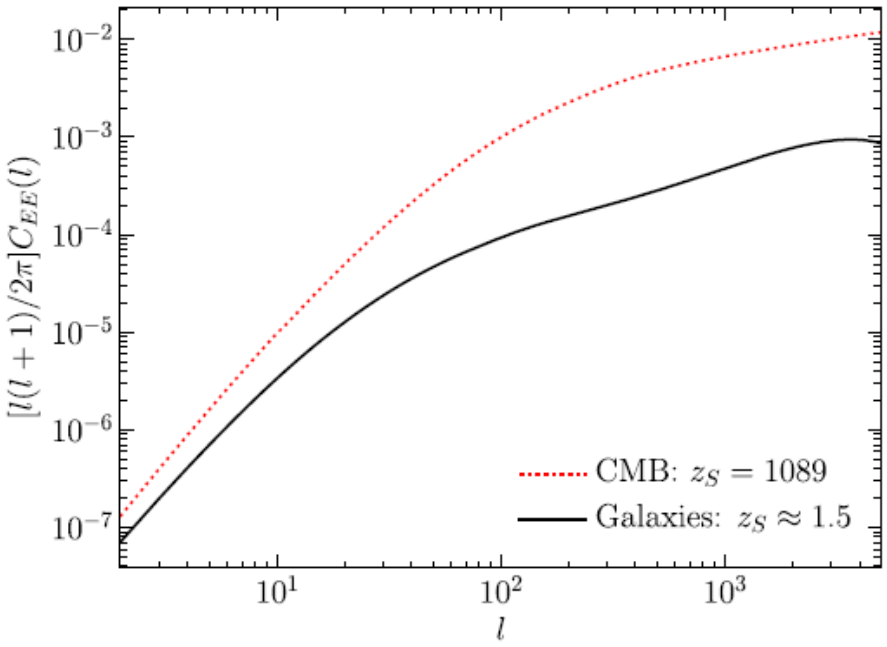


Figure 26: E-mode shear power spectrum

Change shear γ_1, γ_2 to rotation-invariant tangent shear γ_t and cross shear $\gamma_\times$.

Shear correlation functions:

$$
\begin{aligned}
\xi_{+,-}(\theta) &= \langle \gamma_t(\mathbf{0}) \gamma_t(\theta) \rangle \pm \langle \gamma_\times(\mathbf{0}) \gamma_\times(\theta) \rangle \\
&= \langle \gamma_1(\mathbf{0}) \gamma_1(\theta) \rangle \pm \langle \gamma_2(\mathbf{0}) \gamma_2(\theta) \rangle \\
&= \int \frac{l dl}{2\pi} J_{0,4}(l\theta) C_{EE}(l)
\end{aligned}
\tag{198}
$$

19.5 Shear Cross Angular Power Spectrum

Projected galaxy density:

$$
\Delta_g(\theta) = \int_0^\infty d\chi W_g(\chi) \delta_g(\theta, \eta = \eta_0 - \chi) \tag{199}
$$

On large scale, galaxy-matter cross power spectrum satisfies:

$$
P_{gm}(M, k, z) = b_1(M, z) P(k, z), \quad (k \ll 0.1h/Mpc) \tag{200}
$$

Shear cross angular power spectrum:

$$C_{gE}(l) = (\frac{3}{2}\Omega_{m0}H_0^2)\int_0^\infty \frac{d\chi}{\chi}W_g(\chi)a^{-1}(\chi)g_L(\chi)P_{gm}(k=\frac{l}{\chi},\eta=\eta_0-\chi) \tag{201}$$

Shear cross correlation functions

$$\begin{aligned}\xi_{g,t}(\theta) = \langle\Delta_g(\mathbf{0})\gamma_t(\theta)\rangle = -\int \frac{ldl}{2\pi}J_2(l\theta)C_{gE}(l)\\ \xi_{g,\times} = \langle\Delta_g(\mathbf{0})\gamma_\times(\theta)\rangle = 0\end{aligned} \tag{202}$$

19.6 CMB Lensing

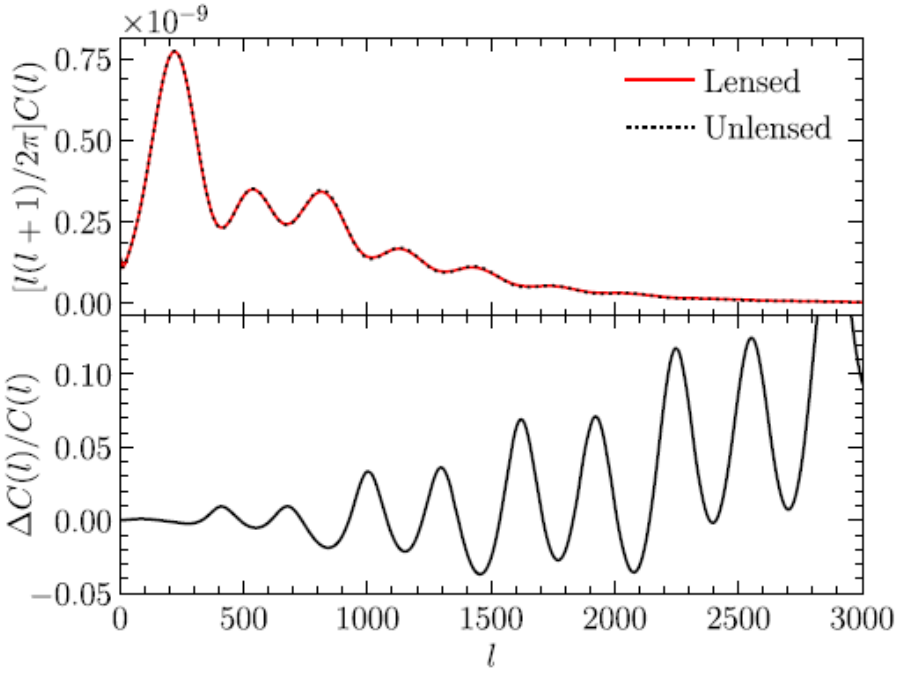


Figure 27: Effect of lensing on the CMB temperature power spectrum

From angle deflection due to gravitational lensing:

$$T_{obs}(\theta) = T(\theta_s) \tag{203}$$

Lensing angular power spectrum of CMB:

$$C_{\phi_L\phi_L}(l) = \frac{4}{l^4}(\frac{3}{2}\Omega_{m0}H_0^2)^2\int_0^{\chi_*} d\chi a^{-2}(\chi)(1-\frac{\chi}{\chi_*})^2 P(k=\frac{l}{\chi},\eta=\eta_0-\chi) \tag{204}$$

Angular power spectrum of CMB lensing in Figure 27.

$$\begin{aligned}C^{obs}(l) &= (1-l^2/l_{lens}^2)C(l) + C^{(22)}(l)\\ l_{lens}^{-2} &= \frac{1}{4\pi}\int_0^\infty dl_1 l_1^3 C_{\phi_L\phi_L}(l_1)\\ C^{(22)}(l) &= \frac{1}{(2\pi)^2}\int_0^{2\pi}d\theta\int_0^\infty dl_1 l_1(ll_1\cos\theta - l_1^2)^2 C_{\phi_L\phi_L}(l_1)C(\sqrt{l^2+l_1^2-2ll_1\cos\theta})\end{aligned} \tag{205}$$

20 Analysis

20.1 Overview: From Raw Data to Parameter Constraints

The process in Figure 28

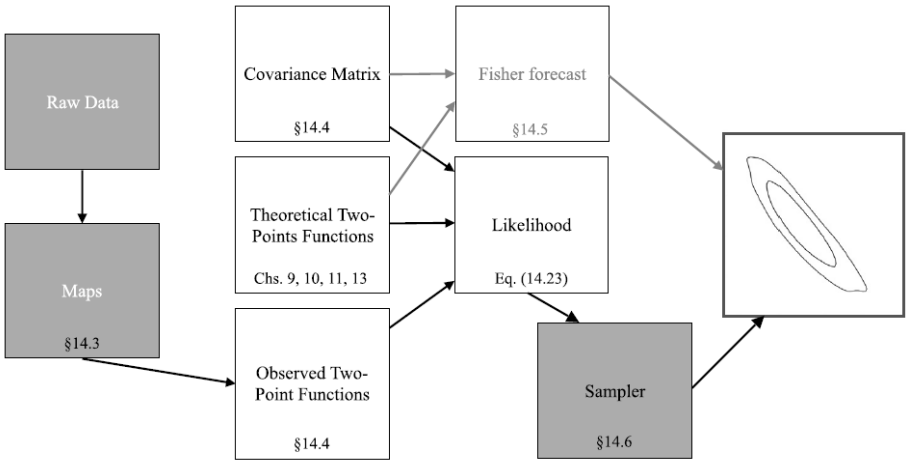


Figure 28: Overview: From Raw Data to Parameter Constraints

20.2 The Likelihood Function

Bayes' Theorem:

$$P(w, \sigma_w|\{d_i\}) = \frac{P(\{d_i\}|w, \sigma_w)P(w, \sigma_w)}{P(\{d_i\})} \tag{206}$$

$$P(w, \sigma_w|\{d_i\}) \propto \mathcal{L}\left(\{d_i\}_{i=1}^m \middle| w, \sigma_w\right) P_{\rm prior}(w, \sigma_w) \tag{207}$$

Marginalized posterior:

$$P(w|\{d_i\}) = \int_0^\infty d\sigma_w P(w, \sigma_w|\{d_i\}) \tag{208}$$

The likelihood function for Gaussian Distribution:

$$\mathcal{L}\left(\{d_i\}_{i=1}^m \middle| w, \sigma_w\right) = P\left(\{d_i\}_{i=1}^m \middle| w, \sigma_w\right) = \frac{1}{(2\pi\sigma_w^2)^{m/2}} \exp\left\{-\frac{\sum_{i=1}^m (d_i - w)^2}{2\sigma_w^2}\right\} \tag{209}$$

To maximum the likelihood function:

$$\frac{\partial \mathcal{L}}{\partial w} = 0, \quad \frac{\partial \mathcal{L}}{\partial \sigma_w^2} = 0 \tag{210}$$

We get:

$$\hat{w} = \frac{1}{m}\sum_{i=1}^m d_i, \quad \widehat{\sigma_w^2} = \frac{1}{m}\sum_{i=1}^m (d_i - w)^2 \tag{211}$$

Expand the likelihood function:

$$\begin{aligned}\ln \mathcal{L}(w) &= \ln \mathcal{L}(\hat{w}) + \left.\frac{\partial \ln \mathcal{L}}{\partial w}\right|_{w=\hat{w}} (w-\hat{w}) + \frac{1}{2}\left.\frac{\partial^2 \ln \mathcal{L}}{\partial w^2}\right|_{w=\hat{w}} (w-\hat{w})^2 \\ &= \ln \mathcal{L}(\hat{w}) - \frac{m}{2\sigma_w^2}(w-\hat{w})^2,\end{aligned} \tag{212}$$

We get:

$$\mathrm{Var}[\hat{w}] = \left\langle (\hat{w}-w)^2 \right\rangle = -\big[\frac{\partial^2 \ln \mathcal{L}}{\partial w^2}\big]^{-1} = \frac{\sigma_w^2}{m} \tag{213}$$

$$\mathrm{Var}(\sigma_w^2) \equiv \left\langle \left(\widehat{\sigma_w^2} - \sigma_w^2\right)^2 \right\rangle = \frac{2}{m}\sigma_w^4 \tag{214}$$

20.3 Example: CMB Power Spectrum

For a CMB experiment, we infer parameters $\{\lambda_\alpha\}$ from observed temperature anisotropy $\hat{C}(l)$

CMB temperature anisotropy we observe:

$$\Delta = \frac{T - T_0}{T_0} \tag{215}$$

Considering beam pattern $B(\hat{\mathbf{n}}, \hat{\mathbf{n}}')$ and noise $\eta(\hat{\mathbf{n}})$

$$\Delta(\hat{\mathbf{n}}) = \int d\Omega' \Theta(\hat{\mathbf{n}}') B(\hat{\mathbf{n}}, \hat{\mathbf{n}}') + \eta(\hat{\mathbf{n}}) \tag{216}$$

In spherical harmonics:

$$a_{lm}^{\mathrm{obs}} = \int d\Omega Y_{lm}^*(\hat{\mathbf{n}}) \Delta(\hat{\mathbf{n}}) = \sum_{l'm'} B_{lm,l'm'} a_{l'm'} + \eta_{lm} \tag{217}$$

If the beam is isotropic $B(\hat{\mathbf{n}}, \hat{\mathbf{n}'}) = B(\hat{\mathbf{n}} \cdot \hat{\mathbf{n}'})$:

$$a_{lm}^{obs} = a_{lm} B_l + \eta_{lm} \tag{218}$$

We assume noise has mean zero with power spectrum:

$$\langle \eta_{lm} \eta_{l'm'}^* \rangle = N(l) \delta_{ll'} \delta_{mm'} \tag{219}$$

From:

$$a_{lm} | C(l) \sim N(\mu = 0, \sigma^2 = C(l)) \tag{220}$$

We know:

$$a_{lm}^{obs} | C(l) \sim N(\mu = 0, \sigma^2 = C(l) B_l^2 + N(l)) \tag{221}$$

The likelihood function:

$$\mathcal{L} \equiv P\left(\{a_{lm}^{\mathrm{obs}}\} | C(l)\right) = \left(2\pi \left[C(l)B_l^2 + N(l)\right]\right)^{-(2l+1)/2} \exp\left\{ -\frac{1}{2} \sum_{m=-l}^{l} \frac{\left|a_{lm}^{\mathrm{obs}}\right|^2}{C(l)B_l^2 + N(l)} \right\}. \tag{222}$$

To maximize $\mathcal{L}$, $d\ln\mathcal{L}/dC(l) = 0$, we get:

$$\hat{C}(l) = B_l^{-2}\left(\frac{1}{2l+1}\sum_{m=-l}^{l}\left|a_{lm}^{\rm obs}\right|^2 - N(l)\right) \tag{223}$$

$$\mathrm{Var}\left[\hat{C}(l)\right] = \left\langle\hat{C}(l)^2\right\rangle - C(l)^2 = \frac{2}{2l+1}[C(l)+N(l)B_l^{-2}]^2 \tag{224}$$

Different l modes are independent:

$$\mathrm{Cov}_{ll'} = \mathrm{Var}\left[\hat{C}(l)\right]\delta_{ll'} = \frac{2}{2l+1}[C(l)+N(l)B_l^{-2}]^2\delta_{ll'} \tag{225}$$

For a Gaussian beam pattern:

$$B_l = e^{-l^2\theta_{beam}^2/2} \tag{226}$$

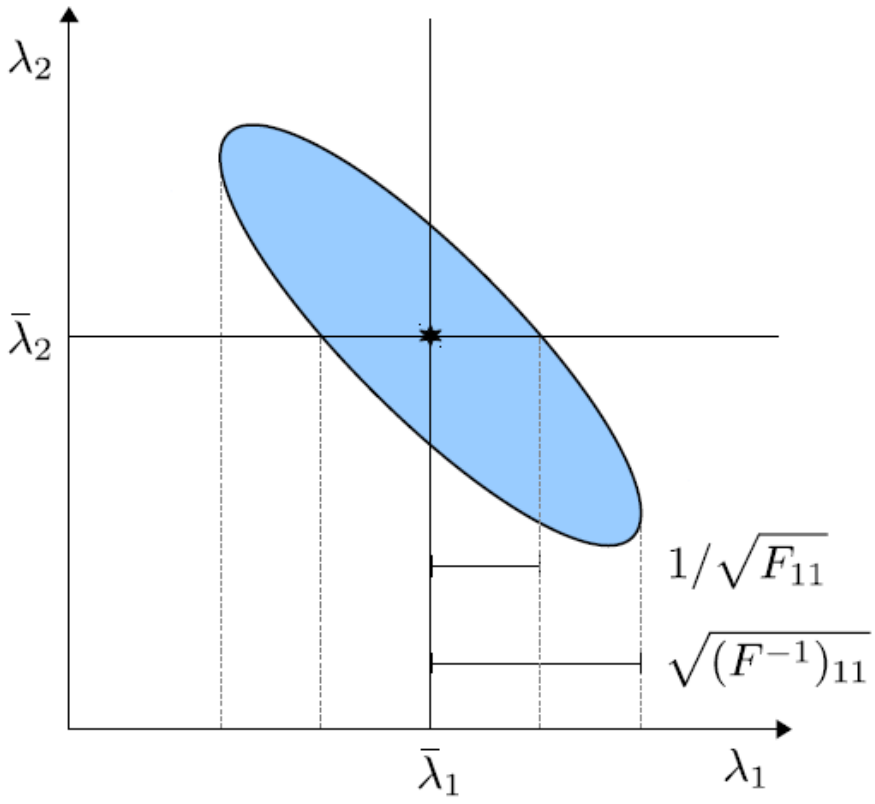


Figure 29: Error Ellipse from Fisher Matrix

The likelihood function for CMB anisotropies:

$$\ln\mathcal{L}(\{\lambda_\gamma\}) = -\frac{1}{2}\sum_{ll'}\left(\hat{C}(l) - C^{\rm theory}(l,\{\lambda_\gamma\})\right)\left(\mathrm{Cov}^{-1}\right)_{ll'}\left(\hat{C}(l') - C^{\rm theory}(l',\{\lambda_\gamma\})\right) \tag{227}$$

So,

$$\ln \mathcal{L}(\{\lambda_\gamma\}) = -\frac{\chi^2(\{\lambda_\gamma\})}{2} \tag{228}$$

$$\chi^2(\{\lambda_\gamma\}) = \sum_l \frac{[\hat{C}(l) - C^{\rm theory}(l, \{\lambda_\gamma\})]^2}{{\rm Var}\Big[\hat{C}(l)\Big]} \tag{229}$$

To maximize χ^2, $d\chi^2/d\lambda_\gamma = 0$, we expand χ^2 to 2nd order:

$$\chi^2(\{\lambda_\gamma\}) = \chi^2(\{\bar{\lambda}_\gamma\}) + \sum_{\alpha\beta} F_{\alpha\beta}(\lambda_\alpha - \bar{\lambda}_\alpha)(\lambda_\beta - \bar{\lambda}_\beta) \tag{230}$$

The Fisher Matrix is defined as:

$$\begin{aligned} F_{\alpha\beta} &= \frac{1}{2}\left\langle \frac{\partial^2 \chi^2}{\partial\lambda_\alpha \partial\lambda_\beta} \right\rangle \Bigg|_{\{\lambda_\gamma\}=\{\bar{\lambda}_\gamma\}} \\ &\equiv -\left\langle \frac{\partial^2 \ln \mathcal{L}}{\partial\lambda_\alpha \partial\lambda_\beta} \right\rangle \Bigg|_{\{\lambda_\gamma\}=\{\bar{\lambda}_\gamma\}} \end{aligned} \tag{231}$$

$$(F^{-1})_{\alpha\beta} = \langle (\lambda_\alpha - \bar{\lambda}_\alpha)(\lambda_\beta - \bar{\lambda}_\beta) \rangle \tag{232}$$

For CMB anisotropies:

$$F_{\alpha\beta} = \sum_l \frac{1}{{\rm Var}\Big[\hat{C}(l)\Big]} \frac{\partial C^{\rm theory}(l, \{\bar{\lambda}_\gamma\})}{\partial\bar{\lambda}_\alpha} \frac{\partial C^{\rm theory}(l, \{\bar{\lambda}_\gamma\})}{\partial\bar{\lambda}_\beta} \tag{233}$$

From Fisher Matrix, we can draw error ellipse for parameters in Figure 29.
Joint posterior for λ_1 and λ_2 :

$$P(\lambda_1, \lambda_2) \propto \exp\left\{-\frac{1}{2}\lambda_\alpha F_{\alpha\beta}\lambda_\beta\right\} \tag{234}$$

So,if λ_2 is known, error of λ_1 is : $\sqrt{\mathbf{Var}(\lambda_1)} = 1/\sqrt{F_{11}}$
Varying λ_2, posterior for λ_1 :

$$\begin{aligned} P(\lambda_1) &= \int d\lambda_2 P(\lambda_1, \lambda_2) \\ &\propto \exp\left\{-\frac{\lambda_1^2}{2}\left(\frac{F_{11}F_{22} - F_{12}F_{21}}{F_{22}}\right)\right\} = \exp\left\{-\frac{\lambda_1^2}{2(F^{-1})_{11}}\right\} \end{aligned} \tag{235}$$

So,if λ_2 is unknown, error of λ_1 is : $\sqrt{\mathbf{Var}(\lambda_1)} = \sqrt{(F^{-1})_{11}}$

20.4 Example: Galaxy Power Spectrum

1. Galaxy Covariance Matrix
The number perturbation of galaxy in pixel i:

$$\begin{aligned} \Delta_i &= \int d^3x \psi_i(\mathbf{x})[\frac{n(\mathbf{x}) - \bar{n}(\mathbf{x})}{\bar{n}(\mathbf{x})}] \\ \psi_i(\mathbf{x}) &= \frac{1}{V} e^{i\mathbf{k}_i\cdot\mathbf{x}} \end{aligned} \tag{236}$$

Covariance Matrix:

$$\langle \Delta_i \Delta_j^* \rangle = (C_S)_{ij} + (C_N)_{ij} \tag{237}$$

Shot noise from Poisson Distribution particles $n_{shot} \sim Pois(\bar{n})$:

$$P(n) = \frac{(\bar{n})^n e^{-\bar{n}}}{n!}, \quad n \in \{0, 1, 2, \dots\} \tag{238}$$

Covariance Matrix:

$$\begin{gathered}(C_N)_{ij} = \langle \Delta_i \Delta_j^* \rangle \Big|_{PoissonNoise} = \delta_{ij} \frac{1}{N} \\ N = nV\end{gathered} \tag{239}$$

$$\begin{gathered}(C_S)_{ij} = \langle \Delta_i \Delta_j^* \rangle \Big|_{NoNoise} = \int_0^\infty \frac{dk}{k} [\frac{k^3 P(k)}{2\pi^2}] W_{ij}(k) \\ W_{ij}(k) = \delta_{ij} \frac{9}{2kk_iR^2} \int_{|k-k_i|}^{(k+k_i)R} \frac{dy}{y} j_1^2(y)\end{gathered} \tag{240}$$

At $k_iR \gg 1$:

$$(C_S)_{ii} = \frac{1}{V} P(k_i) \tag{241}$$

$$\frac{(C_S)_{ii}}{(C_N)_{ii}} = P(k_i)\bar{n} \tag{242}$$

2.Galaxy Power Spectrum

In x space, $V = L^3$.

In k space, we have: number of grids in an axis K_{grid}, edge length of a grid $k_F = 2\pi/L$, total length in half axis $k_{N_y} = K_{grid}k_F/2$.

FFT of perturbation:

$$\begin{gathered}\delta_g(\mathbf{k}) = L^{3/2} \sum_i^{K_{grid}^3} \delta_g(\mathbf{x}_i) e^{-i\mathbf{k}\cdot\mathbf{x}_i} \\ \delta_g(\mathbf{x}) = \frac{1}{K_{grid}^3 L^{3/2}} \sum_{\mathbf{k}}^{k_{N_y}} \delta_g(\mathbf{k}_i) e^{i\mathbf{k}_i\cdot\mathbf{x}}\end{gathered} \tag{243}$$

The number of modes that each volume α contains, for half of Fourier modes ($k_\alpha - \Delta k/2 \leq |\mathbf{k}| \leq k_\alpha + \Delta k/2$):

$$m_{k,\alpha} = \frac{1}{2} \frac{4\pi k_\alpha^2 \Delta k}{k_F^3} = \frac{1}{4\pi^2} V k_\alpha^2 \Delta k \tag{244}$$

Omit beam pattern effects on galaxy $B_l = 1$, we estimate galaxy power spectrum as:

$$\hat{P}_g(k_\alpha) = \frac{1}{m_{k,\alpha}} \sum_{\mathbf{k}}^{||\mathbf{k}|-k_\alpha|<\Delta k/2} |\delta_g(\mathbf{k})|^2 - P_N \tag{245}$$

δ_g is Gaussian, we get standard deviation:

$$\sqrt{\mathbf{Var}[\hat{P}_g(k_\alpha)]} = \sqrt{\langle \hat{P}_g(k_\alpha)^2 \rangle - \langle \hat{P}_g(k_\alpha) \rangle^2} = \sqrt{\frac{2}{m_{k,\alpha}}} [P_g(k_\alpha) + P_N] \tag{246}$$

Covariance matrix:

$$\mathbf{Cov}_{\alpha\beta} = \sqrt{\langle \hat{P}_g(k_\alpha) \hat{P}_g(k_\beta) \rangle - \langle \hat{P}_g(k_\alpha) \rangle \langle \hat{P}_g(k_\beta) \rangle} = \frac{2}{m_{k,\alpha}} [P_g(k_\alpha) + P_N]^2 \delta_{\alpha\beta} \tag{247}$$

For discrete Fourier-space field:

$$\langle \delta_g(\mathbf{k}) \delta_g(\mathbf{k}') \rangle = \delta_{\mathbf{k},-\mathbf{k}'} [P_g(k) + P_N] \tag{248}$$

$$\begin{aligned} \delta_g^{obs}(\mathbf{x}_i) &= \tilde{W}(\mathbf{x}_i) \delta_g(\mathbf{x}_i) \\ \delta_g^{obs}(\mathbf{k}_i) &= \sum_{\mathbf{k}_j} \tilde{W}(\mathbf{k}_j) \delta_g(\mathbf{k}_i - \mathbf{k}_j) \end{aligned} \tag{249}$$

$$\tilde{W}(\mathbf{x}_i) = \begin{cases} 1, & \mathbf{x}_i \, in \, surveyed \, volume \\ 0, & otherwise. \end{cases} \tag{250}$$

A Basic Math

A.1 Special Functions

Legendre:

$$\mathcal{P}_l(\hat{n}\cdot\hat{n}') = \frac{4\pi}{2l+1}\sum_{m=-l}^{l} Y_{lm}(\hat{n})Y^*_{lm}(\hat{n}') \tag{251}$$

$$\int_{-1}^{1}\frac{d\mu}{2}\mathcal{P}_l(\mu)\mathcal{P}_{l'}(\mu) = \frac{1}{2l+1}\delta_{ll'} \tag{252}$$

Spherical Harmonics:

$$\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]Y_{lm}(\theta,\phi) = -l(l+1)Y_{lm}(\theta,\phi) \tag{253}$$

$$\int d\Omega Y_{lm}(\hat{n})Y^*_{l'm'}(\hat{n}) = \delta_{ll'}\delta_{mm'} \tag{254}$$

Spherical Bessel Harmonics:

$$\frac{d^2j_l}{dx^2} + \frac{2}{x}\frac{dj_l}{dx} + \left[1 - \frac{l(l+1)}{x^2}\right]j_l = 0 \tag{255}$$

$$\int_0^\infty drr^2 j_l(kr)j_r(k'r) = \frac{\pi}{2k^2}\delta_D(k-k') \tag{256}$$

$$e^{i\mathbf{k}\cdot\mathbf{x}} = \sum_{l=0}^{\infty} i^l(2l+1)j_l(kx)\mathcal{P}_l(\mu = \hat{k}\cdot\hat{x}) \tag{257}$$

$$\frac{1}{2}\int_{-1}^{1} d\mu\mathcal{P}_l(\mu)e^{iz\mu} = \frac{j_l(z)}{(-i)^l} \tag{258}$$

Bessel Function:

$$J_n(x) = \frac{i^{-n}}{\pi}\int_0^\pi d\theta e^{ix\cos\theta}\cos(n\theta) \tag{259}$$

Cosine and sine integrals:

$$\begin{aligned}\mathrm{Ci}(x) &= -\int_x^\infty \frac{\cos z}{z}dz,\\ \mathrm{Si}(x) &= \int_0^x \frac{\sin z}{z}dz.\end{aligned} \tag{260}$$

A.2 Transformations

Fourier:

$$\begin{cases}\Theta(\mathbf{x}) = \int \dfrac{d^3\mathbf{k}}{(2\pi)^3}\Theta(\mathbf{k})e^{i\mathbf{k}\cdot\mathbf{x}}\\ \Theta(\mathbf{k}) = \int d^3\mathbf{x}\Theta(\mathbf{x})e^{-i\mathbf{k}\cdot\mathbf{x}}\end{cases} \tag{261}$$

Legendre:

$$\begin{cases} \Theta(k,\hat{k},\hat{p}) = \sum_l (-i)^l (2l+1)\mathcal{P}_l(\mu = \hat{k}\cdot\hat{p})\Theta_l(k) \\ \Theta_l(k) = \frac{1}{(-i)^l}\int_{-1}^{1}\frac{d\mu}{2}\mathcal{P}_l(\mu = \hat{k}\cdot\hat{p})\Theta(k,\hat{k},\hat{p}) \end{cases} \tag{262}$$

Harmonic in real space:

$$\begin{cases} \Theta(\mathbf{x},\hat{p}) = \sum_{l=1}^{\infty}\sum_{m=-l}^{l} a_{lm}(\mathbf{x})Y_{lm}(\hat{p}) \\ a_{lm}(\mathbf{x}) = \int d\Omega\Theta(\mathbf{x},\hat{p})Y^*_{lm}(\hat{p}) \end{cases} \tag{263}$$

Harmonic in k space:

$$\begin{cases} \Theta_l(\mathbf{k}) = \frac{1}{(-i)^l(2l+1)\mathcal{P}_l(\mu = \hat{k}\cdot\hat{p})}\sum_{m=-l}^{l} a_{lm}(\mathbf{k})Y_{lm}(\hat{p}) \\ a_{lm}(\mathbf{k}) = \int d^3\mathbf{x}a_{lm}(\mathbf{x})e^{-i\mathbf{k}\cdot\mathbf{x}} \end{cases} \tag{264}$$

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1 Fundamental Particle Properties

This section summarizes the key properties of all elementary particles in the Standard Model. Natural units $\hbar = c = 1$ are used; masses are given in GeV/c^2.

1.1 Quarks

Quarks are spin-1/2 fermions carrying color charge. They are confined into hadrons.

Quark	Flavor	Mass (GeV)	Q/e	I_3	B
u	up	~ 0.002	$+2/3$	$+1/2$	$1/3$
d	down	~ 0.005	$-1/3$	$-1/2$	$1/3$
s	strange	~ 0.095	$-1/3$	0	$1/3$
c	charm	~ 1.275	$+2/3$	0	$1/3$
b	bottom	~ 4.18	$-1/3$	0	$1/3$
t	top	~ 173.0	$+2/3$	0	$1/3$

Note: Quark masses are scheme-dependent and quoted at the $\overline{\text{MS}}$ scale; top quark mass is the pole mass.

1.2 Leptons

Leptons are spin-1/2 fermions that do not carry color charge. Neutrino masses are non-zero but extremely small.

Lepton	Mass (GeV)	Q/e	L	I_3 (L)
e	0.000511	-1	1	$-1/2$
μ	0.10566	-1	1	$-1/2$
τ	1.77686	-1	1	$-1/2$
ν_e	$< 10^{-9}$	0	1	$+1/2$
ν_μ	$< 10^{-9}$	0	1	$+1/2$
ν_τ	$< 10^{-9}$	0	1	$+1/2$

Note: Neutrino masses are not fixed by the Standard Model; the values shown are upper limits from oscillation experiments.

1.3 Gauge Bosons

Gauge bosons are spin-1 particles that mediate the fundamental forces.

Boson	Mass (GeV)	Q/e	Interaction	Spin
γ (photon)	0	0	Electromagnetic	1
g (gluon)	0	0	Strong	1
$W^\pm$	80.379	± 1	Weak (charged)	1
Z^0	91.1876	0	Weak (neutral)	1

1.4 Higgs Boson

The Higgs boson is a spin-0 scalar particle, responsible for spontaneous electroweak symmetry breaking and the generation of fermion masses.

Particle	Mass (GeV)	Q/e	I	J^P
H	125.10	0	0	0^+

1.5 Summary of Generations

The Standard Model contains three generations of fermions:

$$\begin{array}{ccc} \text{Generation I} & \text{Generation II} & \text{Generation III} \\ \begin{pmatrix} \nu_e \\ e \end{pmatrix} & \begin{pmatrix} \nu_\mu \\ \mu \end{pmatrix} & \begin{pmatrix} \nu_\tau \\ \tau \end{pmatrix} \\ \begin{pmatrix} u \\ d \end{pmatrix} & \begin{pmatrix} c \\ s \end{pmatrix} & \begin{pmatrix} t \\ b \end{pmatrix} \end{array} \tag{1}$$

2 Fundamental Quantum Numbers and the Quark Model

2.1 Gell-Mann–Nishijima Charge Formula

2.1.1 General form (all 6 quark flavors)

Relates electric charge, isospin, and additive flavor quantum numbers:

$$\frac{Q}{e} = I_3 + \frac{1}{2}Y \tag{2}$$

$$Y = B + S + C + \tilde{B} + T \tag{3}$$

$$\implies \frac{Q}{e} = \frac{1}{2}\left[B + \left(U + C + T + D + S + \tilde{B}\right)\right] \tag{4}$$

where $\tilde{B}$ denotes the bottom quantum number (distinguished from baryon number B). Note that for down-type quarks d, s, b, the corresponding flavor numbers $D, S, \tilde{B}$ are defined as $-(n_q - n_{\bar{q}})$, so that $I_3 = \frac{1}{2}(U + D)$, where $D = -(n_d - n_{\bar{d}})$; the formula above is self-consistent.

2.1.2 Restricted 3-quark approximation (u, d, s only)

$$\frac{Q}{e} = I_3 + \frac{1}{2}(B + S) \tag{5}$$

2.1.3 Proof via fundamental quark charges

Quark flavor	Q/e	B	$U + C + T + D + S + \tilde{B}$
$u/c/t$	$+2/3$	$1/3$	$+1$
$d/s/b$	$-1/3$	$1/3$	-1

The formula holds for all elementary quarks and all composite hadrons.

2.2 Definitions of Core Quantum Numbers

1. **Electric charge** Q: Charge of the particle, in units of the electron charge e.
2. **Third component of isospin** I_3:
   - Total isospin: $I = \frac{n-1}{2}$, where n is the number of particles with nearly identical mass
   - Allowed values: $I_3 = -I,\ -I+1,\ \ldots,\ I-1,\ I$
   - Relation to up/down quark number: $I_3 = \frac{1}{2}(U + D)$, where $D = -(n_d - n_{\bar{d}})$
3. **Hypercharge** Y: Sum of baryon number and all flavor quantum numbers.
4. **Baryon number** B:
$$B = \frac{1}{3}\left(n_q - n_{\bar{q}}\right) \tag{6}$$
where n_q = number of quarks, $n_{\bar{q}}$ = number of antiquarks.

5. **Flavor quantum numbers**:

   - Up number: $U = n_u - n_{\bar{u}}$
   - Down number: $D = -(n_d - n_{\bar{d}})$
   - Strangeness: $S = -(n_s - n_{\bar{s}})$
     - 1 strange quark s: $S = -1$; 1 anti-strange quark $\bar{s}$: $S = +1$
     - u, d, c, b, t quarks carry $S = 0$
   - Charm: $C = n_c - n_{\bar{c}}$
   - Bottom: $\tilde{B} = -(n_b - n_{\bar{b}})$
   - Top: $T = n_t - n_{\bar{t}}$

6. **Lepton number** L_l:

$$L_l = n_l - n_{\bar{l}} + n_{\nu_l} - n_{\bar{\nu}_l}, \quad l = e, \mu, \tau \tag{7}$$

   The total lepton number $L = L_e + L_\mu + L_\tau$ is conserved in all Standard Model interactions, while individual lepton flavors (L_e, L_μ, L_τ) are not conserved due to neutrino oscillations (which require neutrino masses and PMNS mixing).

2.3 Isospin Values for Quarks

Quark	u	d	$\bar{u}$	$\bar{d}$	$c, s, t, b, \bar{c}, \bar{s}, \bar{t}, \bar{b}$
I	$1/2$	$1/2$	$1/2$	$1/2$	0
I_3	$+1/2$	$-1/2$	$-1/2$	$+1/2$	0

Gauge bosons and the Higgs boson carry zero weak isospin. For leptons, left-handed leptons form $SU(2)_L$ doublets with $I = 1/2$ (e.g. $\begin{pmatrix} \nu_{eL} \\ e_L \end{pmatrix}$ with $I_3 = +1/2$ for ν_L and $I_3 = -1/2$ for e_L), while right-handed leptons are singlets with $I = 0$.

2.4 Worked Examples

2.4.1 Example 1: Ω^- (sss)

$$B = 1,\ S = -3 \implies Y = B + S = -2, \quad n = 1 \implies I = 0 \implies \frac{Q}{e} = -1 \tag{8}$$

2.4.2 Example 2: Δ^{++} (uuu)

$$B = 1,\ S = 0 \implies Y = B + S = 1 \tag{9}$$

For the isospin multiplet $\Delta^-, \Delta^0, \Delta^+, \Delta^{++}$, $n = 4$:

$$I = \frac{n-1}{2} = \frac{3}{2} \implies I_3 = \frac{3}{2} \quad (\text{for } \Delta^{++}) \implies \frac{Q}{e} = 2 \tag{10}$$

3 Relativistic Field Equations (by Spin)

Natural units $\hbar = c = k_B = 1$ are used by default; non-natural units are noted explicitly.

3.1 Spin-0: Klein–Gordon Equation (Scalar Bosons)

Energy-momentum dispersion relation:

$$E^2 = \mathbf{p}^2 c^2 + m^2 c^4 \tag{11}$$

Quantum operator substitution: $E \to i\hbar \frac{\partial}{\partial t},\ \mathbf{p} \to -i\hbar \nabla$

Covariant form:

$$\begin{cases} \left(\Box + \frac{m^2 c^2}{\hbar^2}\right) \psi = 0 \\ \Box = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \end{cases} \tag{12}$$

Natural unit simplification:

$$\left(\Box + m^2\right) \phi = 0 \tag{13}$$

3.2 Spin-1/2: Dirac Equation (Fermionic Spinors)

3.2.1 Time-dependent form

$$i\hbar \frac{\partial \psi}{\partial t} = \left(-i\hbar c \alpha \cdot \nabla + \beta m c^2\right) \psi \tag{14}$$

Anticommutator identities:

- Anticommutator definition: $\{A, B\} = AB + BA$
- $\{\alpha^i, \alpha^j\} = 2\delta^{ij} I,\ \{\alpha^i, \beta\} = 0,\ \beta^2 = I$

3.2.2 Covariant form (natural units)

$$(i\gamma^\mu \partial_\mu - m)\, \psi = 0, \quad \not{\partial} = \gamma^\mu \partial_\mu \tag{15}$$

with $\gamma^0 = \beta,\ \gamma^i = \beta \alpha^i$ (Dirac gamma matrices).

3.2.3 Standard Dirac matrix representation

$$\gamma^0 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \quad I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{16}$$

3.2.4 Pauli matrices

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{17}$$

3.2.5 Chiral structure

- Chiral matrix (governs the $V - A$ structure of weak interactions):

$$\gamma^5 = i\gamma^0 \gamma^1 \gamma^2 \gamma^3 \tag{18}$$

- Chiral projection operators (project onto left/right-handed spinors):

$$P_{L,R} = \frac{1 \mp \gamma^5}{2} \tag{19}$$

3.2.6 Massless Weyl fermions

$$\begin{cases} \text{Right-handed: } \bar{\sigma}^\mu \partial_\mu \psi_R = 0 \\ \text{Left-handed: } \sigma^\mu \partial_\mu \psi_L = 0 \end{cases} \tag{20}$$

where $\sigma^\mu = (1, \sigma)$, $\bar{\sigma}^\mu = (1, -\sigma)$.

3.3 Spin-1: Vector Field Equations

3.3.1 Abelian (massless photon, QED): Maxwell equations

$$\begin{cases} \partial_\mu F^{\mu\nu} = j^\nu \\ \partial_\mu F_{\nu\rho} + \partial_\nu F_{\rho\mu} + \partial_\rho F_{\mu\nu} = 0 \end{cases} \tag{21}$$

3.3.2 Non-Abelian (gluons, W/Z bosons): Yang–Mills equations

$$\begin{cases} D_\mu F_a^{\mu\nu} = 0 \\ D_\mu F_{a\nu\rho} + D_\nu F_{a\rho\mu} + D_\rho F_{a\mu\nu} = 0 \end{cases} \tag{22}$$

3.4 Spin-$3/2$: Rarita–Schwinger Equation

$$\begin{cases} (i \not{\partial} - m)\psi^\mu = 0 \\ \gamma^\mu \psi_\mu = \partial_\mu \psi^\mu = 0 \end{cases} \tag{23}$$

3.5 Spin-2: Vacuum Einstein Field Equation (Gravity)

$$R_{\mu\nu} = 0 \tag{24}$$

where $R_{\mu\nu}$ is the Ricci curvature tensor. The intrinsic parity of the massless graviton is $+1$.

4 Perturbative Calculation Framework

4.1 Fermi's Golden Rule

Computes the transition rate for an interaction process:

$$\Gamma_{fi} = \frac{2\pi}{\hbar} |T_{fi}|^2 \rho(E_f) \tag{25}$$

Definitions:

- Γ_{fi}: Number of transitions per unit time from initial state $|i\rangle$ to final state $|f\rangle$. The lifetime τ satisfies $\Gamma = 1/\tau$.
- T_{fi}: Transition matrix element, $T_{fi} = \langle f|\hat{H}_I|i\rangle$
- $|T_{fi}|^2$: Squared probability amplitude for the process
- $\rho(E_f)$: Density of final states, counting available phase-space states
- T_{fi} can be recast as the relativistically invariant amplitude $\mathcal{M}_{fi}$, calculated via Feynman Rules.

4.2 Feynman Path Integral Formalism

$$\langle x_f|e^{-iHt/\hbar}|x_i\rangle = \int \mathcal{D}x(t)\ e^{\frac{i}{\hbar}S[x(t)]} \tag{26}$$

Definitions:

- $\langle x_f|e^{-iHt/\hbar}|x_i\rangle$: Quantum transition amplitude
- $\int \mathcal{D}x(t)$: Functional integral over all paths connecting x_i and x_f
- $S[x(t)] = \int_0^t L(x, \dot{x})dt$: Classical action functional
- $e^{iS/\hbar}$: Phase weight factor for each path
- Hamiltonian: $H = \frac{\mathbf{p}^2}{2m} + V(x)$

4.3 Feynman Rules

4.3.1 Position-space ϕ^4 Theory

- Each propagator:

$$D_F(x-y) = \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\varepsilon} e^{-ip\cdot(x-y)} \tag{27}$$

- Each vertex: $(-i\lambda)\int d^4z$
- Each external leg: 1
- Divide the final result by the symmetry factor.

4.3.2 Momentum-space General Rules

- Enforce four-momentum conservation at every vertex.
- Integrate over all undetermined loop momenta:

$$\int \frac{d^4p}{(2\pi)^4} \tag{28}$$

- Divide the result by the symmetry factor.

4.3.3 External Line Factors (asymptotic states)

1. **Spin-0 (scalar particles)**: Factor = 1
2. **Spin-**$1/2$ **(spinor particles)**
   - p: four-momentum, s: spin index, u, v: spinor wavefunctions
   - Incoming electron: $u^{(s)}(p)$
   - Outgoing electron: $\bar{u}^{(s)}(p) = u^\dagger\gamma^0$
   - Incoming positron: $\bar{v}^{(s)}(p)$
   - Outgoing positron: $v^{(s)}(p)$
3. **Spin-1 (vector particles)**
   - λ: polarization index, ε: polarization four-vector
   - Incoming photon: $\varepsilon_\mu^{(\lambda)}(p)$
   - Outgoing photon: $\varepsilon_\mu^{(\lambda)*}(p)$

4.3.4 Internal Line Propagators (virtual particles)

- Scalar particle (mass m):

$$\frac{i}{p^2 - m^2 + i\varepsilon} \tag{29}$$

- Spinor fermion (mass m):

$$\frac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon}, \quad \not{p} = \gamma^\mu p_\mu \tag{30}$$

- Massless photon:

$$\frac{-ig_{\mu\nu}}{p^2 + i\varepsilon} \tag{31}$$

- Massive vector boson (W, Z):

$$\frac{-i\left(g_{\mu\nu} - \frac{p_\mu p_\nu}{m^2}\right)}{p^2 - m^2 + i\varepsilon} \tag{32}$$

  High-energy approximation: $\dfrac{-ig_{\mu\nu}}{p^2 - m^2}$

- The $i\varepsilon$ prescription enforces causality; it is usually omitted in cross-section calculations.

4.3.5 Vertex Factors

- ϕ^4 theory: $-i\lambda$
- QED electron-photon vertex: $-ie\gamma^\mu$
- QCD quark-gluon vertex: $-ig_s T^a \gamma^\mu$
  - T^a: $SU(3)$ fundamental generators; g_s: strong coupling constant
- Weak interaction vertices
  - W-fermion vertex: $-i\dfrac{g_W}{\sqrt{2}}\gamma^\mu \dfrac{(1-\gamma^5)}{2}$
  - Z-fermion vertex: $-i\dfrac{g_W}{\cos\theta_W}\gamma^\mu \dfrac{(c_V - c_A\gamma^5)}{2}$

  where $c_V = T_3 - 2Q\sin^2\theta_W$ and $c_A = T_3$ depend on the specific fermion.

4.3.6 Supplementary Rules

- **Fermion lines**: Arrows indicate the direction of particle number flow; forward = particle, backward = antiparticle.
  - Closed fermion loops acquire a global factor of -1.
- **Relativistic normalization**: Additional factors from external wavefunction normalizations.
- **Symmetry factors**: For diagrams with identical particles, divide by a symmetry factor (e.g. 1/2) to avoid overcounting.
- **Loop diagrams**: Integrate over the undetermined loop momentum k:

$$\int \frac{d^4k}{(2\pi)^4} \tag{33}$$

4.3.7 Example: Muon-electron scattering

Process: $e^-(p_1)+\mu^-(p_2) \to e^-(p_3)+\mu^-(p_4)$, with momentum transfer $q = p_1 - p_3 = p_4 - p_2$

$$i\mathcal{M} = [\bar{u}(p_3)(-ie\gamma^\mu)u(p_1)] \cdot \left(\frac{-ig_{\mu\nu}}{q^2}\right) [\bar{u}(p_4)(-ie\gamma^\nu)u(p_2)] \tag{34}$$

- $\mathcal{M}$: Lorentz-invariant matrix element (equivalent to T in Fermi's Golden Rule)
- γ^μ: Dirac gamma matrix

5 Standard Model Gauge Group Structure

5.1 Conservation Laws Summary for All Interactions

Before delving into the individual gauge groups and their associated interactions, it is useful to have a global picture of which quantum numbers are conserved by which force. The table below summarizes the conservation laws for the electromagnetic, strong, and weak interactions.

Quantum Number	Electromagnetic	Strong	Weak
Electric Charge Q	✓	✓	✓
Baryon Number B	✓	✓	✓
Total Lepton Number L	✓	✓	✓
Color Charge	✓	✓	✓
Flavor Quantum Number	✓	✓	× (CKM/PMNS mixing)
Charge Conjugation C	✓	✓	×
Parity P	✓	✓	×
Time Reversal T	✓	✓	×
Combined CPT	✓	✓	✓

Note: Individual lepton flavors (L_e, L_μ, L_τ) are not conserved in weak interactions due to neutrino oscillations, but total lepton number $L = L_e + L_\mu + L_\tau$ remains conserved.

5.2 Full Standard Model Gauge Symmetry

$$G_{SM} = \underbrace{SU(3)_c}_{\text{Strong}} \times \underbrace{SU(2)_L \times U(1)_Y}_{\text{Electroweak}} \tag{35}$$

5.3 $SU(3)_c$ – Strong Interaction (QCD)

- c: Color charge quantum number
- **Role**: Describes quantum chromodynamics, the theory of strong interactions
- **Matter content**:
  - Quarks carry 3 color charges: red, green, blue
  - 8 massless gluons are the gauge bosons
- **Key features**:
  - Color confinement: Isolated free quarks cannot exist in nature
  - Asymptotic freedom: Interaction strength weakens at high energy / short distance
  - Leptons are $SU(3)_c$ color singlets (colorless) and do not feel the strong force
- **Mathematical definition**: Special Unitary group of degree 3; all 3×3 unitary matrices with determinant 1
- **Generators**: $T^a = \lambda^a/2,\ a = 1, \ldots, 8$
  - λ^a: Gell-Mann matrices; $8 = 3^2 - 1$ independent generators
- **Irreducible representations**: **3** (quarks), **3** (antiquarks), **8** (gluons)

5.4 $SU(2)_L$ – Weak Interaction (Left-Handed Chiral Sector)

- L: Left-handed chiral projection
- **Role**: Governs charged weak interactions; only couples to left-handed fermions
- **Matter content**:
  - Left-handed fermions form $SU(2)_L$ doublets, e.g.

$$\begin{pmatrix} \nu_e \\ e_L \end{pmatrix}, \quad \begin{pmatrix} u_L \\ d_L \end{pmatrix} \tag{36}$$

  - Right-handed fermions are $SU(2)_L$ singlets and have no charged weak coupling
- **Key feature**: The chiral nature is the origin of parity violation in weak interactions
- **Gauge bosons**: W^1, W^2, W^3. After electroweak symmetry breaking, these become massive $W^\pm$ and mix with the $U(1)_Y$ B boson to form the neutral Z^0.
- **Mathematical definition**: Special Unitary group of degree 2; all 2×2 unitary matrices with determinant 1
- **Generators**: $T^a = \sigma^a/2,\ a = 1, 2, 3$; $3 = 2^2 - 1$ generators
- **Lie algebra**:

$$[T^a, T^b] = i\varepsilon^{abc} T^c \tag{37}$$

  where ε^{abc} is the fully antisymmetric Levi-Civita symbol.
- **Representations**:
  - **2** (fundamental): Left-handed fermion doublets
  - **1** (singlet): Right-handed fermions
  - **3** (adjoint): W boson weak isospin triplet

5.5 $U(1)_Y$ – Hypercharge Gauge Symmetry

- Y: Weak hypercharge quantum number
- **Role**: Part of electroweak unification
- **Gauge boson**: B_μ
- **Relation to electric charge**: Gell-Mann–Nishijima formula

$$Q = I_3 + \frac{Y}{2} \tag{38}$$

- **Mathematical definition**: Degree-1 unitary group; all 1×1 unitary phase factors $e^{i\theta}$
- Single generator: 1; trivial abelian Lie algebra
- **Representation**: Hypercharge Y; field transformation: $\psi \to e^{iY\theta}\psi$

5.6 Electroweak Symmetry Breaking

$$SU(2)_L \times U(1)_Y \xrightarrow{\text{symmetry breaking}} U(1)_{EM} \tag{39}$$

The original gauge multiplet (W^1, W^2, W^3, B) mixes into physical electroweak bosons $(W^\pm, Z^0, \gamma)$ via the Weinberg angle θ_W:

$$\begin{cases} W^\pm = \frac{1}{\sqrt{2}}(W^1 \mp iW^2) \\ \gamma = \sin\theta_W W^3 + \cos\theta_W B \\ Z^0 = \cos\theta_W W^3 - \sin\theta_W B \end{cases} \tag{40}$$

6 Full Standard Model Lagrangian

$$\mathcal{L}_{SM} = \mathcal{L}_{gauge} + \mathcal{L}_{fermion} + \mathcal{L}_{Higgs} + \mathcal{L}_{Yukawa} \tag{41}$$

6.1 Gauge Field (Yang–Mills) Sector

$$\mathcal{L}_{gauge} = -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu} - \frac{1}{4} W^i_{\mu\nu} W^{i\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} \tag{42}$$

Field strength tensors:

1. Gluon ($SU(3)_c$, non-abelian, self-interacting):

$$G^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu + g_s f^{abc} G^b_\mu G^c_\nu \tag{43}$$

2. Weak W boson ($SU(2)_L$, non-abelian, self-interacting):

$$W^i_{\mu\nu} = \partial_\mu W^i_\nu - \partial_\nu W^i_\mu + g \varepsilon^{ijk} W^j_\mu W^k_\nu \tag{44}$$

3. Hypercharge B boson ($U(1)_Y$, abelian, no self-interactions):

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu \tag{45}$$

6.2 Fermion Kinetic (Matter) Sector

$$\mathcal{L}_{fermion} = \sum_{fermions} i \bar{\psi} \gamma^\mu D_\mu \psi \tag{46}$$

Covariant derivative (includes all three gauge interactions):

$$D_\mu = \partial_\mu - i g_s G^a_\mu T^a - i g W^i_\mu t^i - i g' \frac{Y}{2} B_\mu \tag{47}$$

- γ^μ: Dirac matrices
- g_s, g, g': Coupling constants for strong, weak, and hypercharge interactions
- T^a: $SU(3)_c$ generators
- $t^i = \sigma^i/2$: $SU(2)_L$ generators
- Y: $U(1)_Y$ hypercharge operator

6.3 Higgs Sector

$$\mathcal{L}_{Higgs} = (D_\mu \Phi)^\dagger (D^\mu \Phi) - V(\Phi) \tag{48}$$

- Higgs field Φ: $SU(2)_L$ complex doublet with hypercharge $Y = +1$

$$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} \tag{49}$$

(The hypercharge assignment $Y = +1$ follows from $Q = I_3 + Y/2$ and the fact that the neutral component ϕ^0 has $Q = 0$ and $I_3 = -1/2$.)

- Higgs covariant derivative:

$$D_\mu \Phi = \left(\partial_\mu - igW_\mu^i \frac{\sigma^i}{2} - ig' \frac{Y}{2} B_\mu \right) \Phi \tag{50}$$

- Higgs potential (drives spontaneous symmetry breaking):

$$V(\Phi) = -\mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2 \quad (\mu^2 > 0,\ \lambda > 0) \tag{51}$$

  The vacuum expectation value is $v = \sqrt{\mu^2/\lambda} \approx 246$ GeV.

6.4 Yukawa Coupling (Fermion Mass Generation)

$$\mathcal{L}_{Yukawa} = -y_e \bar{L}_e \Phi e_R - y_d \bar{Q}_L \Phi d_R - y_u \bar{Q}_L \tilde{\Phi} u_R + \text{h.c.} \tag{52}$$

Charge-conjugated Higgs doublet: $\tilde{\Phi} = i\sigma^2 \Phi^*$
Fermion masses after electroweak symmetry breaking (VEV v):

$$m_e = y_e \frac{v}{\sqrt{2}}, \quad m_u = y_u \frac{v}{\sqrt{2}}, \quad m_d = y_d \frac{v}{\sqrt{2}} \tag{53}$$

7 Quantum Electrodynamics (QED) – Electromagnetic Interaction

7.1 QED Action and Lagrangian Density

Action definition:

$$S = \int d^4x \; \mathcal{L}(\phi, \partial_\mu \phi) \tag{54}$$

Euler–Lagrange equation of motion:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0 \tag{55}$$

7.1.1 Component Lagrangians

1. Klein–Gordon scalar (spin-0):

$$\mathcal{L}_{KG} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2} m^2 \phi^2 \tag{56}$$

2. Dirac fermion (spin-1/2):

$$\mathcal{L}_{Dirac} = \bar{\psi} \left(i\gamma^\mu \partial_\mu - m \right) \psi \tag{57}$$

3. Maxwell photon (spin-1):

$$\mathcal{L}_{Maxwell} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \tag{58}$$

   Maxwell field equations: $\partial_\mu F^{\mu\nu} = j^\nu$

7.1.2 Full QED Lagrangian

$$\mathcal{L}_{QED} = \underbrace{i\bar{\psi}\gamma^\mu \partial_\mu \psi - m\bar{\psi}\psi}_{\text{Free Dirac electron}} - \underbrace{\frac{1}{4} F_{\mu\nu} F^{\mu\nu}}_{\text{Free Maxwell photon}} - \underbrace{e(\bar{\psi}\gamma^\mu \psi) A_\mu}_{e-\gamma \text{ local invariant interaction}} \tag{59}$$

Supporting identities:

- Dirac adjoint: $\bar{\psi} = \psi^\dagger \gamma^0$
- QED covariant derivative: $D_\mu = \partial_\mu + ieA_\mu$
- Field strength tensor: $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

7.2 Gauge Invariance

$U(1)$ gauge transformation operator: $U = e^{-i\theta}$

1. **Global gauge transformation** (θ = constant, spacetime-independent):

$$\psi \to \tilde{\psi} = e^{-i\theta} \psi \tag{60}$$

   $\mathcal{L}_{QED}$ remains invariant.

2. **Local gauge transformation** ($\theta = \lambda(x)$, spacetime-dependent):

$$\psi \to \tilde{\psi} = e^{-ie\lambda(x)} \psi, \quad D_\mu \tilde{\psi} = e^{-ie\lambda} D_\mu \psi \tag{61}$$

   Gauge field shift: $A_\mu \to \tilde{A}_\mu = A_\mu + \partial_\mu \lambda$
   The full QED Lagrangian remains invariant under local $U(1)$ transformations.

7.3 QED Vertex and Feynman Rules

The QED interaction vertex is:

$$-ie\gamma^{\mu} \tag{62}$$

For a generic QED process, the amplitude is constructed by:

- Assigning fermion propagators: $\dfrac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon}$
- Assigning photon propagators: $\dfrac{-ig_{\mu\nu}}{p^2 + i\varepsilon}$
- Multiplying by vertex factors $-ie\gamma^{\mu}$ at each interaction point
- Enforcing momentum conservation at each vertex

8 Quantum Chromodynamics (QCD) – Strong Interaction

8.1 Color Charge and Gluon States

- Quark colors: Red R, Green G, Blue B
- Antiquark colors: Anti-red $\bar{R}$, Anti-green $\bar{G}$, Anti-blue $\bar{B}$

Color tensor product decomposition:

$$3 \otimes \bar{3} = \mathbf{8} \oplus \mathbf{1} \tag{63}$$

8.1.1 8 physical gluon octet states

1. 6 off-diagonal (edge) states: $R\bar{G}, G\bar{R}, R\bar{B}, B\bar{R}, G\bar{B}, B\bar{G}$
2. 2 traceless diagonal (center) states:

$$\frac{1}{\sqrt{2}}(R\bar{R} - G\bar{G}), \quad \frac{1}{\sqrt{6}}(R\bar{R} + G\bar{G} - 2B\bar{B}) \tag{64}$$

8.1.2 1 unobservable color singlet

$$\frac{1}{\sqrt{3}}(R\bar{R} + G\bar{G} + B\bar{B}) \tag{65}$$

Colorless; does not participate in strong interactions.

8.2 Color Confinement

All observable hadrons are color singlets (color-neutral):

- Baryons: $R + G + B$
- Antibaryons: $\bar{R} + \bar{G} + \bar{B}$
- Mesons: $R\bar{R}$ or $G\bar{G}$ or $B\bar{B}$

8.3 Asymptotic Freedom

At high energy / short inter-quark distance, the strong coupling constant α_s becomes small, permitting perturbative calculations.

The running of the strong coupling is governed by the QCD β-function:

$$\beta(\alpha_s) = \mu^2 \frac{d\alpha_s}{d\mu^2} = -\frac{11N_c - 2N_f}{12\pi}\alpha_s^2 + \mathcal{O}(\alpha_s^3) \tag{66}$$

For $N_c = 3$ and $N_f \leq 16$, the first coefficient is negative, leading to asymptotic freedom.

8.3.1 QCD static quark potential

$$V_{\text{QCD}}(r) = -\frac{4}{3}\frac{\alpha_s}{r} + kr \tag{67}$$

Parameters:

- Strong coupling: $\alpha_s \approx 0.3$
- String tension: $k = 1$ GeV/fm
- Short distance: Coulomb-like term dominates (asymptotic freedom)
- Long distance: Linear confining term dominates (color confinement)

8.4 QCD Vertex and Feynman Rules

The QCD quark-gluon vertex is:

$$-ig_s T^a \gamma^\mu \tag{68}$$

where T^a are the $SU(3)$ generators ($a = 1, \ldots, 8$).

QCD also features:

- Gluon self-interactions (3-gluon and 4-gluon vertices) due to the non-abelian nature of $SU(3)$.
- Ghost fields in covariant gauges to maintain unitarity.

8.5 Experimental Evidence: R Ratio in e^+e^- Annihilation

Definition: Ratio of total hadronic cross section to muon pair cross section

$$R = \frac{\sigma_{\text{hadrons}}}{\sigma_{\mu^+\mu^-}} = \sum_{\text{active quarks}} Q_q^2 \tag{69}$$

With 3 color degrees of freedom per quark:

$$R = 3\sum Q_q^2 \tag{70}$$

Numerical predictions at different energy thresholds:

- u, d, s active:

$$R = 3\left[\left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(-\frac{1}{3}\right)^2\right] = 2 \tag{71}$$

- u, d, s, c active:

$$R = 3\left[\left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2\right] = \frac{10}{3} \tag{72}$$

- u, d, s, c, b active:

$$R = 3\left[\left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2\right] = \frac{11}{3} \tag{73}$$

Center-of-mass energy: $\sqrt{s} = E_{e^-} + E_{e^+}$
Experimental measurements match the factor-of-3 color enhancement, confirming 3 color charges per quark.

8.6 Quark Model: Baryon Flavor Wavefunctions

Total baryon wavefunction factorization:

$$\Psi = \Psi(\text{space})\,\Psi(\text{spin})\,\Psi(\text{flavor})\,\Psi(\text{color}) \tag{74}$$

8.6.1 $SU(3)_f$ Three-Quark Decomposition

$$3 \otimes 3 \otimes 3 = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1} \tag{75}$$

Symmetry classes:

- S Fully symmetric: **10** decuplet; $\Psi(a,b,c) = \Psi(b,a,c)$ for all permutations
- A Fully antisymmetric: **1** singlet; $\Psi(a,b,c) = -\Psi(b,a,c)$ for all swaps
- M Mixed symmetry: two **8** octets; antisymmetric under specific quark pair swaps

8.6.2 Fully Symmetric Decuplet (10)

Normalized basis states:

- Pure-flavor states: $ddd,\ uuu,\ sss$
- Two-identical-flavor states:

$$\frac{1}{\sqrt{3}}(dds + dsd + sdd), \quad \frac{1}{\sqrt{3}}(uus + usu + suu) \tag{76}$$

$$\frac{1}{\sqrt{3}}(udd + udu + duu), \quad \frac{1}{\sqrt{3}}(dss + sds + ssd) \tag{77}$$

$$\frac{1}{\sqrt{3}}(uss + sus + ssu) \tag{78}$$

- Three-different-flavor symmetric state:

$$\frac{1}{\sqrt{6}}(uds + usd + dus + dsu + sud + sdu) \tag{79}$$

8.6.3 Fully Antisymmetric Singlet (1)

$$\frac{1}{\sqrt{6}}\big(uds - usd + sud - sdu + dsu - dus\big) \tag{80}$$

8.6.4 Mixed-Symmetry Octets $(\mathbf{8},\mathbf{8})$

1. Ψ_{12}: Antisymmetric under quark $1 \leftrightarrow 2$ exchange
   Building blocks: $\dfrac{ud-du}{\sqrt{2}},\ \dfrac{ds-sd}{\sqrt{2}},\ \dfrac{us-su}{\sqrt{2}}$
   Linear combinations:

$$\frac{1}{2}\big[(us-su)d + (ds-sd)u\big], \quad \frac{1}{\sqrt{12}}\big[2(ud-du)s + (us-su)d - (ds-sd)u\big] \tag{81}$$

2. Ψ_{23}: Antisymmetric under quark $2 \leftrightarrow 3$ exchange
   Building blocks: $d\dfrac{ud-du}{\sqrt{2}},\ d\dfrac{ds-sd}{\sqrt{2}},\ u\dfrac{ud-du}{\sqrt{2}},\ u\dfrac{us-su}{\sqrt{2}},\ s\dfrac{ds-sd}{\sqrt{2}},\ s\dfrac{us-su}{\sqrt{2}}$
   Linear combinations:

$$\frac{1}{2}\big[d(us-su)+u(ds-sd)\big], \quad \frac{1}{\sqrt{12}}\big[2s(ud-du)+d(us-su)-u(ds-sd)\big] \tag{82}$$

Linear dependence relation:

$$\Psi_{13} = \Psi_{12} + \Psi_{23} \tag{83}$$

Ψ_{13} is not an independent basis state.

8.6.5 Clebsch–Gordan Coefficients

Angular momentum coupling expansion:

$$|j_1 m_1, j_2 m_2\rangle = \sum_{\substack{j \\ m=m_1+m_2}} |jm\rangle\langle jm|j_1 m_1, j_2 m_2\rangle \tag{84}$$

Allowed ranges:

- Total angular momentum: $j = |j_1 - j_2|,\ |j_1 - j_2|+1,\ \ldots,\ j_1 + j_2$
- Magnetic projection: $m = m_1 + m_2$
- Individual projections: $m_1 = -j_1, \ldots, j_1$; $m_2 = -j_2, \ldots, j_2$; $m = -j, \ldots, j$

9 Weak Interaction

The weak interaction is described by the $SU(2)_L \times U(1)_Y$ sector of the Standard Model. It is characterized by:

- **Chirality**: Only left-handed fermions participate in charged current interactions.
- **Parity violation**: The $V - A$ structure of weak currents maximally violates parity.
- **Massive mediators**: $W^{\pm}$ and Z^0 bosons are massive due to electroweak symmetry breaking.

9.1 Charged Current Interaction

The charged current interaction couples left-handed fermion doublets via $W^{\pm}$ bosons.

For a generic fermion doublet (f_1, f_2), the charged current Lagrangian is:

$$\mathcal{L}_{CC} = -\frac{g}{\sqrt{2}} \bar{f}_1 \gamma^\mu P_L f_2 W_\mu^+ + \text{h.c.} \tag{85}$$

where $P_L = (1 - \gamma^5)/2$ is the left-handed projection operator.

This mediates processes such as:

- β-decay: $d \to u + e^- + \bar{\nu}_e$
- Muon decay: $\mu^- \to e^- + \bar{\nu}_e + \nu_\mu$
- Neutrino scattering: $\nu_\mu + e^- \to \mu^- + \nu_e$

9.2 Neutral Current Interaction

The neutral current interaction is mediated by the Z^0 boson:

$$\mathcal{L}_{NC} = -\frac{g}{2\cos\theta_W} \bar{\psi} \gamma^\mu (c_V - c_A \gamma^5) \psi Z_\mu \tag{86}$$

where:

$$c_V = T_3 - 2Q \sin^2\theta_W, \qquad c_A = T_3 \tag{87}$$

depend on the specific fermion.

Key properties:

- The Z^0 couples to both left- and right-handed fermions (unlike the $W^{\pm}$).
- Neutral currents do not change fermion flavor (e.g., $Z^0 \to e^+e^-$ but not $Z^0 \to e^{\pm}\mu^{\mp}$).

9.3 Weak Interaction Vertex Factors

- W-fermion vertex: $-i\frac{g_W}{\sqrt{2}} \gamma^\mu \frac{(1 - \gamma^5)}{2}$
- Z-fermion vertex: $-i\frac{g_W}{\cos\theta_W} \gamma^\mu \frac{(c_V - c_A \gamma^5)}{2}$

9.4 Flavor Mixing in Weak Interactions

The weak interaction eigenstates are not the same as the mass eigenstates for quarks and neutrinos.

9.4.1 Quark Mixing: CKM Matrix

Up-type weak eigenstates match mass eigenstates exactly:

$$\begin{pmatrix} u' \\ c' \\ t' \end{pmatrix} = \begin{pmatrix} u \\ c \\ t \end{pmatrix} \tag{88}$$

Down-type weak eigenstates mix via the Cabibbo–Kobayashi–Maskawa matrix V_{CKM}:

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = V_{CKM} \begin{pmatrix} d \\ s \\ b \end{pmatrix} \tag{89}$$

Wolfenstein parametrization:

$$V_{CKM} = \begin{pmatrix} 1 - \frac{\lambda^2}{2} & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1 - \frac{\lambda^2}{2} & A\lambda^2 \\ A\lambda^3(1 - \rho - i\eta) & -A\lambda^2 & 1 \end{pmatrix} \tag{90}$$

Parameters: $A \approx 0.83,\ \rho \approx 0.13,\ \eta \approx 0.35,\ \lambda = \sin\theta_C = 0.225.$

9.4.2 Lepton Mixing: PMNS Matrix

Charged lepton weak eigenstates equal mass eigenstates:

$$\begin{pmatrix} e' \\ \mu' \\ \tau' \end{pmatrix} = \begin{pmatrix} e \\ \mu \\ \tau \end{pmatrix} \tag{91}$$

Neutrino flavor $\leftrightarrow$ mass eigenstate mixing via the Pontecorvo–Maki–Nakagawa–Sakata matrix U_{PMNS}:

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = U_{PMNS} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix} \tag{92}$$

Standard factorized parametrization:

$$U_{PMNS} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} e^{i\alpha_1/2} & 0 & 0 \\ 0 & e^{i\alpha_2/2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{93}$$

where $c_{ij} = \cos\theta_{ij},\ s_{ij} = \sin\theta_{ij}$, δ is the Dirac CP-violating phase, and α_1, α_2 are Majorana phases.

Measured mixing angles:

$$\theta_{12} \approx 34^\circ, \quad \theta_{23} \approx 45^\circ, \quad \theta_{13} \approx 9^\circ \tag{94}$$

9.5 Neutrino Physics

9.5.1 Helicity

$$h = \frac{\mathbf{S} \cdot \hat{P}}{|\mathbf{S}|} \tag{95}$$

- $\mathbf{S}$: Spin vector; $\hat{P}$: Unit momentum vector
- $h < 0$: Left-handed (LH), spin antiparallel to momentum
- $h > 0$: Right-handed (RH), spin parallel to momentum

Standard Model chiral rules:

- Photons are produced in equal LH and RH helicity states
- All active neutrinos are left-handed: ν_L
- All active antineutrinos are right-handed: $\bar{\nu}_R$

CP transformation of neutrino helicity:

$$\nu_L \xrightarrow{\hat{C}} \bar{\nu}_L \xrightarrow{\hat{P}} \bar{\nu}_R \tag{96}$$

$\bar{\nu}_L$ does not exist in nature; massless neutrinos violate CP intrinsically.

9.5.2 Neutrino Detection

Charged-current detection reactions:

$$\nu_e + n \to p + e^-, \quad \nu_\mu + n \to p + \mu^- \tag{97}$$

Typical mean free path in matter: $d_p \sim 1.5 \times 10^{16}$ m

9.5.3 Neutrino Sources

1. Accelerator neutrino beams
2. Solar proton-proton fusion chain:

$$p + p \to d + e^+ + \nu_e \tag{98}$$
$$d + p \to {}^3\mathrm{He} + \gamma \tag{99}$$
$${}^3\mathrm{He} + {}^3\mathrm{He} \to {}^4\mathrm{He} + p + p \tag{100}$$

3. Atmospheric neutrinos (cosmic ray air showers)
4. Reactor electron antineutrinos (nuclear beta decay):

$$(Z, A) \to (Z + 1, A) + e^- + \bar{\nu}_e \tag{101}$$

where Z = proton number, A = mass number, N = neutron number.

9.5.4 Neutrino Oscillations

Oscillation requires non-zero neutrino mass: $m_\nu > 0$
Flavor states oscillate via PMNS mixing: $\nu_e \leftrightarrow \nu_\mu \leftrightarrow \nu_\tau$

10 Discrete Symmetries and CP Violation

10.1 Charge Conjugation Operator $\hat{C}$

State transformation:

$$\hat{C}|\text{particle}\rangle = \eta_C|\text{antiparticle}\rangle \tag{102}$$

- For self-conjugate neutral particles (e.g. π^0), $\eta_C = \pm 1$ is an observable phase.
- For non-self-conjugate particles (e.g. e^-), $\hat{C}$ simply maps to the antiparticle state without an additional phase factor.

Transformation rules:

- Electric charge: $Q \to -Q$
- Baryon number: $B \to -B$
- Lepton number: $L \to -L$
- All flavor quantum numbers flip sign
- Electromagnetic fields: $\mathbf{E} \to -\mathbf{E},\ \mathbf{B} \to -\mathbf{B}$
- Mass m, energy E, momentum $\mathbf{p}$, spin $\mathbf{S}$: Unchanged

10.2 Parity Operator $\hat{P}$

State transformation:

$$\hat{P}|\psi(\mathbf{x}, t)\rangle = \eta_P|\psi(-\mathbf{x}, t)\rangle \tag{103}$$

10.2.1 Intrinsic parity assignments

1. Fermions
   - Particle: $\eta_P = +1$
   - Antiparticle: $\eta_P = -1$
2. Bosons
   - Spin-0 pseudoscalar ($\pi^0, \pi^\pm$): $\eta_P = -1$
   - Spin-0 scalar (Higgs H): $\eta_P = +1$
   - Spin-1 vector ($\gamma, g, W^\pm, Z^0$): $\eta_P = -1$
   - Spin-2 graviton: $\eta_P = +1$

10.2.2 Hadron parity

$$\begin{cases} \eta_{P,\text{meson}} = (-1)^L \eta_{q_1}\eta_{\bar{q}_2} \\ \eta_{P,\text{baryon}} = (-1)^L \eta_{q_1}\eta_{q_2}\eta_{q_3} \end{cases} \tag{104}$$

where L is the total orbital angular momentum between constituent quarks.

10.3 Time Reversal Operator $\hat{T}$

State transformation:

$$\hat{T}|\psi(\mathbf{x},t)\rangle = \eta_T|\psi^*(\mathbf{x},-t)\rangle \tag{105}$$

Quantum number flips:

- Momentum: $\mathbf{P} \to -\mathbf{P}$
- Spin: $\mathbf{S} \to -\mathbf{S}$

10.4 CPT Theorem

Unitary condition:

$$(\hat{C}\hat{P}\hat{T})^\dagger(\hat{C}\hat{P}\hat{T}) = \mathbb{I} \tag{106}$$

For any quantum state:

$$\hat{C}\hat{P}\hat{T}|\Psi\rangle = e^{i\theta}|\Psi\rangle \tag{107}$$

CPT symmetry is exact in all local Lorentz-invariant quantum field theories.

10.5 CP Violation in the Neutral Kaon System

10.5.1 Flavor eigenstates

- $|K^0\rangle = |d\bar{s}\rangle,\ S = +1$
- $|\bar{K}^0\rangle = |s\bar{d}\rangle,\ S = -1$

10.5.2 Pure CP eigenstates (theoretical)

$$\begin{cases} |K_1\rangle = \dfrac{1}{\sqrt{2}}\big(|K^0\rangle + |\bar{K}^0\rangle\big), & CP = +1 \quad \text{(decays to } 2\pi\text{)} \\ |K_2\rangle = \dfrac{1}{\sqrt{2}}\big(|K^0\rangle - |\bar{K}^0\rangle\big), & CP = -1 \quad \text{(decays to } 3\pi\text{)} \end{cases} \tag{108}$$

10.5.3 Physical mass eigenstates

If CP were exact: $|K_S\rangle = |K_1\rangle$ (short-lived), $|K_L\rangle = |K_2\rangle$ (long-lived).

Experimentally, the rare decay $K_L \to \pi^+\pi^-$ is observed, proving CP violation. The physical long-lived state is a mixed state:

$$|K_L\rangle = \frac{1}{\sqrt{1+|\varepsilon|^2}}\big(|K_2\rangle + \varepsilon|K_1\rangle\big) \tag{109}$$

- $\varepsilon \approx 2.2 \times 10^{-3}$: CP violation parameter
- Semileptonic decay asymmetry:

$$\frac{\Gamma(K_L \to e^+\nu_e\pi^-)}{\Gamma(K_L \to e^-\bar{\nu}_e\pi^+)} \approx 1.00648 \tag{110}$$

By the CPT theorem, CP violation implies time reversal symmetry violation.

11 Beyond the Standard Model

11.1 Fundamental Coupling Constants and Interaction Ranges

Effective range generic formula: $R = \frac{\hbar}{mc}$ (Compton wavelength form)

Force	Fine-structure constant	Effective Range
Electromagnetic	$\alpha = \frac{e^2}{4\pi\hbar c} \approx \frac{1}{137}$	∞
Strong	$\alpha_s = \frac{g_s^2}{4\pi} \approx 1$	1.4×10^{-15} m
Weak	$\alpha_W = \frac{g^2}{4\pi} \approx \frac{1}{30}$	2.5×10^{-18} m
Gravitation	$\alpha_G = \frac{G_N m_p^2}{\hbar c} \approx 10^{-39}$	∞

G_N: Newtonian gravitational constant

11.2 Limitations of the Standard Model

1. Original formulation predicts massless neutrinos, conflicting with oscillation data
2. Many undetermined free parameters: fermion masses, coupling constants, mixing angles
3. Strong and electroweak forces are not unified (requires Grand Unified Theory, GUT)
4. No description of gravity (requires Theory of Everything, ToE)
5. Lepton anomalous magnetic moment g-factor: 3σ discrepancy between theory and experiment

11.2.1 Anomalous magnetic moment

Magnetic moment operator:

$$\mu_l = g_l \frac{e\hbar}{2mc}\mathbf{S} \tag{111}$$

Theoretical predictions:

- Non-relativistic Schrödinger: $g_l = 1$
- Relativistic Dirac: $g_l = 2$
- QFT radiative corrections: $g_l = 2(1 + a_l)$

Measured values:

- Electron: $a_e \approx 1.15965 \times 10^{-3}$
- Muon: $a_\mu \approx 1.16592 \times 10^{-3}$

11.3 Gauge Coupling Unification

The coupling constants of strong, weak, and electromagnetic forces converge to a common value at the grand unification scale $\sim 10^{16}$ GeV.

11.4 Dark Matter: WIMP Candidate

Weakly Interacting Massive Particle (WIMP):

- Mass: $M_{\text{WIMP}} \sim 100 \sim 200$ GeV/c^2
- Electrically neutral, cosmologically stable

11.5 Supersymmetry (SUSY)

Core symmetry: Maps Standard Model fermions $\leftrightarrow$ boson superpartners, and vice versa.

SM Particle (spin)	Superpartner (spin)	Name
Quark q (1/2)	Squark $\tilde{q}$ (0)	Squark
Electron e (1/2)	Selectron $\tilde{e}$ (0)	Selectron
Muon μ (1/2)	Smuon $\tilde{\mu}$ (0)	Smuon
Tau τ (1/2)	Stau $\tilde{\tau}$ (0)	Stau
Neutrino ν (1/2)	Sneutrino $\tilde{\nu}$ (0)	Sneutrino
W boson (1)	Wino $\tilde{W}$ (1/2)	Wino
Z boson (1)	Zino $\tilde{Z}$ (1/2)	Zino
Photon γ (1)	Photino $\tilde{\gamma}$ (1/2)	Photino
Gluon g (1)	Gluino $\tilde{g}$ (1/2)	Gluino

Neutral gaugino-higgsino mixing produces the neutralino, a leading WIMP dark matter candidate.

11.6 Seesaw Mechanism for Neutrino Masses

In the Standard Model, neutrino masses can be generated via the seesaw mechanism, which introduces heavy right-handed neutrinos N_R:

$$\mathcal{L}_{\text{seesaw}} = -y_\nu \bar{L}\tilde{\Phi}N_R - \frac{1}{2}M_N \bar{N}_R^c N_R + \text{h.c.} \tag{112}$$

For $M_N \gg y_\nu v$, the light neutrino mass is suppressed:

$$m_\nu \sim \frac{(y_\nu v)^2}{M_N} \tag{113}$$

This naturally explains the smallness of neutrino masses.

A Appendix 1: Lie Group Notation Reference

Symbol	Full Name	Definition
$GL(n)$	General Linear Group	All invertible $n \times n$ complex matrices
$SL(n)$	Special Linear Group	$GL(n)$ matrices with determinant = 1
$O(n)$	Orthogonal Group	Real matrices preserving real vector norm: $R^T R = I$
$SO(n)$	Special Orthogonal Group	$O(n)$ matrices with determinant = 1; describes rotations
$U(n)$	Unitary Group	Complex matrices preserving complex vector norm: $U^\dagger U = I$
$SU(n)$	Special Unitary Group	$U(n)$ matrices with determinant = 1
$Sp(2n)$	Symplectic Group	Matrices preserving symplectic bilinear form

Key letter mnemonics:

- S = Special (determinant 1)
- O = Orthogonal (real norm-preserving)
- U = Unitary (complex norm-preserving)

B Appendix 2: Matrix Definitions

- Unitary matrix U: $U^\dagger U = UU^\dagger = I$
- Symplectic matrix M: $M^T \Omega M = \Omega$, with symplectic metric

$$\Omega = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix} \tag{114}$$

for $2n$-dimensional phase space.

Gavin's Lectures on Physics - Quantum Field Theory



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1 Invitation

We work in natural units $\hbar = c = k_B = 1$, with metric $g_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1) = \eta_{\mu\nu}$. This chapter lays out the general classical field–theoretic framework that applies to all fields studied below, then gives a concrete warm–up with the photon polarization vector.

1.1 Polarization Vector

A free electromagnetic plane wave with polarization vector.

$$A^\mu(x) = \epsilon^\mu e^{-ip\cdot x}, \tag{1}$$

where ϵ^μ is the polarization vector.
In Lorenz gauge, the four–divergence vanishes.

$$0 = \partial_\mu A^\mu = -ip_\mu \epsilon^\mu e^{-ip\cdot x},$$

so the Lorenz condition is

$$p_\mu \epsilon^\mu = 0. \tag{2}$$

For a photon moving along the z–axis ($\mathbf{p} = (0, 0, p_z)$),

$$p^\mu = (E_p, 0, 0, p_z), \qquad E_p = |\mathbf{p}| = p_z, \tag{3}$$

so $p^2 = 0$. The Lorenz condition then gives

$$E_p \epsilon^0 - p_z \epsilon^3 = 0.$$

Choosing transverse/Coulomb gauge $\epsilon^0 = \epsilon^3 = 0$,

$$\epsilon^\mu = (0, \epsilon_x, \epsilon_y, 0).$$

The transverse complex amplitudes form the Jones vector.

$$\mathbf{J} = \begin{pmatrix} \epsilon_x \\ \epsilon_y \end{pmatrix}, \tag{4}$$

often normalized as $\mathbf{J}^\dagger \mathbf{J} = 1$. The physical electric field is the real part of the complex amplitude. The formal quantization of the vector field is treated in Chapter 4.

1.2 $e^+e^- \to \mu^+\mu^-$ as a Preview

The virtual photon's right–handed circular polarization vector for a photon propagating along the z–axis is $\epsilon^\mu = (0, 1, i, 0)$.

$$\langle \gamma | H_I | e^+e^- \rangle^\mu \propto e\epsilon^\mu = e(0, 1, i, 0) \tag{5}$$

Rotate angle θ in the xz-plane.

$$\langle \gamma | H_I | e^+e^- \rangle^\mu \propto e(0, \cos\theta, i, -\sin\theta) \tag{6}$$

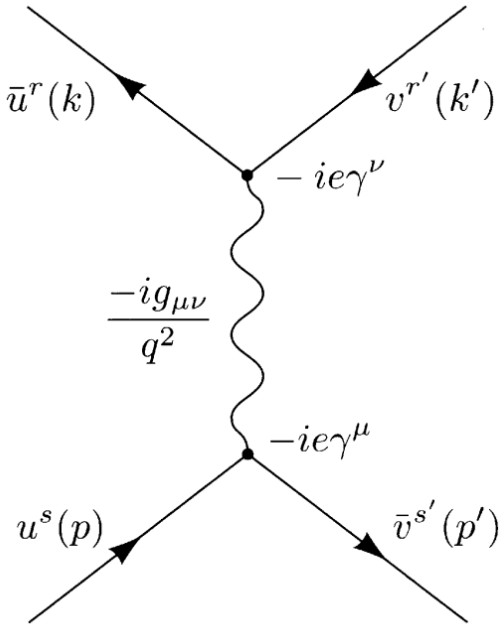


Figure 1: Tree–level Feynman diagram for $e^+e^- \to \mu^+\mu^-$ via a virtual photon.

Applying the Feynman rules to the tree–level diagram below, the invariant amplitude factors into the electron current, the photon propagator, and the muon current. The Feynman rules directly give $i\mathcal{M}$:

$$\begin{aligned} i\mathcal{M} &= \bar{v}^{s'}(p')(-ie\gamma^\mu)u^s(p)\left(\frac{-ig_{\mu\nu}}{q^2}\right)\bar{u}^r(k)(-ie\gamma^\nu)v^{r'}(k') \\ &= \frac{ie^2}{q^2}\left(\bar{v}^{s'}(p')\gamma^\mu u^s(p)\right)\left(\bar{u}^r(k)\gamma_\mu v^{r'}(k')\right), \end{aligned} \tag{7}$$

so the invariant amplitude itself is

$$\mathcal{M} = \frac{e^2}{q^2}\left(\bar{v}^{s'}(p')\gamma^\mu u^s(p)\right)\left(\bar{u}^r(k)\gamma_\mu v^{r'}(k')\right). \tag{8}$$

Here $q = p + p' = k + k'$. Choosing the Weyl spinors for specific helicities (R/L = right/left–handed) and evaluating the bilinears $\bar{v}\gamma^\mu u$ yields the four helicity amplitudes. By way of illustration, for $RL \to RL$:

$$\mathcal{M}(RL \to RL) \sim \langle\mu^+\mu^-|H_I|\gamma\rangle^\mu\langle\gamma|H_I|e^+e^-\rangle_\mu = -e^2(1+\cos\theta) \tag{9}$$

The remaining three follow by the same substitution:

$$\mathcal{M}(LR \to LR) = -e^2(1+\cos\theta) \tag{10}$$

$$\mathcal{M}(LR \to RL) = -e^2(1-\cos\theta) \tag{11}$$

$$\mathcal{M}(RL \to LR) = -e^2(1-\cos\theta) \tag{12}$$

Average for square of the 4 conditions:

$$|\mathcal{M}|^2 = e^4(1 + \cos^2\theta) \tag{13}$$

Fine-structure constant:

$$\alpha = \frac{e^2}{4\pi} \approx \frac{1}{137} \tag{14}$$

E_{cm} is energy of the system in center of mass frame.
Differential cross section:

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 E_{cm}^2} \cdot |\mathcal{M}|^2 = \frac{\alpha^2}{4E_{cm}^2}(1 + \cos^2\theta) \tag{15}$$

Total cross section,

$$\sigma = \frac{4\pi\alpha^2}{3E_{cm}^2} \tag{16}$$

1.3 Action Principle

The action is the spacetime integral of the Lagrangian density.

$$S = \int \mathrm{d}^4x\, \mathcal{L}(\phi, \partial_\mu\phi) \tag{17}$$

Under a coordinate and field variation,

$$\delta\mathcal{L} = \mathcal{L}'(x') - \mathcal{L}(x), \qquad \delta(\,\mathrm{d}^4x) = \,\mathrm{d}^4x\,(\partial_\mu \delta x^\mu) \tag{18}$$

Varying the action and dropping boundary terms,

$$\begin{aligned} 0 = \delta S = \int \mathrm{d}^4x \Bigg\{ &\left[\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right] \bar{\delta}\phi \\ &+\partial_\mu \left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \cdot \bar{\delta}\phi + \mathcal{L}\,\delta x^\mu\right]\Bigg\} \end{aligned} \tag{19}$$

The Lagrangian and the field variation split as follows.

$$L(t) = \int \mathrm{d}^3x\, \mathcal{L}(\phi, \partial_\mu\phi), \qquad \delta\phi = \bar{\delta}\phi + \partial_\mu\phi\, \delta x^\mu \tag{20}$$

1.3.1 Euler–Lagrange Equation

Stationarity of the action gives the field equation.

$$\partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} - \frac{\partial\mathcal{L}}{\partial\phi} = 0 \tag{21}$$

1.3.2 Hamiltonian

Canonical momentum and Hamiltonian density.

$$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}}, \qquad \mathcal{H} = \pi\dot{\phi} - \mathcal{L}, \qquad H = \int \mathrm{d}^3x\, \mathcal{H} \tag{22}$$

Canonical equations of motion.

$$\dot{\pi} = -\frac{\partial \mathcal{H}}{\partial \phi}, \qquad \dot{\phi} = \frac{\partial \mathcal{H}}{\partial \pi} \tag{23}$$

1.3.3 Noether Theorem

Every continuous symmetry yields a conserved current. (Convention: $\bar{\delta}\phi$ is the change at a fixed spacetime point, δx^μ the coordinate shift; overall sign depends on this choice.)

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\, \bar{\delta}\phi + \mathcal{L}\, \delta x^\mu, \qquad \partial_\mu j^\mu = 0 \tag{24}$$

Divergence theorem converts local conservation into a boundary statement.

$$0 = \int \mathrm{d}^3x\, \partial_\mu j^\mu = \int \mathrm{d}^3x\, \frac{\partial j^0}{\partial t} + \int \mathrm{d}^3x\, \nabla \cdot \mathbf{j} \tag{25}$$

The conserved charge, time independent by Gauss's law.

$$Q = \int \mathrm{d}^3x\, j^0, \qquad \frac{dQ}{dt} = -\int \mathrm{d}^3x\, (\nabla \cdot \mathbf{j}) = -\int^{\infty} \mathbf{j} \cdot\, \mathrm{d}\sigma = 0 \tag{26}$$

The formalism above applies to any field; we apply it first to the scalar field in Chapter 2, then reuse it for spinor and vector fields.

2 Scalar Field

We begin with the simplest field: the real scalar field, which describes spin–0 particles. Throughout, $E_p = \sqrt{\mathbf{p}^2 + m^2}$ and $p^\mu = (E_p, \mathbf{p})$.

2.1 Klein–Gordon Equation

The free scalar Lagrangian.

$$\mathcal{L} = \frac{1}{2}\left(\partial_\mu\phi\,\partial^\mu\phi - m^2\phi^2\right) \tag{27}$$

Canonical momentum and Hamiltonian density.

$$\pi = \frac{\partial\mathcal{L}}{\partial\dot{\phi}} = \dot{\phi}, \qquad \mathcal{H} = \frac{1}{2}\left[\pi^2 + (\nabla\phi)^2 + m^2\phi^2\right] \tag{28}$$

The Klein–Gordon equation and the d'Alembertian.

$$(\partial^2 + m^2)\phi = 0, \qquad \partial^2 = \partial_\mu\partial^\mu = \partial_0^2 - \nabla^2 \tag{29}$$

2.1.1 Plane Wave Solutions

Mode expansion in the standard relativistic normalization.

$$\phi(x) = \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\left(a_p e^{-ip\cdot x} + a_p^\dagger e^{ip\cdot x}\right) \tag{30}$$

Klein–Gordon inner product (orthogonality).

$$\int \mathrm{d}^3x\, f_p^*(x)\, i\overleftrightarrow{\partial}_0\, f_{p'}(x) = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{p}'), \qquad f_p(x) = \frac{e^{-ip\cdot x}}{\sqrt{2E_p}}, \quad A\overleftrightarrow{\partial}_\mu B = A\partial_\mu B - (\partial_\mu A)B \tag{31}$$

Relativistic phase–space measure identity.

$$\int \frac{\mathrm{d}^3p}{2E_p} = \int \mathrm{d}^4p\,\delta(p^2 - m^2)\,\theta(p^0) \tag{32}$$

2.2 Canonical Quantization

Equal–time canonical commutation relations.

$$[\phi(\mathbf{x},t),\phi(\mathbf{x}',t)] = 0, \qquad [\pi(\mathbf{x},t),\pi(\mathbf{x}',t)] = 0, \qquad [\phi(\mathbf{x},t),\pi(\mathbf{x}',t)] = i\delta^{(3)}(\mathbf{x}-\mathbf{x}') \tag{33}$$

Creation and annihilation operators satisfy harmonic oscillator algebra.

$$[a_p, a_{p'}] = 0, \qquad [a_p^\dagger, a_{p'}^\dagger] = 0, \qquad [a_p, a_{p'}^\dagger] = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{p}') \tag{34}$$

2.2.1 Box Normalization

Replace continuum integrals by discrete sums in a box $V = L^3$:

$$\int \frac{\mathrm{d}^3 p}{(2\pi)^3} \to \sum_{\mathbf{p}} \frac{1}{V}, \qquad \delta^{(3)}(\mathbf{p} - \mathbf{p}') \to \frac{V}{(2\pi)^3} \delta_{\mathbf{p}\mathbf{p}'} \tag{35}$$

Mode expansion in finite volume.

$$\phi(x) = \sum_{\mathbf{p}} \frac{1}{\sqrt{2E_p V}} \left(a_p e^{-ip\cdot x} + a_p^\dagger e^{ip\cdot x} \right) \tag{36}$$

2.2.2 Particle Number Representation

In box normalization the commutator becomes discrete: $[a_p, a_{p'}^\dagger] = \delta_{pp'}$, which makes N_p a discrete number operator with integer eigenvalues.

Number operator and its eigenstates.

$$N_p = a_p^\dagger a_p, \qquad N_p \left| n_p \right\rangle = n_p \left| n_p \right\rangle \tag{37}$$

Raising and lowering on the number–state ladder.

$$a_p^\dagger \left| n_p \right\rangle = \sqrt{n_p + 1} \left| n_p + 1 \right\rangle, \qquad a_p \left| n_p \right\rangle = \sqrt{n_p} \left| n_p - 1 \right\rangle \tag{38}$$

Vacuum condition and normalized number states.

$$a_p \left| 0 \right\rangle = 0, \qquad \left| n_p \right\rangle = \frac{(a_p^\dagger)^{n_p}}{\sqrt{n_p!}} \left| 0 \right\rangle \tag{39}$$

2.2.3 Energy–Momentum Tensor

The symmetric Noether current for spacetime translations.

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial^\nu \phi - \mathcal{L} g^{\mu\nu}, \qquad \partial_\mu T^{\mu\nu} = 0 \tag{40}$$

Total four–momentum as the spatial integral of $T^{0\nu}$.

$$P^\nu = \int \mathrm{d}^3 x \, T^{0\nu}, \qquad T^{00} = \mathcal{H} = \frac{1}{2} \left[\pi^2 + (\nabla\phi)^2 + m^2 \phi^2 \right], \qquad P^0 = H = \int \mathrm{d}^3 x \, \mathcal{H} \tag{41}$$

Mode expansion of the four–momentum (in box normalization; continuum version follows by replacing $\sum_{\mathbf{p}} \to \int \frac{\mathrm{d}^3 p}{(2\pi)^3} V$).

$$P^\mu = \int \mathrm{d}^3 x \, T^{0\mu} = \sum_{\mathbf{p}} p^\mu \left(N_p + \frac{1}{2} \right) \tag{42}$$

2.2.4 Single–Particle Wave Function

One–particle wave function as the field's matrix element.

$$\psi_{\mathbf{p}} = \frac{1}{\sqrt{2E_p}} \langle \mathbf{p}|\psi\rangle = \langle 0|\, a_p\, |\psi\rangle\,, \qquad \psi(\mathbf{x},t) = \langle 0|\, \phi(x,t)\, |\psi\rangle = \int \frac{\mathrm{d}^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} e^{-ip\cdot x} \psi_{\mathbf{p}} \tag{43}$$

2.2.5 Zero–Point Energy

Vacuum energy summed over all modes, divergent.

$$E_0 = \frac{V}{(2\pi)^3} \int \mathrm{d}^3p \frac{1}{2} E_p = \frac{V}{16\pi^3} \int_0^{K_c} 4\pi p^2 \sqrt{p^2+m^2}\, \mathrm{d}p \sim K_c^4 \tag{44}$$

This diverges as K_c^4 (ultraviolet divergence). It is removed by normal ordering: $H \to H - E_0 = \sum_{\mathbf{p}} N_p E_p$.

2.2.6 Fock Space

Vacuum and relativistically normalized one–particle states.

$$|\mathbf{p}\rangle = \sqrt{2E_p}\, a_p^\dagger\, |0\rangle\,, \qquad \langle \mathbf{p}|\mathbf{q}\rangle = 2E_p (2\pi)^3 \delta^{(3)}(\mathbf{p}-\mathbf{q}) \tag{45}$$

Two–particle states are symmetric (bosons). The number operator is $N = \int \frac{\mathrm{d}^3p}{(2\pi)^3} a_p^\dagger a_p$.

2.3 Complex Scalar Field

Lagrangian for a charged (complex) scalar.

$$\mathcal{L} = \partial_\mu \phi^\dagger\, \partial^\mu \phi - m^2 \phi^\dagger \phi \tag{46}$$

U(1) symmetry gives the conserved current and charge.

$$j^\mu = i(\phi^\dagger \partial^\mu \phi - \phi \partial^\mu \phi^\dagger), \qquad Q = \int \mathrm{d}^3x\, j^0 \tag{47}$$

Mode expansion with particle and antiparticle operators.

$$\phi(x) = \int \frac{\mathrm{d}^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \left(a_p e^{-ip\cdot x} + b_p^\dagger e^{ip\cdot x} \right) \tag{48}$$

Hamiltonian and conserved charge in modes (already normal ordered; zero-point energy subtracted).

$$H = \int \frac{\mathrm{d}^3p}{(2\pi)^3} E_p (a_p^\dagger a_p + b_p^\dagger b_p), \qquad Q = \int \frac{\mathrm{d}^3p}{(2\pi)^3} (a_p^\dagger a_p - b_p^\dagger b_p) = N_a - N_b \tag{49}$$

where $N_a = \int \frac{\mathrm{d}^3p}{(2\pi)^3} a_p^\dagger a_p$ and $N_b = \int \frac{\mathrm{d}^3p}{(2\pi)^3} b_p^\dagger b_p$.

The negative–frequency solutions correspond to antiparticles. For a real scalar field the particle is its own antiparticle; for a complex scalar field they form particle–antiparticle pairs.

2.4 Causality and Feynman Propagator

2.4.1 Causality and Spacelike Separation

Spacelike separation criterion.

$$(x-y)^2 = (t_1-t_2)^2 - |\mathbf{x}_1-\mathbf{x}_2|^2 \tag{50}$$

For spacelike separations $(x-y)^2 < 0$, signals cannot propagate; for timelike or lightlike separations $(x-y)^2 \geq 0$, signals can propagate.

2.4.2 Vacuum Two–Point Function

Wightman function, positive–frequency part only.

$$D(x-y) = \langle 0|\phi(x)\phi(y)|0\rangle = \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{2E_p}e^{-ip\cdot(x-y)} \tag{51}$$

Commutator vanishes outside the light cone by cancellation.

$$[\phi(x),\phi(y)] = D(x-y) - D(y-x) \tag{52}$$

2.4.3 Feynman Propagator

Time–ordered product definition.

$$T\phi(x)\phi(y) = \theta(x^0-y^0)\phi(x)\phi(y) + \theta(y^0-x^0)\phi(y)\phi(x) \tag{53}$$

The Feynman propagator in momentum space.

$$D_F(x-y) = \langle 0|T\phi(x)\phi(y)|0\rangle = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\frac{i}{p^2-m^2+i\epsilon}e^{-ip\cdot(x-y)} \tag{54}$$

Green's function for the Klein–Gordon operator.

$$(\partial^2+m^2)D_F(x-y) = -i\delta^{(4)}(x-y) \tag{55}$$

The $i\epsilon$ prescription shifts the poles off the real axis to make the integral well–defined; one takes $i\epsilon \to 0$ at the end.

3 Spinor Field

We now turn to the Dirac field, which describes spin–$\frac{1}{2}$ particles such as the electron.

3.1 Dirac Equation and Gamma Matrices

Factorize the KG operator into first–order derivatives.

$$0 = (\eta^{\mu\nu}\partial_\mu\partial_\nu + m^2)\psi = [-\eta^{\mu\nu}(i\partial_\mu)(i\partial_\nu) + m^2]\psi \tag{56}$$

Introducing the Clifford algebra $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$,

$$0 = -(i\gamma^\mu\partial_\mu + m)(i\gamma^\nu\partial_\nu - m)\psi \tag{57}$$

The Dirac equation in slashed notation.

$$\text{Def:} \quad \not{\partial} = \gamma^\mu\partial_\mu, \qquad (i\not{\partial} - m)\psi = 0 \tag{58}$$

3.1.1 Dirac Matrices in Chiral Representation

Pauli matrices (building blocks of γ^i).

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{59}$$

Gamma matrices in the chiral (Weyl) representation.

$$\gamma^0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} \tag{60}$$

Chiral gamma matrix.

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -I_2 & 0 \\ 0 & I_2 \end{pmatrix} \tag{61}$$

Chiral projection operators.

$$P_L = \frac{1-\gamma^5}{2} = \begin{pmatrix} I_2 & 0 \\ 0 & 0 \end{pmatrix}, \qquad P_R = \frac{1+\gamma^5}{2} = \begin{pmatrix} 0 & 0 \\ 0 & I_2 \end{pmatrix} \tag{62}$$

Properties: $\gamma^{0\dagger} = \gamma^0$, $\gamma^{i\dagger} = -\gamma^i$, and $(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0$.

3.2 Plane Wave Solutions

Adjoint spinor and Dirac Lagrangian.

$$\bar{\psi} = \psi^\dagger\gamma^0, \qquad \mathcal{L} = \bar{\psi}(i\not{\partial} - m)\psi \tag{63}$$

Positive–frequency (particle) solutions.

$$\psi(x) = u(p)e^{-ip\cdot x}, \qquad (\not{p} - m)u(p) = 0 \tag{64}$$

Negative–frequency (antiparticle) solutions.

$$\psi(x) = v(p)e^{ip\cdot x}, \qquad (\not{p} + m)v(p) = 0 \tag{65}$$

Spinor normalization conventions (spinor indices contracted).

$$u^\dagger u = v^\dagger v = 2E_p, \qquad \bar{u}u = -\bar{v}v = 2m \tag{66}$$

Completeness relations over spin states.

$$\sum_s u^s(p)\bar{u}^s(p) = \not{p} + m, \qquad \sum_s v^s(p)\bar{v}^s(p) = \not{p} - m \tag{67}$$

Lorentz generator in the spinor representation.

$$S^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu] \tag{68}$$

3.3 Quantization and Spin–Statistics

Canonical momentum and mode expansion.

$$\psi(x) = \int \frac{\mathrm{d}^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \sum_{s=1,2} \left[a_p^s u^s(p) e^{-ip\cdot x} + b_p^{s\dagger} v^s(p) e^{ip\cdot x}\right] \tag{69}$$

Hamiltonian in modes (already normal ordered; the fermion vacuum energy has been subtracted).

$$H = \int \frac{\mathrm{d}^3p}{(2\pi)^3} E_p \sum_s (a_p^{s\dagger} a_p^s + b_p^{s\dagger} b_p^s) \tag{70}$$

Canonical anticommutation relations.

$$\{a_p^r, a_q^{s\dagger}\} = \{b_p^r, b_q^{s\dagger}\} = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q})\delta^{rs} \tag{71}$$

Equal–time anticommutators for the field itself.

$$\{\psi_a(x), \psi_b^\dagger(y)\} = \delta^{(3)}(\mathbf{x} - \mathbf{y})\delta_{ab}, \qquad \{\psi, \psi\} = \{\psi^\dagger, \psi^\dagger\} = 0 \tag{72}$$

Scalar fields satisfy $[a_p^\dagger, a_q^\dagger] = 0$ (symmetric under exchange, bosons); spinor fields satisfy $\{a_p^\dagger, a_q^\dagger\} = 0$ (antisymmetric under exchange, fermions).

3.4 Fermion Propagator

Time–ordered product with fermionic minus sign.

$$T\psi(x)\bar{\psi}(y) = \theta(x^0 - y^0)\psi(x)\bar{\psi}(y) - \theta(y^0 - x^0)\bar{\psi}(y)\psi(x) \tag{73}$$

The Feynman propagator for the Dirac field.

$$S_F(x - y) = \langle 0|T\psi(x)\bar{\psi}(y)|0\rangle = \int \frac{\mathrm{d}^4p}{(2\pi)^4} \frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon} e^{-ip\cdot(x-y)} \tag{74}$$

Green's function for the Dirac operator.

$$(i\ \not{\partial} - m)S_F(x - y) = i\delta^{(4)}(x - y) \tag{75}$$

4 Vector Field

The vector field describes spin–1 particles. We first treat the massless case (the electromagnetic field), then note the massive (Proca) case for comparison.

4.1 Maxwell Field and Gauge Invariance

Field strength tensor (upper–index convention, Peskin & Schroeder).

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu = \begin{pmatrix} 0 & E^1 & E^2 & E^3 \\ -E^1 & 0 & B^3 & -B^2 \\ -E^2 & -B^3 & 0 & B^1 \\ -E^3 & B^2 & -B^1 & 0 \end{pmatrix} \tag{76}$$

Maxwell Lagrangian.

$$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} = \frac{1}{2}(E^2 - B^2) \tag{77}$$

Gauge redundancy $A_\mu \to A_\mu + \partial_\mu \Lambda$. In Coulomb gauge $\nabla \cdot \mathbf{A} = 0$. Source–free Maxwell equations.

$$\partial_\mu F^{\mu\nu} = 0 \tag{78}$$

4.2 Proca Equation: Massive Vector Field

Lagrangian for a massive vector field.

$$\mathcal{L}_{\text{Proca}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} m^2 A_\mu A^\mu. \tag{79}$$

The Proca equation of motion.

$$\partial_\mu F^{\mu\nu} + m^2 A^\nu = 0. \tag{80}$$

Taking the divergence yields the Lorenz condition automatically.

$$m^2 \, \partial_\nu A^\nu = 0 \quad \Rightarrow \quad \partial_\mu A^\mu = 0 \quad (m \neq 0). \tag{81}$$

Thus for $m \neq 0$ the Lorenz condition holds automatically and there is no gauge freedom; a massive vector has **three** physical polarization degrees of freedom. For $m = 0$ (photon), gauge invariance remains and only **two** transverse polarizations are physical.

4.2.1 Polarization Vectors

Massive vector polarization basis.

$$p \cdot \epsilon_\lambda = 0, \qquad \epsilon_\lambda \cdot \epsilon_{\lambda'} = -\delta_{\lambda\lambda'}. \tag{82}$$

Completeness relation.

$$\sum_{\lambda=1}^{3} \epsilon_\lambda^\mu(p) \epsilon_\lambda^{\nu *}(p) = -g^{\mu\nu} + \frac{p^\mu p^\nu}{m^2}. \tag{83}$$

For a massless photon, only two transverse polarizations $\lambda = 1, 2$ remain physical; the longitudinal and scalar modes are gauge artifacts.

4.3 Mode Expansion

Photon mode expansion with two physical transverse polarizations (Coulomb gauge). In covariant quantization one includes all four polarizations and removes the unphysical longitudinal/scalar modes via the Gupta–Bleuler / BRST condition; the result is equivalent to the two–polarization expansion below.

$$A^\mu(x) = \int \frac{\mathrm{d}^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \sum_{\lambda=1,2} \left[a_p^\lambda \epsilon^{\lambda\mu}(p) e^{-ip\cdot x} + a_p^{\lambda\dagger} \epsilon^{\lambda *\mu}(p) e^{ip\cdot x} \right] \tag{84}$$

Photon commutation relations.

$$[a_p^\lambda, a_{p'}^{\lambda'\dagger}] = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') \delta^{\lambda\lambda'} \tag{85}$$

4.4 Covariant Gauge and Propagator

Gauge–fixing term.

$$\mathcal{L}_{\mathrm{gf}} = -\frac{1}{2\xi} (\partial^\mu A_\mu)^2 \tag{86}$$

The R_ξ gauge propagator (where q denotes the propagator momentum, distinct from external particle momentum p).

$$D_F^{\mu\nu}(q) = \frac{-i}{q^2 + i\epsilon} \left[g^{\mu\nu} - (1-\xi) \frac{q^\mu q^\nu}{q^2 + i\epsilon} \right] \tag{87}$$

In Feynman gauge $\xi = 1$: $D_F^{\mu\nu} = -ig^{\mu\nu}/(q^2 + i\epsilon)$. The Ward identity $q^\mu M_\mu = 0$ ensures that the unphysical longitudinal/scalar modes cancel in any physical amplitude.

5 Interacting Field

We now turn to interactions. The free–field propagators derived in Chapters 2–4 are the building blocks; interactions introduce vertices and connect them.

5.1 Interaction Lagrangians

5.1.1 ϕ^4 Theory

The ϕ^4 interaction and its equation of motion. (Here λ is the coupling constant, not the polarization index used in Chapter 4.)

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{4!}\phi^4, \qquad (\partial^2 + m^2)\phi = -\frac{\lambda}{3!}\phi^3 \tag{88}$$

5.1.2 QED

QED interaction and covariant derivative.

$$\mathcal{L}_{\text{int}} = -e\bar{\psi}\gamma^\mu\psi A_\mu, \qquad D_\mu = \partial_\mu + ieA_\mu(x) \tag{89}$$

Full QED Lagrangian.

$$\mathcal{L}_{\text{QED}} = \bar{\psi}(i\not{D} - m)\psi - \frac{1}{4}(F_{\mu\nu})^2 \tag{90}$$

Equations of motion: $(i\not{D} - m)\psi = 0$, and $\partial_\mu F^{\mu\nu} = ej^\nu$ with $j^\nu = \bar{\psi}\gamma^\nu\psi$.

5.2 Interaction Picture

Three pictures compared (assuming operators have no explicit time dependence).

Picture	State evolution driven by	Operator evolution driven by
Schrödinger	H (full Hamiltonian)	none (O_s time–independent)
Heisenberg	none (state time–independent)	H (full Hamiltonian)
Interaction	H_I (interaction Hamiltonian)	H_0 (free Hamiltonian)

Interaction Hamiltonian.

$$H_I = e^{iH_0 t} H_{\text{int}} e^{-iH_0 t}, \qquad H_{\text{int}} = -\int \mathrm{d}^3x \mathcal{L}_{\text{int}} \tag{91}$$

5.3 Dyson Series and Gell–Mann–Low Formula

Schrödinger equation in the interaction picture.

$$i\frac{\partial \left|\psi(t)\right\rangle_I}{\partial t} = H_I(t)\left|\psi(t)\right\rangle_I, \qquad \left|\psi(t)\right\rangle_I = U_I(t,t_0)\left|\psi(t_0)\right\rangle_I \tag{92}$$

Dyson series (time–ordered exponential).

$$U_I(t,t_0) = \mathcal{T}\left\{\exp\left[-i\int_{t_0}^{t}\mathrm{d}t'\, H_I(t')\right]\right\} \tag{93}$$

$|0\rangle$: free vacuum (ground state of H_0); $|\Omega\rangle$: interacting vacuum (ground state of H).

Gell–Mann–Low formula.

$$\langle\Omega|T\{\phi(x_1)\cdots\phi(x_n)\}|\Omega\rangle = \lim_{T\to\infty(1-i\epsilon)} \frac{\langle 0|T\{\phi_I(x_1)\cdots\phi_I(x_n)\, e^{-i\int_{-T}^{T}\mathrm{d}tH_I(t)}\}|0\rangle}{\langle 0|T\{e^{-i\int_{-T}^{T}\mathrm{d}tH_I(t)}\}|0\rangle} \tag{94}$$

5.4 The S–Matrix

The S–matrix element is the overlap between asymptotically free initial and final states.

$$S_{fi} = \langle f|S|i\rangle. \tag{95}$$

Taking $t_0 \to -\infty$ and $t \to +\infty$ with adiabatic switching of the interaction, the S–matrix is the limit of the Dyson series of Section 5.3:

$$S = U_I(\infty,-\infty) = \mathcal{T}\exp\left[-i\int_{-\infty}^{\infty}\mathrm{d}t\, H_I(t)\right], \tag{96}$$

where $H_I(t)$ is the interaction Hamiltonian in the interaction picture and $\mathcal{T}$ denotes time ordering.

5.4.1 T–Matrix Decomposition

Separate the identity (no–scattering) part from the genuine transition part.

$$S_{fi} = \delta_{fi} + iT_{fi}. \tag{97}$$

The T–matrix factorizes out overall four–momentum conservation, leaving the Lorentz–invariant amplitude $\mathcal{M}_{fi}$:

$$T_{fi} = (2\pi)^4\delta^{(4)}(p_f - p_i)\,\mathcal{M}_{fi}, \tag{98}$$

so that

$$S_{fi} = \delta_{fi} + i(2\pi)^4\delta^{(4)}(p_f - p_i)\,\mathcal{M}_{fi}. \tag{99}$$

For $i \neq f$, the δ_{fi} term vanishes, leaving the transition amplitude. The invariant amplitude $\mathcal{M}_{fi}$ is obtained from the Feynman rules of Section 5.6.

5.4.2 Transition Probability: Discrete vs. Continuum

For discrete, finite–volume normalized final states, the transition probability is

$$P_{i\to f} = |S_{fi}|^2 \qquad (i \neq f). \tag{100}$$

For continuum momentum states, S_{fi} contains $(2\pi)^4\delta^{(4)}(p_f - p_i)$, so $|S_{fi}|^2$ contains $[\delta^{(4)}]^2$ and is not an ordinary probability. It must be regularized:

$$\left|(2\pi)^4\delta^{(4)}(p_f - p_i)\right|^2 \;\longrightarrow\; VT\,(2\pi)^4\delta^{(4)}(p_f - p_i), \tag{101}$$

where V is the spatial volume and T the time interval. In practice, for continuum states one uses the transition rate $\mathrm{d}\Gamma$, the probability element $\mathrm{d}P$, or the cross section $\mathrm{d}\sigma$ (defined in Section 5.7), integrated over phase space.

5.5 Wick's Theorem

Normal ordering N: move all $a^\dagger$ to the left and all a to the right. Split the field into positive/negative frequency parts.

$$\phi_I(x) = \phi_I^+(x) + \phi_I^-(x), \qquad \phi_I^+(x) = \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \frac{a_p}{\sqrt{2E_p}} e^{-ip\cdot x} \tag{102}$$

Contraction equals the time–ordered product's vacuum expectation.

$$D_F(x-y) = \underbrace{\phi(x)\phi(y)} = \langle 0|T\phi(x)\phi(y)|0\rangle \tag{103}$$

Wick's theorem (schematic; the actual expansion sums over all possible complete sets of contractions). For fermions, each exchange of fermionic fields picks up a minus sign.

$$T\{\phi(x_1)\cdots\phi(x_m)\} = N\left\{\phi(x_1)\cdots\phi(x_m) + \sum_{\text{all complete contraction sets}} \underbrace{\phi(x_i)\phi(x_j)}\ \phi(x_k)\cdots\right\} \tag{104}$$

5.6 Feynman Rules

Wick's theorem (Section 5.5) reduces the time–ordered products in the Gell–Mann–Low formula to contractions, i.e. propagators, joined at interaction vertices. The resulting Feynman rules are:

Position–space (ϕ^4): propagator $D_F(x-y)$; vertex $(-i\lambda)\int \mathrm{d}^4 z$; external leg 1; divide by symmetry factor.

Momentum–space (ϕ^4):

- Propagator: $\dfrac{i}{p^2 - m^2 + i\epsilon}$
- Vertex: $-i\lambda$
- External leg: 1 (scalar; replaced by spinors/polarizations for other fields)
- Momentum conservation: $(2\pi)^4\delta^{(4)}(\sum p)$
- Loop integral: $\int \frac{\mathrm{d}^4 p}{(2\pi)^4}$
- Divide by symmetry factor.

5.7 Cross Sections and Decay Rates

With the Feynman rules of Section 5.6, we can compute $\mathcal{M}$ for any process and extract physical observables.

5.7.1 Differential Cross Section

Starting from the transition probability $P_{i\to f} = |S_{fi}|^2$ of Section 5.4.2, dividing by the flux factor and the time interval T, and using the regularization $|(2\pi)^4\delta^{(4)}|^2 \to VT(2\pi)^4\delta^{(4)}$, one obtains the differential cross section for $2 \to n$ scattering:

$$\mathrm{d}\sigma = \frac{1}{4E_1E_2|v_1 - v_2|}|\mathcal{M}|^2\,(2\pi)^4\delta^{(4)}(p_f - p_i)\;\mathrm{d}\Phi_n, \tag{105}$$

where the final–state Lorentz–invariant phase space is

$$\mathrm{d}\Phi_n = \prod_f \frac{\mathrm{d}^3p_f}{(2\pi)^3\,2E_f}. \tag{106}$$

For collinear beams, the flux factor can also be written in Lorentz–invariant form:

$$4E_1E_2|v_1 - v_2| = 4\sqrt{(p_1\cdot p_2)^2 - m_1^2m_2^2}. \tag{107}$$

If initial spins or polarizations are not measured, average over them:

$$|\mathcal{M}|^2 \;\longrightarrow\; \frac{1}{(2s_1+1)(2s_2+1)}\sum_{\text{spins}}|\mathcal{M}|^2. \tag{108}$$

For identical particles in the final state, include the appropriate symmetry factor (e.g. $1/n!$ for n identical particles).

5.7.2 Total Cross Section and Decay Rate

The total cross section is the integral over final–state phase space.

$$\sigma = \int \mathrm{d}\sigma. \tag{109}$$

The decay rate Γ is the probability per unit time for an unstable particle to decay:

$$\Gamma = \frac{\text{decays per unit time}}{\text{number of } A \text{ particles}}. \tag{110}$$

5.7.3 Center–of–Mass Frame for $2 \to 2$ Scattering

For $2 \to 2$ scattering in the CM frame,

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\text{cm}} = \frac{1}{64\pi^2 s}\frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}\,|\mathcal{M}|^2, \qquad s = E_{\text{cm}}^2, \tag{111}$$

where $E_{\text{cm}} = \sqrt{s}$ is the **total** center–of–mass energy.

For massless initial and final states (or equal incoming/outgoing masses), $|\mathbf{p}_f|/|\mathbf{p}_i| \to 1$, so

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\text{cm}} = \frac{|\mathcal{M}|^2}{64\pi^2 s} = \frac{|\mathcal{M}|^2}{64\pi^2 E_{\text{cm}}^2}. \tag{112}$$

This is the formula used in the $e^+e^- \to \mu^+\mu^-$ calculation in Chapter 1.

5.7.4 Breit–Wigner Resonance

Near a resonance energy E_R, the cross section follows the Breit–Wigner form.

$$f(E) \propto \frac{1}{E - E_R + i\Gamma/2}, \qquad \sigma \propto |f(E)|^2 \propto \frac{1}{(E - E_R)^2 + \Gamma^2/4} \tag{113}$$

5.8 Path Integral Formulation

The time–evolution operator generates the propagator kernel.

$$|\psi(t)\rangle = U(t, t_0)\,|\psi(t_0)\rangle\,, \qquad U(t, t_0) = e^{-iH(t-t_0)} \tag{114}$$

Propagator kernel as a matrix element.

$$\psi(q, t) = \langle q|\psi(t)\rangle = \int \mathrm{d}q_0\, K(q, t; q_0, t_0)\, \psi(q_0, t_0). \tag{115}$$

Feynman's central insight: the kernel is a sum over all paths, weighted by e^{iS}.

$$K(q, t; q_0, t_0) = \langle q|U(t, t_0)|q_0\rangle = \mathcal{N} \int \mathcal{D}q\, e^{iS} \tag{116}$$

The action and path integral measure.

$$S = \int_{t_0}^{t} \mathrm{d}t\, L(q, \dot{q}), \qquad \int \mathcal{D}q = \lim_{\substack{n\to\infty \\ \Delta t\to 0}} \prod_{m=1}^{n-1} \int \mathrm{d}q(t_m) \tag{117}$$

Normalization factor for a free particle of mass m.

$$\mathcal{N} = \lim_{n\to\infty} \left(\frac{m}{2\pi i \Delta t}\right)^{n/2} \tag{118}$$

In field theory the same idea lifts to functional integrals over field configurations; the vacuum expectation values of time–ordered products are then generated by

$$\langle \Omega|T\phi(x_1)\cdots\phi(x_n)|\Omega\rangle = \frac{\int \mathcal{D}\phi\, \phi(x_1)\cdots\phi(x_n)\, e^{i\int \mathrm{d}^4x \mathcal{L}}}{\int \mathcal{D}\phi\, e^{i\int \mathrm{d}^4x \mathcal{L}}} \tag{119}$$

This provides an alternative derivation of the Feynman rules already obtained via Wick's theorem in Section 5.5 and used in Section 5.6.

6 Gauge Theory and Yang–Mills Fields

The guiding principle of the Standard Model is local gauge symmetry: the dynamics of all fundamental interactions is uniquely dictated by requiring invariance under a continuous local Lie group. This chapter follows Peskin & Schroeder Ch. 15–17.

6.1 General Framework

6.1.1 Lie Algebra Conventions

Hermitian generators T^a ($a = 1, \ldots, N$, where N is the dimension of the gauge group) satisfy the Lie algebra

$$[T^a, T^b] = i\, f^{abc}\, T^c, \tag{120}$$

where f^{abc} are the totally antisymmetric structure constants. They vanish for Abelian groups (e.g. U(1)) and are non–zero for non–Abelian groups (e.g. SU(N)). The fundamental–representation normalization (Peskin convention) is

$$\mathrm{tr}\left[T^a T^b\right] = \frac{1}{2}\delta^{ab}. \tag{121}$$

6.1.2 Covariant Derivative and Field Strength

The covariant derivative is constructed so that $D_\mu\psi(x)$ transforms covariantly under a local gauge transformation $\psi(x) \to U(x)\psi(x)$:

$$D_\mu = \partial_\mu - ig\, T^a A^a_\mu(x), \tag{122}$$

where g is the gauge coupling and A^a_μ are the gauge fields.
The field strength tensor is the commutator of covariant derivatives:

$$F_{\mu\nu} = \frac{i}{g}[D_\mu, D_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu], \tag{123}$$

or in components,

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g\, f^{abc} A^b_\mu A^c_\nu. \tag{124}$$

The last term is the self–interaction, absent in the Abelian case.

6.1.3 Yang–Mills Lagrangian and Equations of Motion

The pure gauge–field Lagrangian is Lorentz–invariant, gauge–invariant, and renormalizable:

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4} F^a_{\mu\nu} F^{a\mu\nu}. \tag{125}$$

Varying the action gives the Yang–Mills equation of motion:

$$D_\mu F^{a\mu\nu} = j^{a\nu}, \tag{126}$$

where $j^{a\nu}$ is the conserved matter current associated with the gauge group. The Bianchi identity follows from the definition of $F_{\mu\nu}$:

$$D_\mu F^a_{\nu\rho} + D_\nu F^a_{\rho\mu} + D_\rho F^a_{\mu\nu} = 0, \tag{127}$$

the non–Abelian generalization of the homogeneous Maxwell equations.

6.2 U(1): Quantum Electrodynamics

QED is the Abelian special case. The gauge group is U(1), with $f^{abc} = 0$ and a single generator (the electric charge Q).

- Gauge field: photon $A_\mu(x)$.
- Covariant derivative (for electrons, $Q = -1$): $D_\mu = \partial_\mu + ieA_\mu$.
- Field strength: $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ (no self–interaction).
- Full Lagrangian:

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m)\psi. \tag{128}$$

The photon carries no charge, the field equations are linear, and vacuum polarization screens charge (the coupling increases at high energy).

6.3 SU(2)$_L$: Weak Isospin

The weak interaction acts only on left–handed fermions, which form SU(2)$_L$ doublets.

- Structure constants: $f^{abc} = \varepsilon^{abc}$ (Levi–Civita symbol).
- Generators: $T^a = \sigma^a/2$, $a = 1, 2, 3$.
- Gauge fields: $W^1_\mu, W^2_\mu, W^3_\mu$.
- Covariant derivative:

$$D_\mu = \partial_\mu - ig\,\frac{\sigma^a}{2} W^a_\mu. \tag{129}$$

- Field strength:

$$W^a_{\mu\nu} = \partial_\mu W^a_\nu - \partial_\nu W^a_\mu + g\,\varepsilon^{abc} W^b_\mu W^c_\nu. \tag{130}$$

The Pauli matrices (already used in the Dirac representation, Ch. 3) are

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{131}$$

After spontaneous symmetry breaking via the Higgs mechanism, W^1_μ, W^2_μ mix into the charged $W^\pm$ bosons, and W^3_μ mixes with the hypercharge field into the Z boson and the photon.

6.4 $\mathrm{SU}(3)_c$: Quantum Chromodynamics

The strong interaction is described by SU(3) color gauge theory.

- Generators: $T^a = \lambda^a/2$, $a = 1, \ldots, 8$, where λ^a are the Gell–Mann matrices.
- Gauge fields: eight gluons $G^a_\mu(x)$.
- Covariant derivative:
$$D_\mu = \partial_\mu - i g_s \frac{\lambda^a}{2} G^a_\mu. \tag{132}$$
- Field strength:
$$G^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu + g_s\, f^{abc} G^b_\mu G^c_\nu. \tag{133}$$
- Full QCD Lagrangian:
$$\mathcal{L}_{\mathrm{QCD}} = -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu} + \sum_q \bar{q}_i (i\gamma^\mu D_\mu - m_q) q_i, \tag{134}$$
where i is the color index and q runs over quark flavors.

The eight Gell–Mann matrices are the 3×3 Hermitian generators of SU(3):

$$\lambda^1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda^2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \tag{135}$$

$$\lambda^4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \lambda^5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \tag{136}$$

$$\lambda^6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \lambda^7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \lambda^8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \tag{137}$$

Gluons carry color charge, so they interact among themselves (three– and four–gluon vertices). QCD exhibits asymptotic freedom (the coupling weakens at high energy, so quarks and gluons are nearly free) and color confinement (at low energy the coupling is strong and colored particles cannot exist in isolation).

6.5 $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$: Electroweak Unification

The electroweak theory is based on the product group $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, with three weak–isospin generators T^a and one weak–hypercharge generator Y.

- Gauge fields: W^a_μ $(a = 1, 2, 3)$ and B_μ.
- Covariant derivative:
$$D_\mu = \partial_\mu - i g \frac{\sigma^a}{2} W^a_\mu - i \frac{g'}{2} Y\, B_\mu. \tag{138}$$

After Higgs symmetry breaking, the physical gauge fields are linear combinations. The charged bosons are

$$W_\mu^\pm = \frac{1}{\sqrt{2}} \left(W_\mu^1 \mp i W_\mu^2 \right), \tag{139}$$

and the neutral gauge bosons mix through the weak mixing angle θ_W, with $\tan\theta_W = g'/g$:

$$\begin{cases} Z_\mu = \cos\theta_W \, W_\mu^3 - \sin\theta_W \, B_\mu, \\ A_\mu = \sin\theta_W \, W_\mu^3 + \cos\theta_W \, B_\mu. \end{cases} \tag{140}$$

Here A_μ is the massless photon and Z_μ is the massive neutral weak boson.

6.6 Summary

Gauge group	Interaction	# generators	Self–interaction	Key feature
U(1)	Electromagnetic	1	No	Linear, long–range, charge sc
$SU(2)_L$	Weak	3	Yes	Chiral, spontaneously broken, sh
$SU(3)_c$	Strong (QCD)	8	Yes	Asymptotic freedom, color con
$SU(2)_L \times U(1)_Y$	Electroweak	$3+1$	Yes	Higgs mixing, $W^\pm, Z, \gamma$